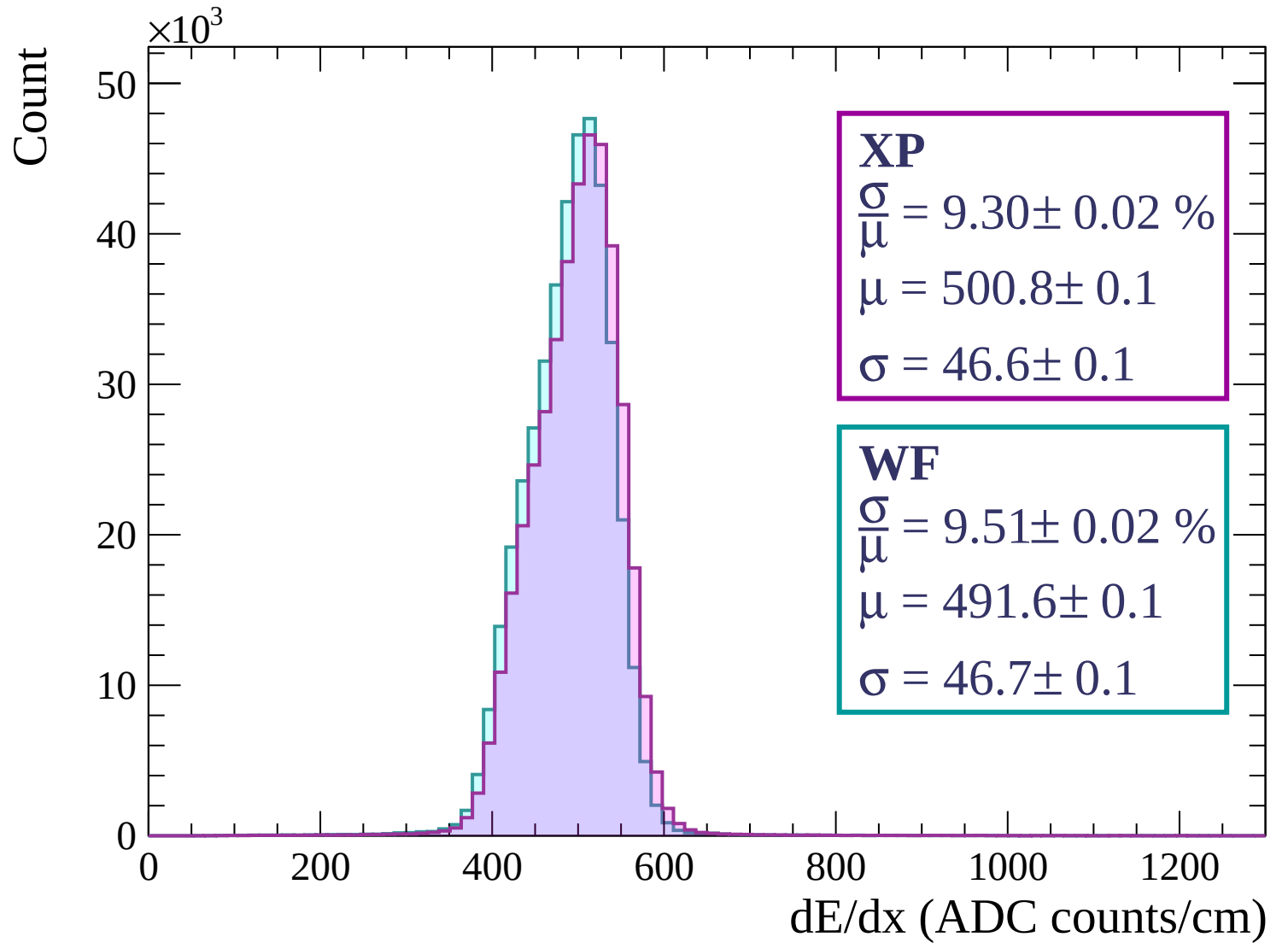
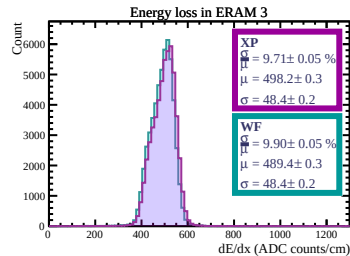
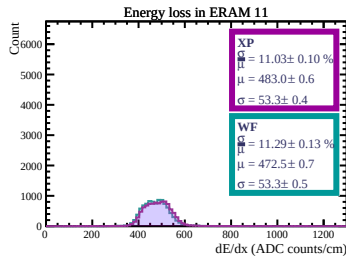
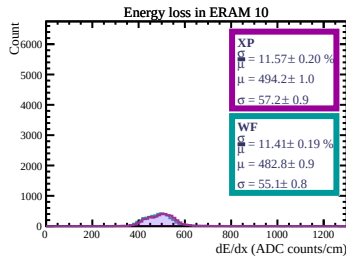
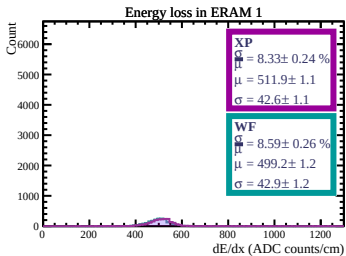
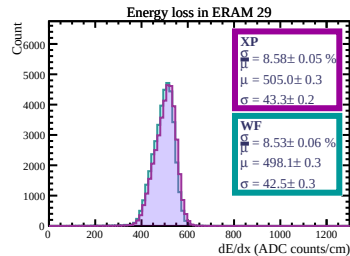
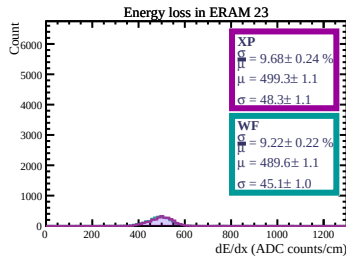
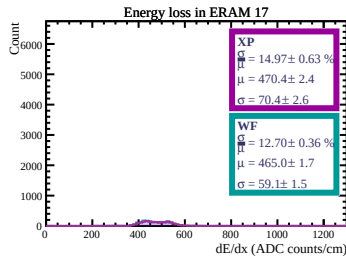
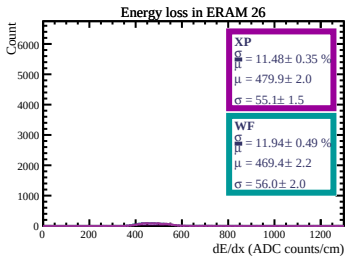
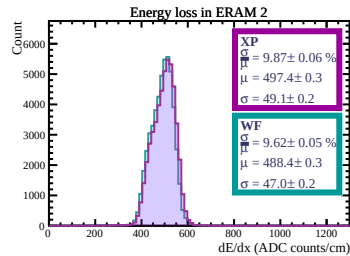
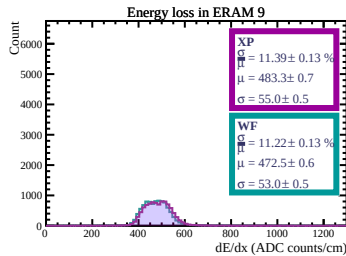
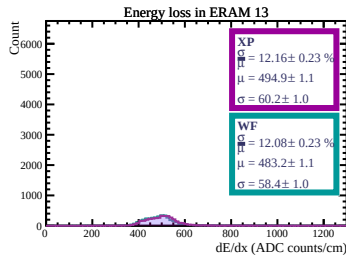
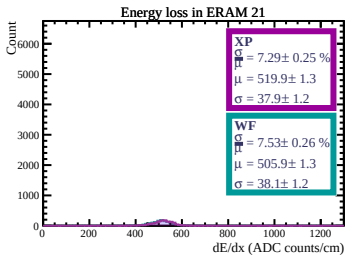
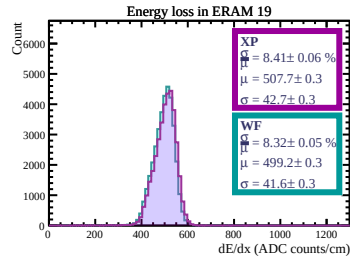
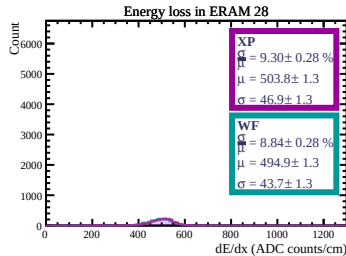
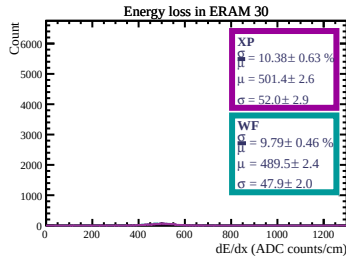
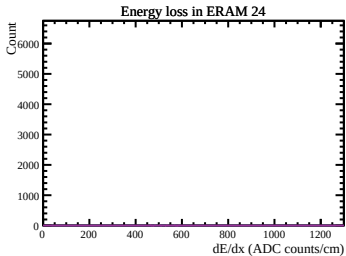
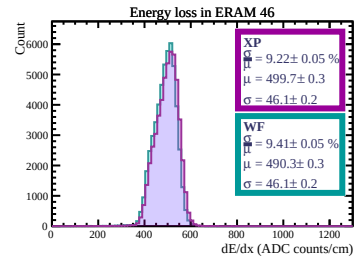
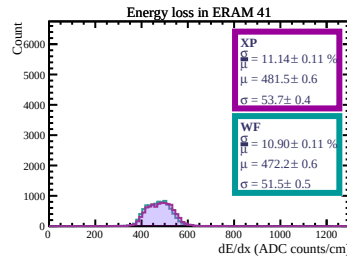
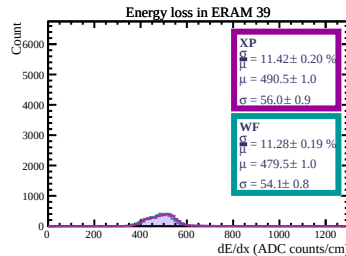
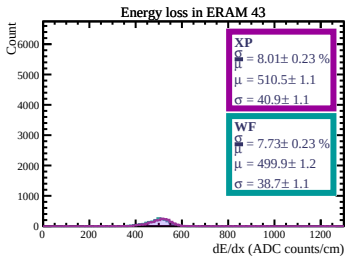
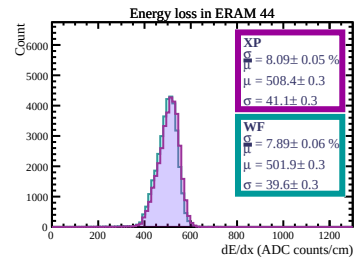
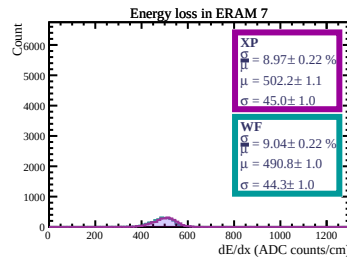
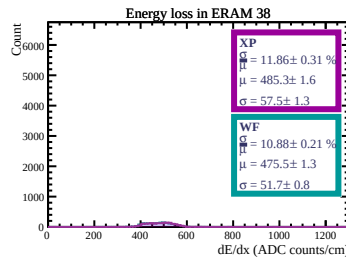
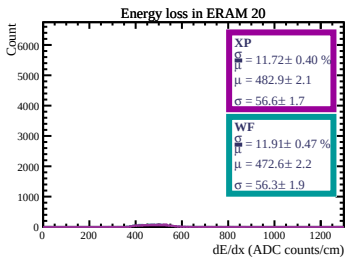
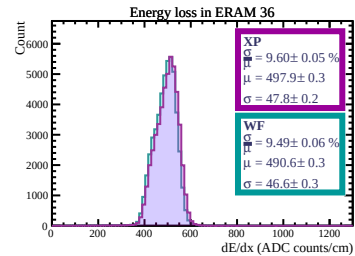
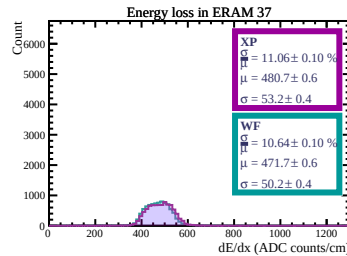
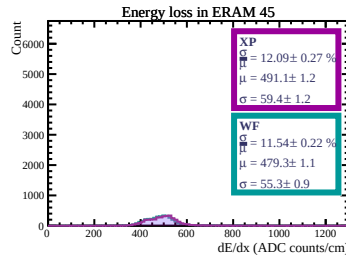
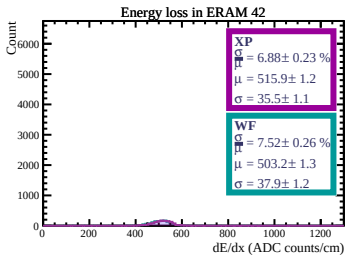
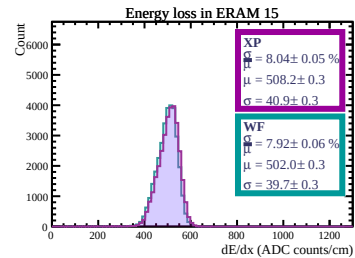
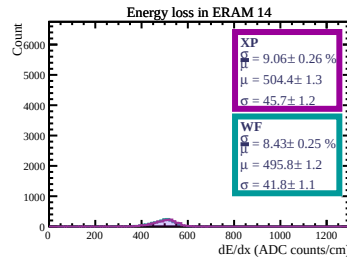
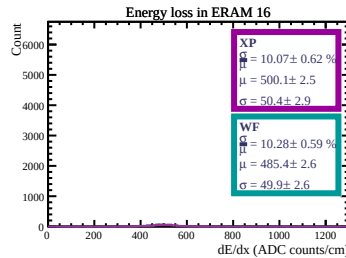
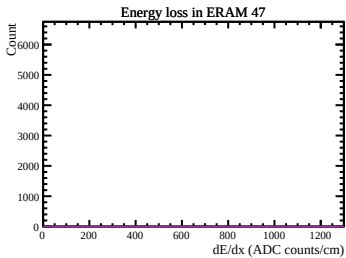
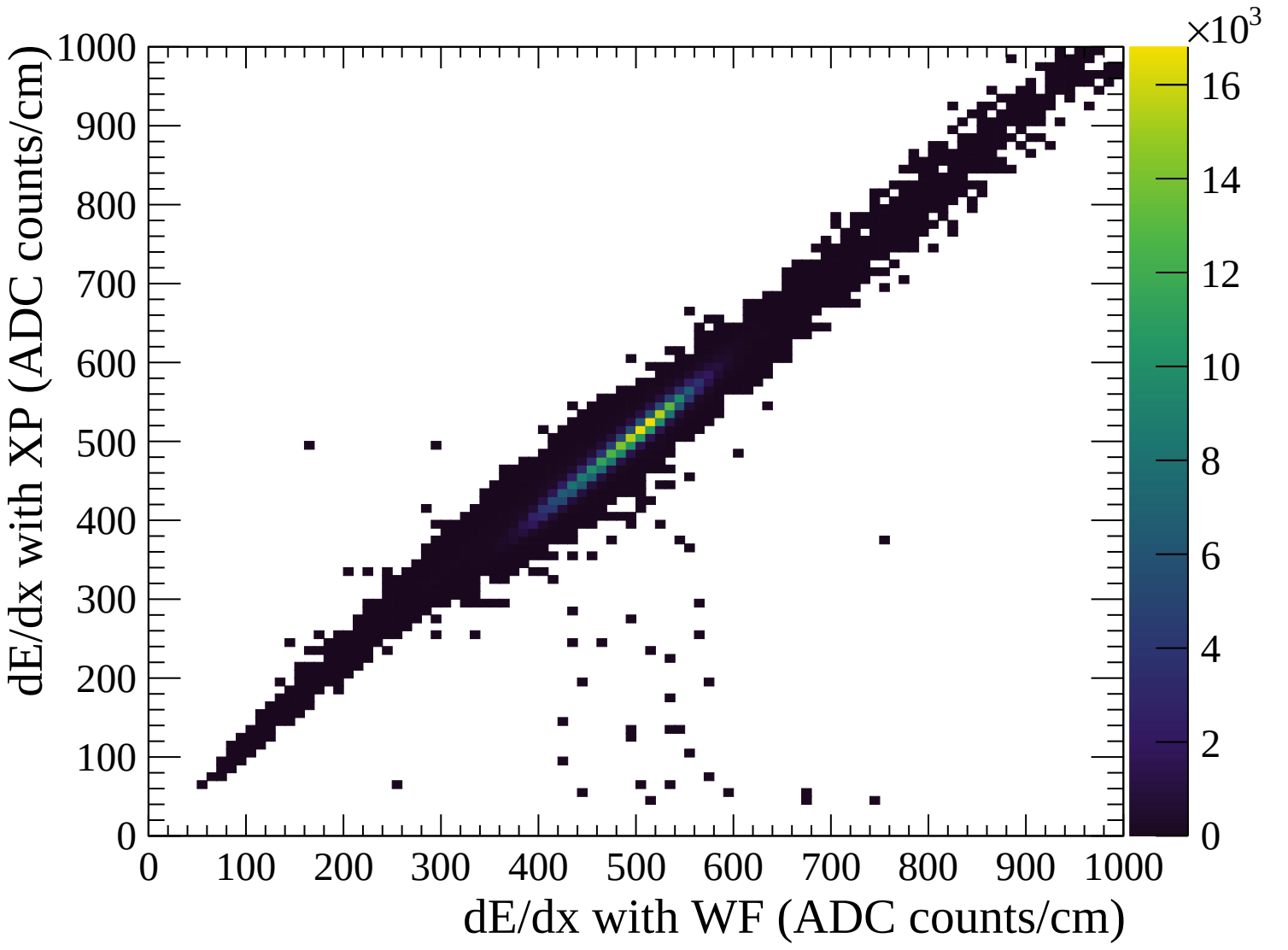
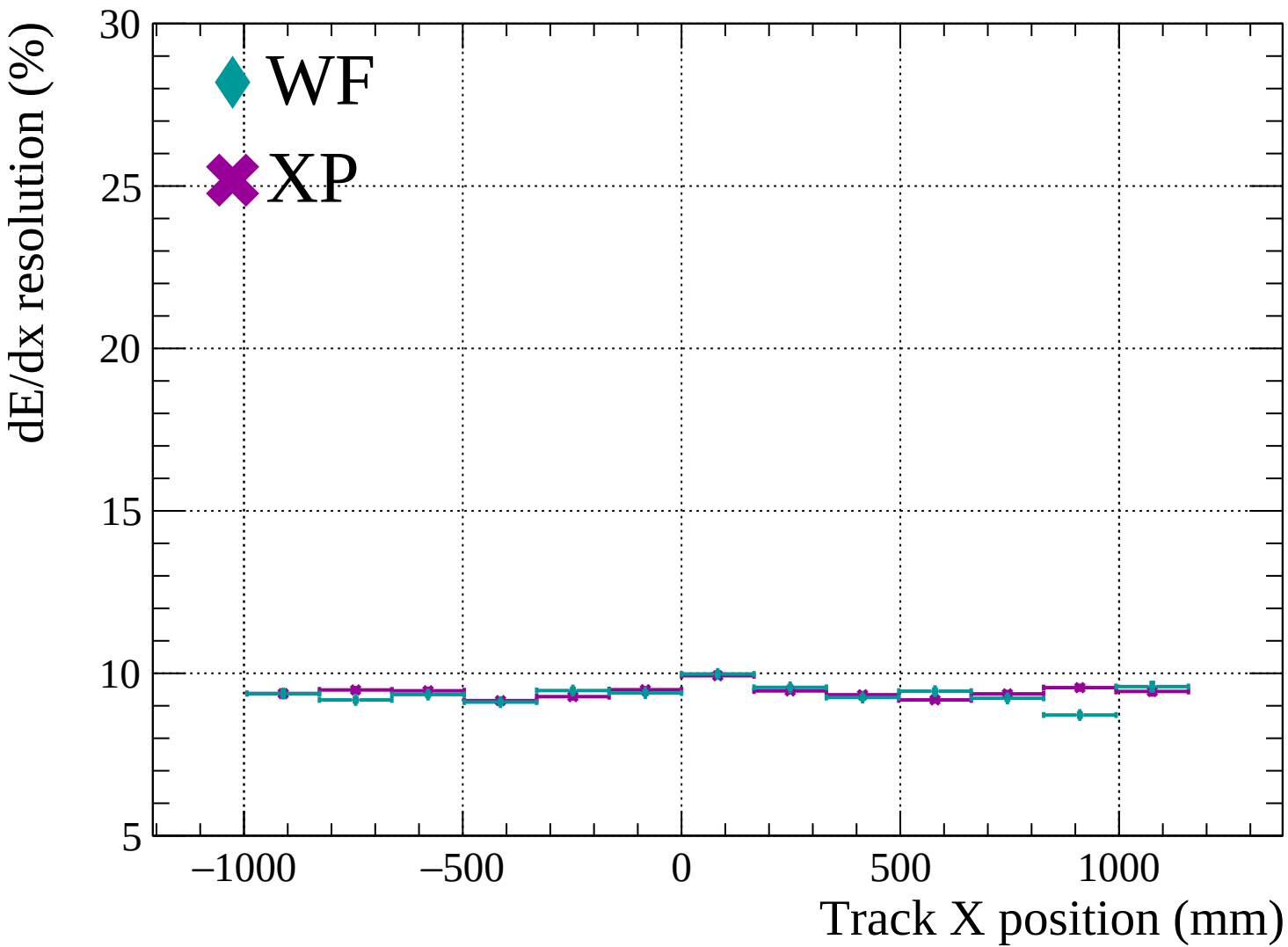


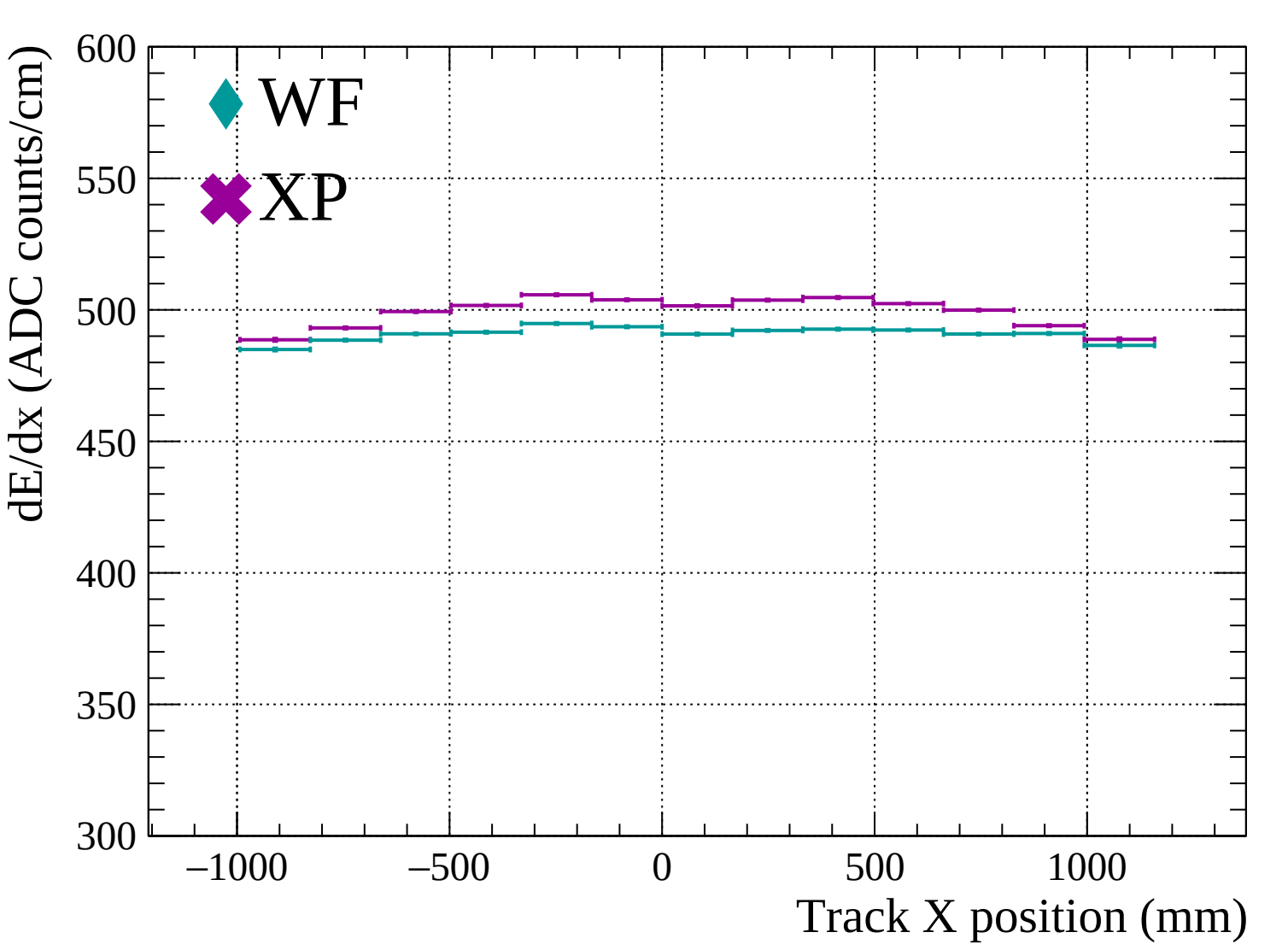


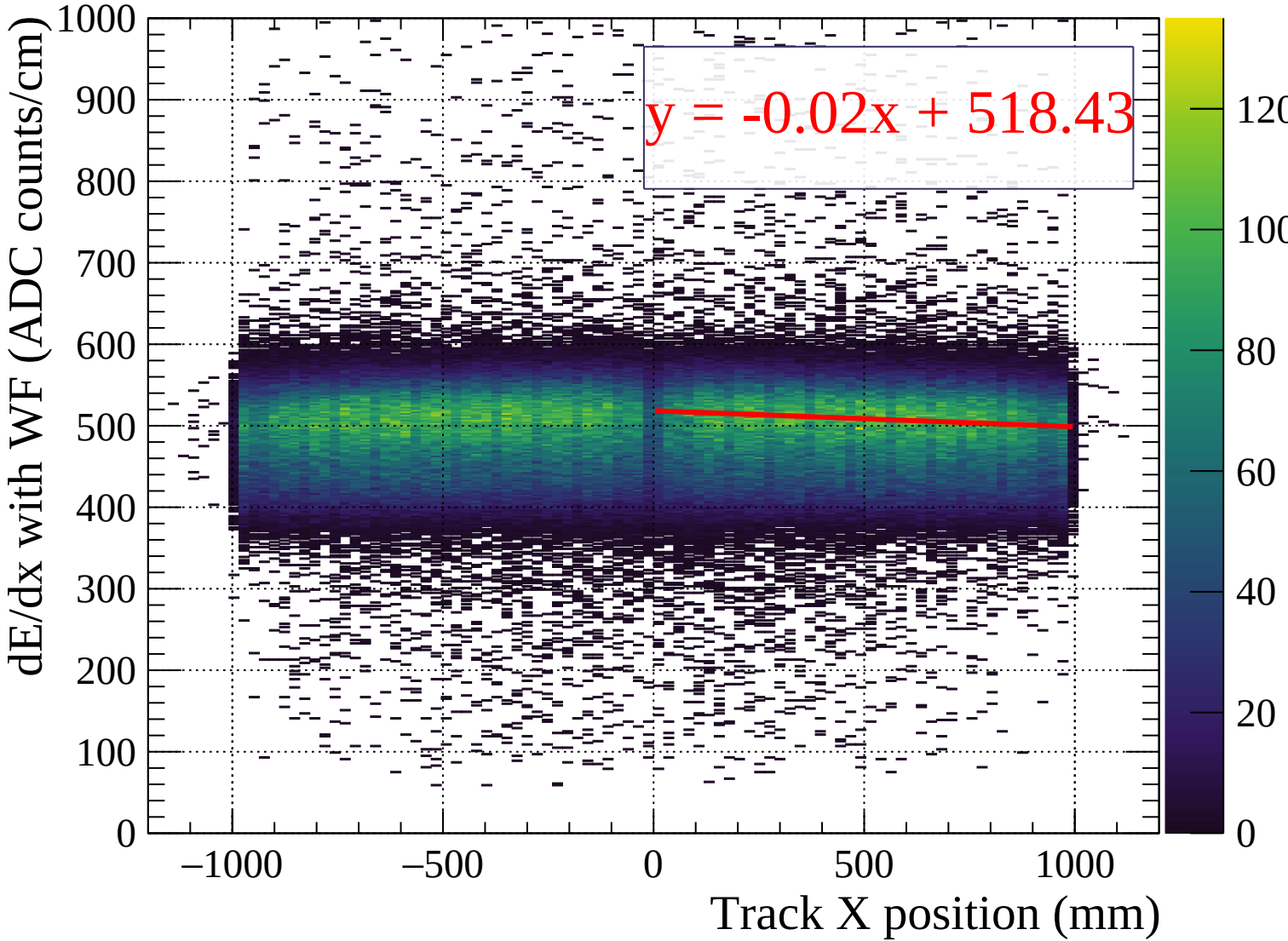


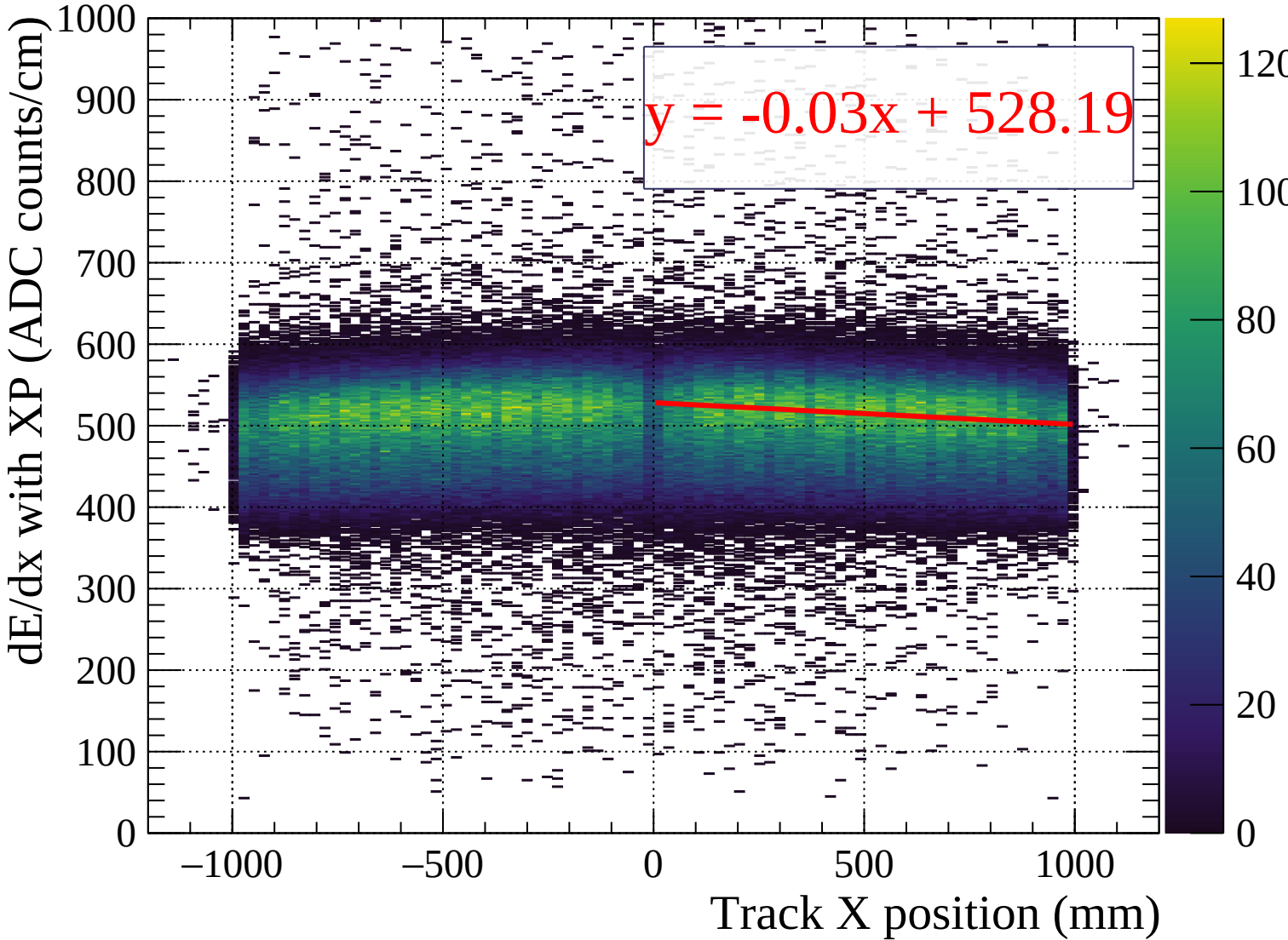


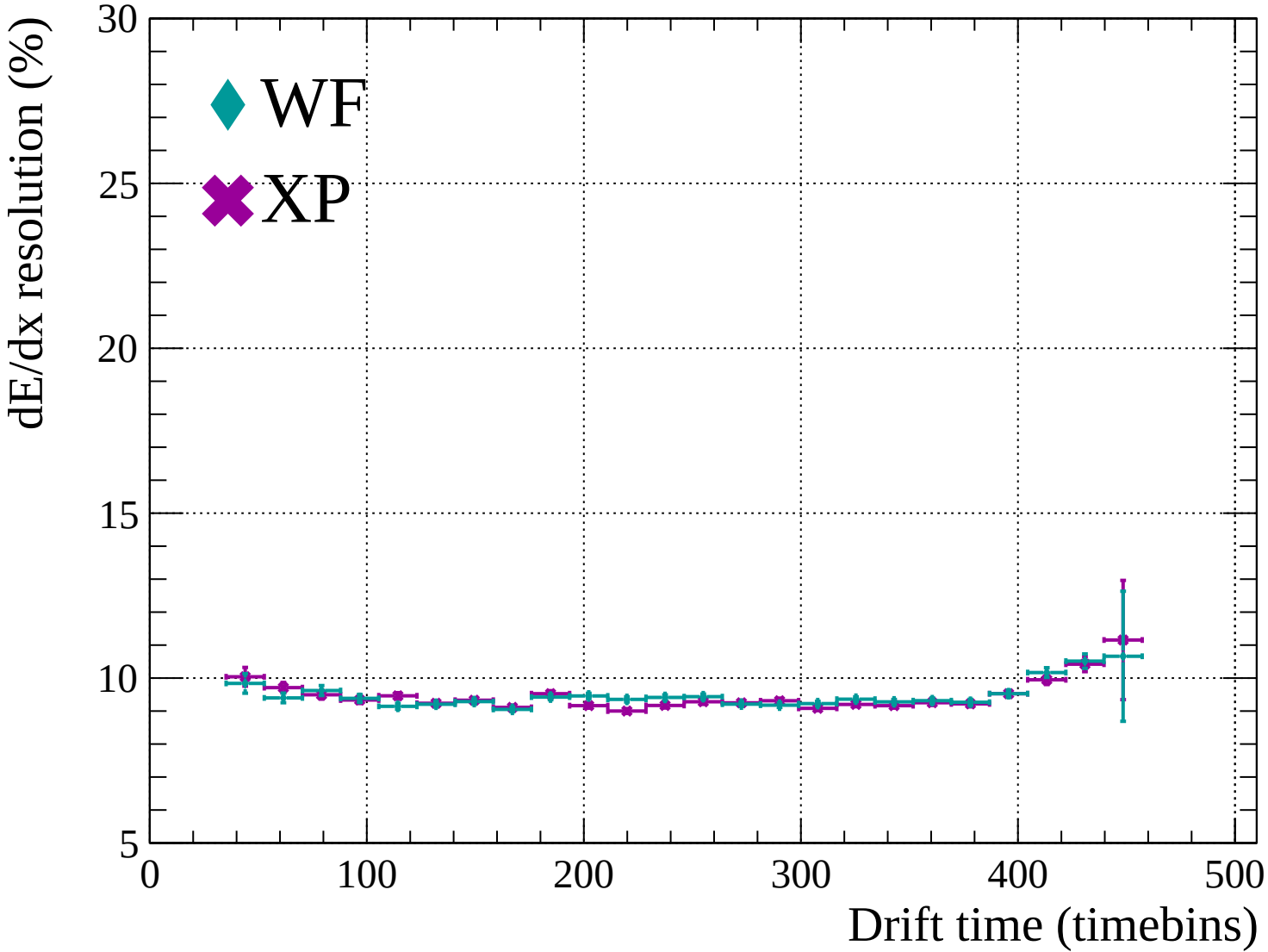


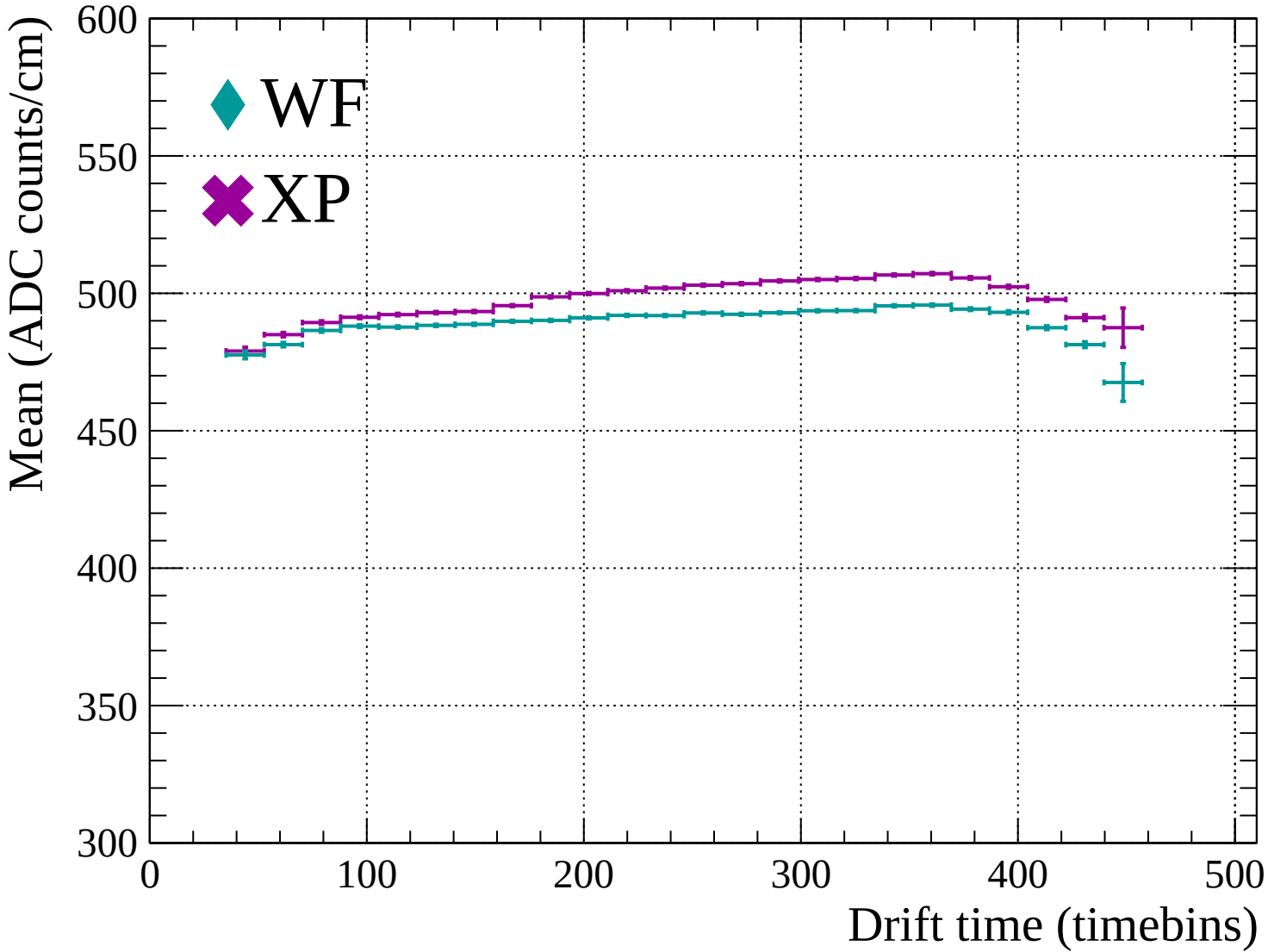


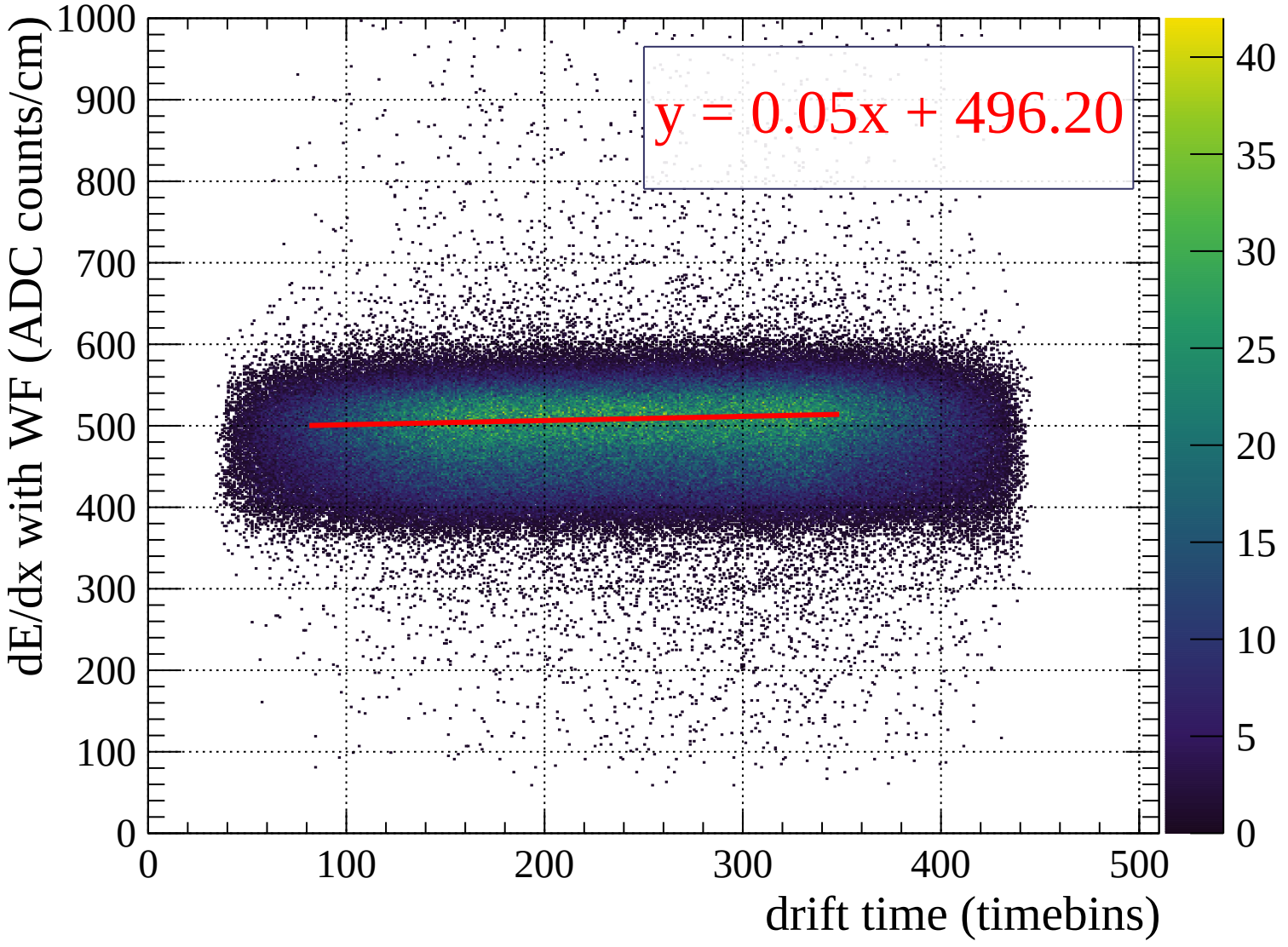


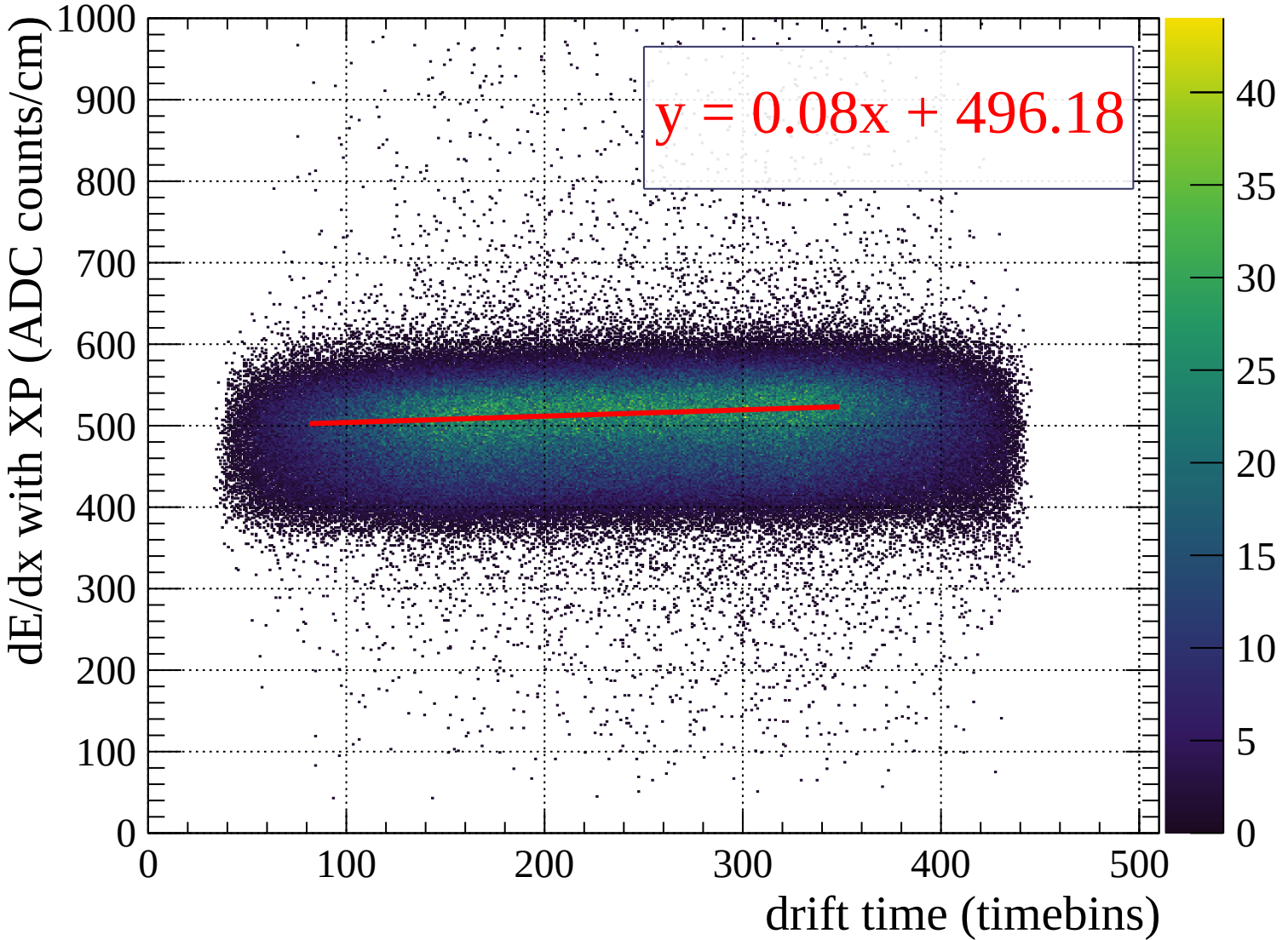


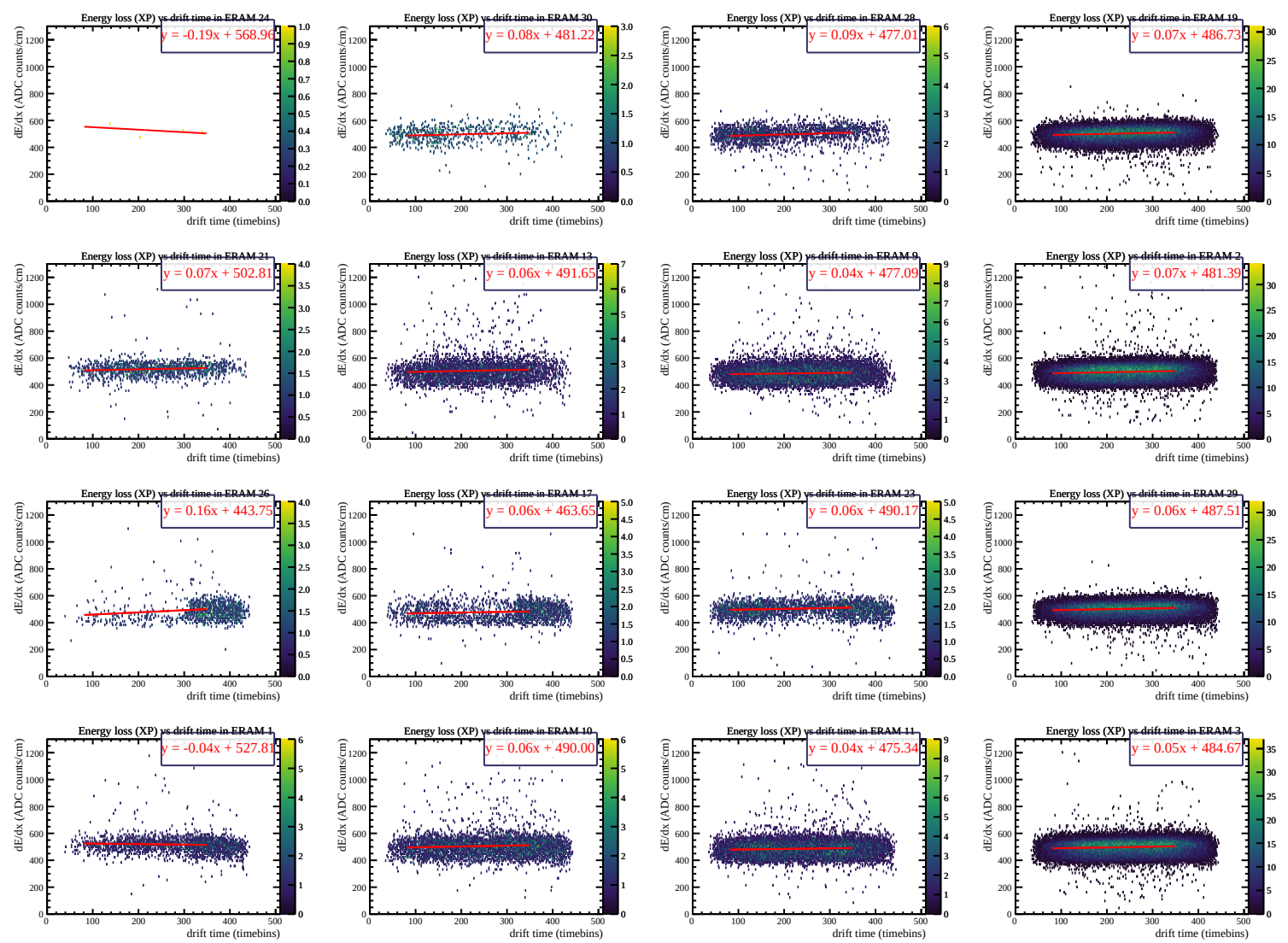


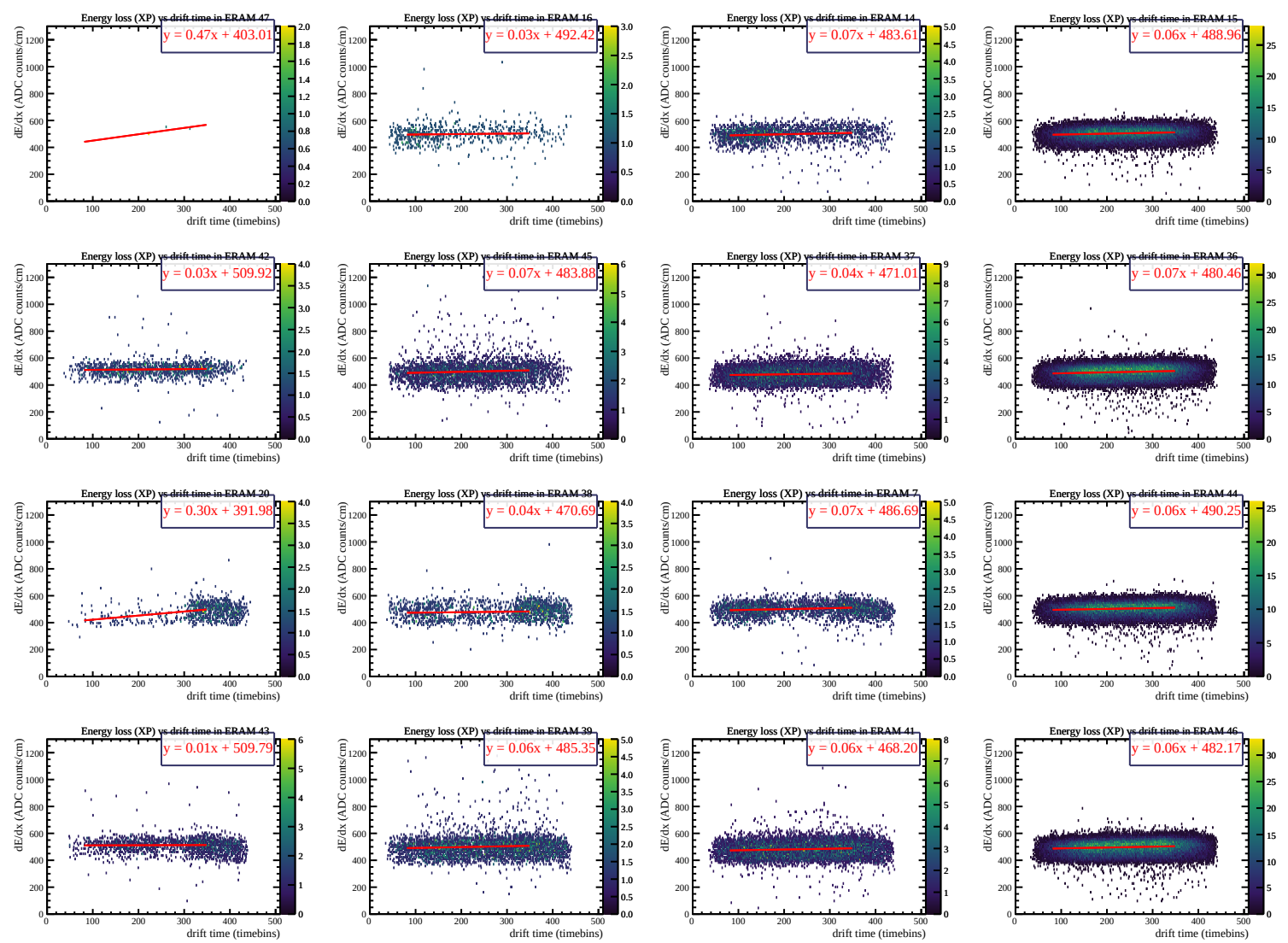


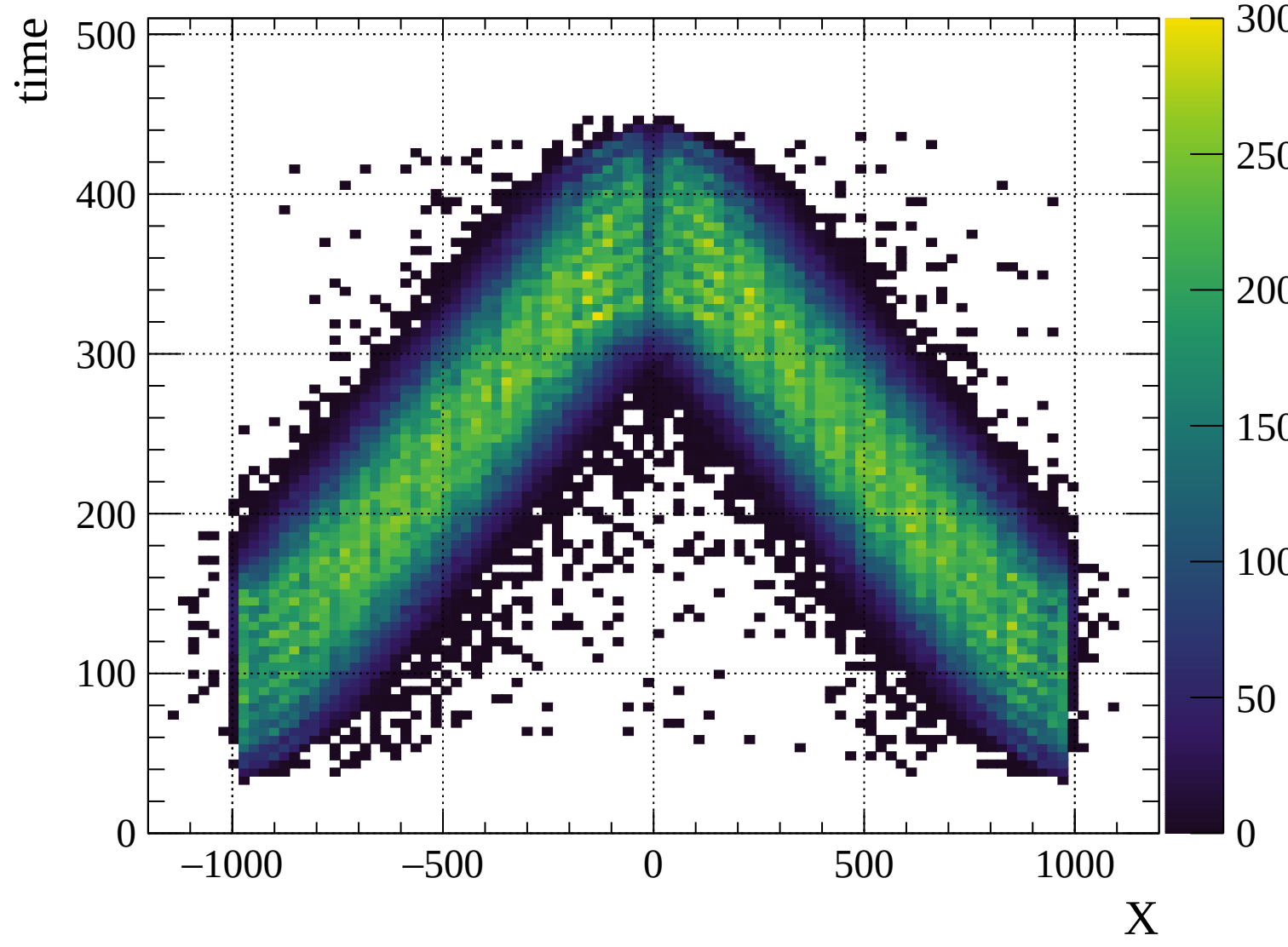


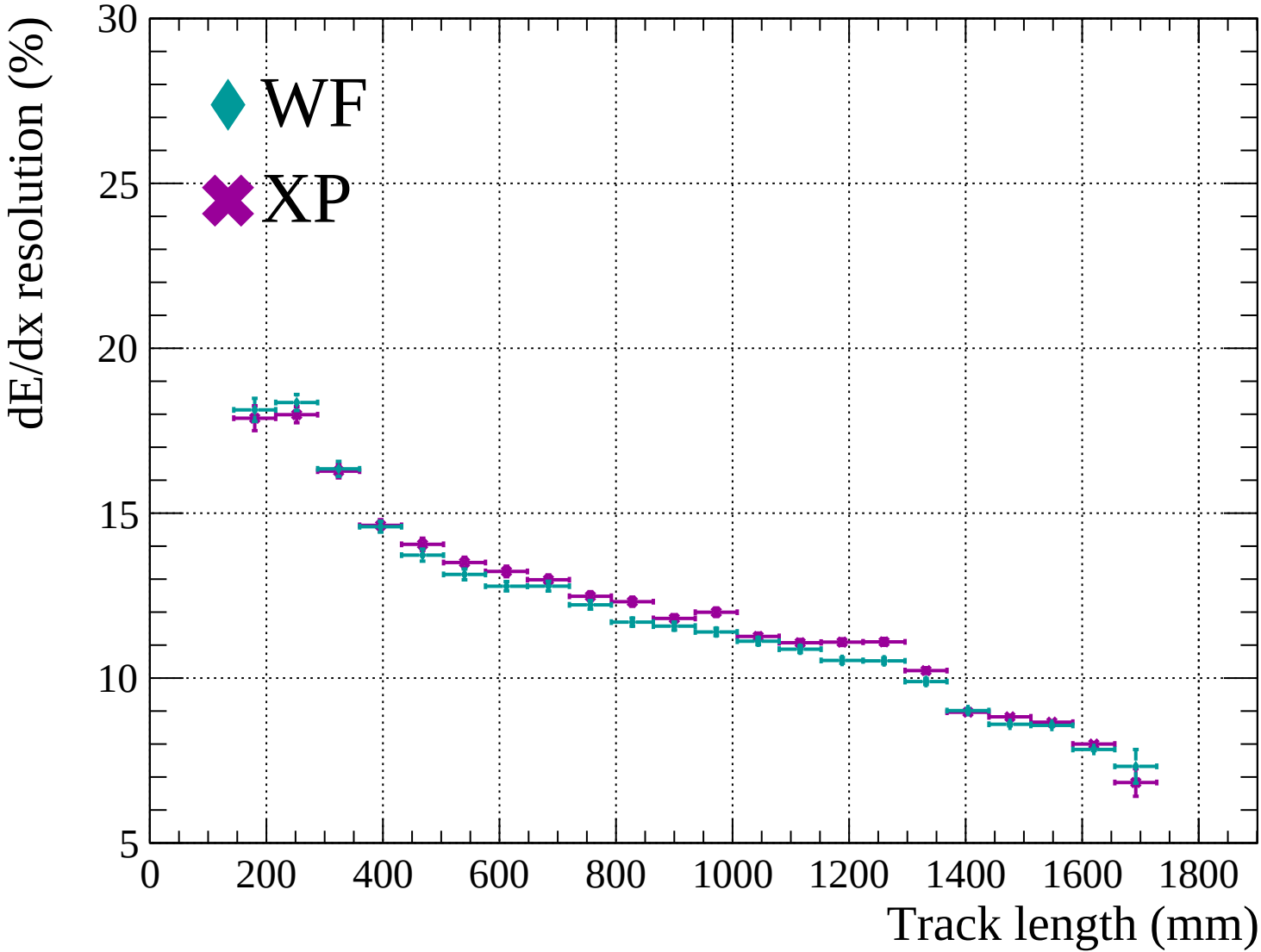


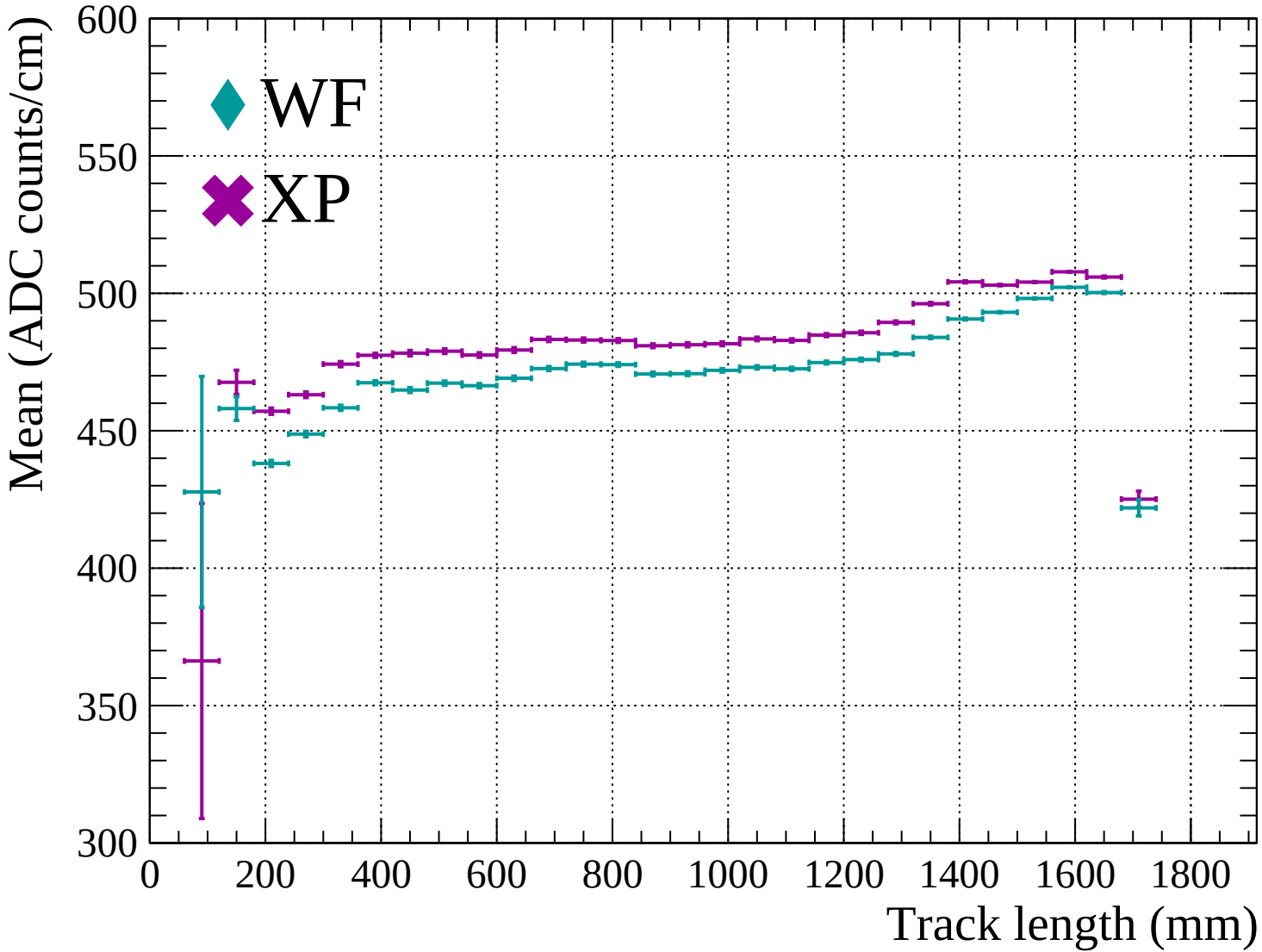


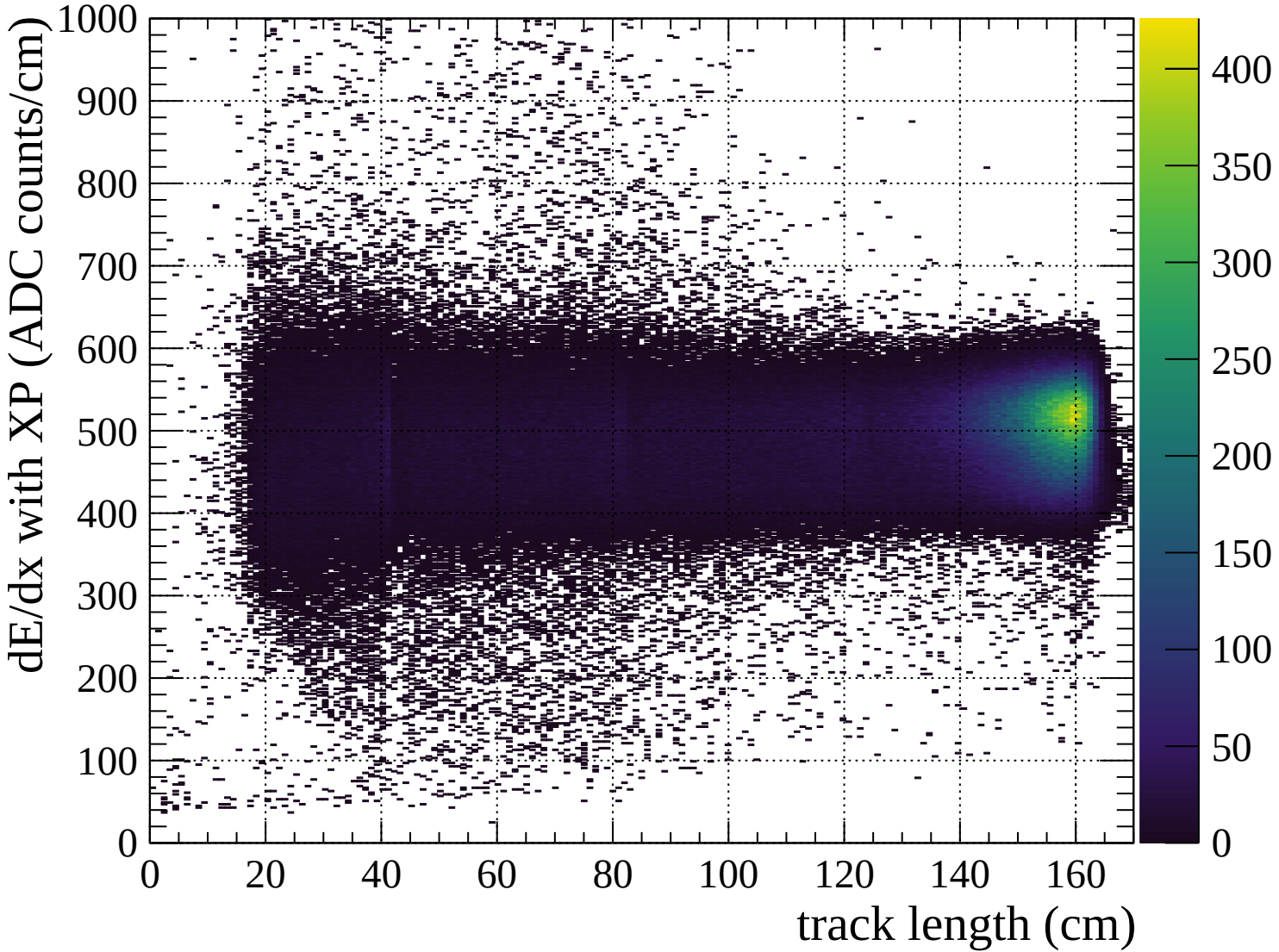


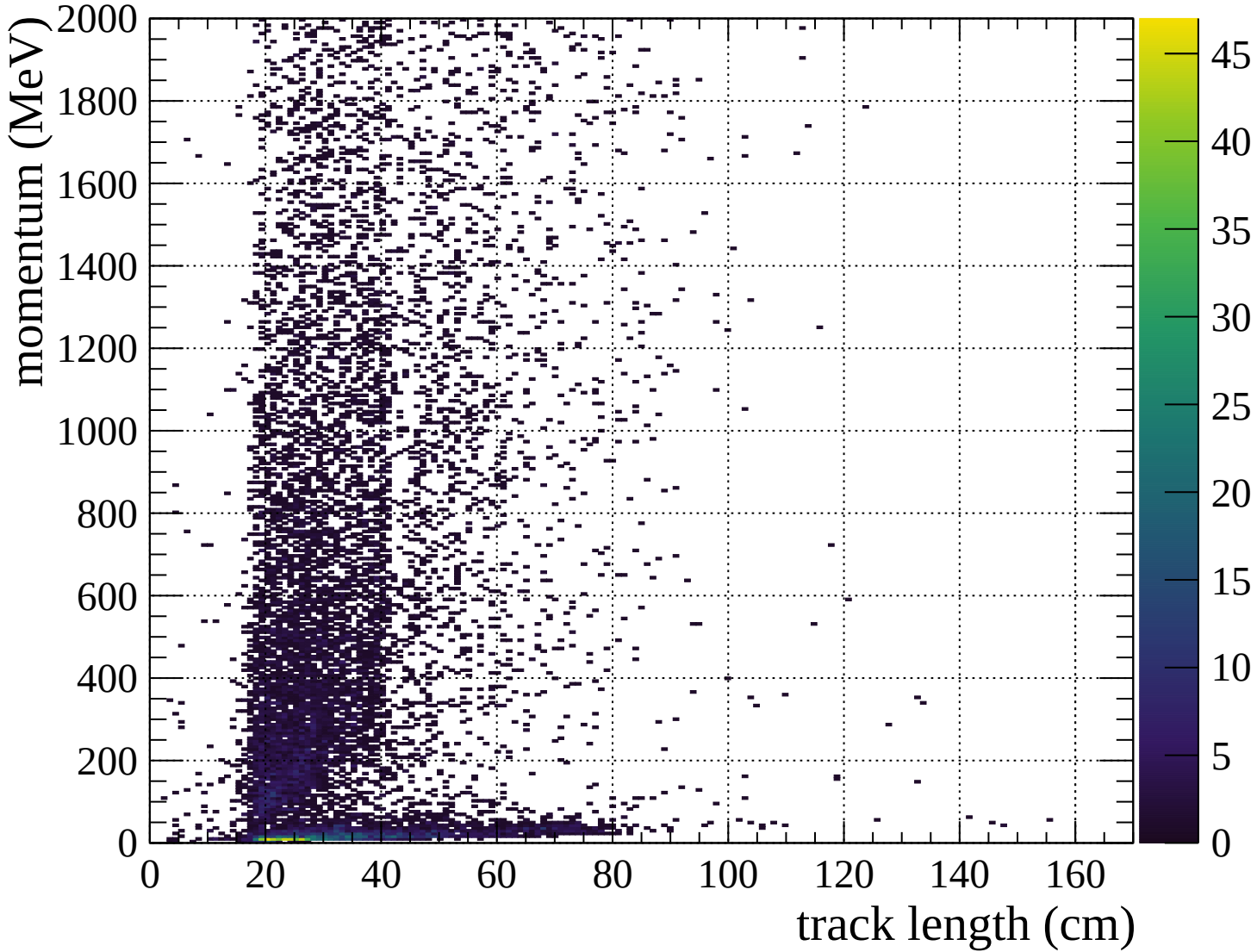


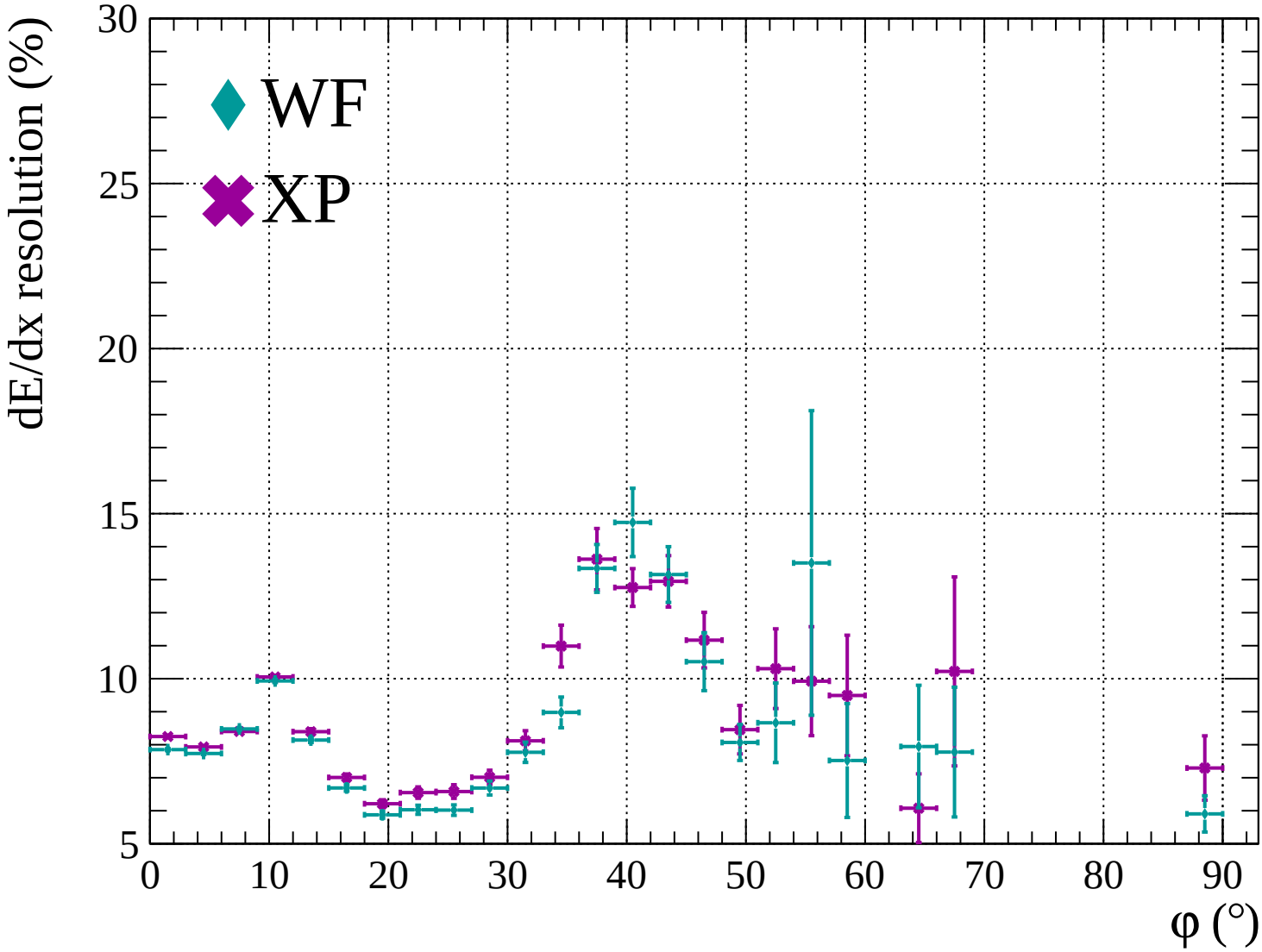


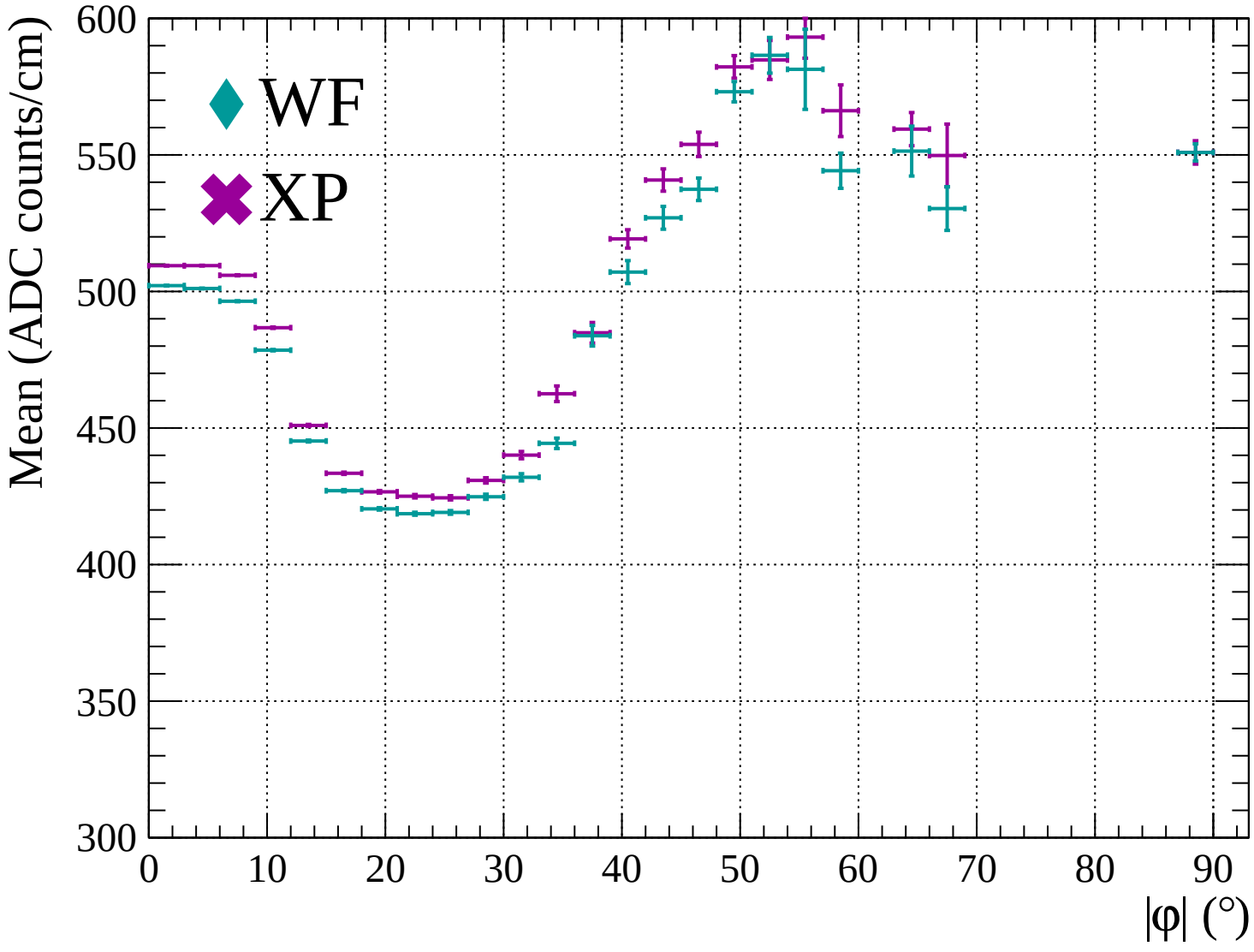


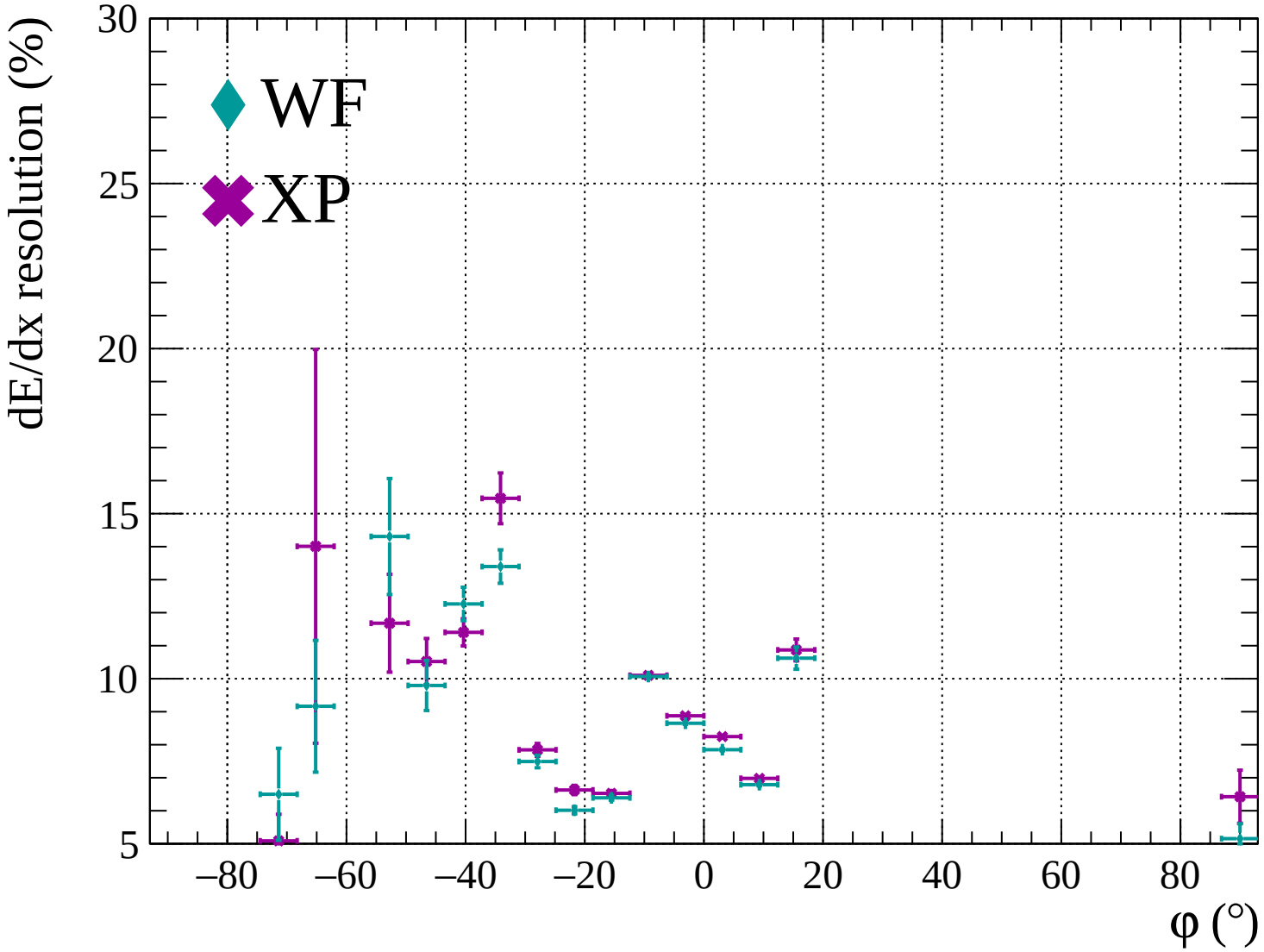


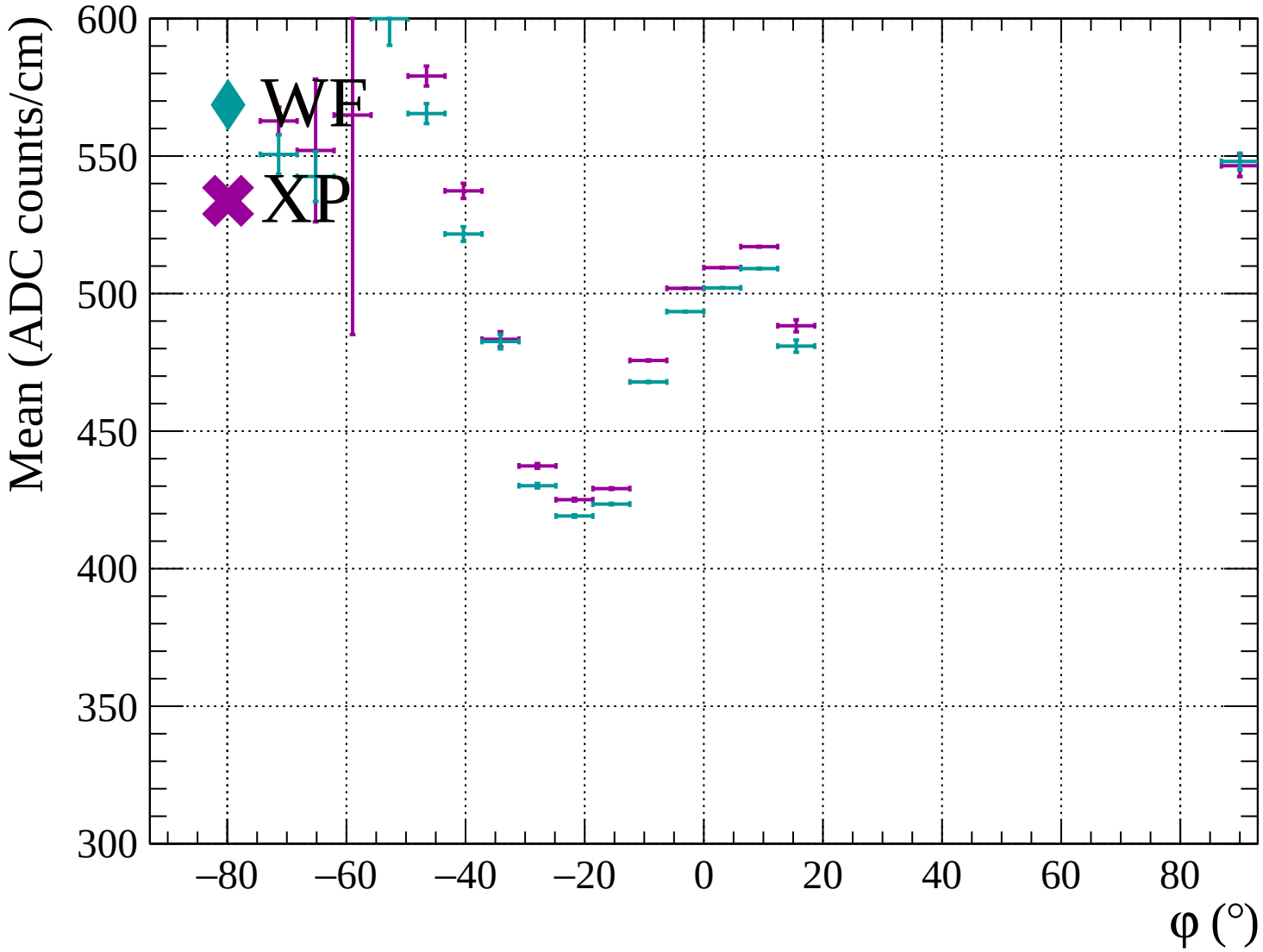


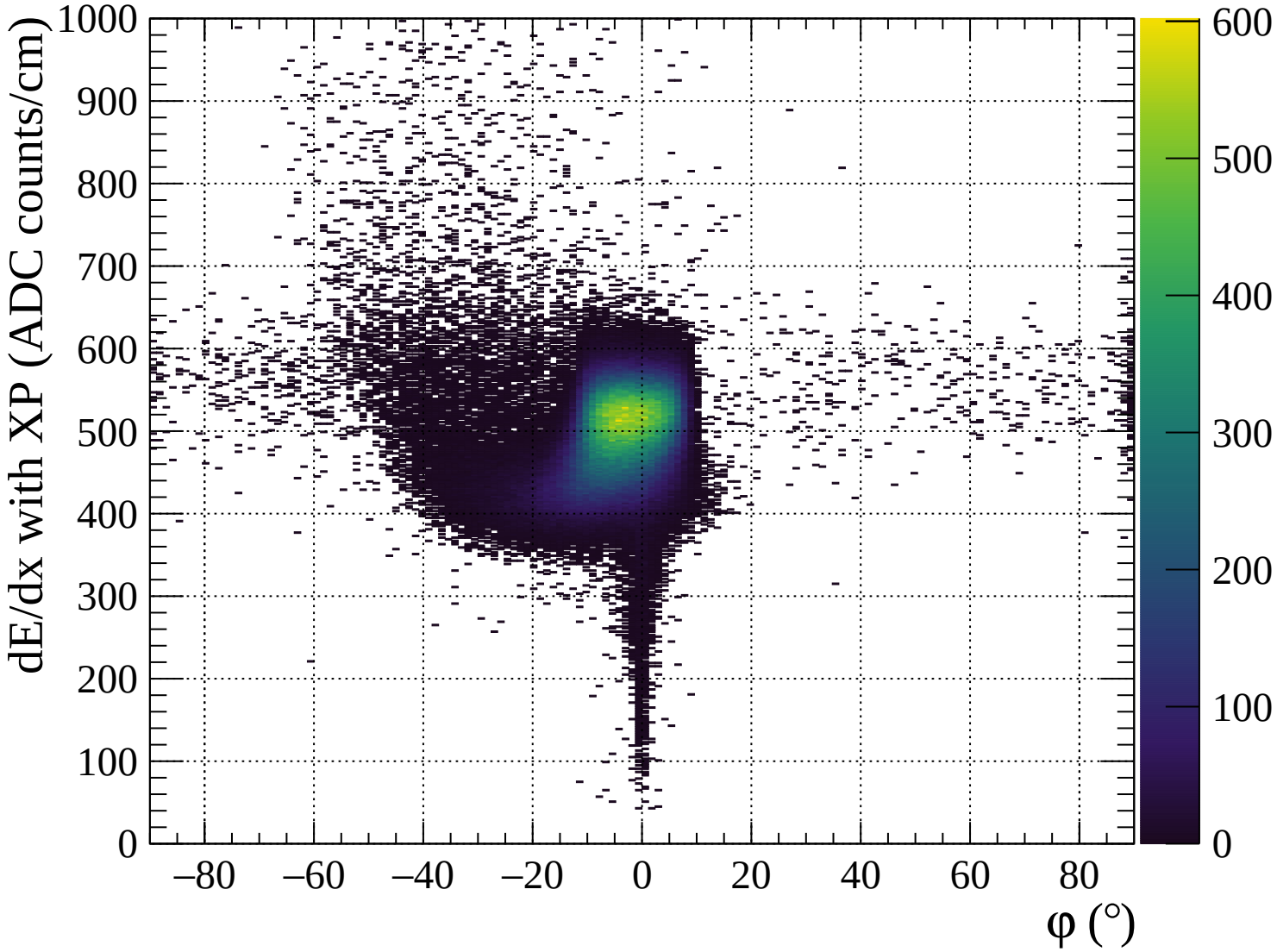


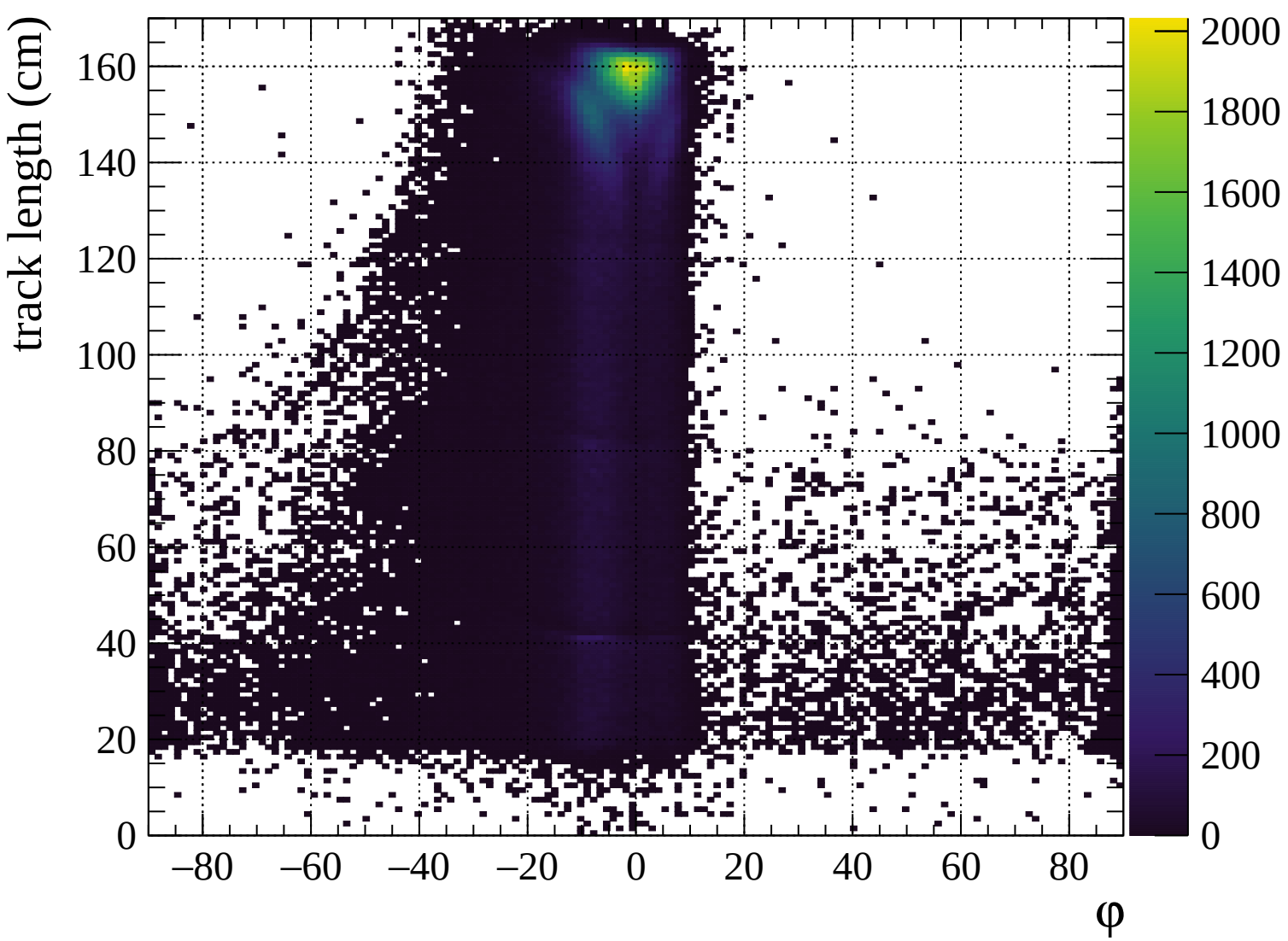


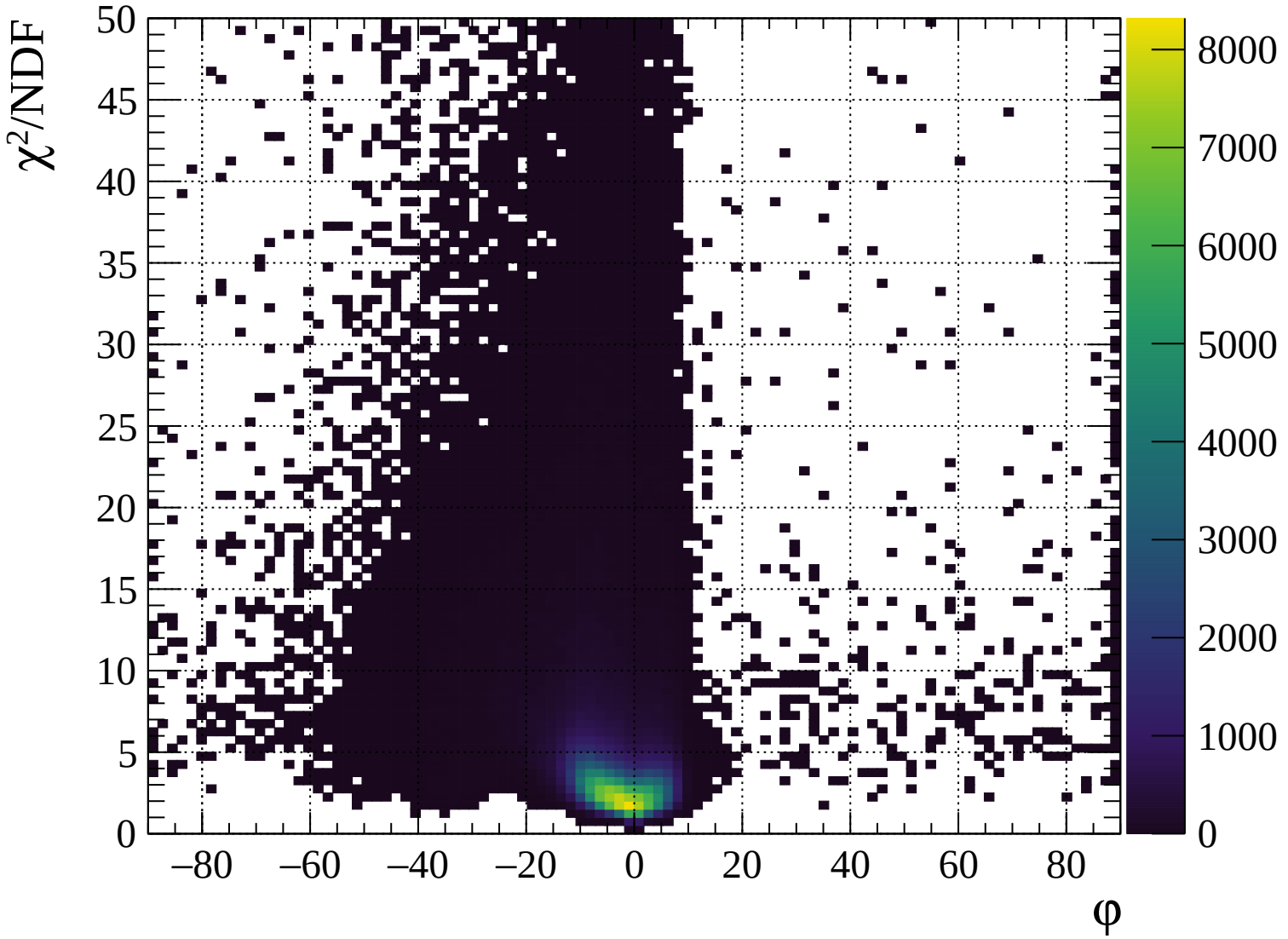


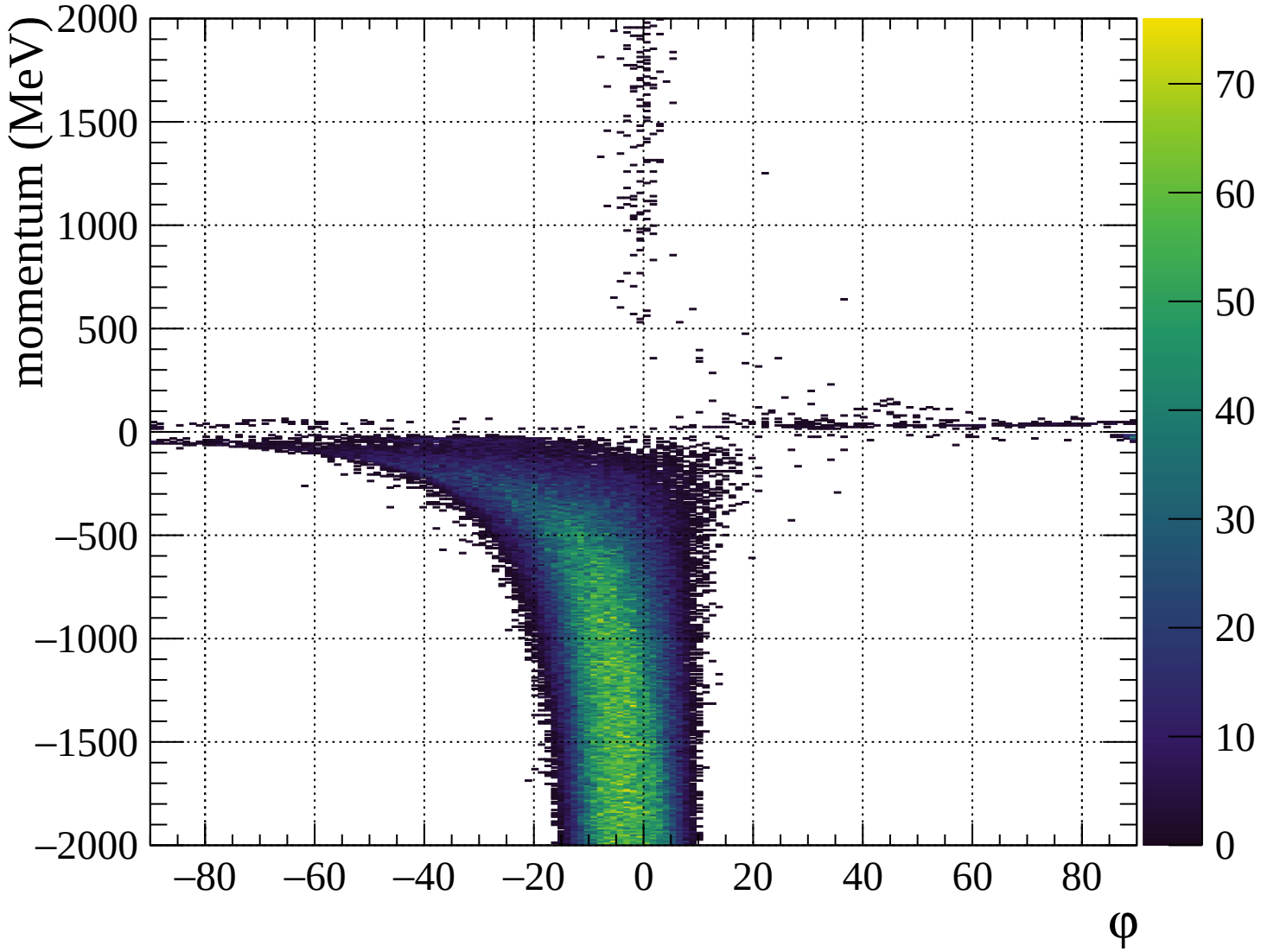


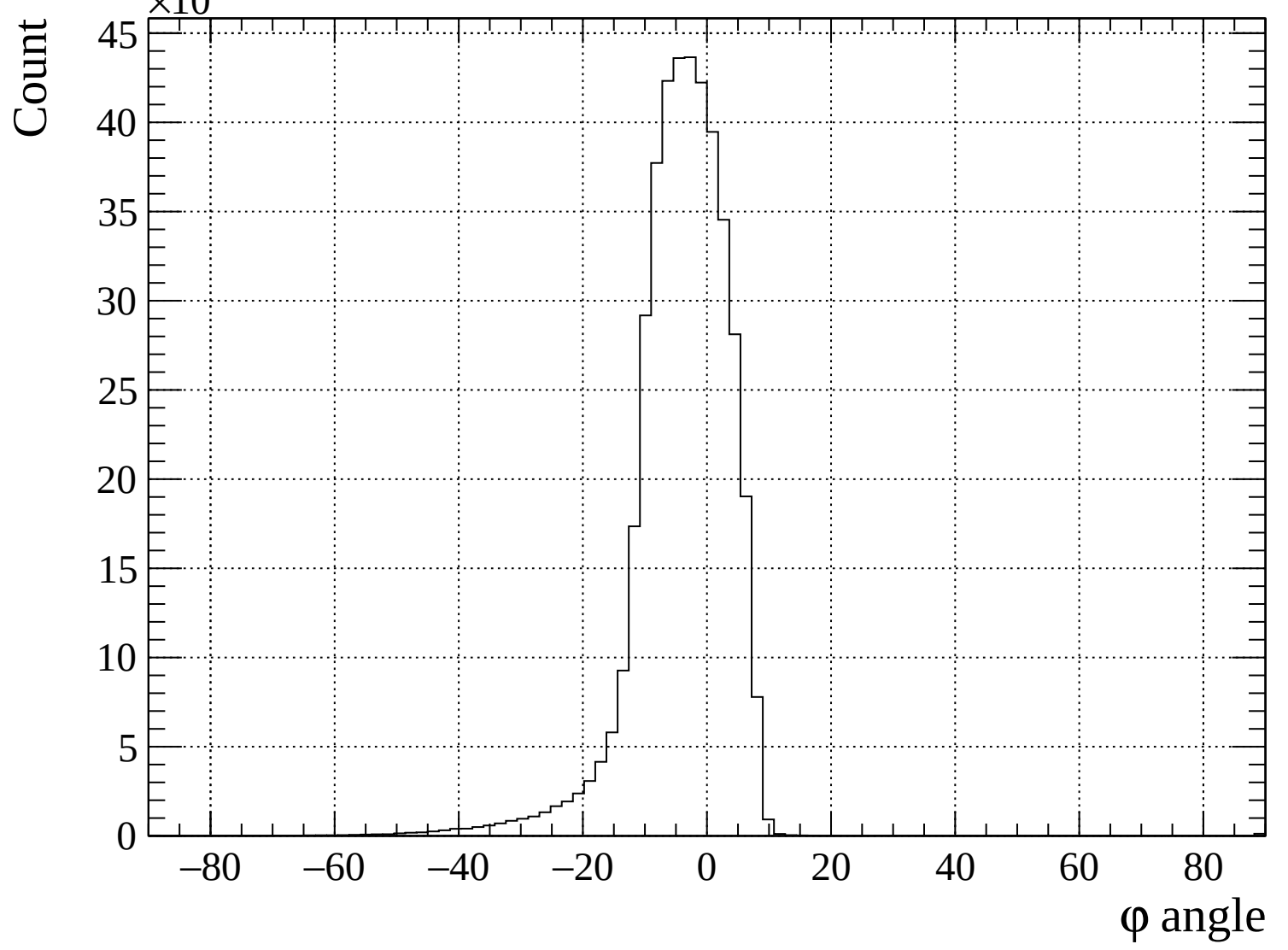


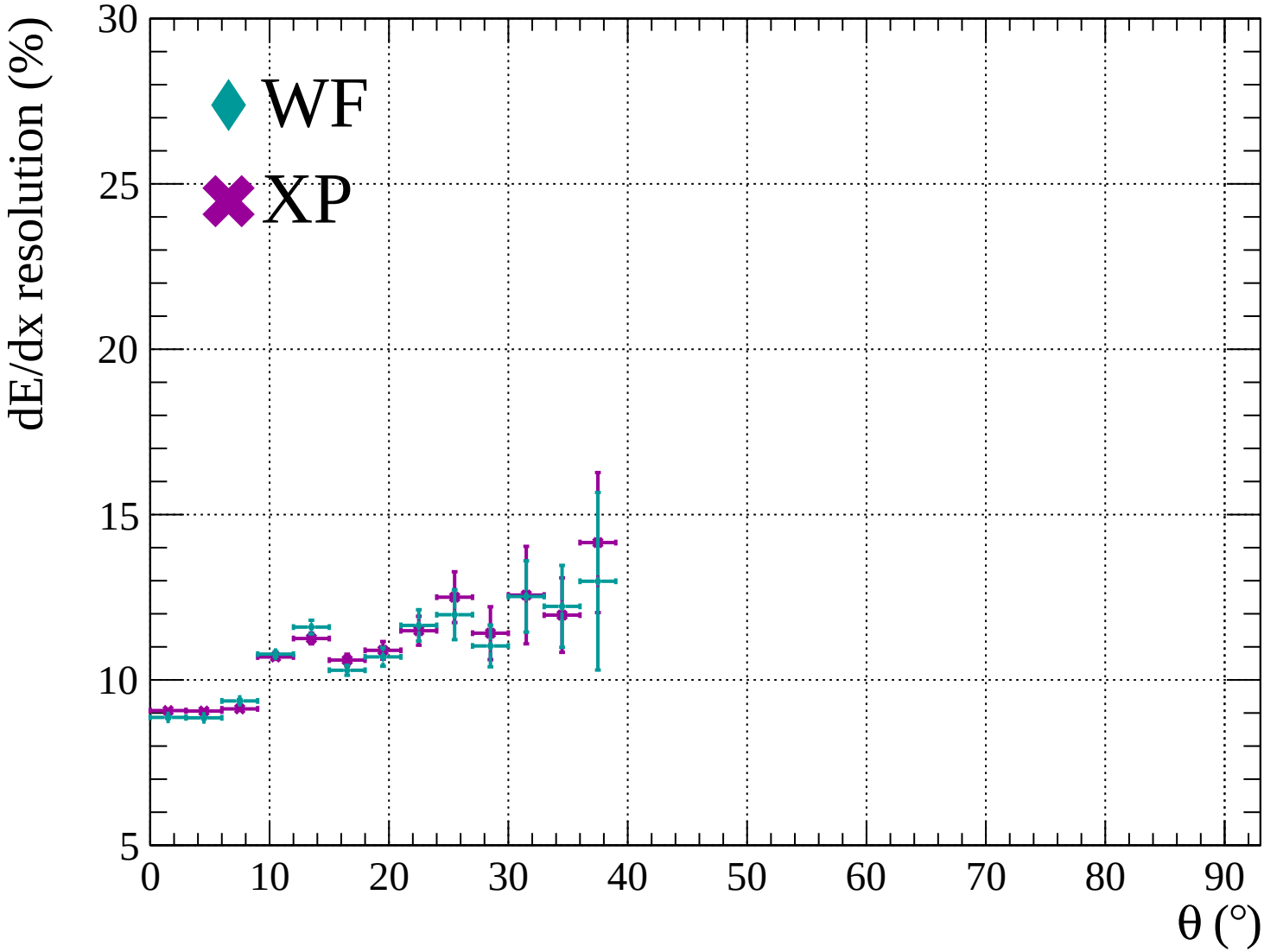


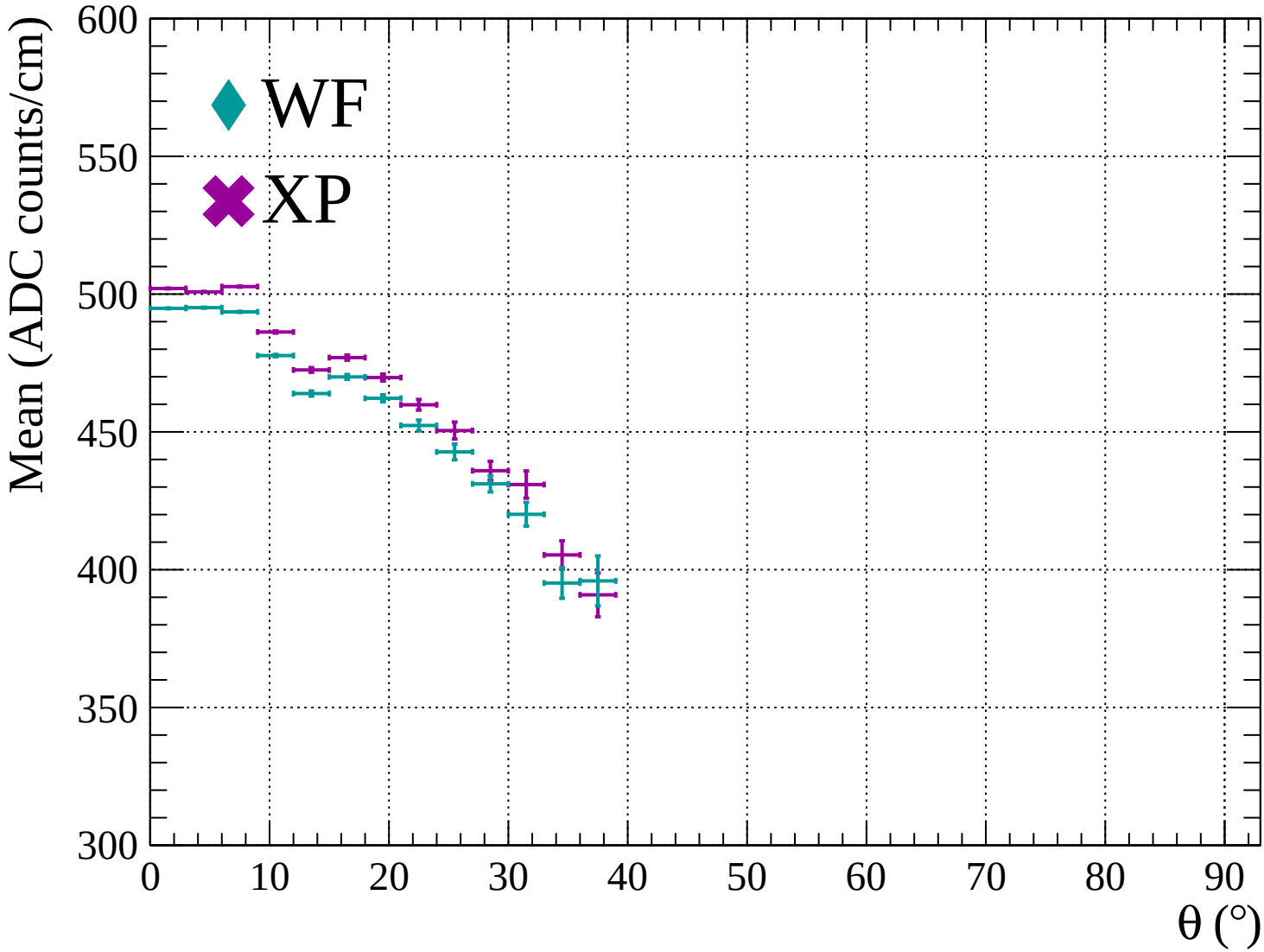


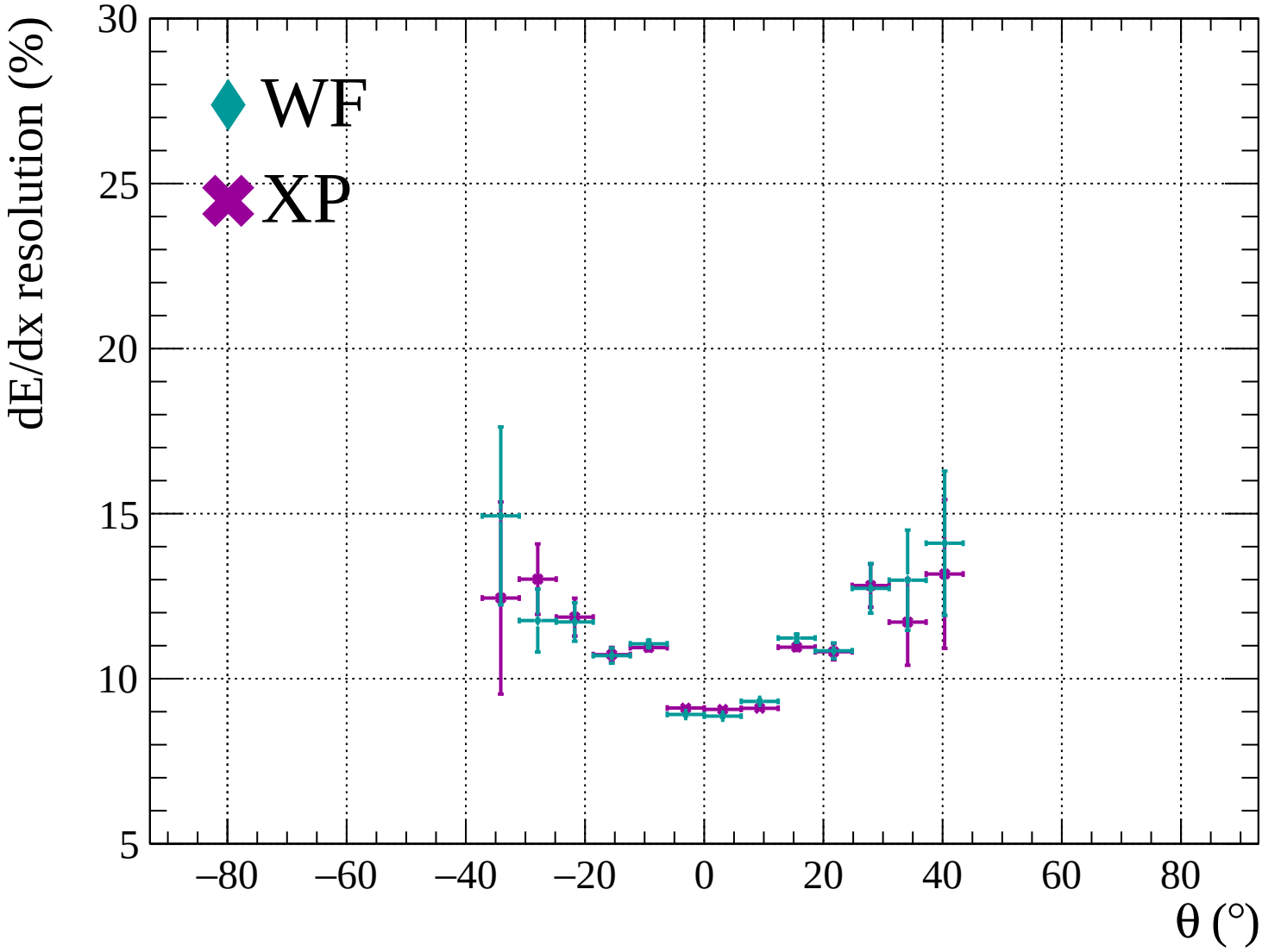


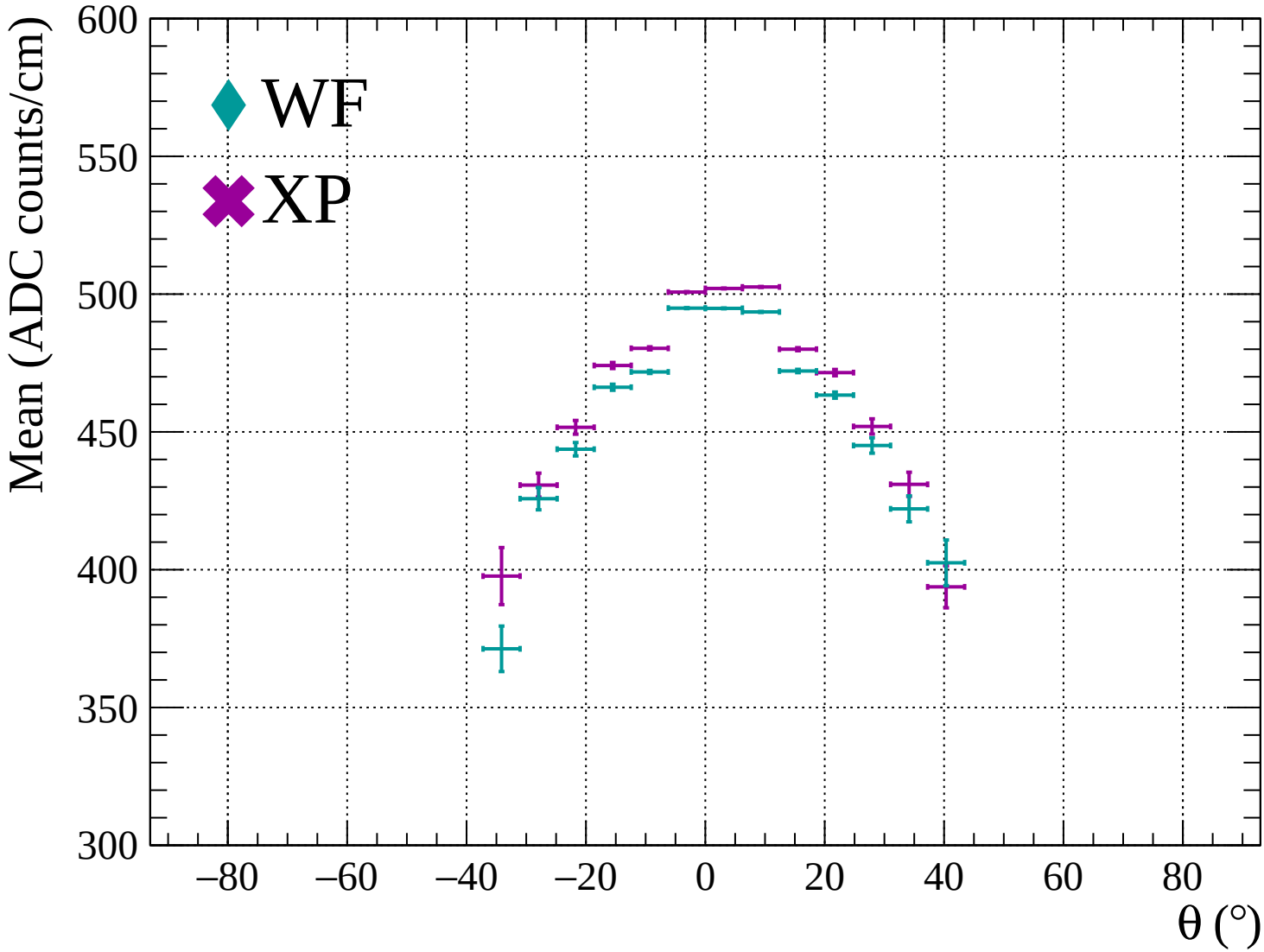


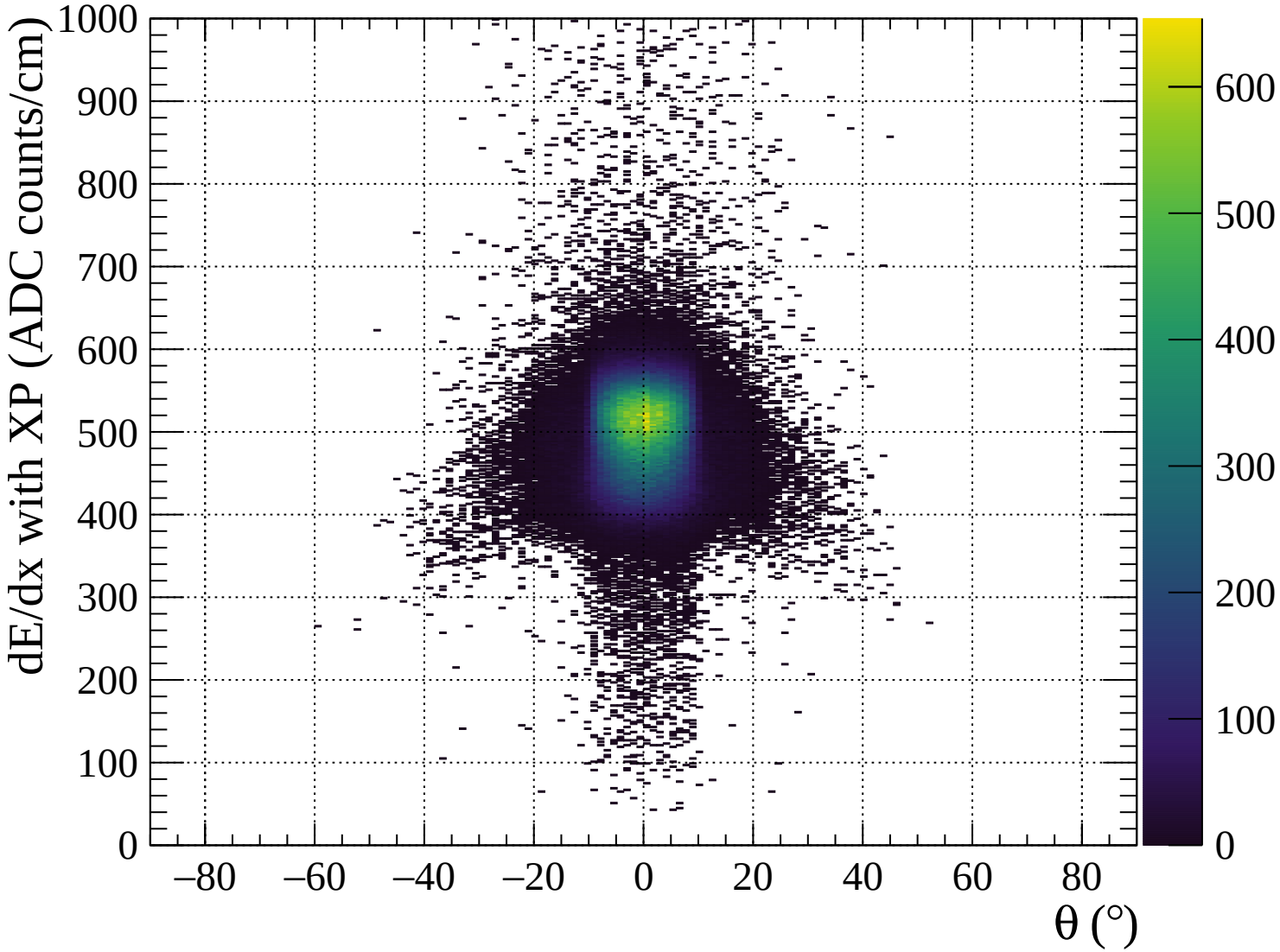


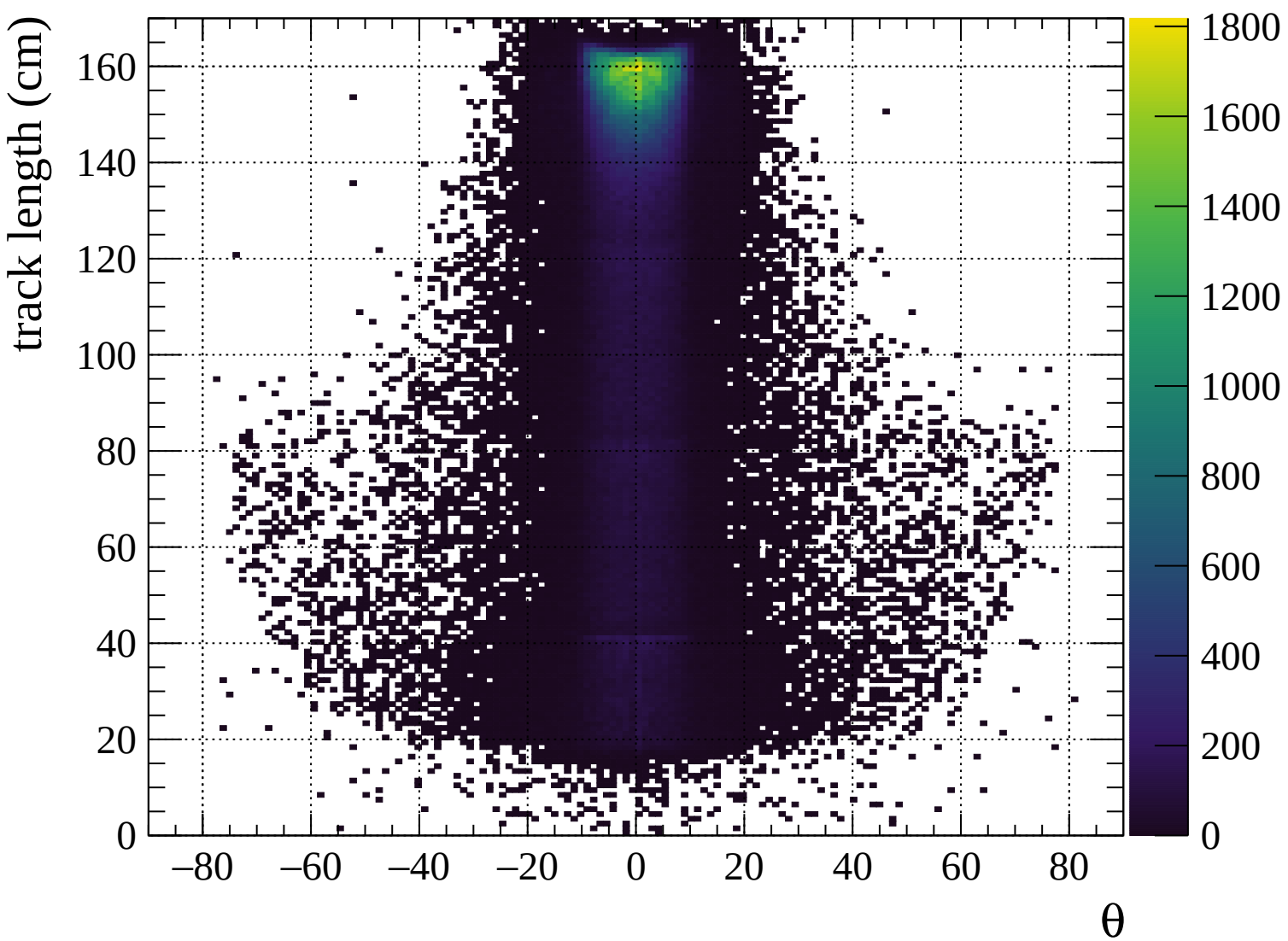


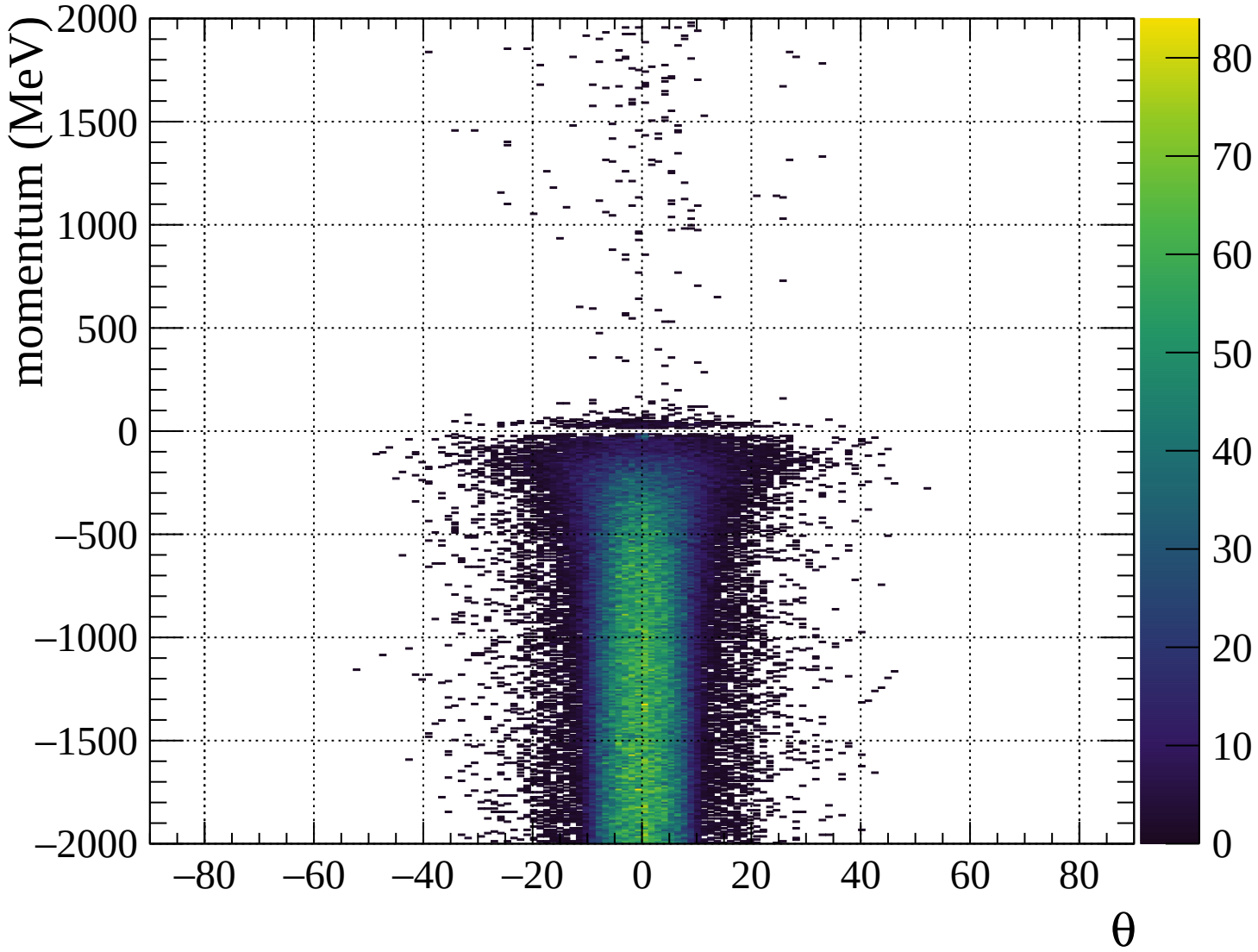


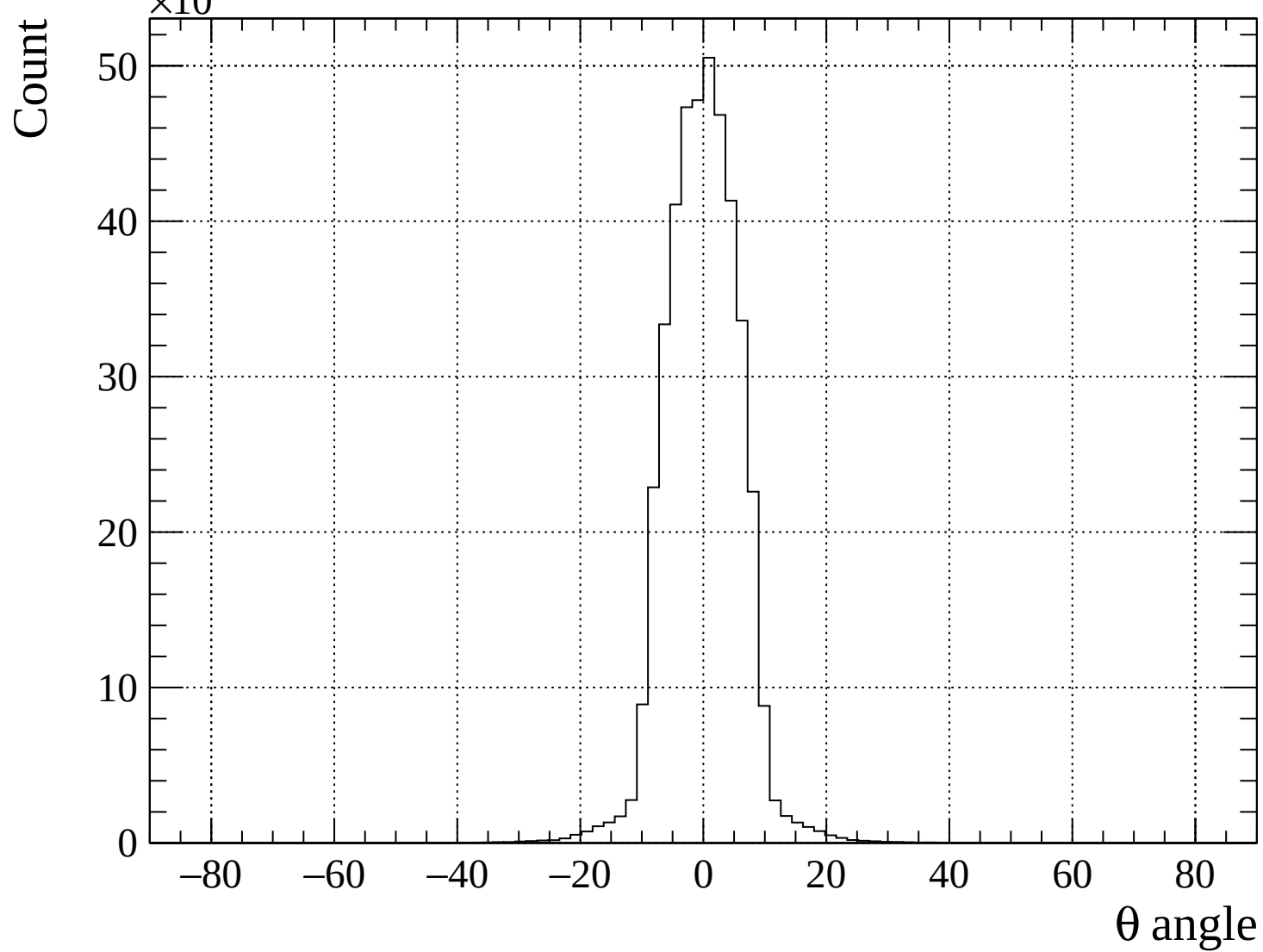


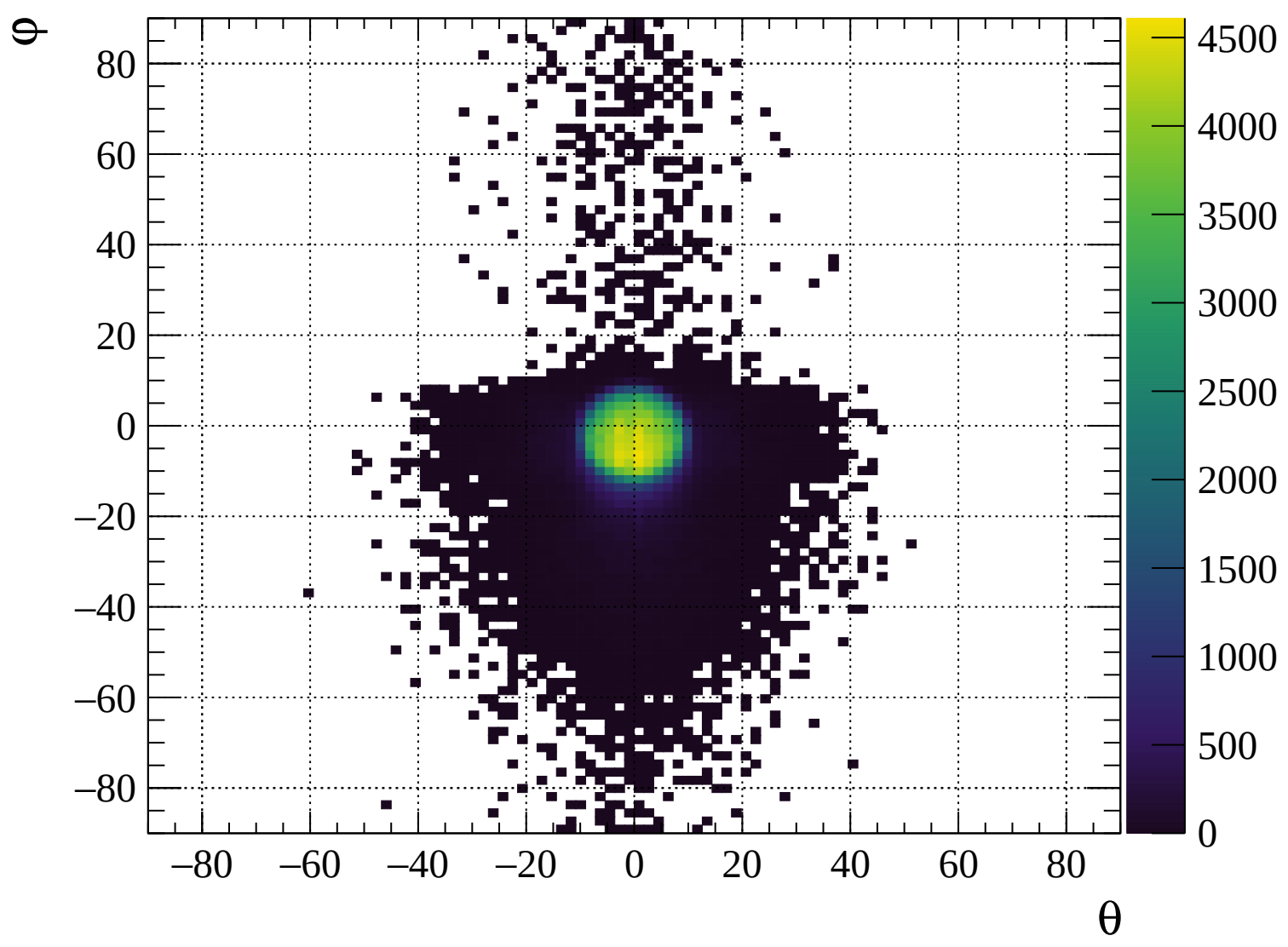


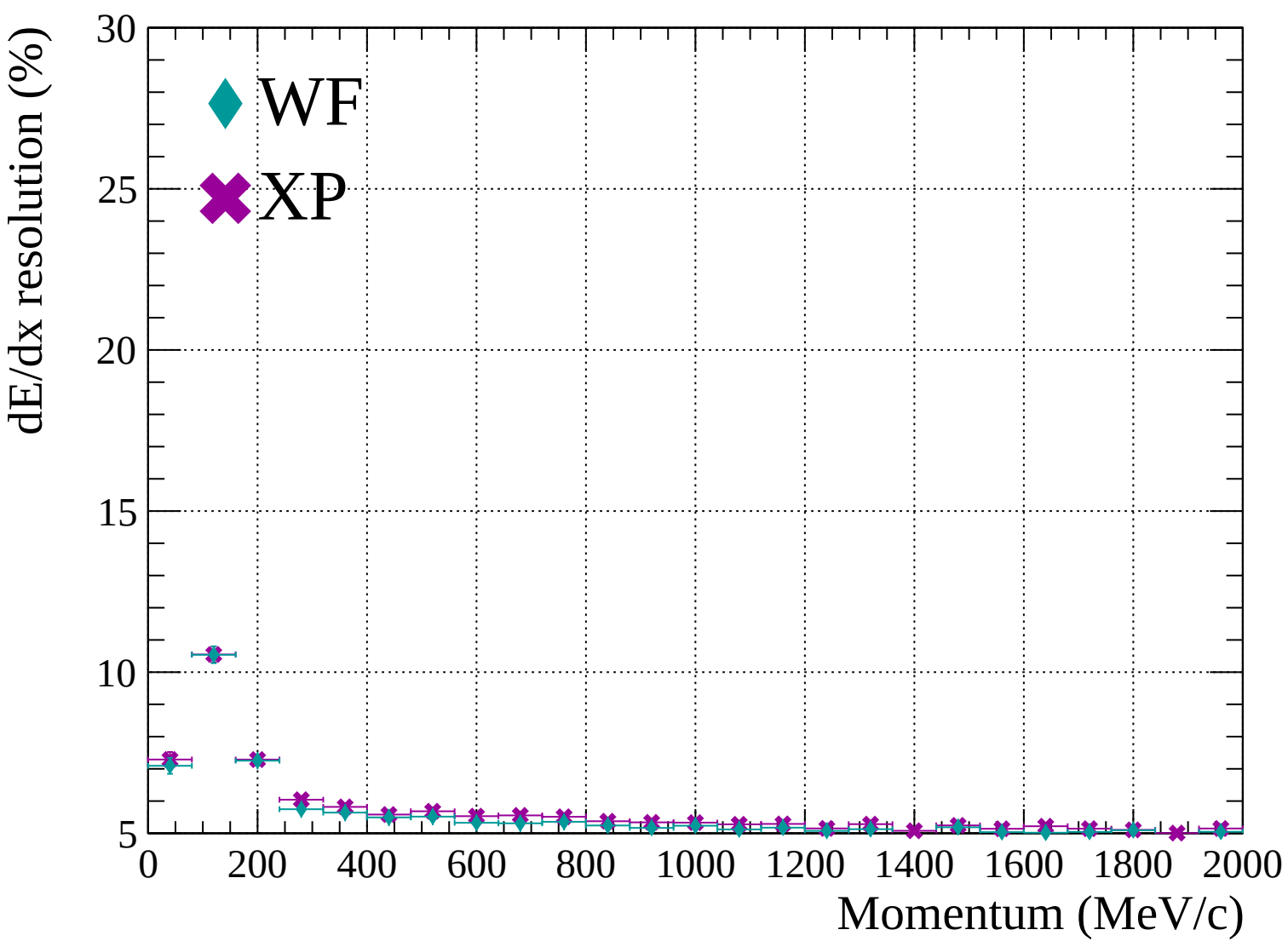


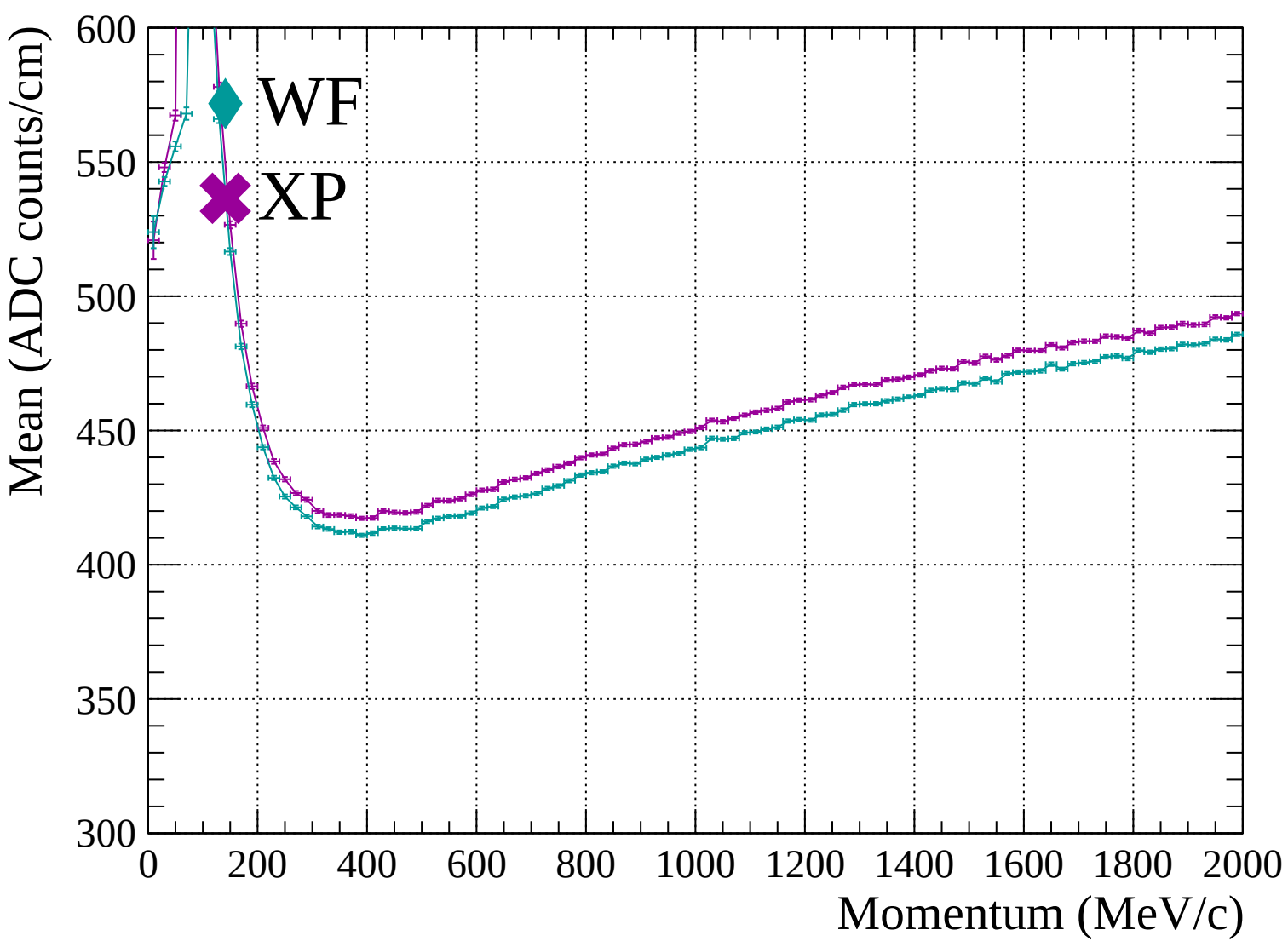


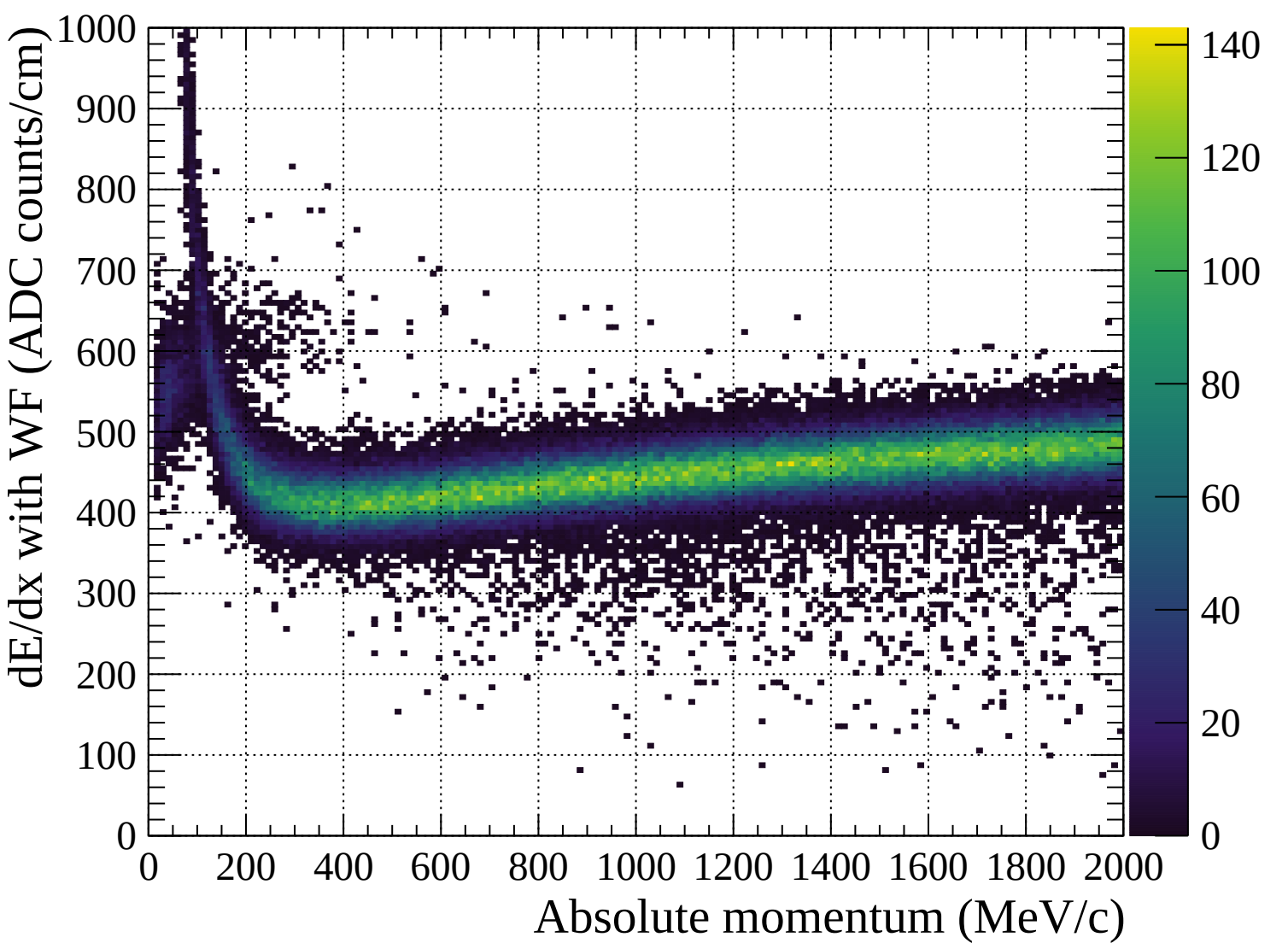


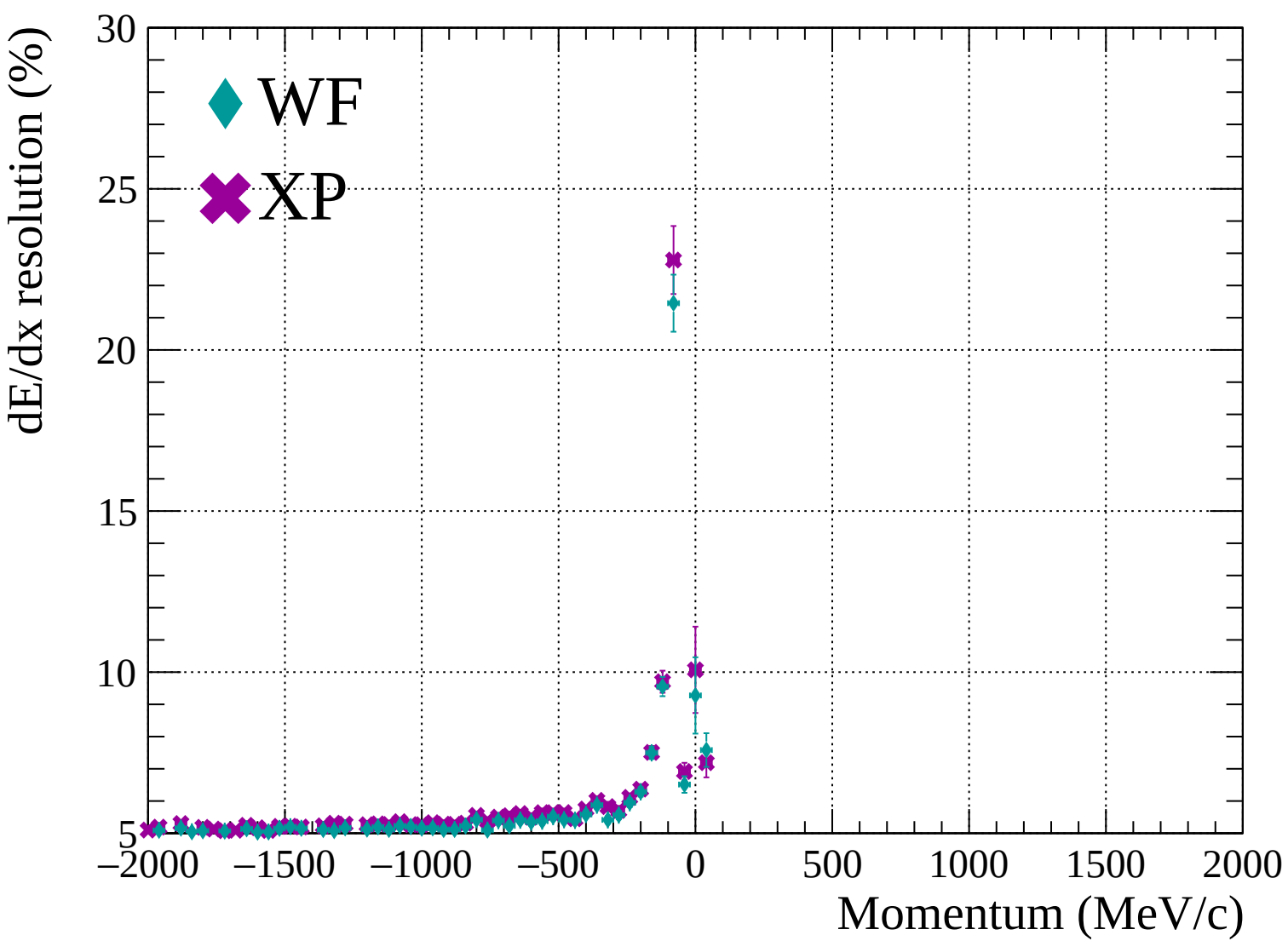


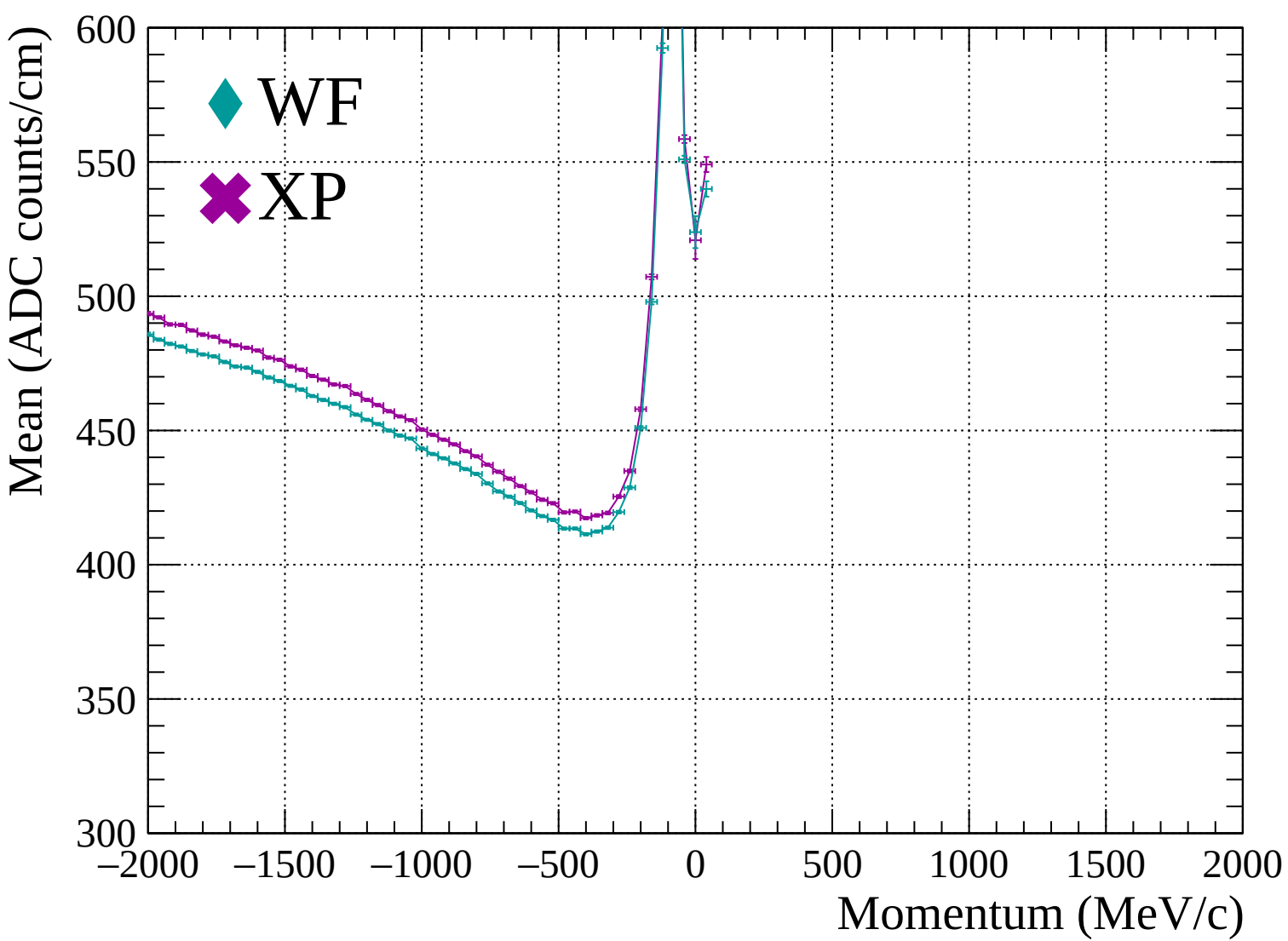


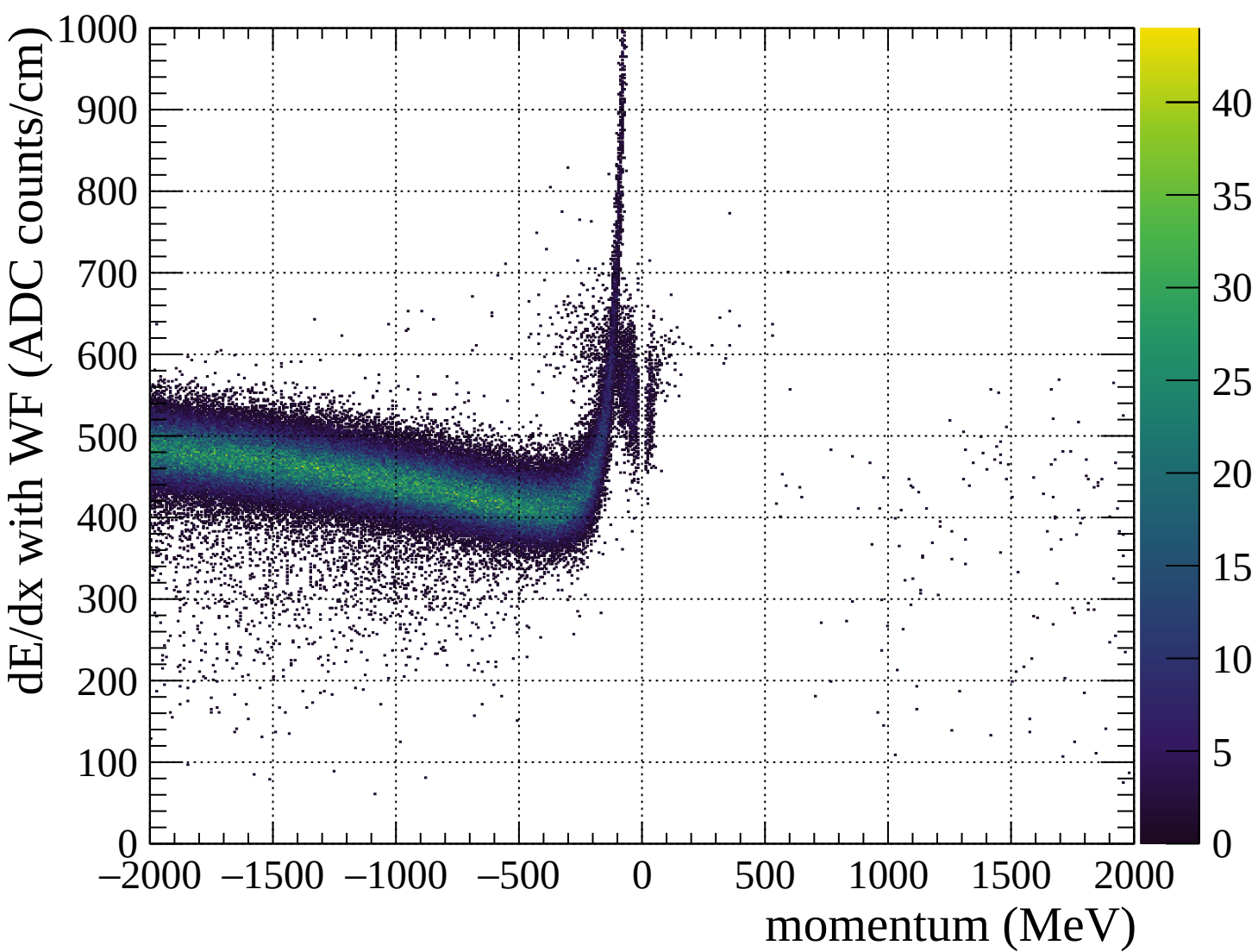


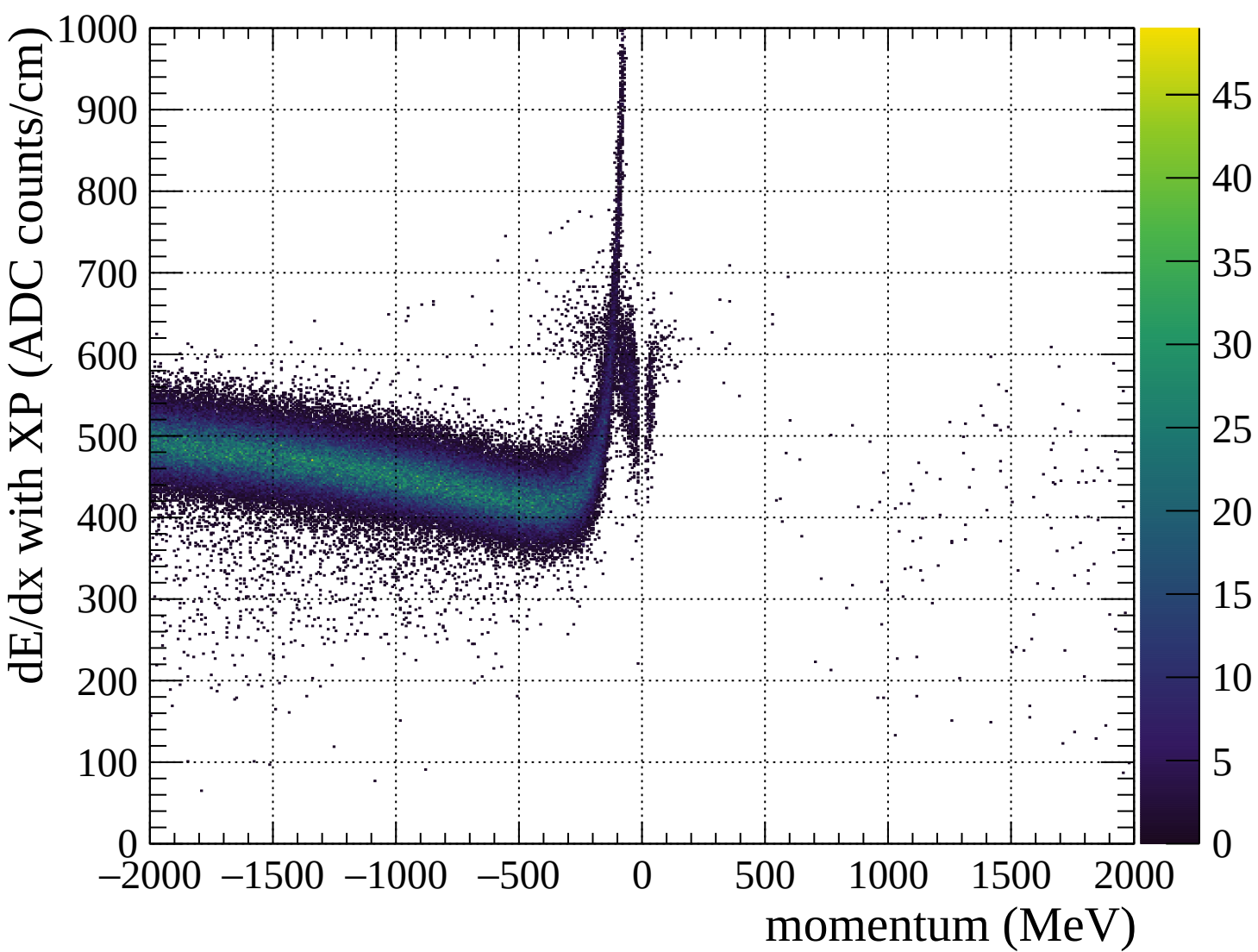


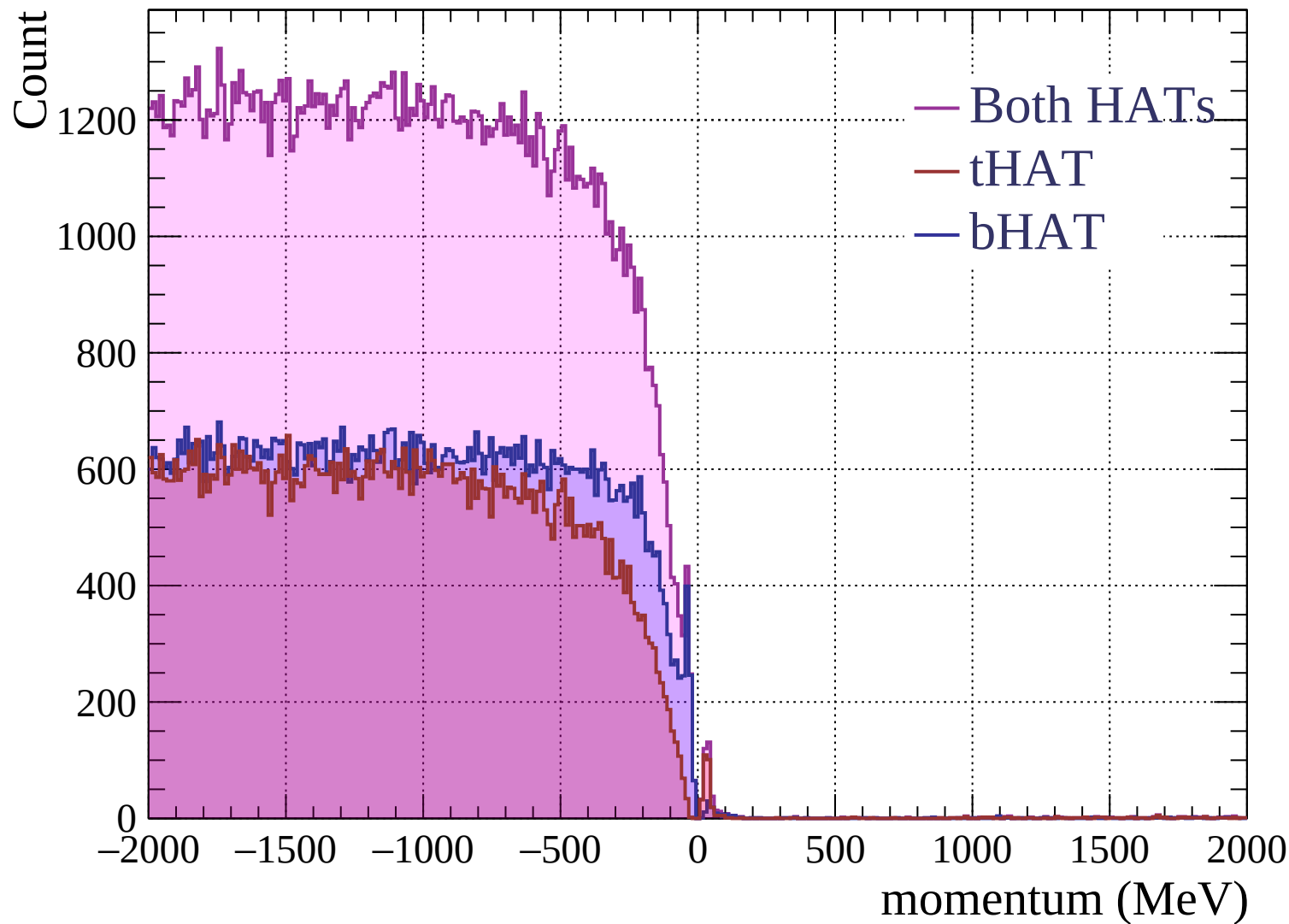


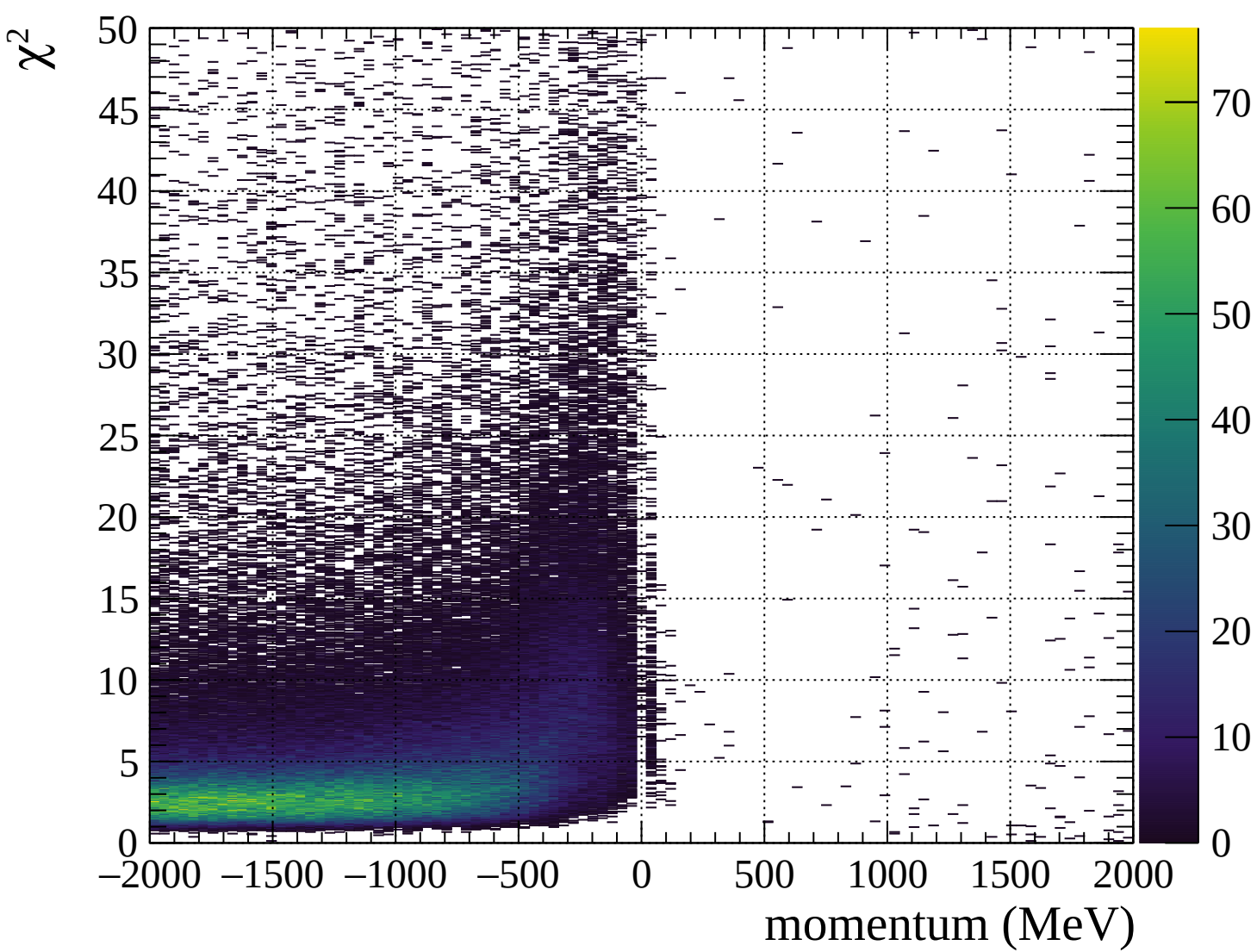




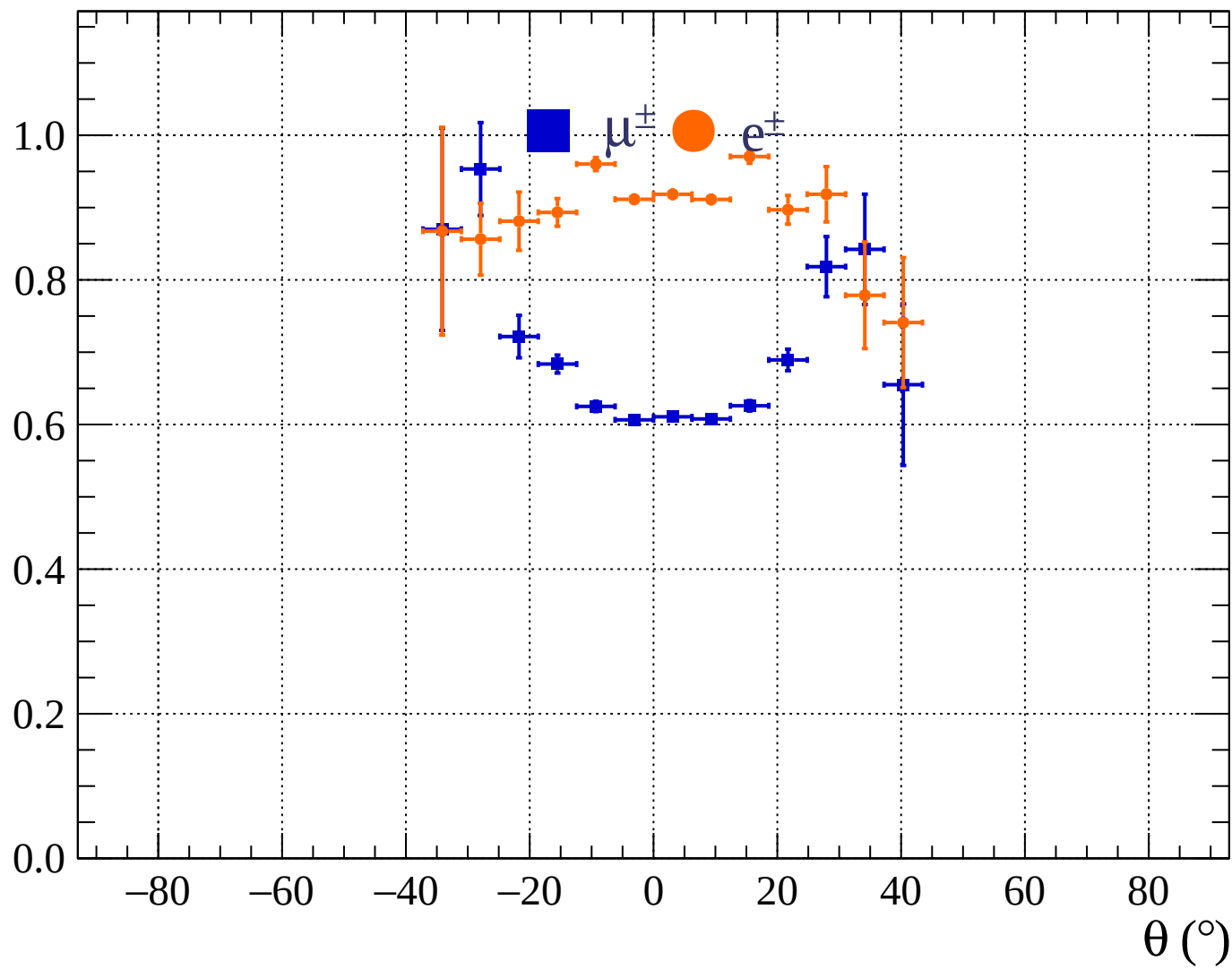




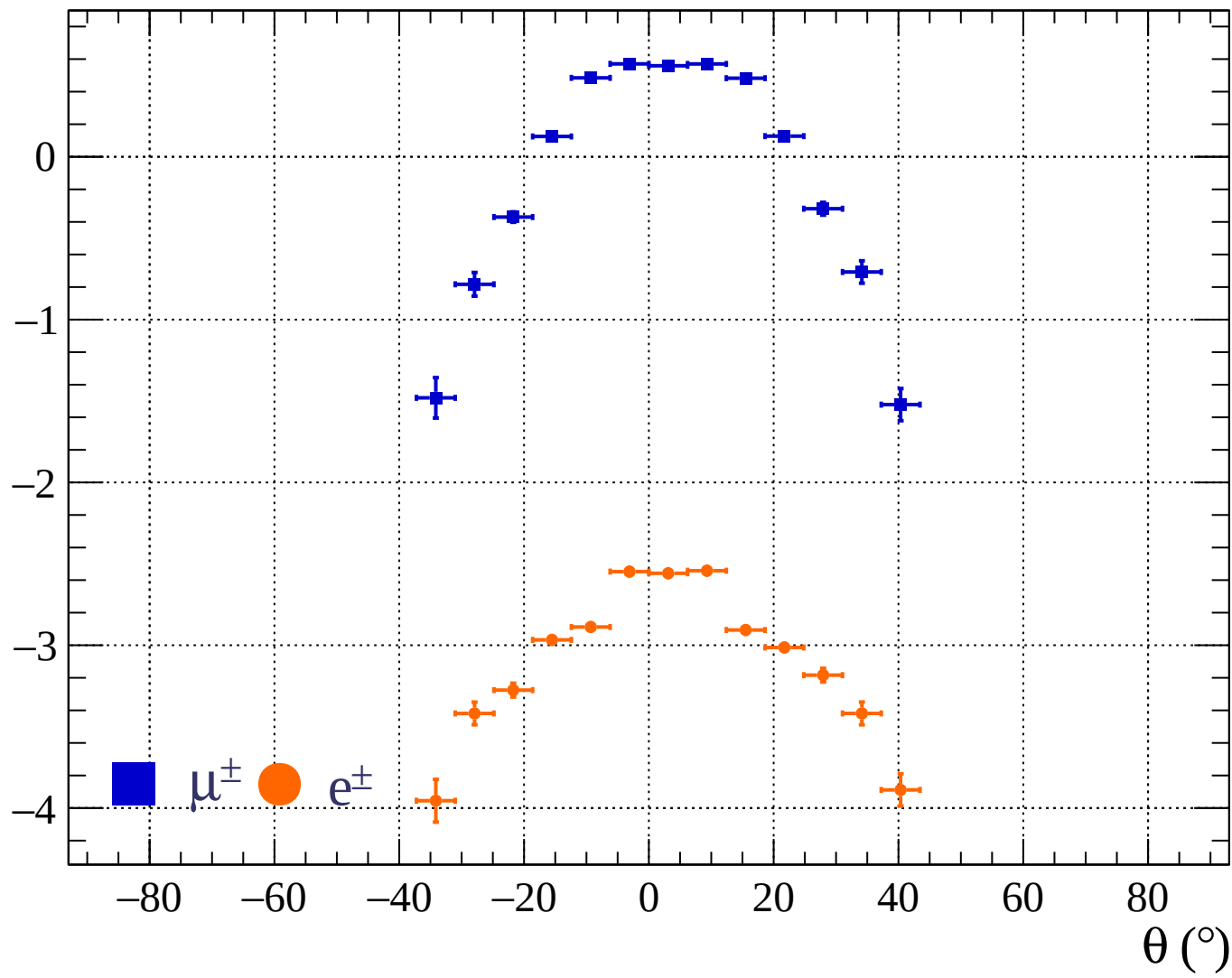




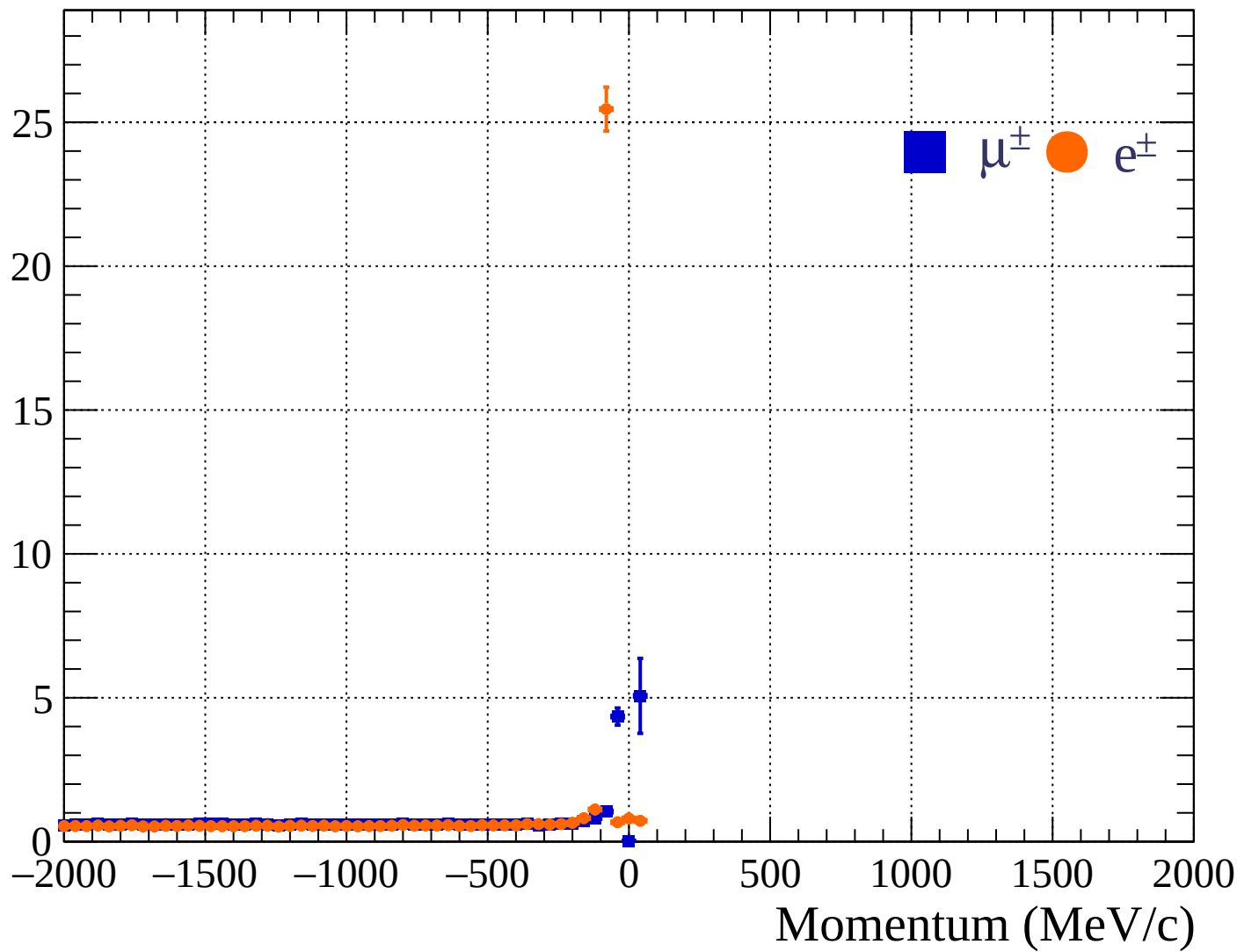
Pulls standard deviation



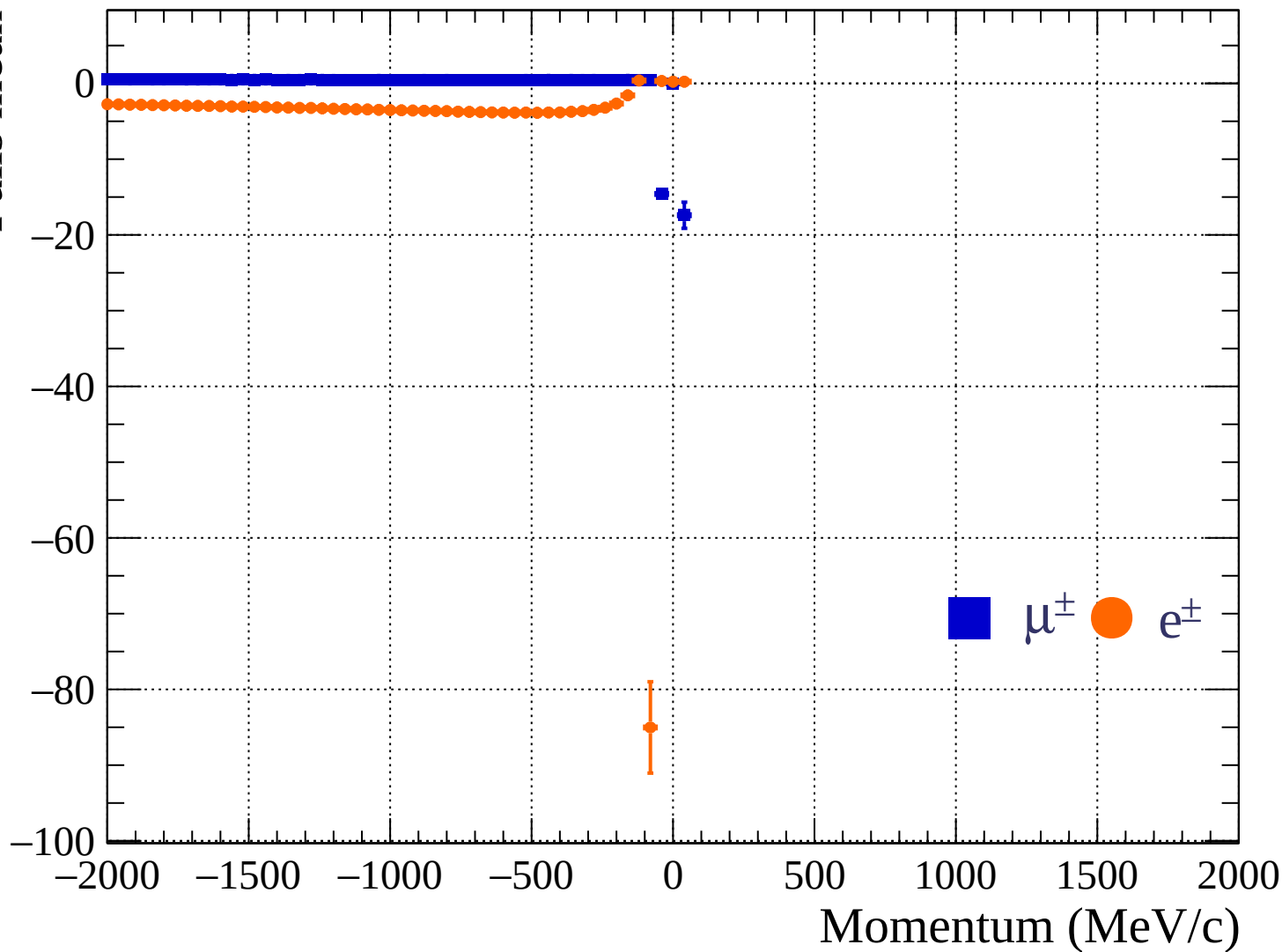
Pulls mean

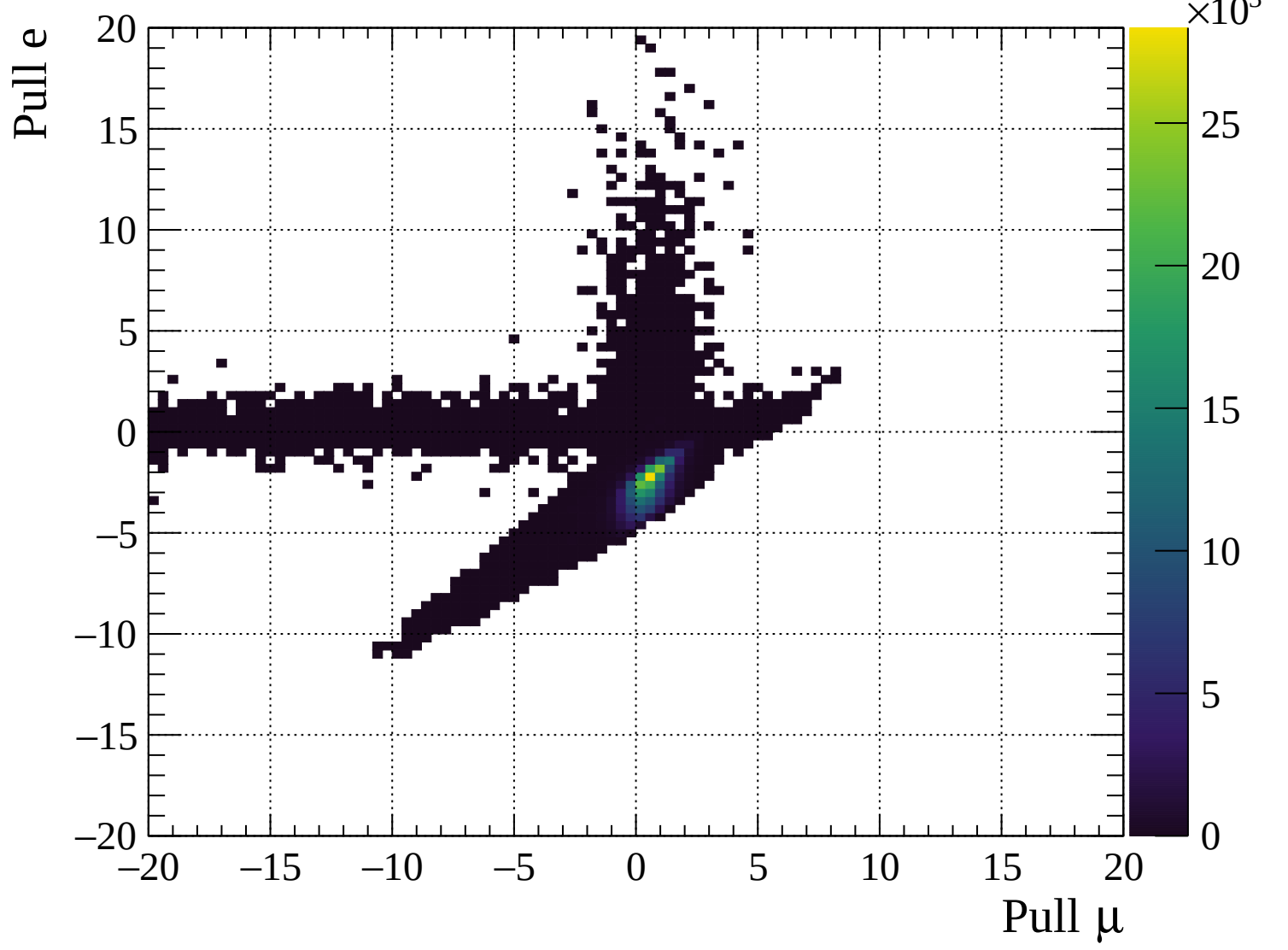


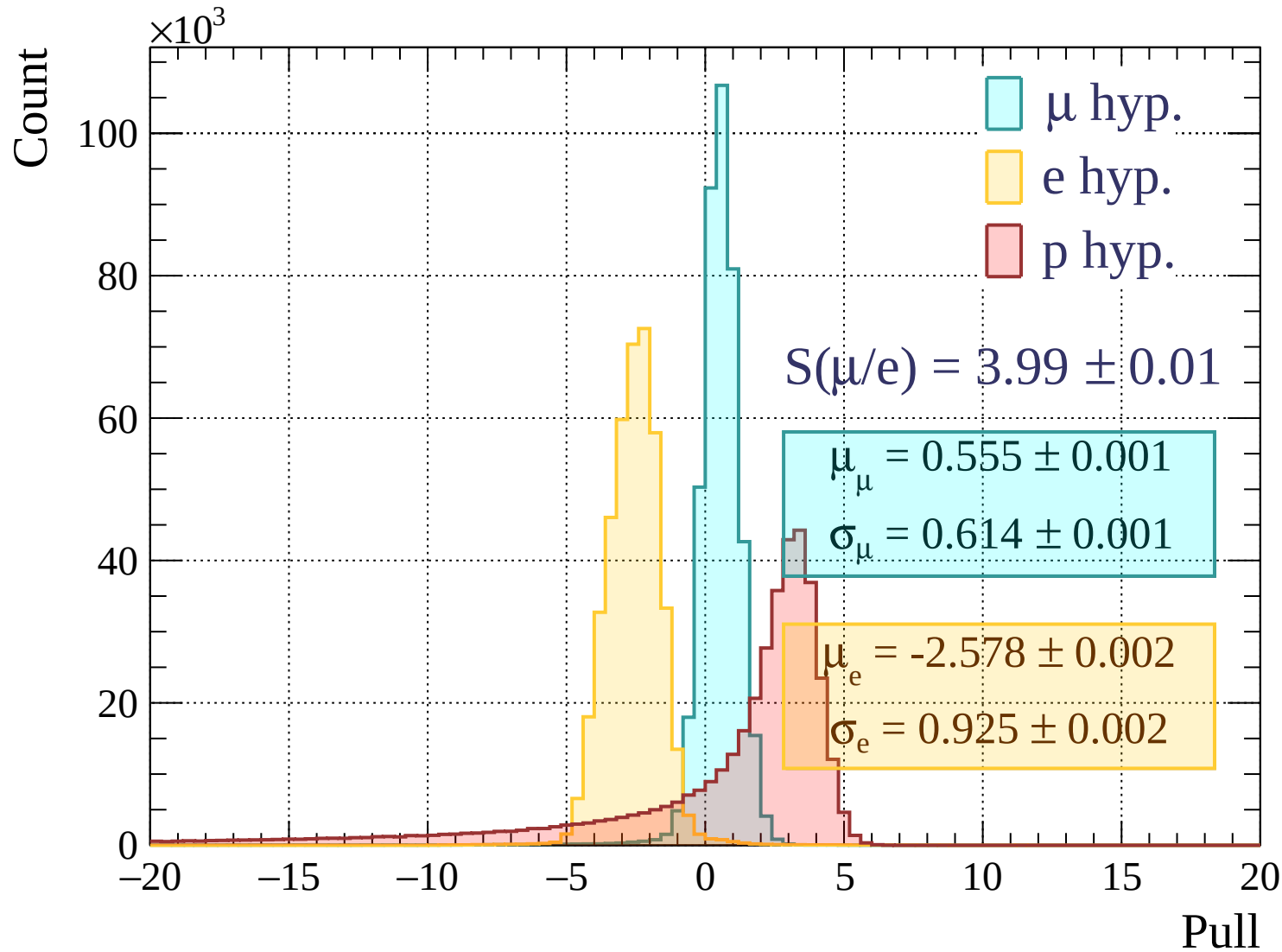
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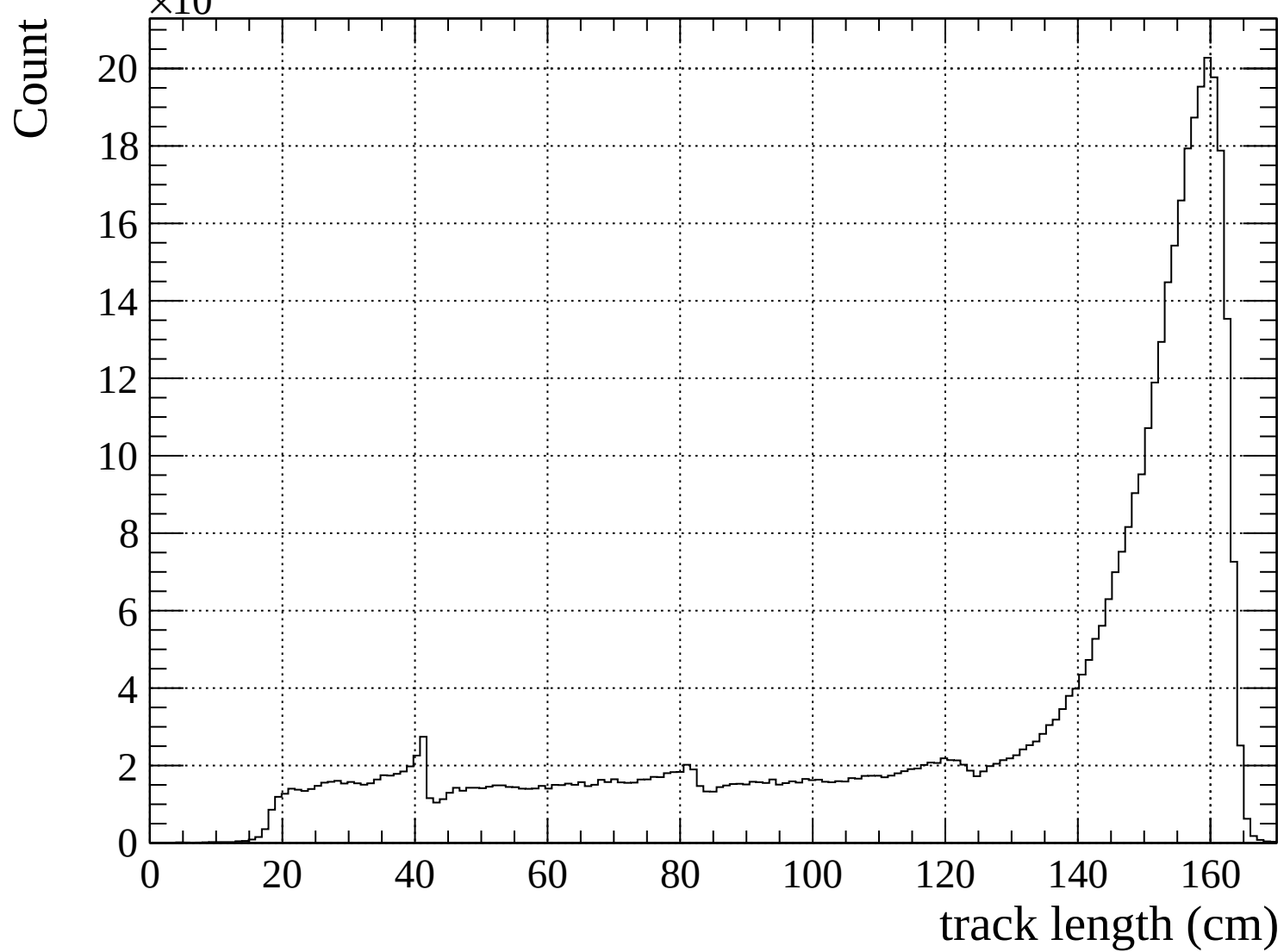


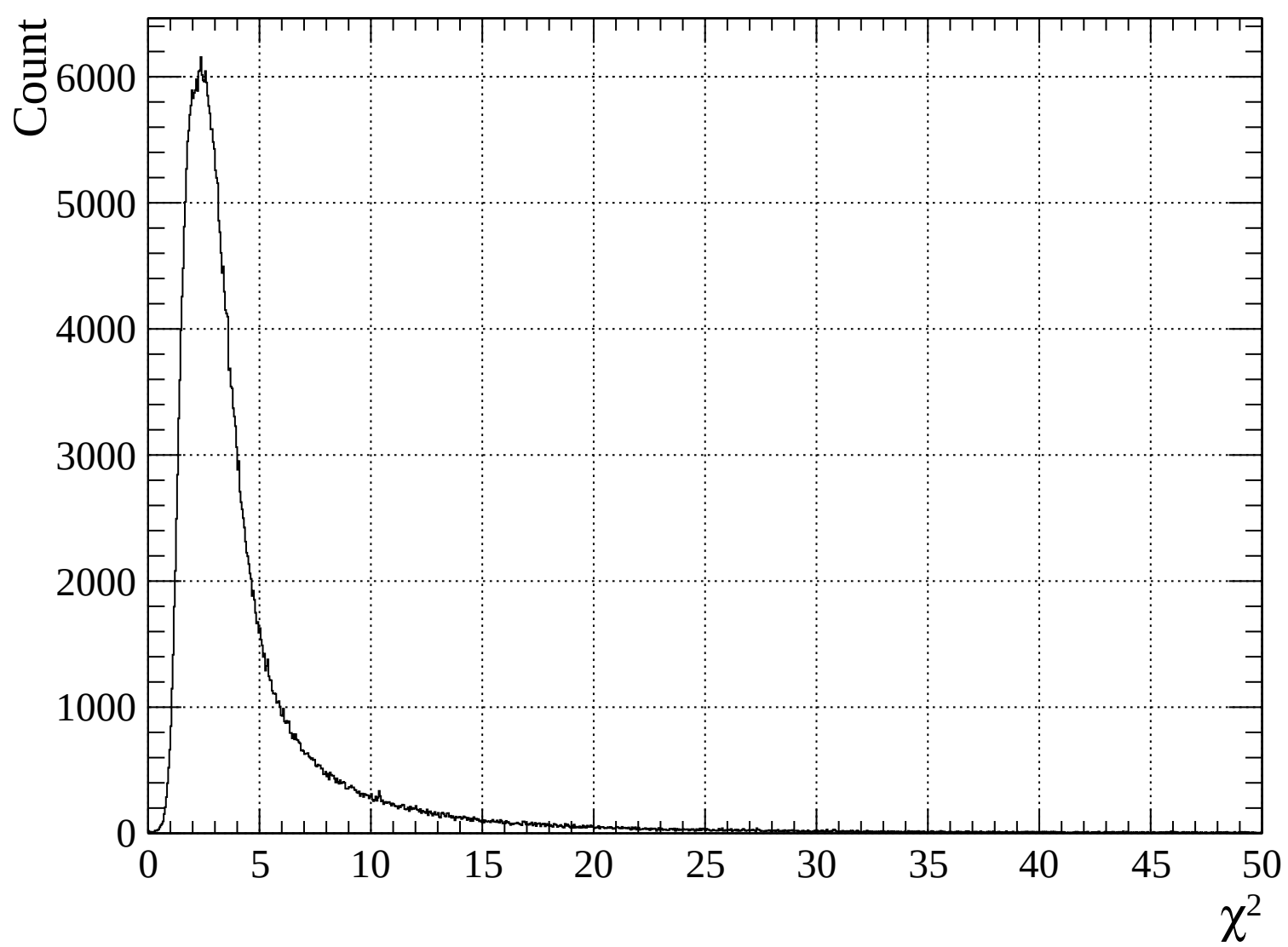
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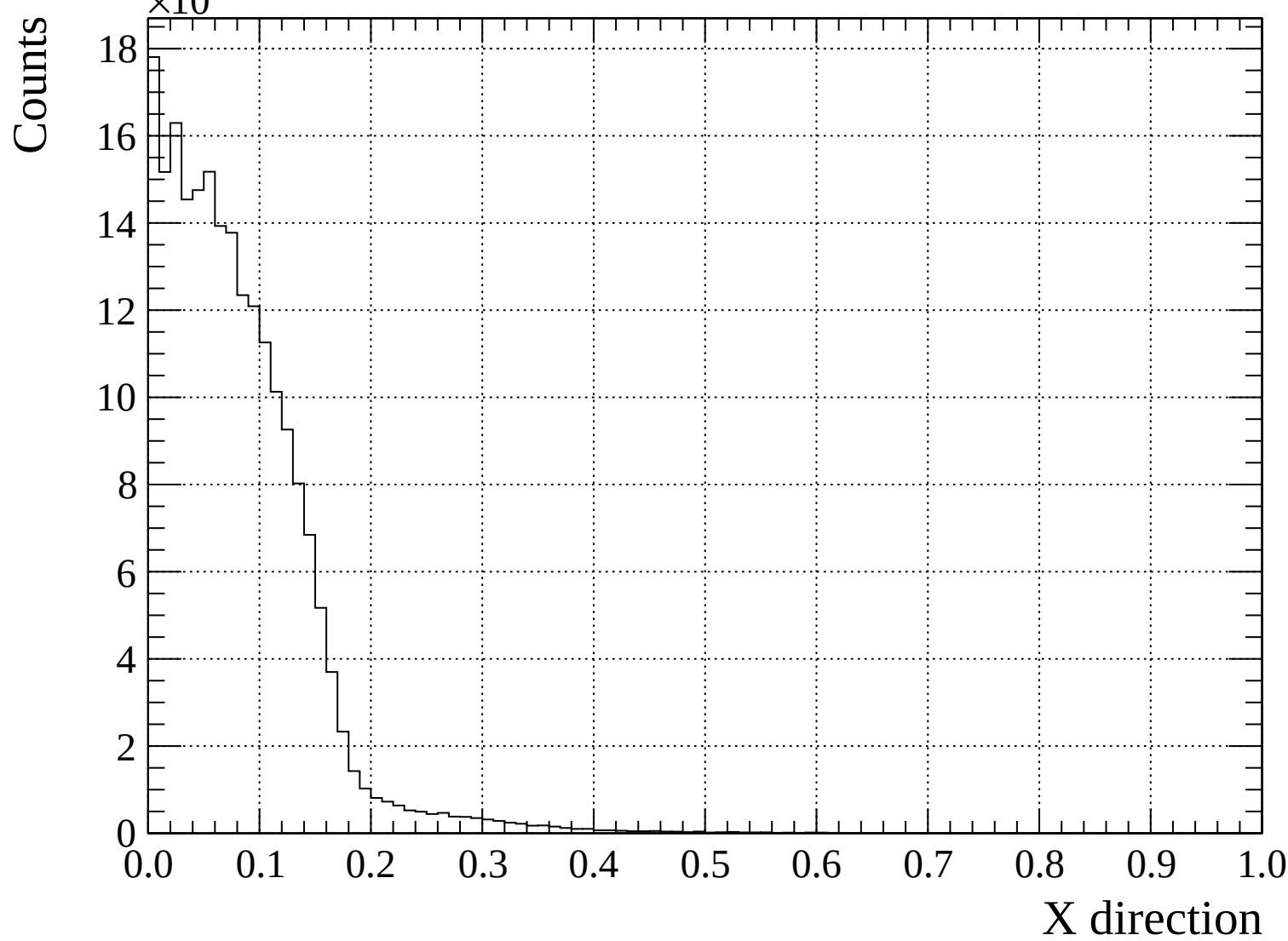


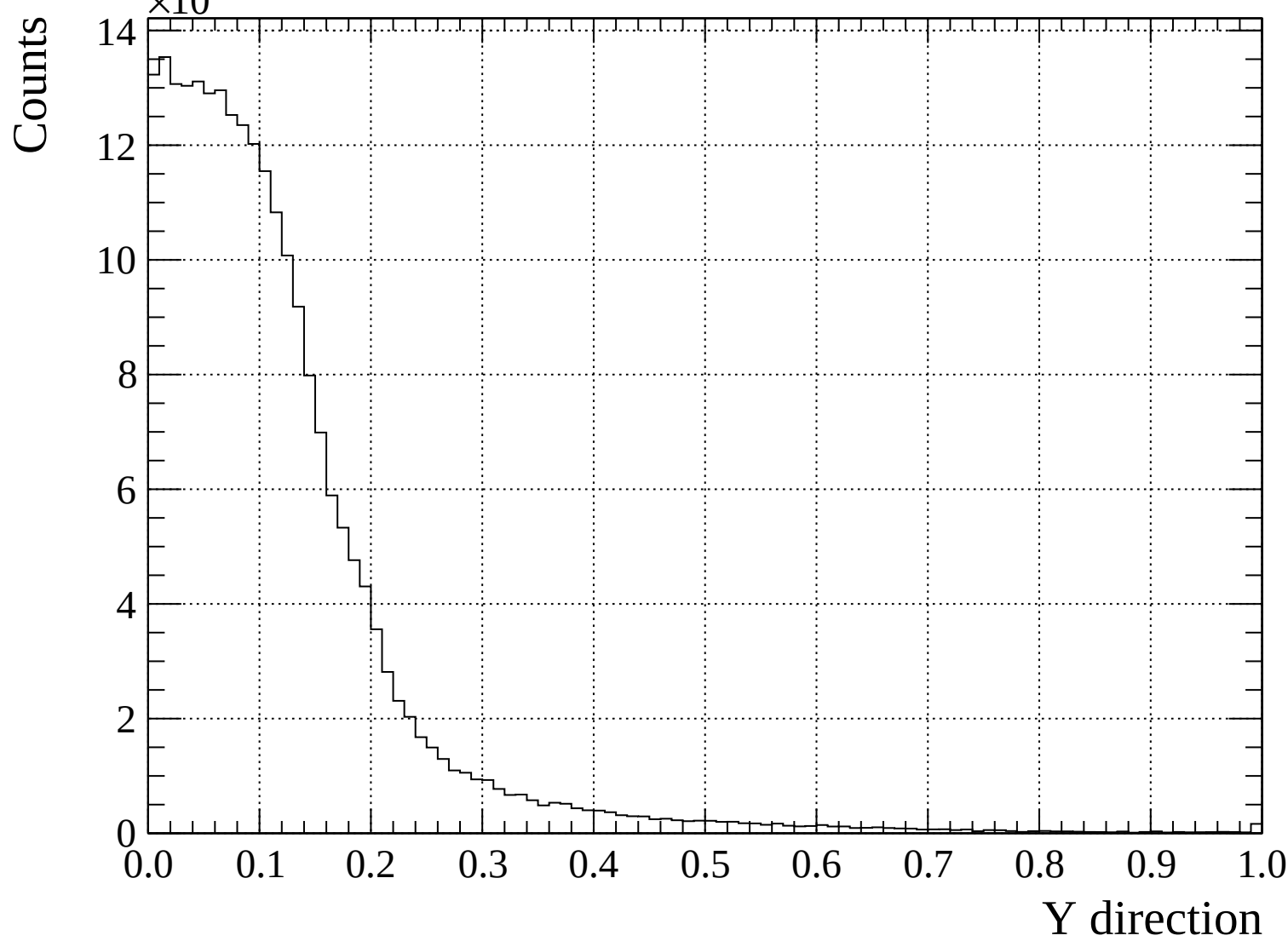


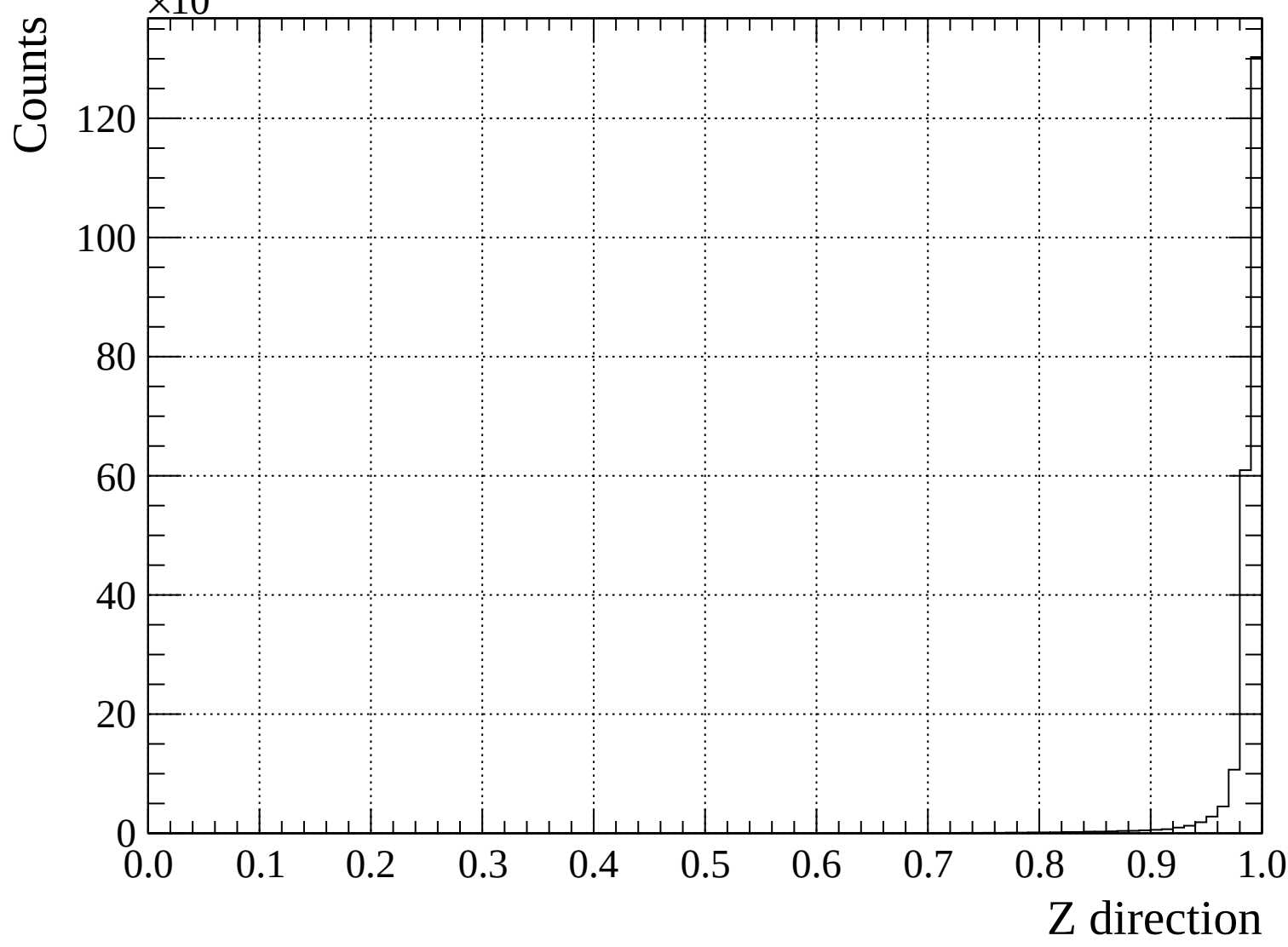


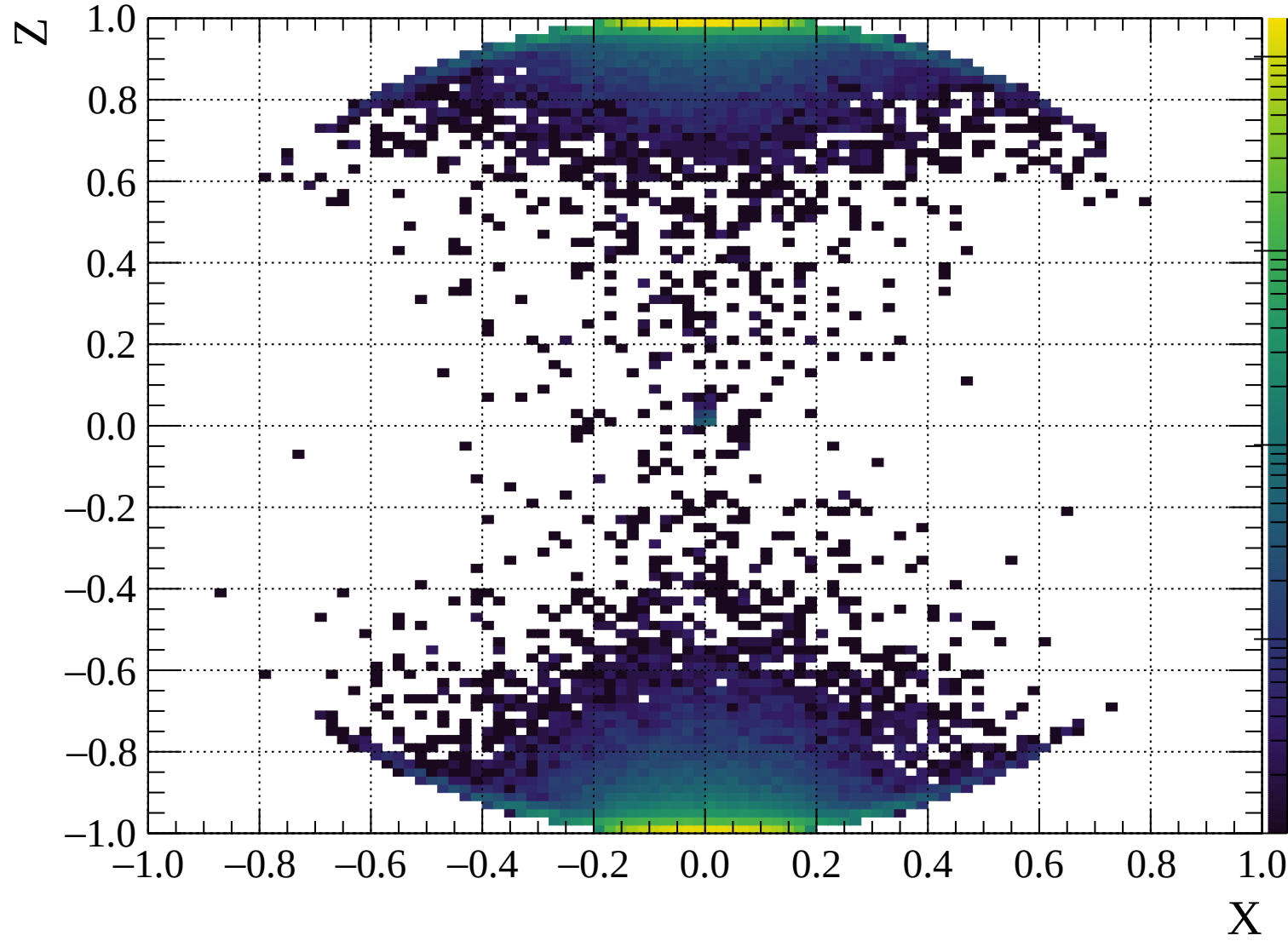


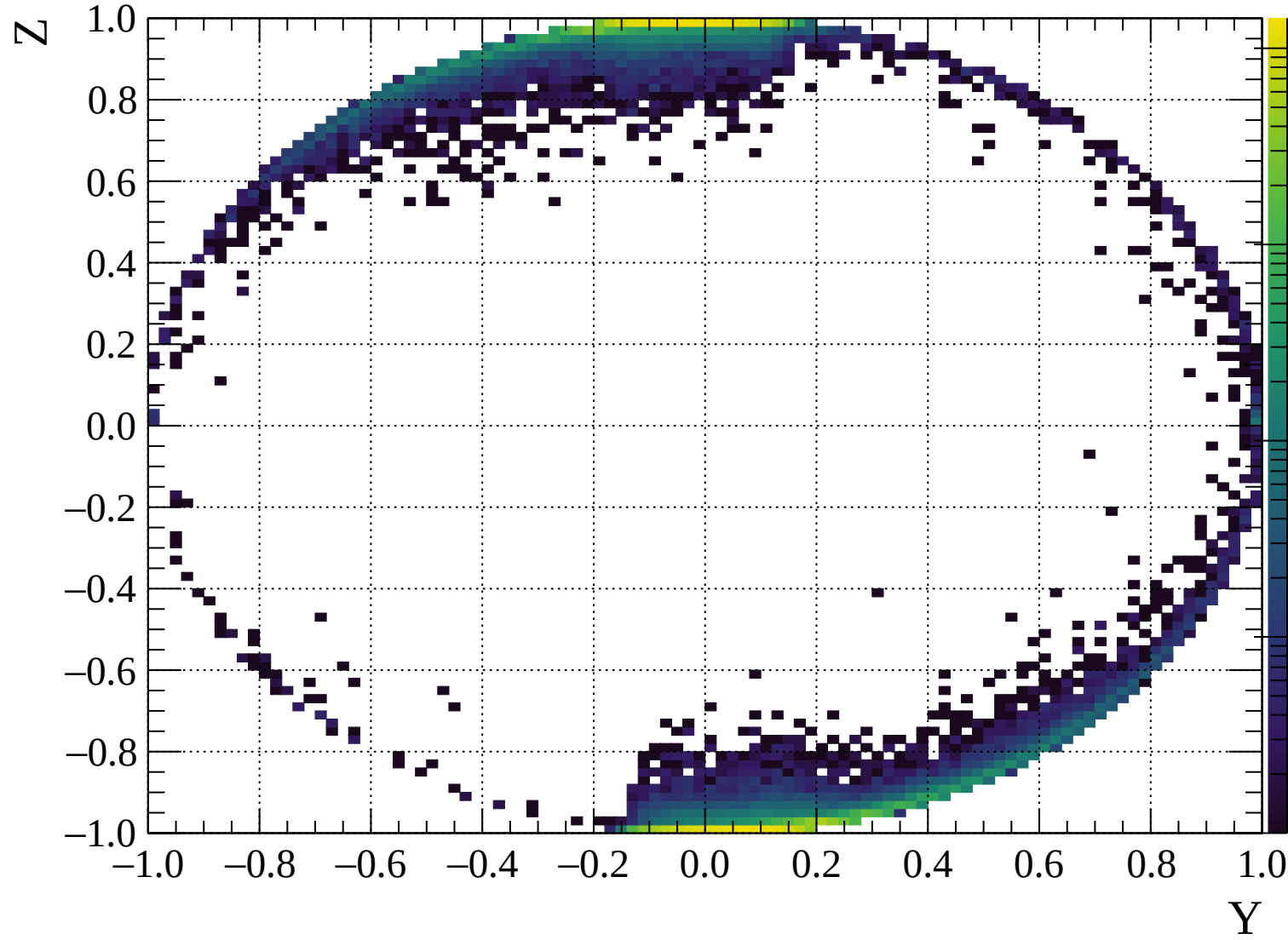


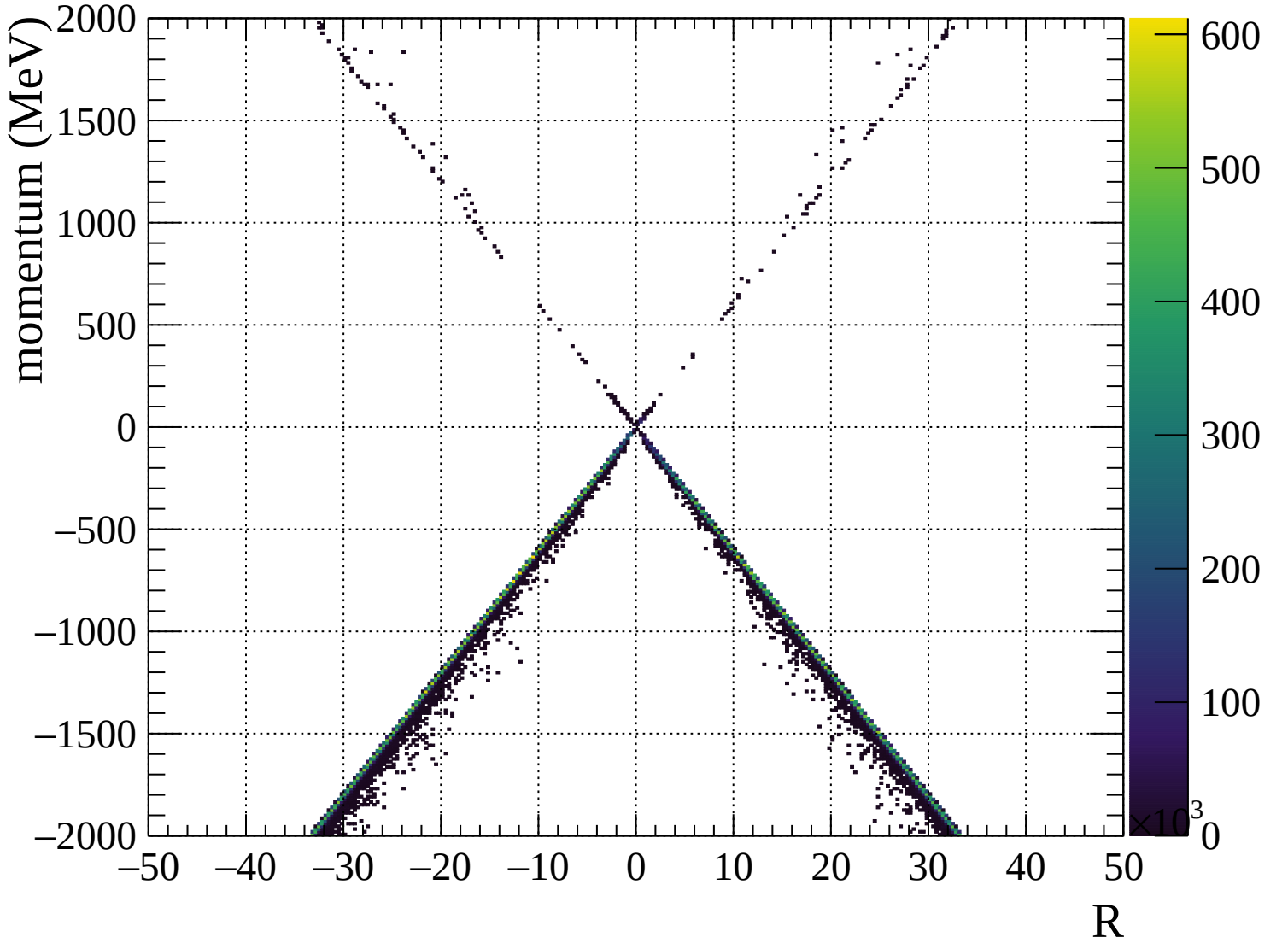


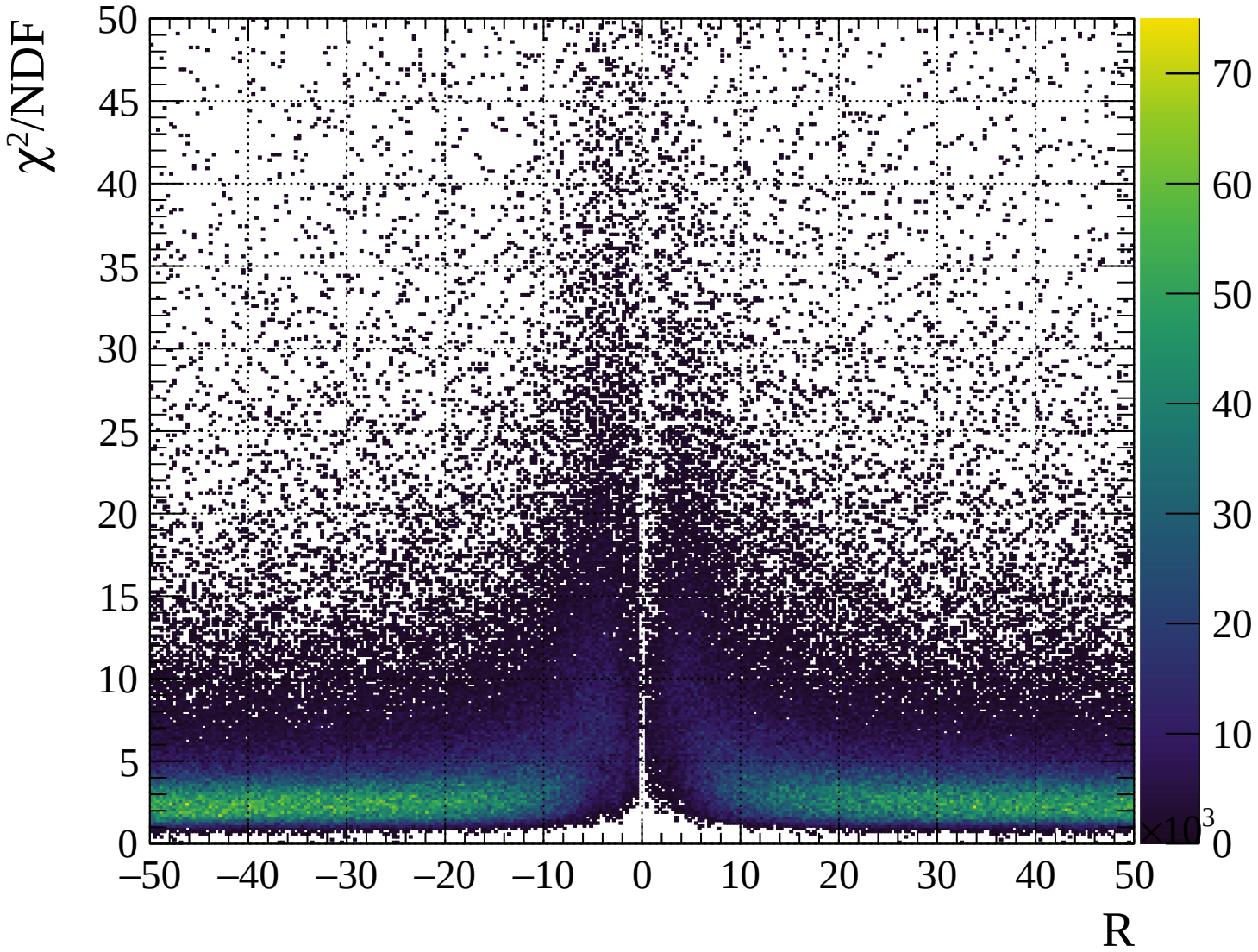




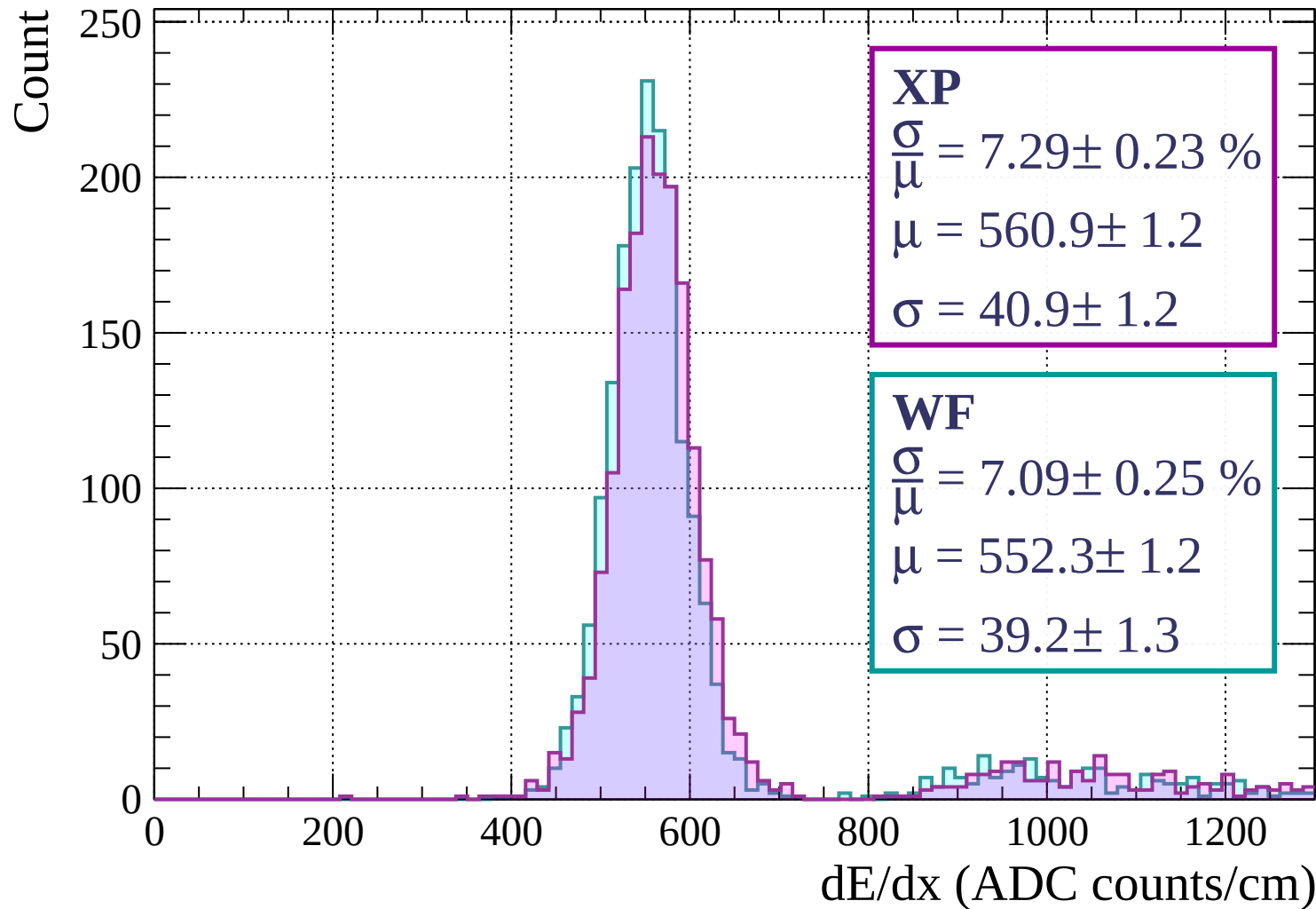






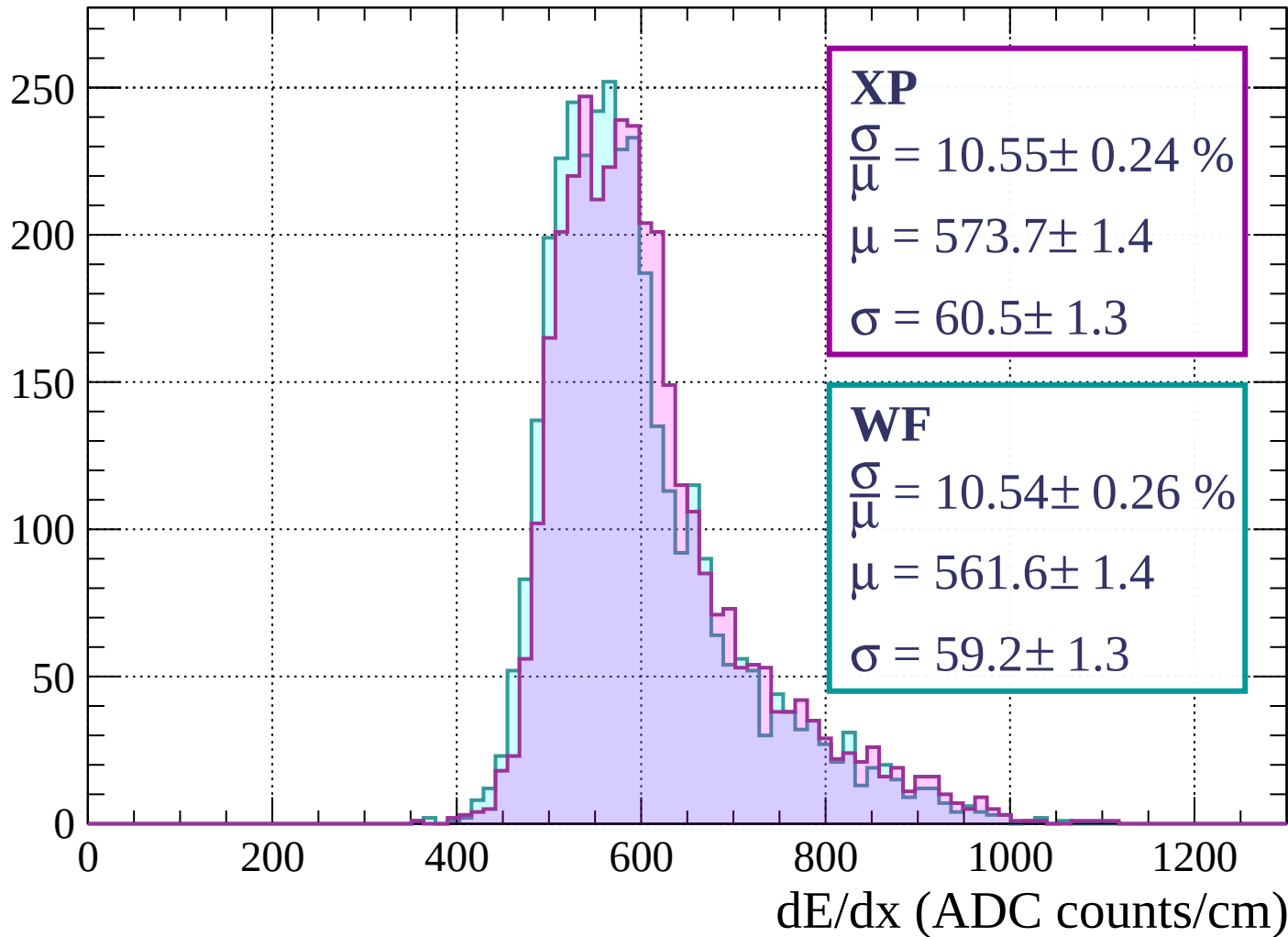


Energy loss | $0 < p < 80$

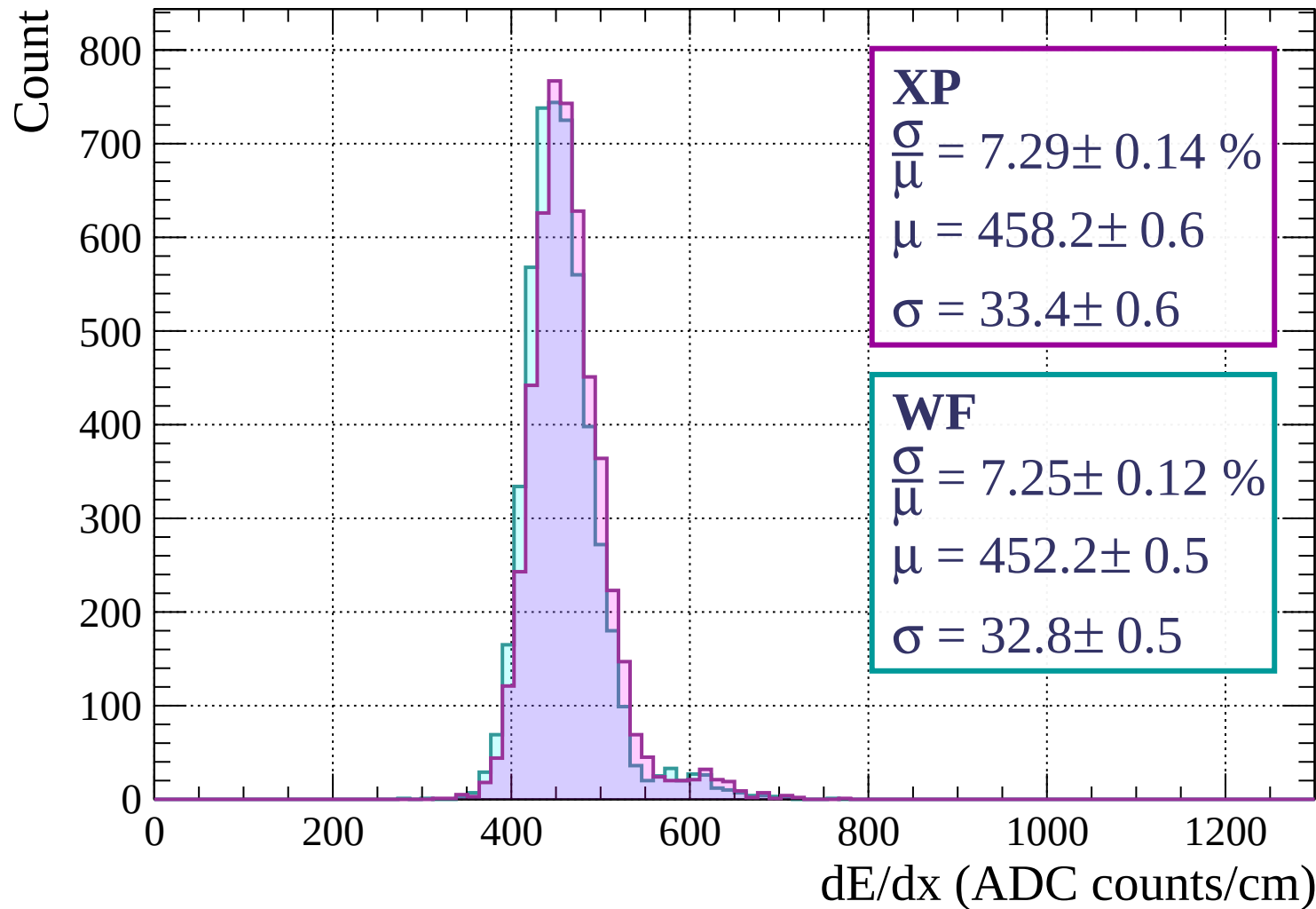


Energy loss | $80 < p < 160$

Count

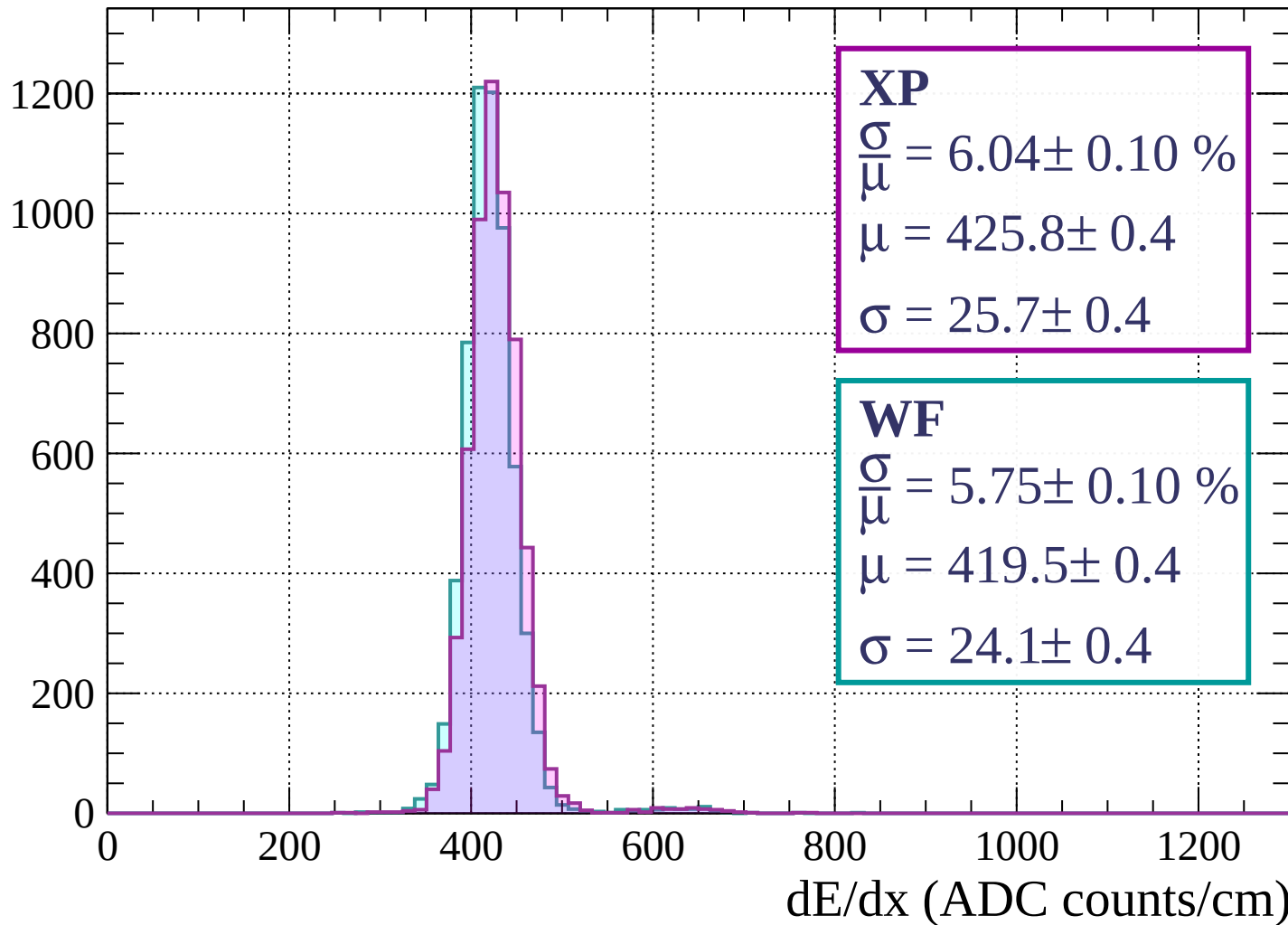


Energy loss | $160 < p < 240$

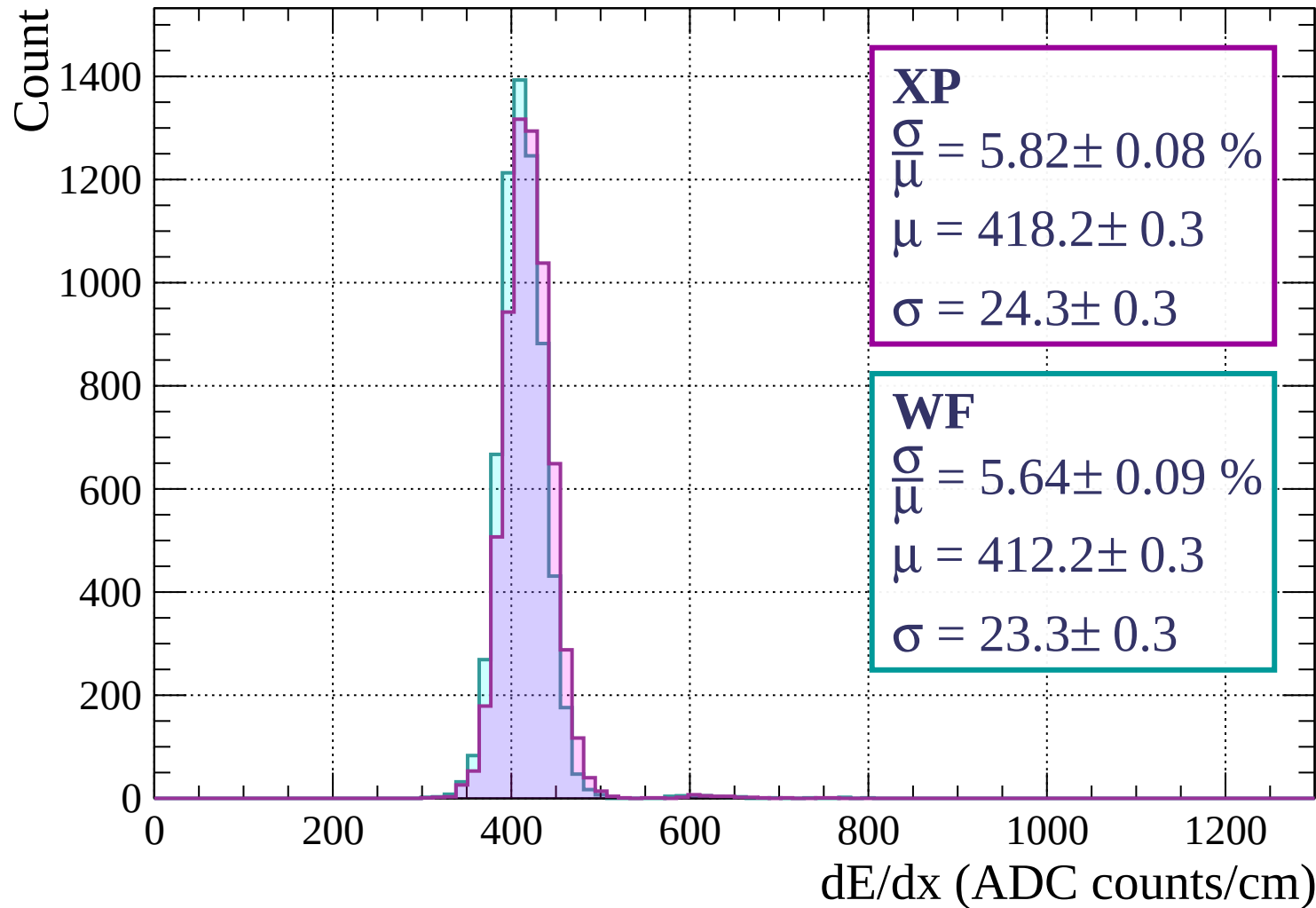


Energy loss | $240 < p < 320$

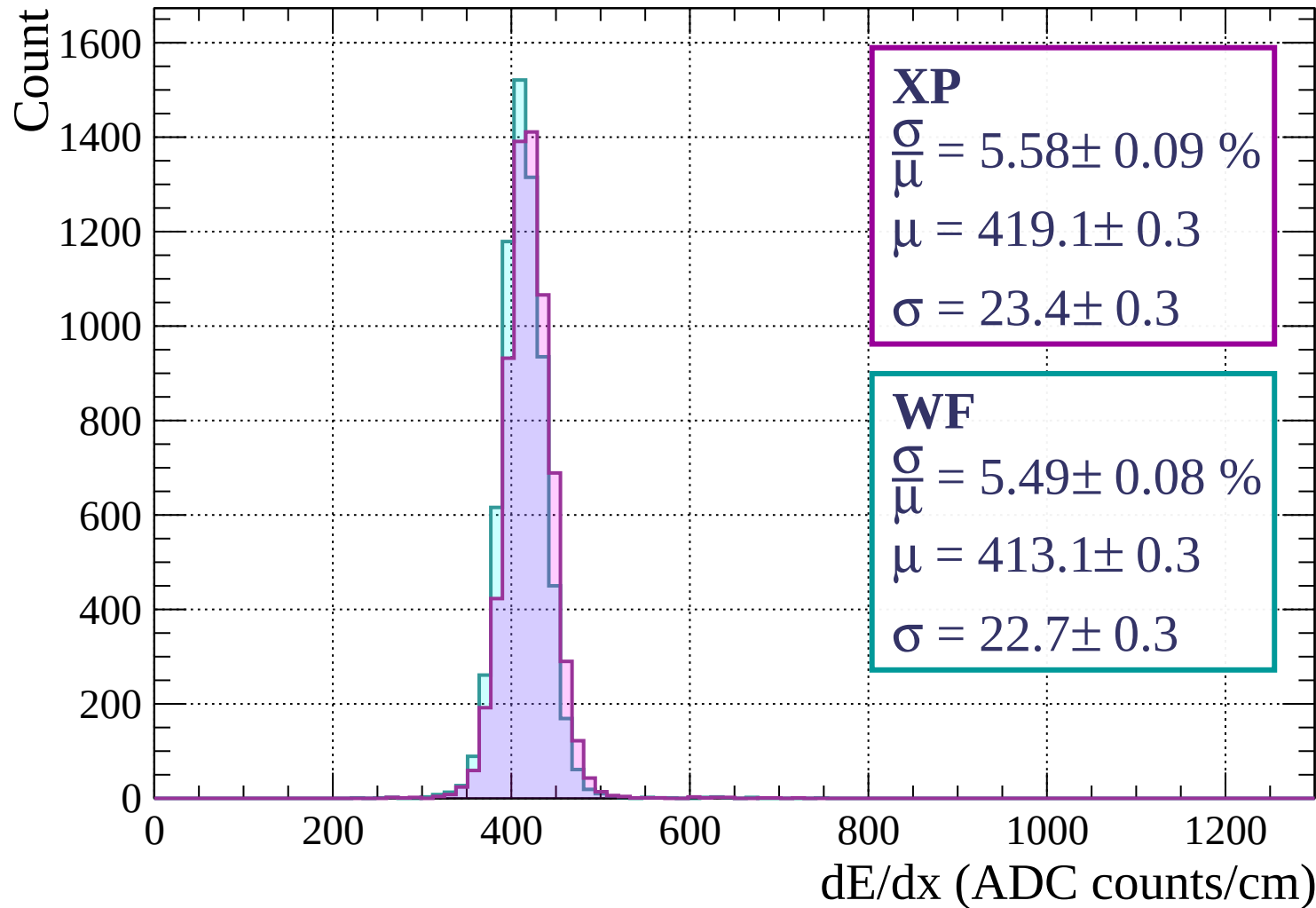
Count



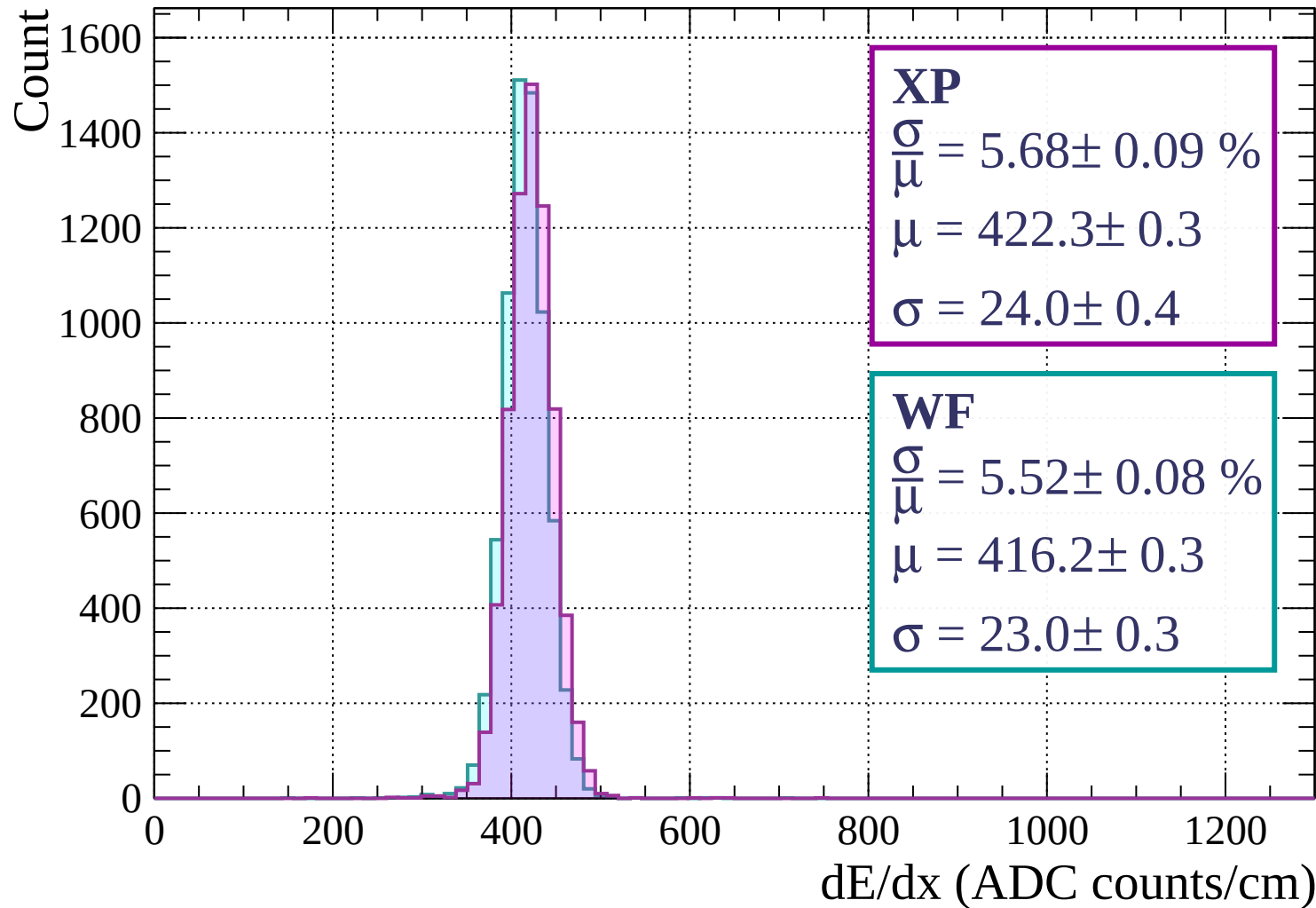
Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$



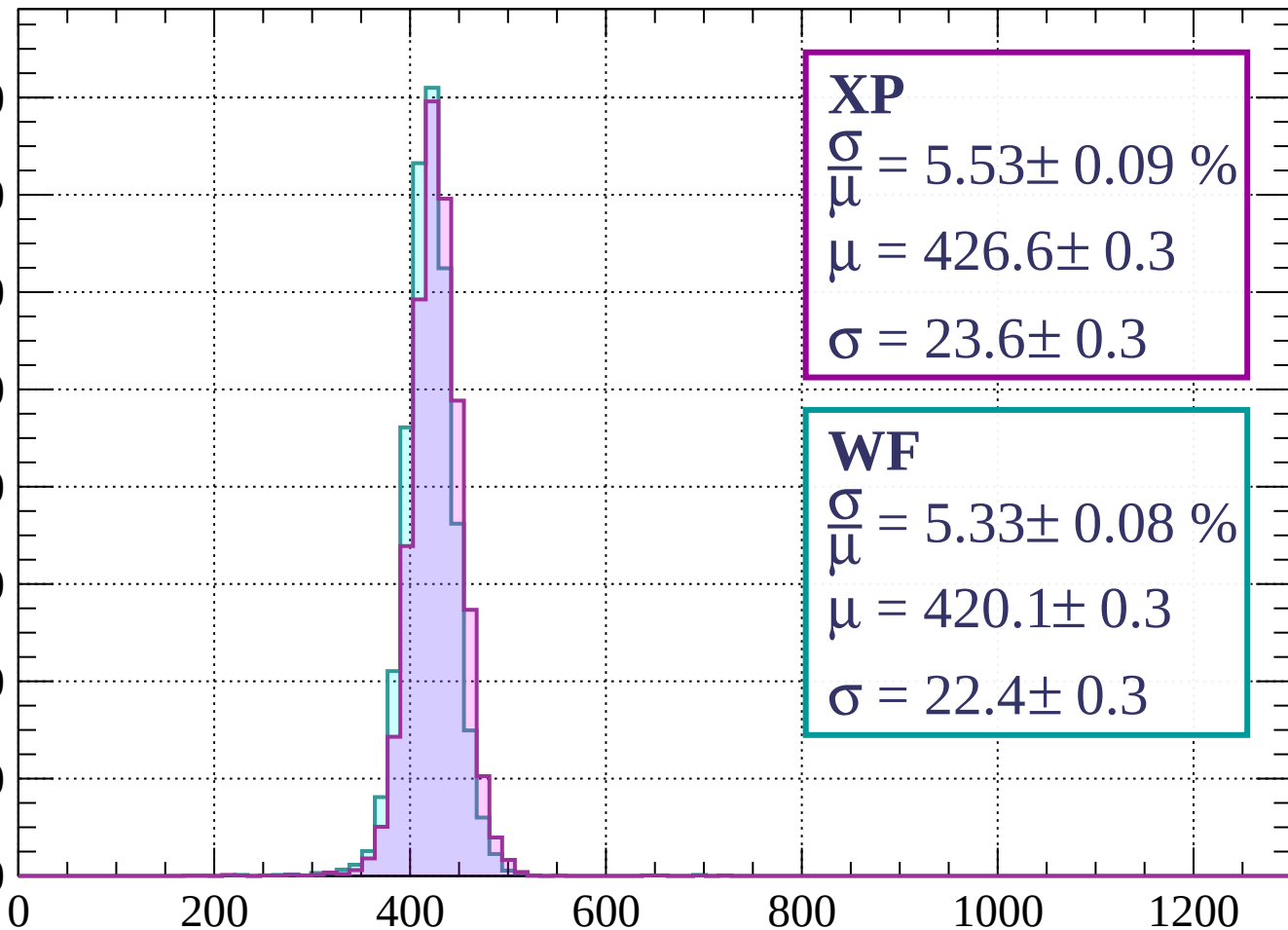
Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.53 \pm 0.09 \%$$

$$\mu = 426.6 \pm 0.3$$

$$\sigma = 23.6 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 5.33 \pm 0.08 \%$$

$$\mu = 420.1 \pm 0.3$$

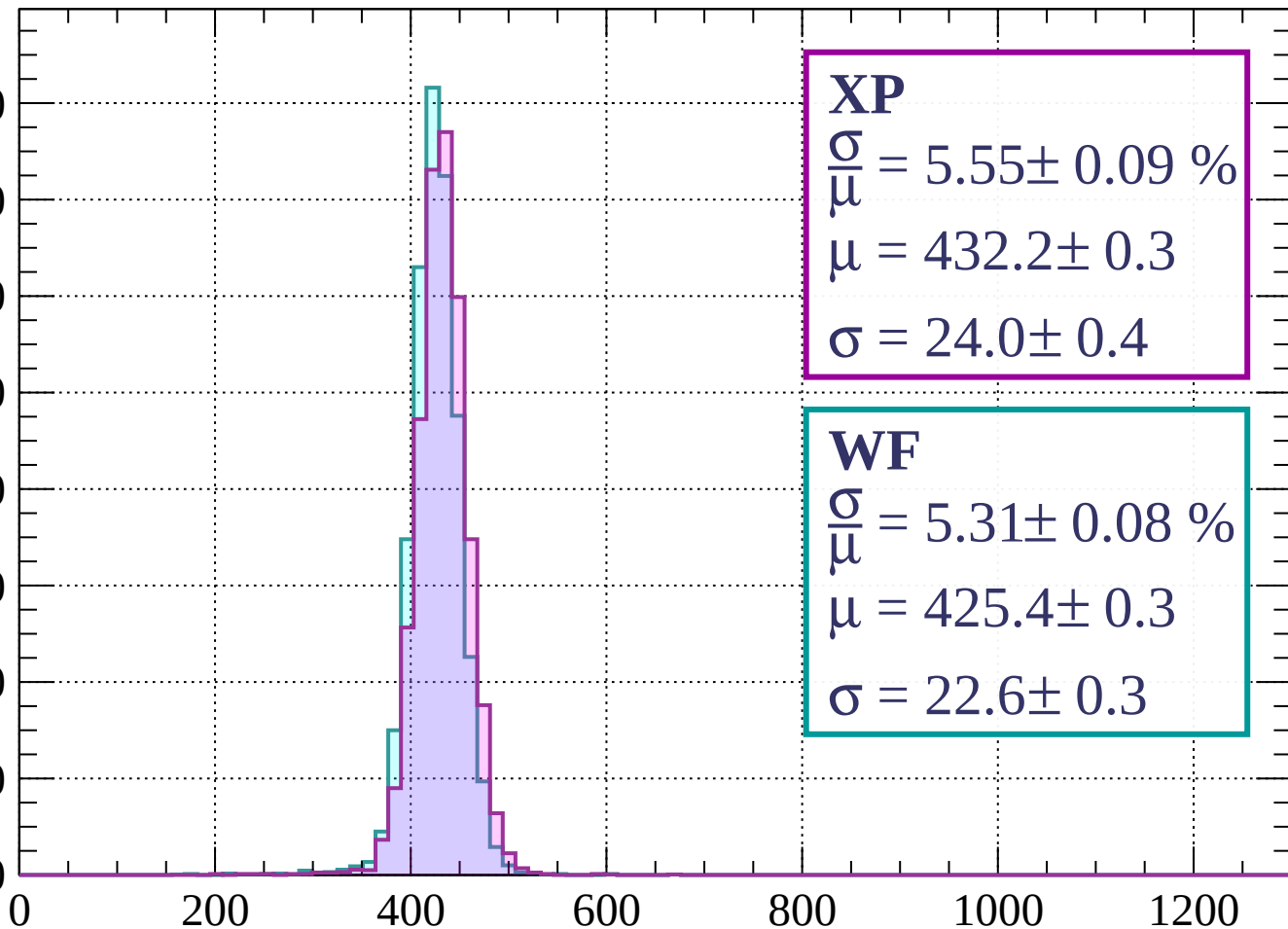
$$\sigma = 22.4 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.55 \pm 0.09 \%$$

$$\mu = 432.2 \pm 0.3$$

$$\sigma = 24.0 \pm 0.4$$

WF

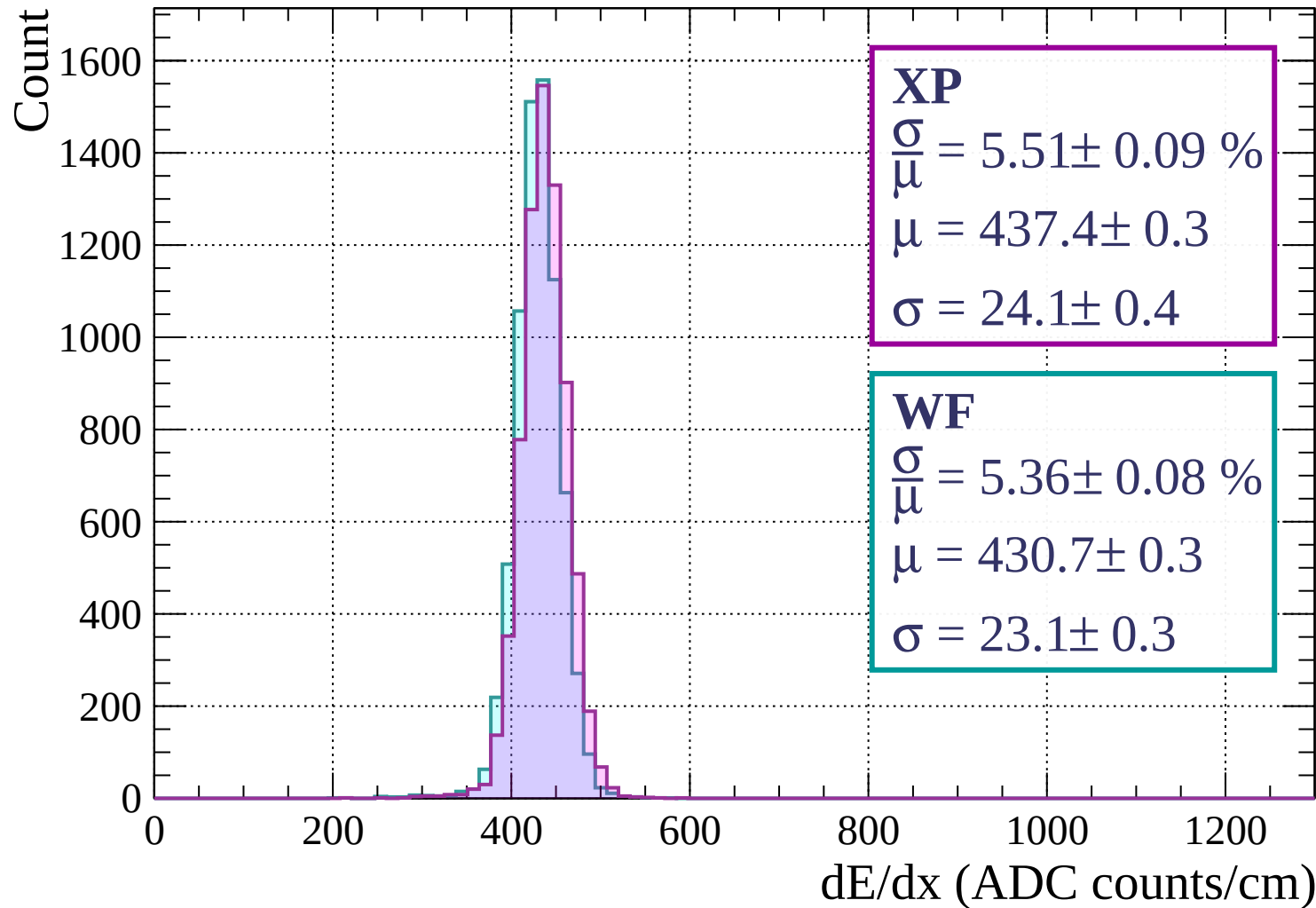
$$\frac{\sigma}{\mu} = 5.31 \pm 0.08 \%$$

$$\mu = 425.4 \pm 0.3$$

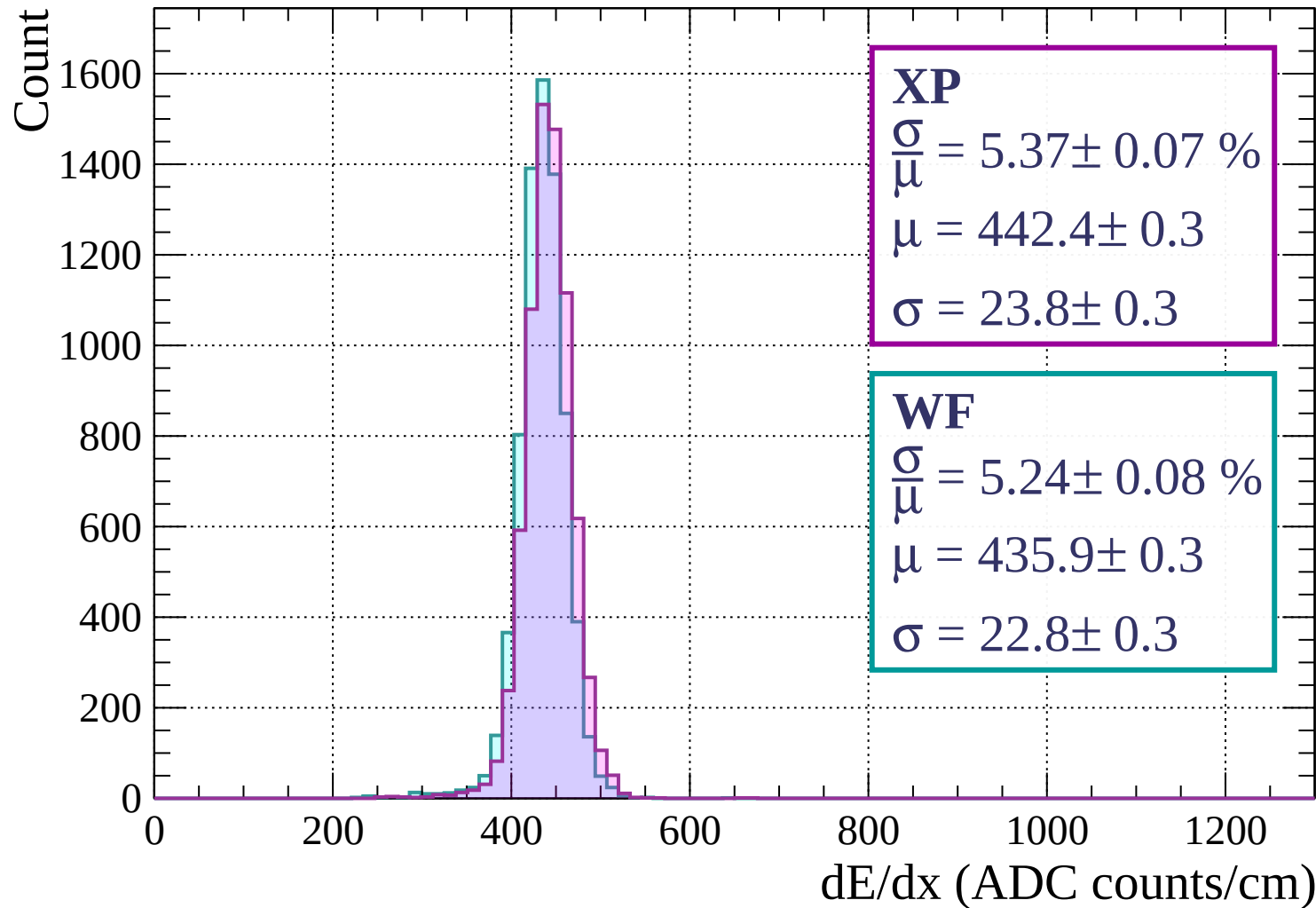
$$\sigma = 22.6 \pm 0.3$$

dE/dx (ADC counts/cm)

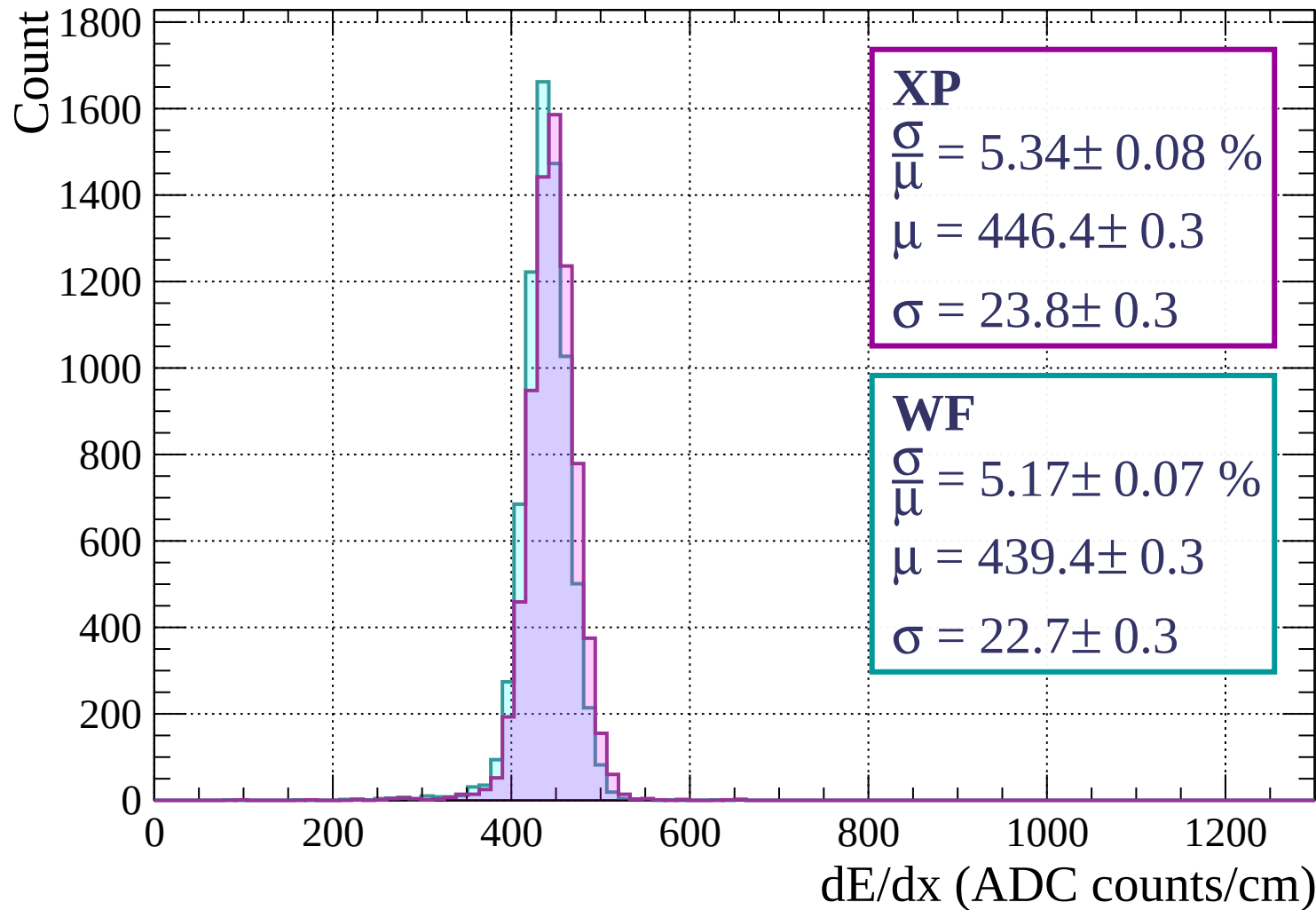
Energy loss | $720 < p < 800$



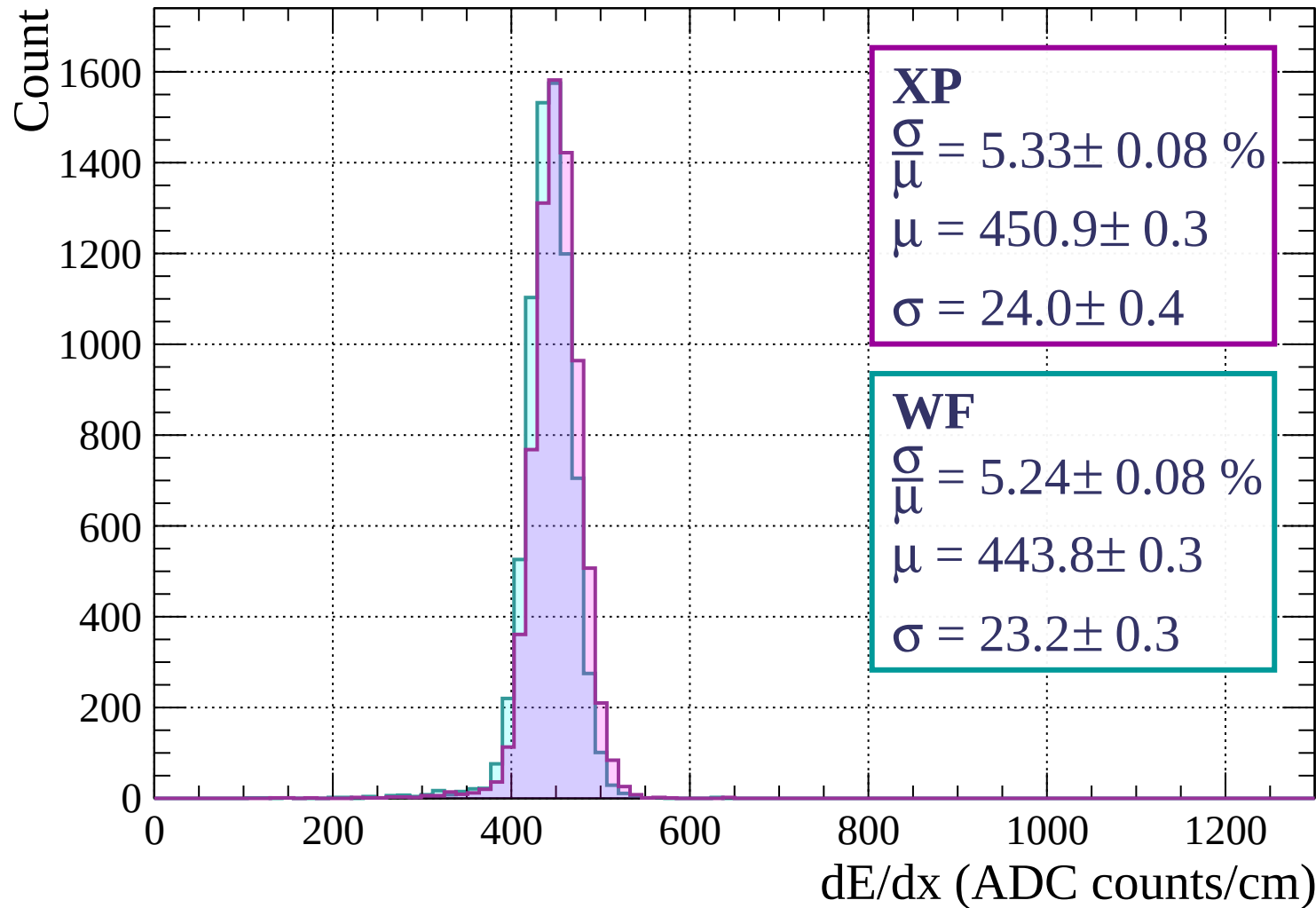
Energy loss | $800 < p < 880$



Energy loss | $880 < p < 960$



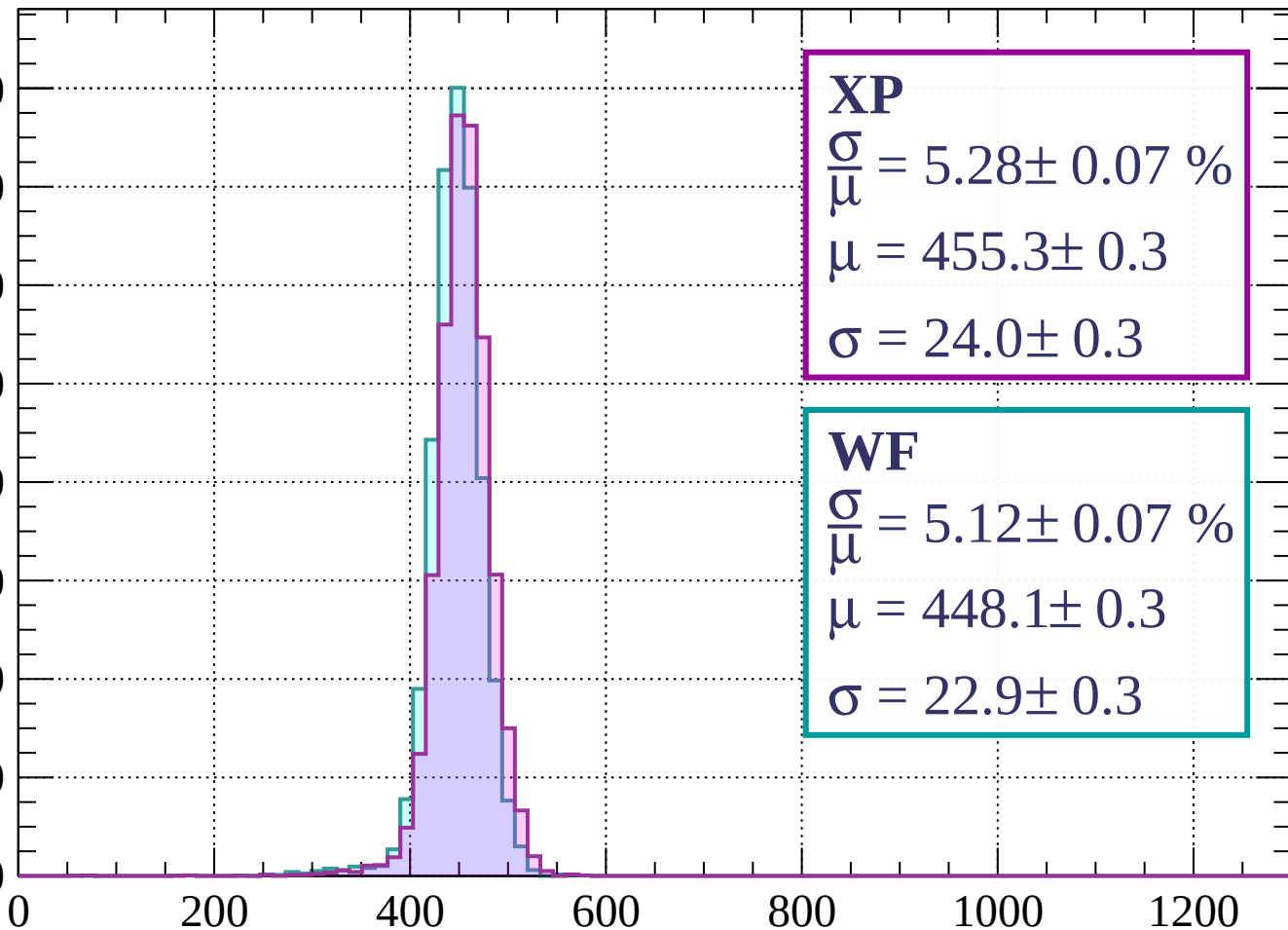
Energy loss | $960 < p < 1040$



Energy loss | $1040 < p < 1120$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.28 \pm 0.07 \%$$

$$\mu = 455.3 \pm 0.3$$

$$\sigma = 24.0 \pm 0.3$$

WF

$$\frac{\sigma}{\mu} = 5.12 \pm 0.07 \%$$

$$\mu = 448.1 \pm 0.3$$

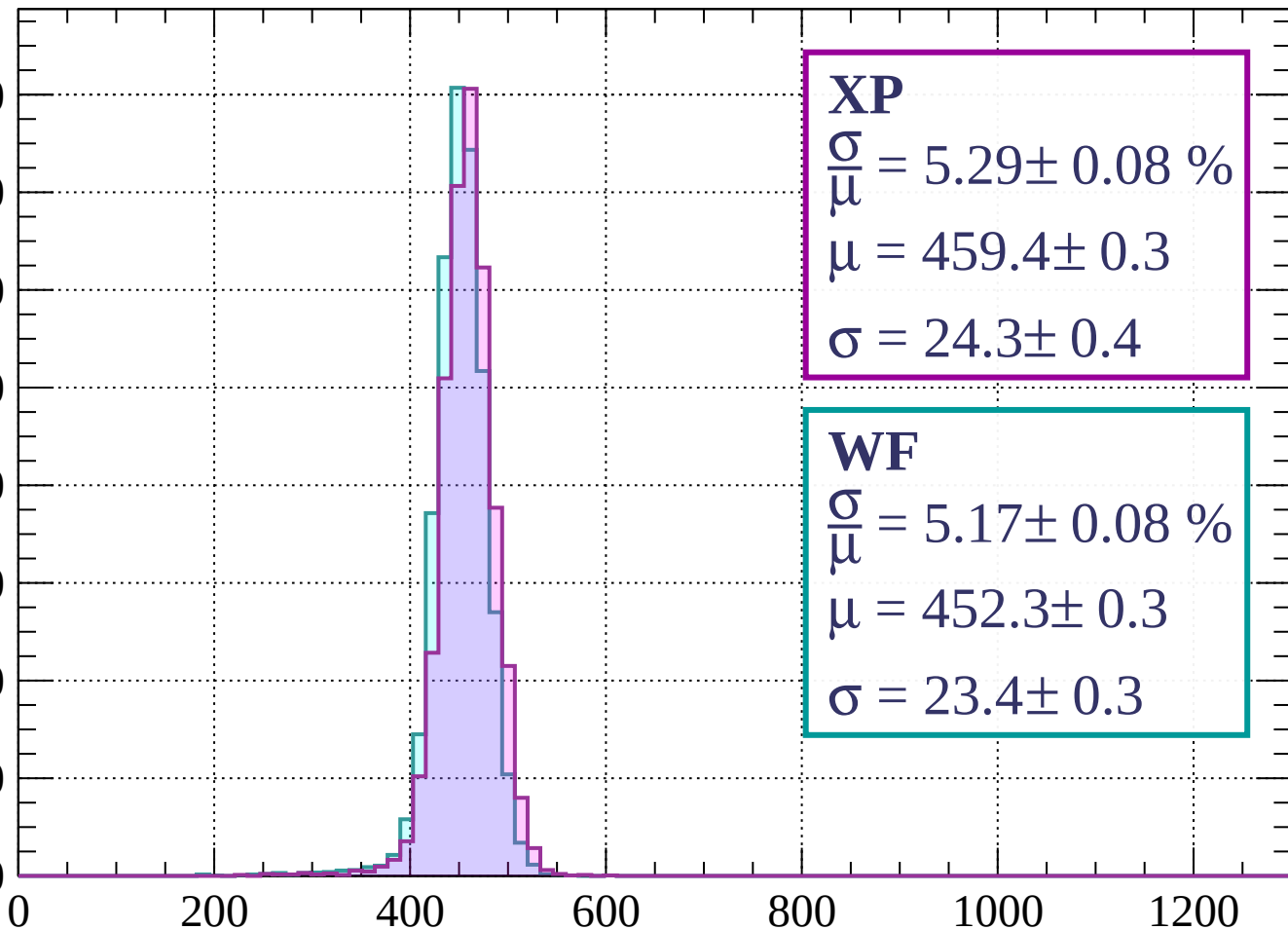
$$\sigma = 22.9 \pm 0.3$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count

1600
1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 5.29 \pm 0.08 \%$$

$$\mu = 459.4 \pm 0.3$$

$$\sigma = 24.3 \pm 0.4$$

WF

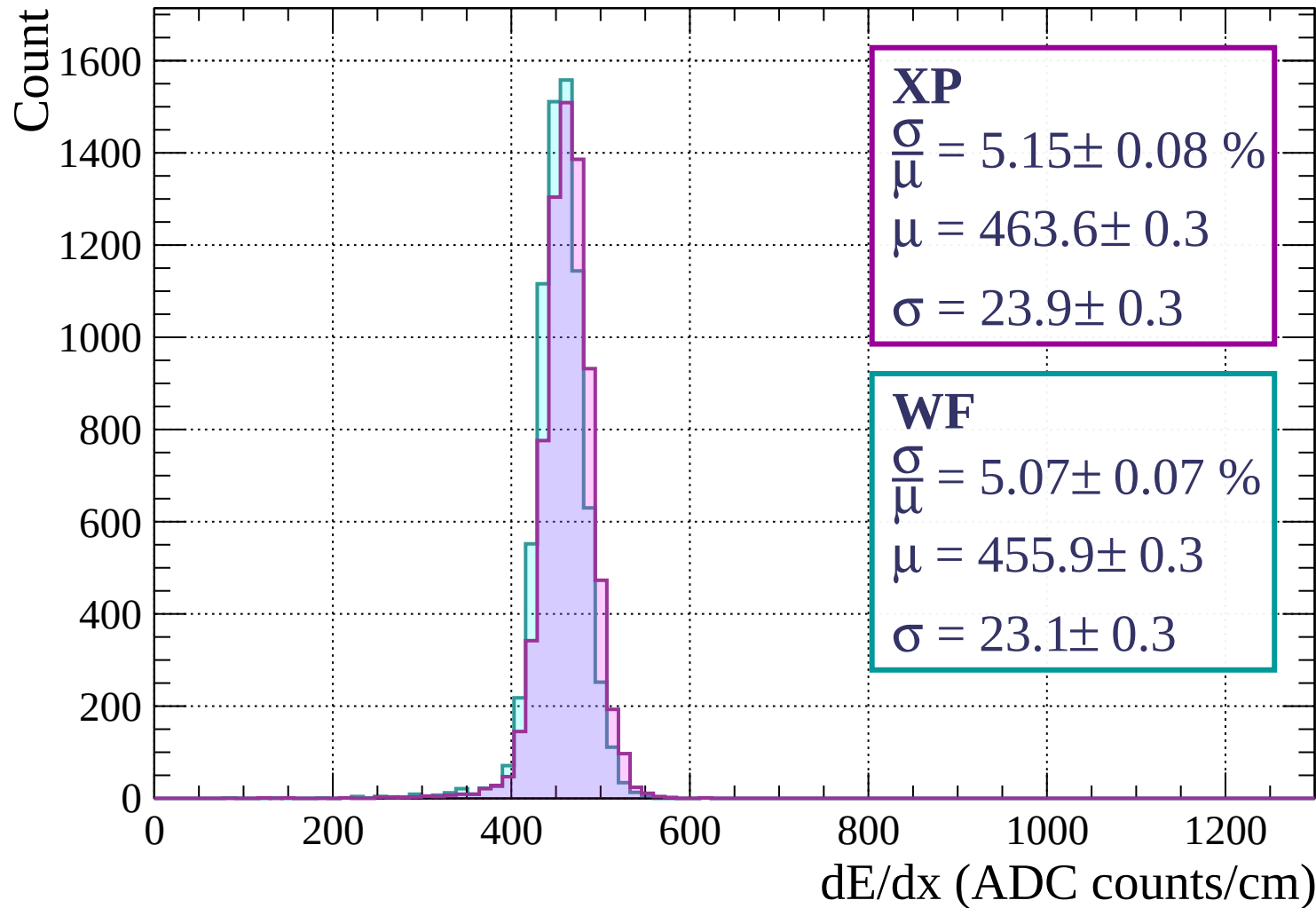
$$\frac{\sigma}{\mu} = 5.17 \pm 0.08 \%$$

$$\mu = 452.3 \pm 0.3$$

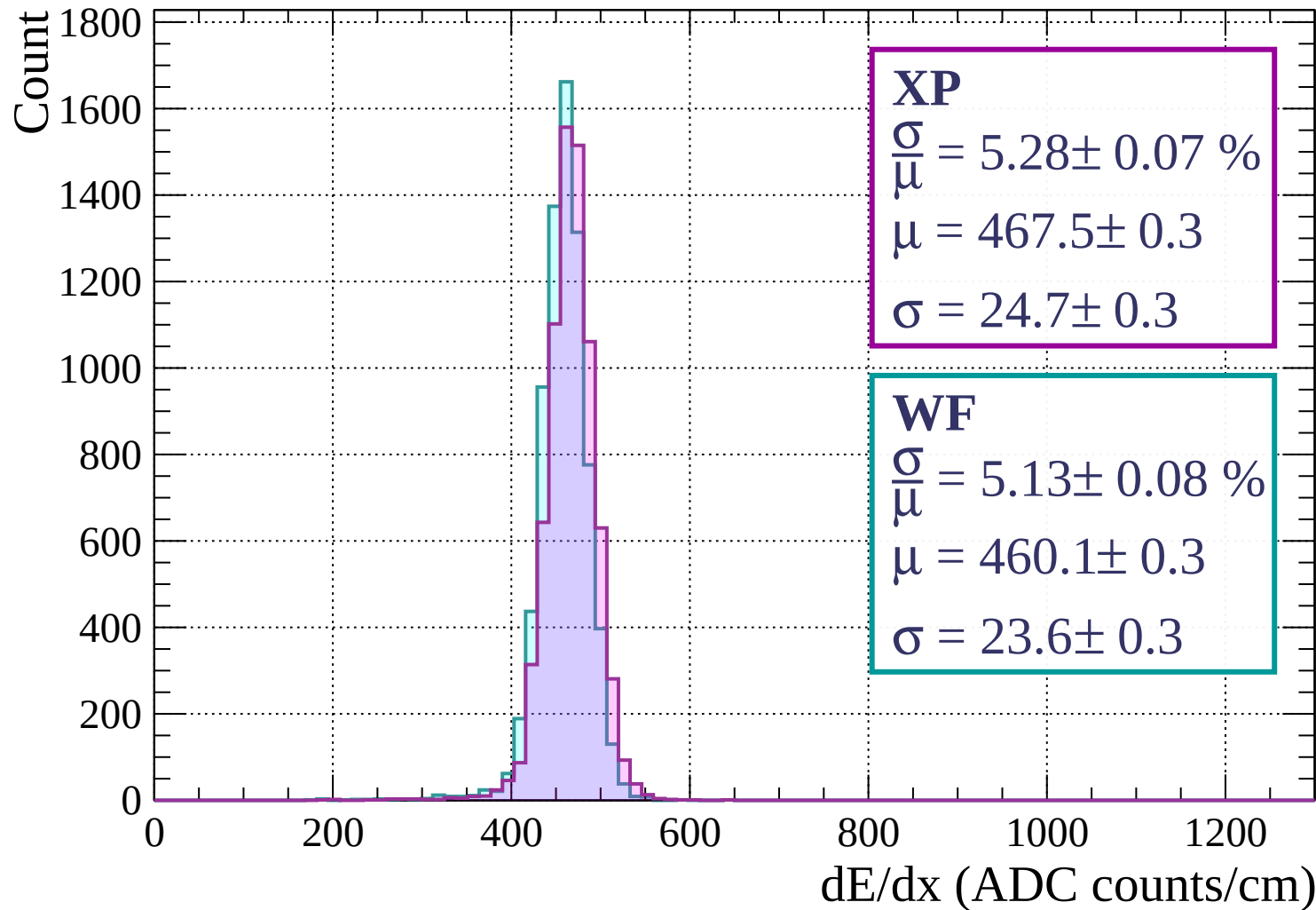
$$\sigma = 23.4 \pm 0.3$$

dE/dx (ADC counts/cm)

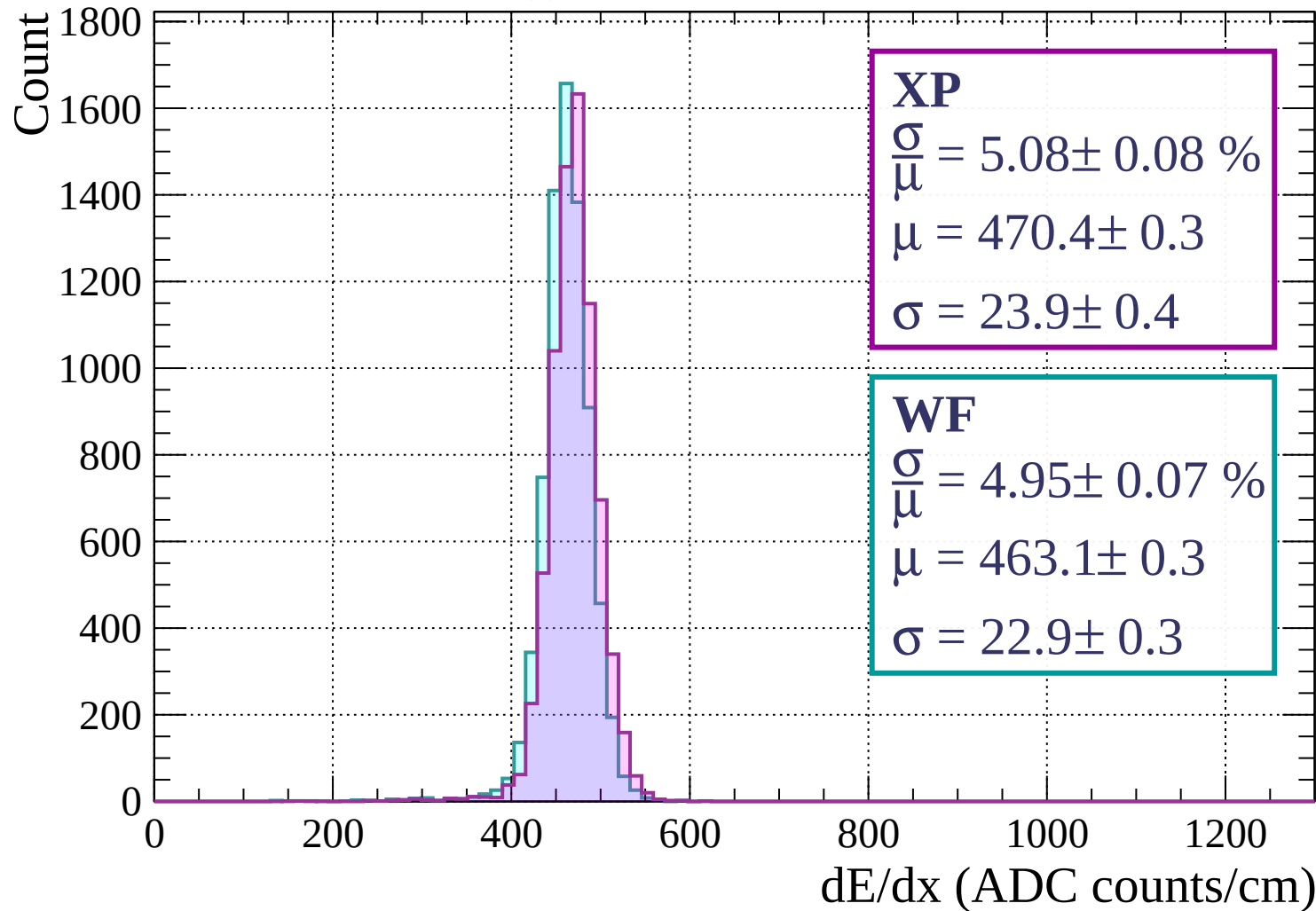
Energy loss | $1200 < p < 1280$



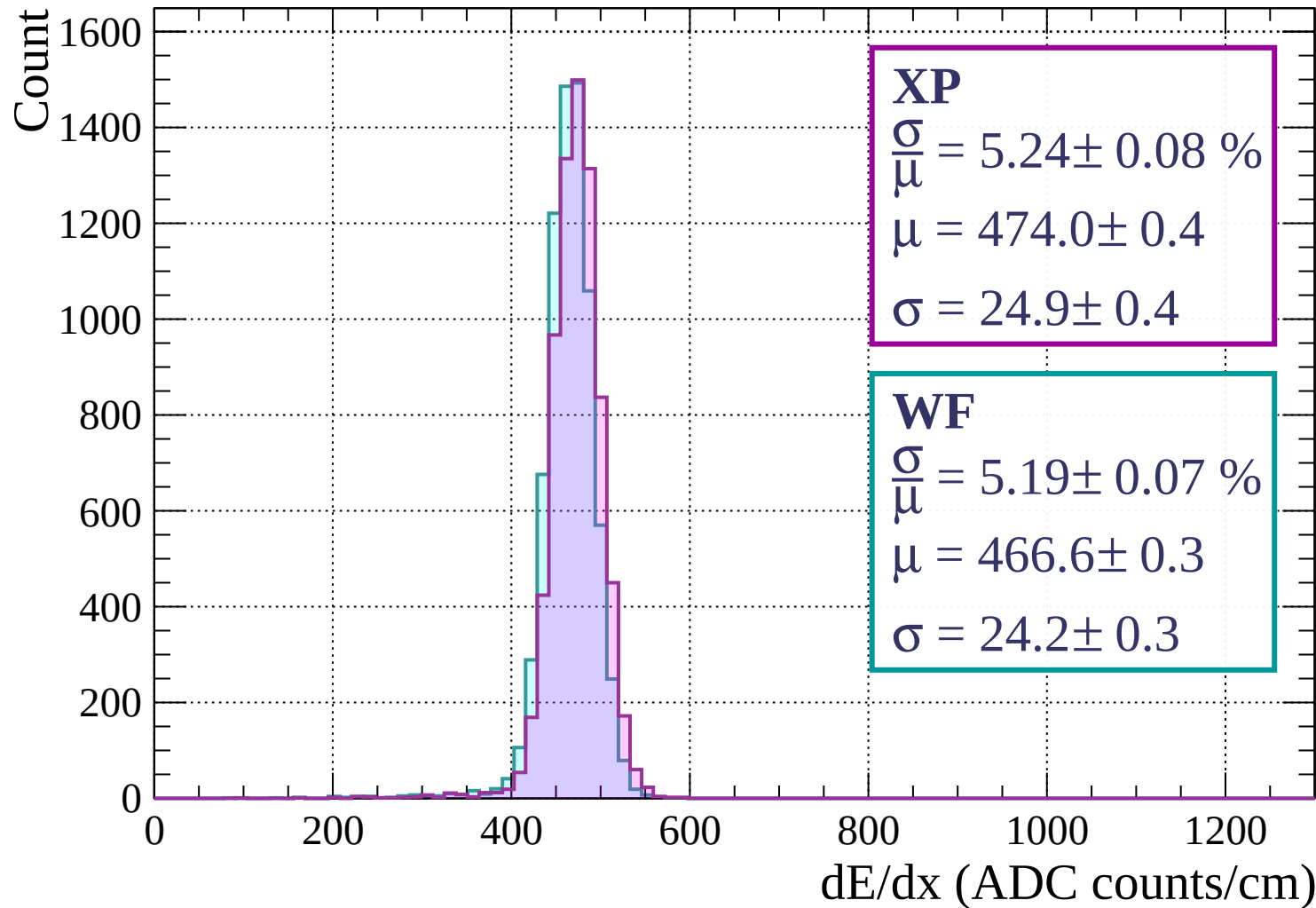
Energy loss | $1280 < p < 1360$



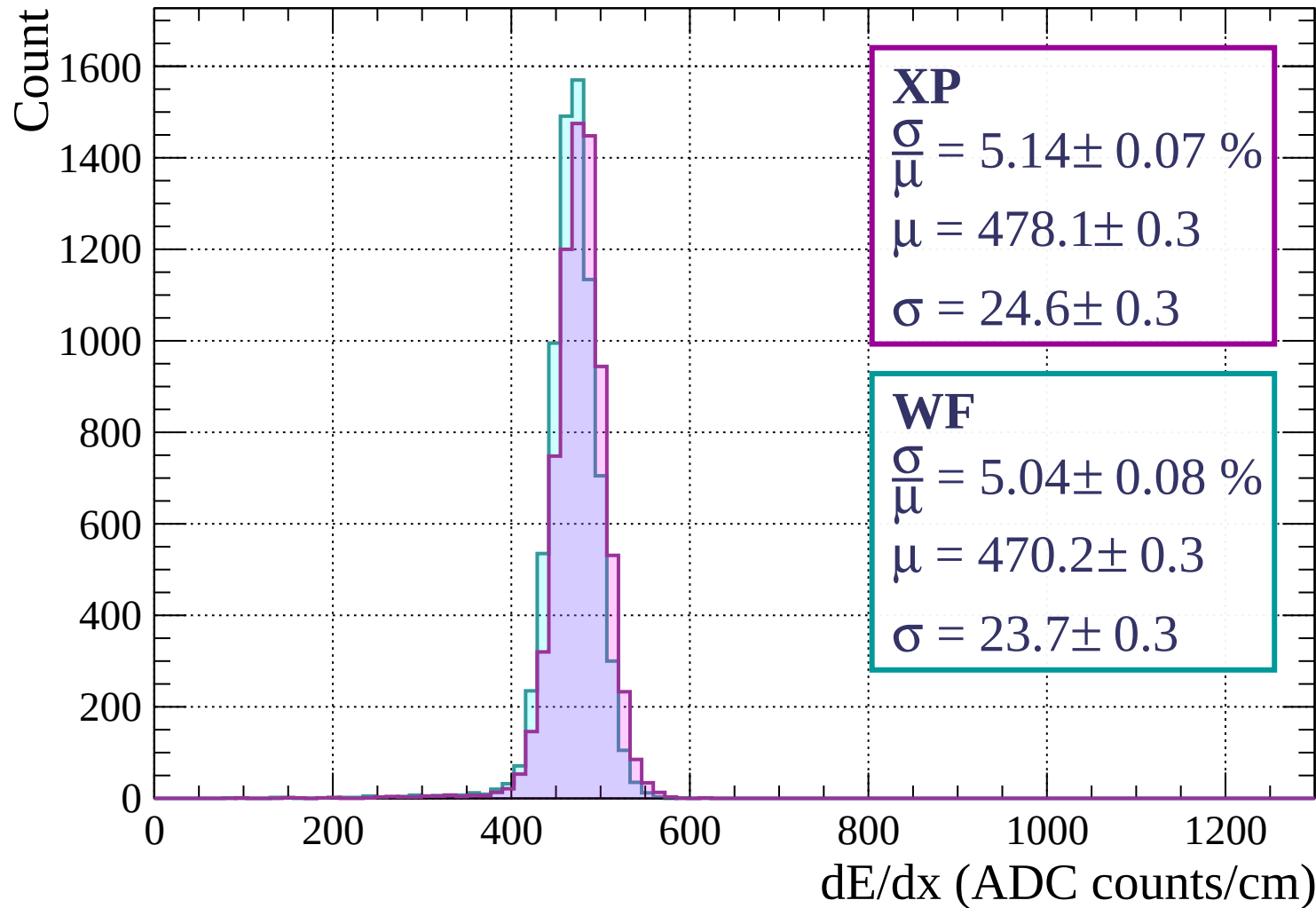
Energy loss | $1360 < p < 1440$



Energy loss | $1440 < p < 1520$

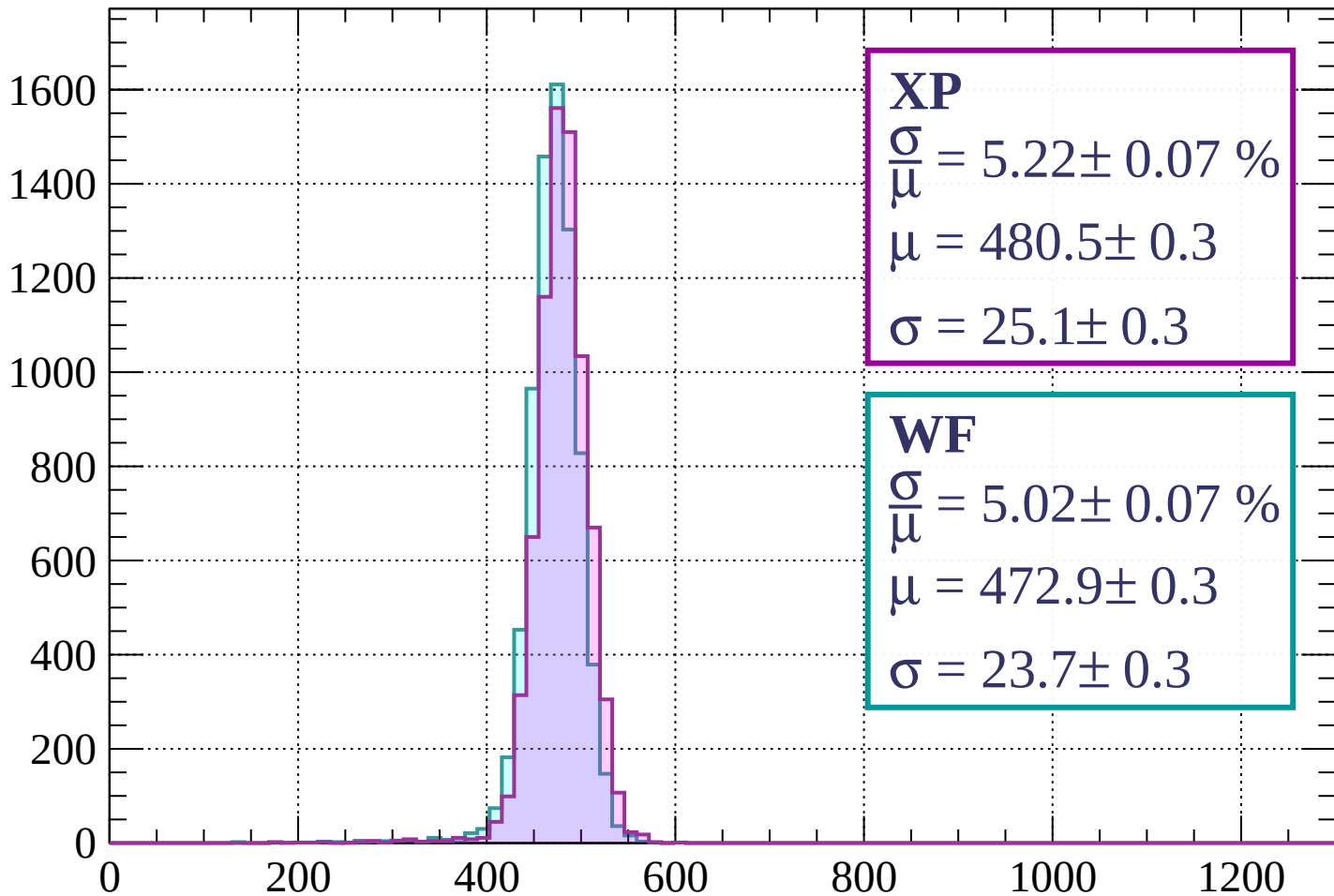


Energy loss | $1520 < p < 1600$



Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 5.22 \pm 0.07 \%$$

$$\mu = 480.5 \pm 0.3$$

$$\sigma = 25.1 \pm 0.3$$

WF

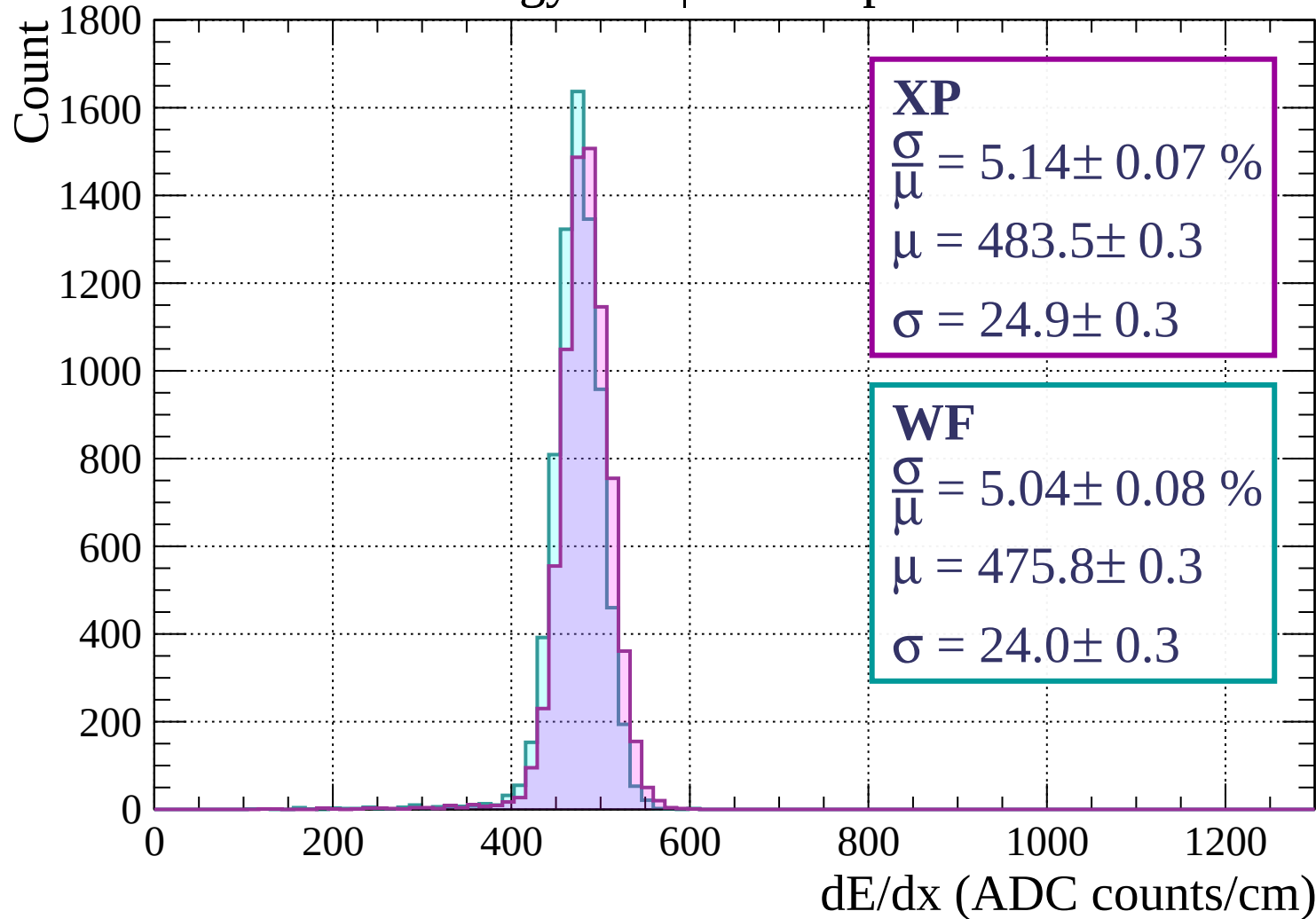
$$\frac{\sigma}{\mu} = 5.02 \pm 0.07 \%$$

$$\mu = 472.9 \pm 0.3$$

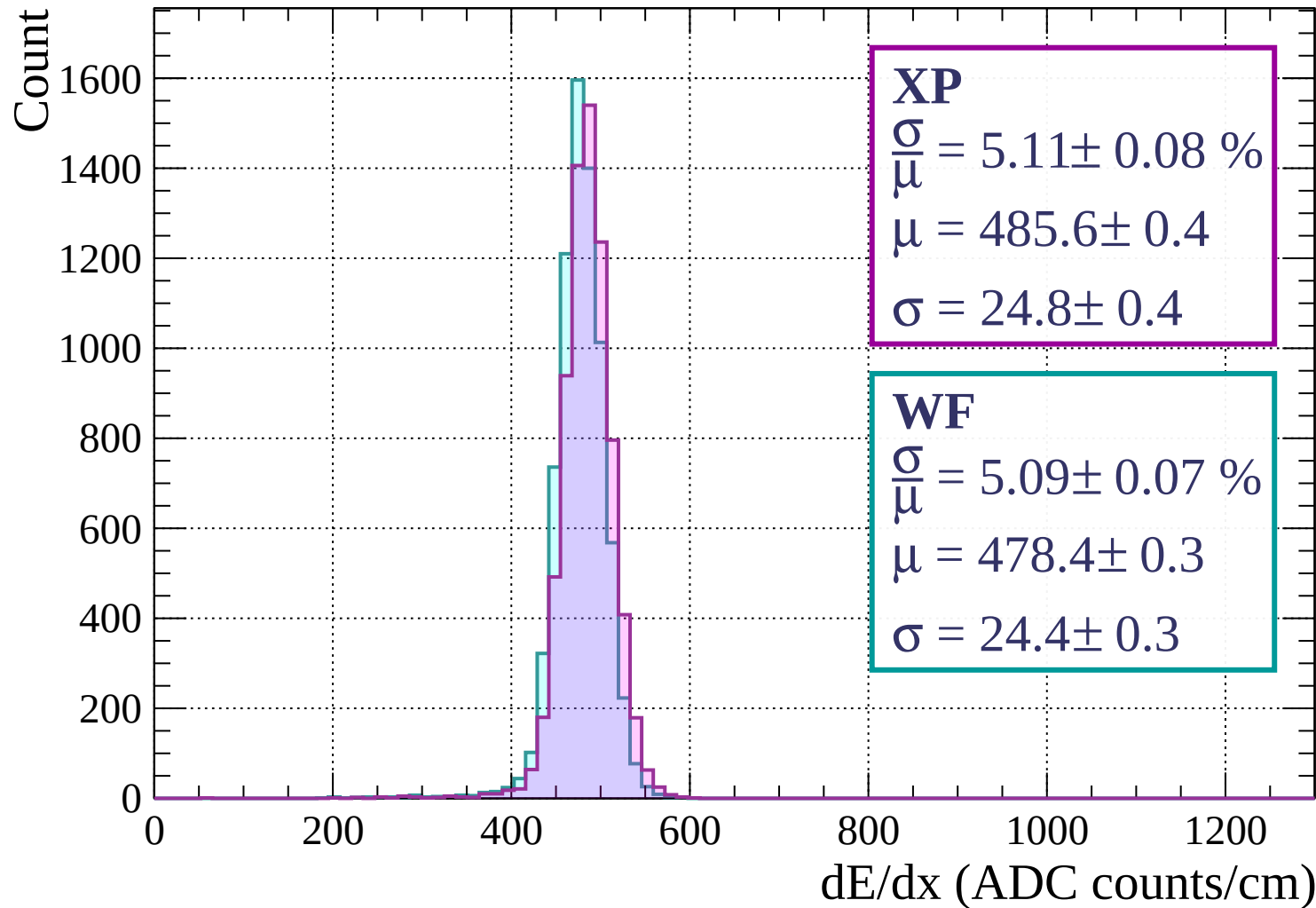
$$\sigma = 23.7 \pm 0.3$$

dE/dx (ADC counts/cm)

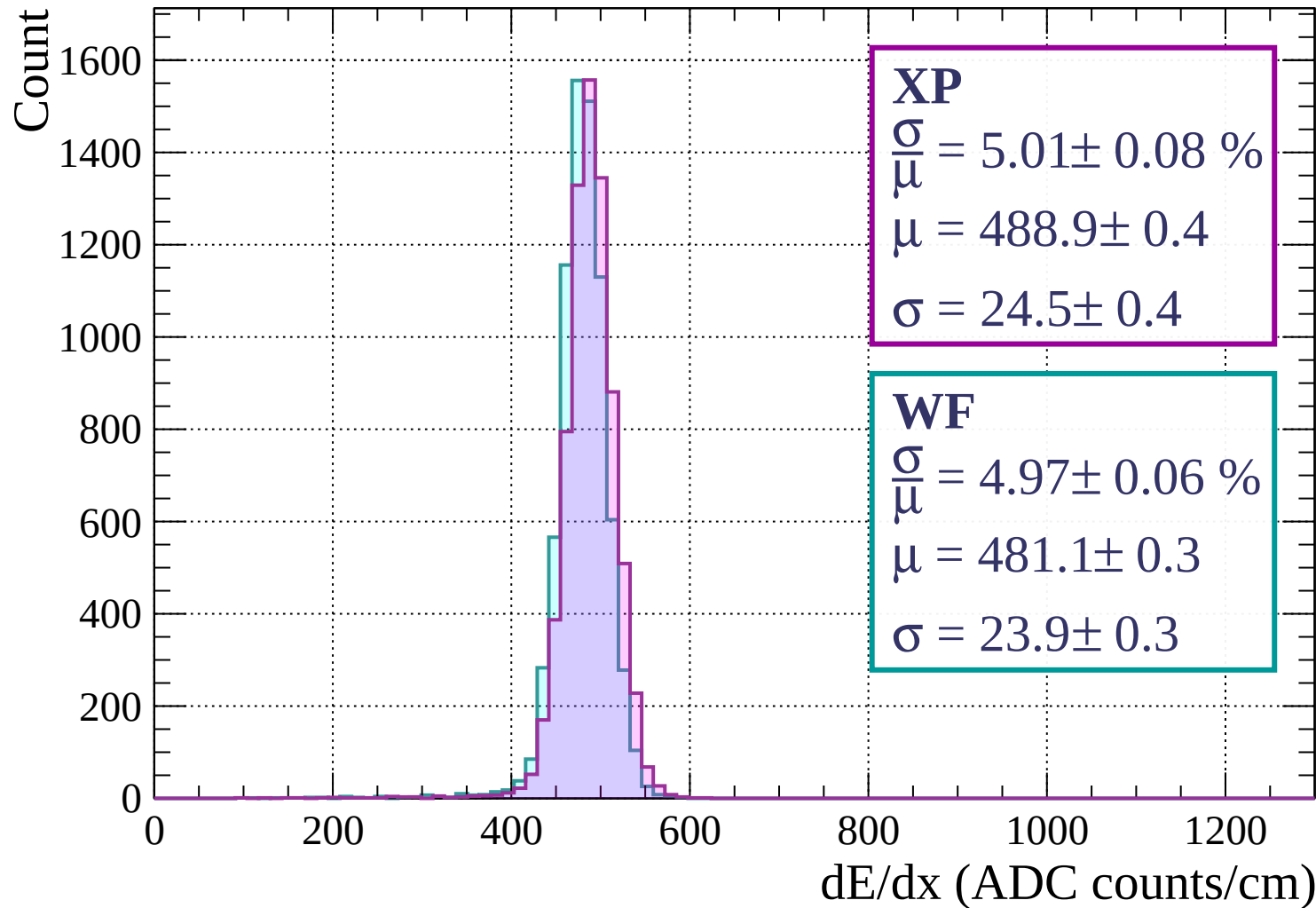
Energy loss | $1680 < p < 1760$



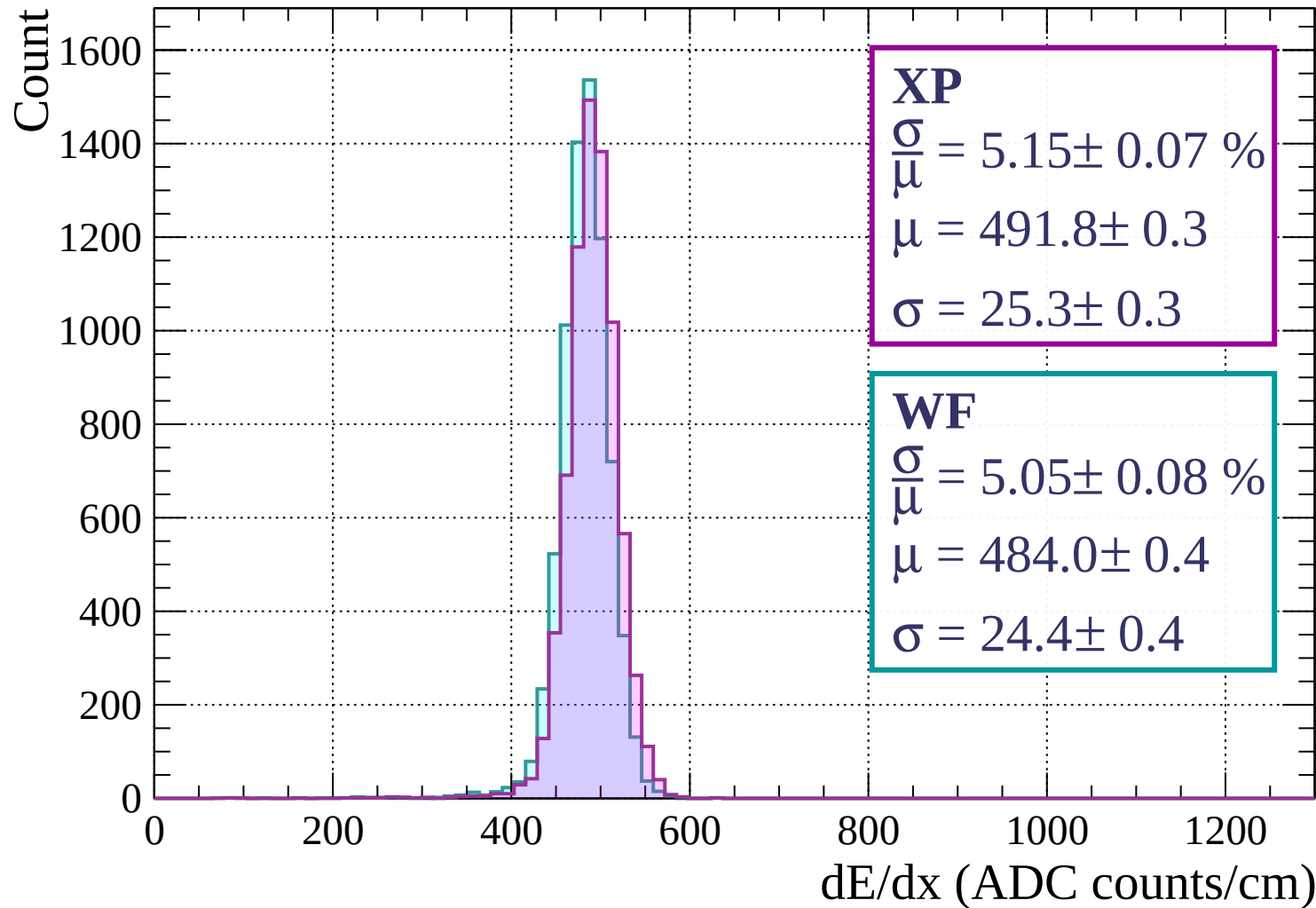
Energy loss | $1760 < p < 1840$



Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$



Energy loss | $0 < \phi < 3$

Count

$\times 10^3$

XP

$$\frac{\sigma}{\mu} = 8.25 \pm 0.03 \%$$

$$\mu = 509.4 \pm 0.1$$

$$\sigma = 42.0 \pm 0.2$$

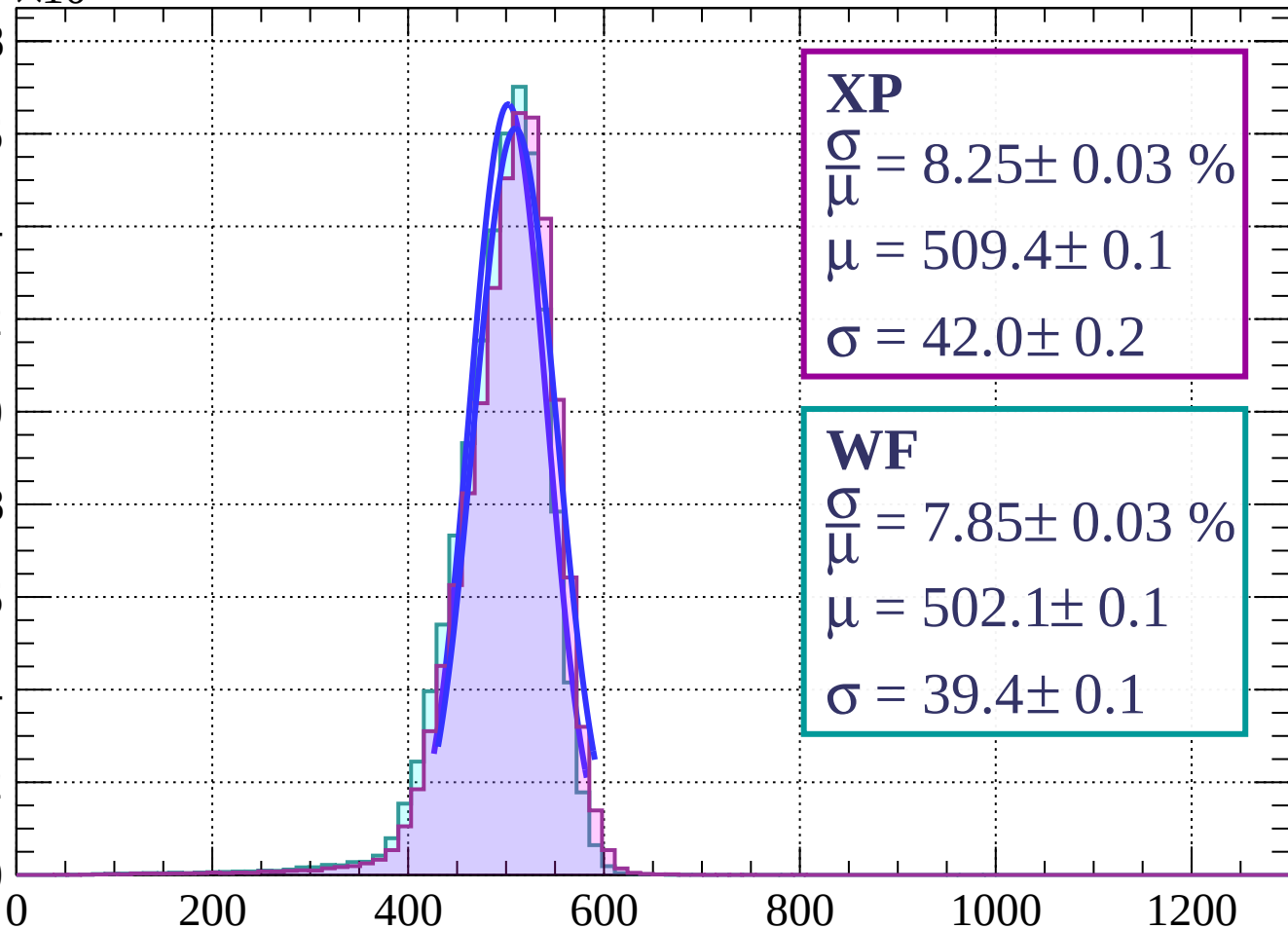
WF

$$\frac{\sigma}{\mu} = 7.85 \pm 0.03 \%$$

$$\mu = 502.1 \pm 0.1$$

$$\sigma = 39.4 \pm 0.1$$

18
16
14
12
10
8
6
4
2
0

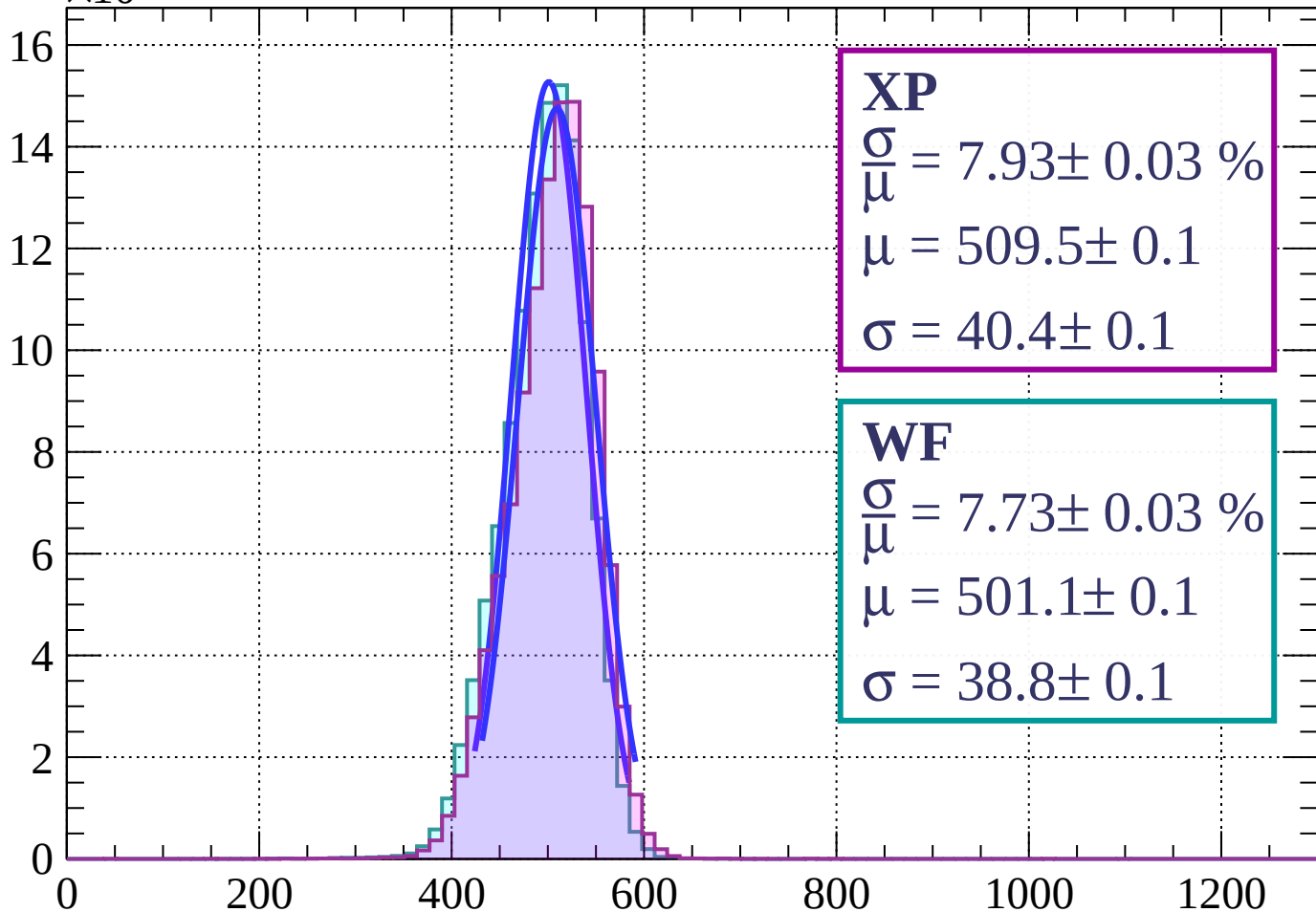


dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$

Count

$\times 10^3$



XP

$$\frac{\sigma}{\mu} = 7.93 \pm 0.03 \%$$

$$\mu = 509.5 \pm 0.1$$

$$\sigma = 40.4 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 7.73 \pm 0.03 \%$$

$$\mu = 501.1 \pm 0.1$$

$$\sigma = 38.8 \pm 0.1$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \phi < 9$

Count

$\times 10^3$

XP

$$\frac{\sigma}{\mu} = 8.40 \pm 0.04 \%$$

$$\mu = 505.9 \pm 0.2$$

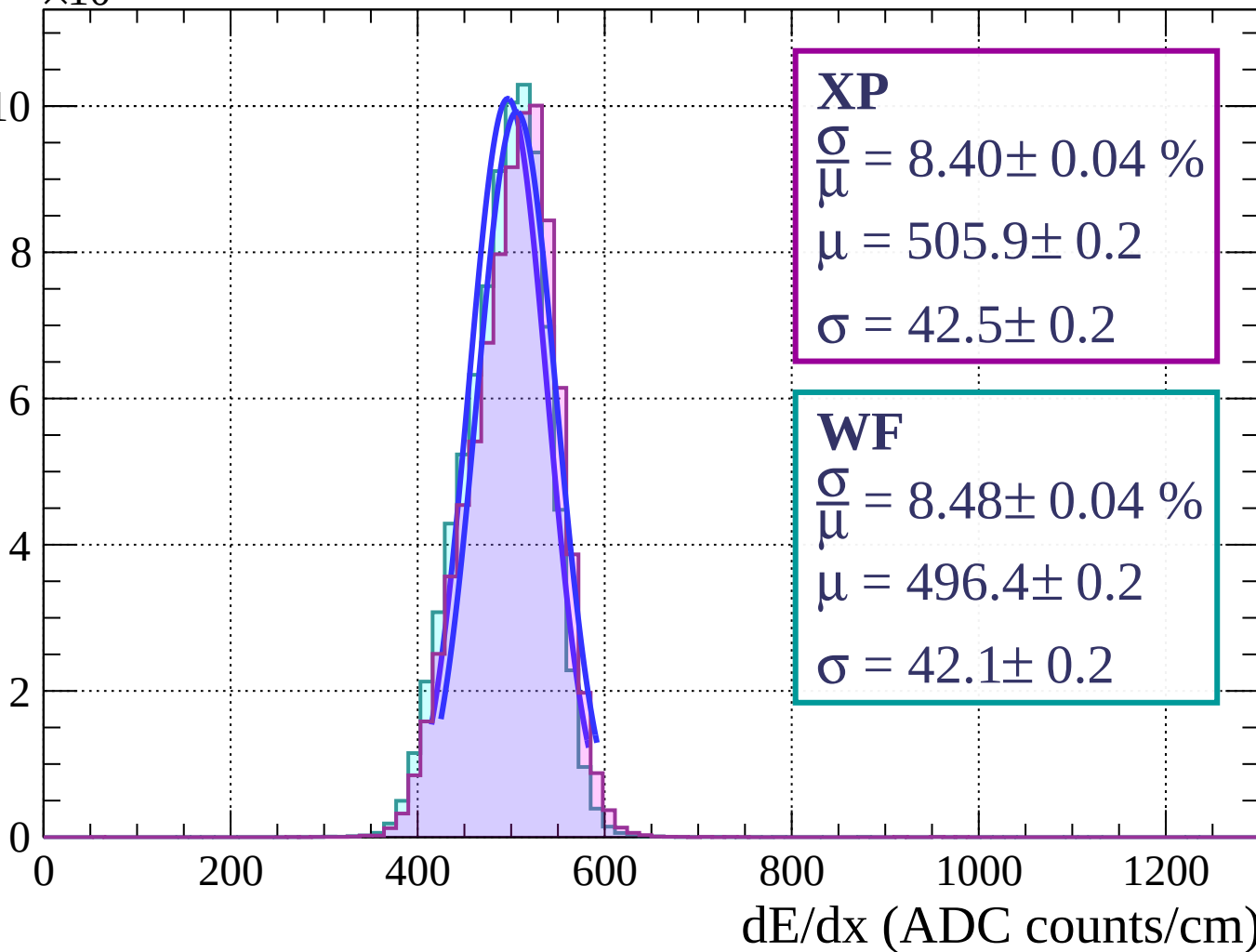
$$\sigma = 42.5 \pm 0.2$$

WF

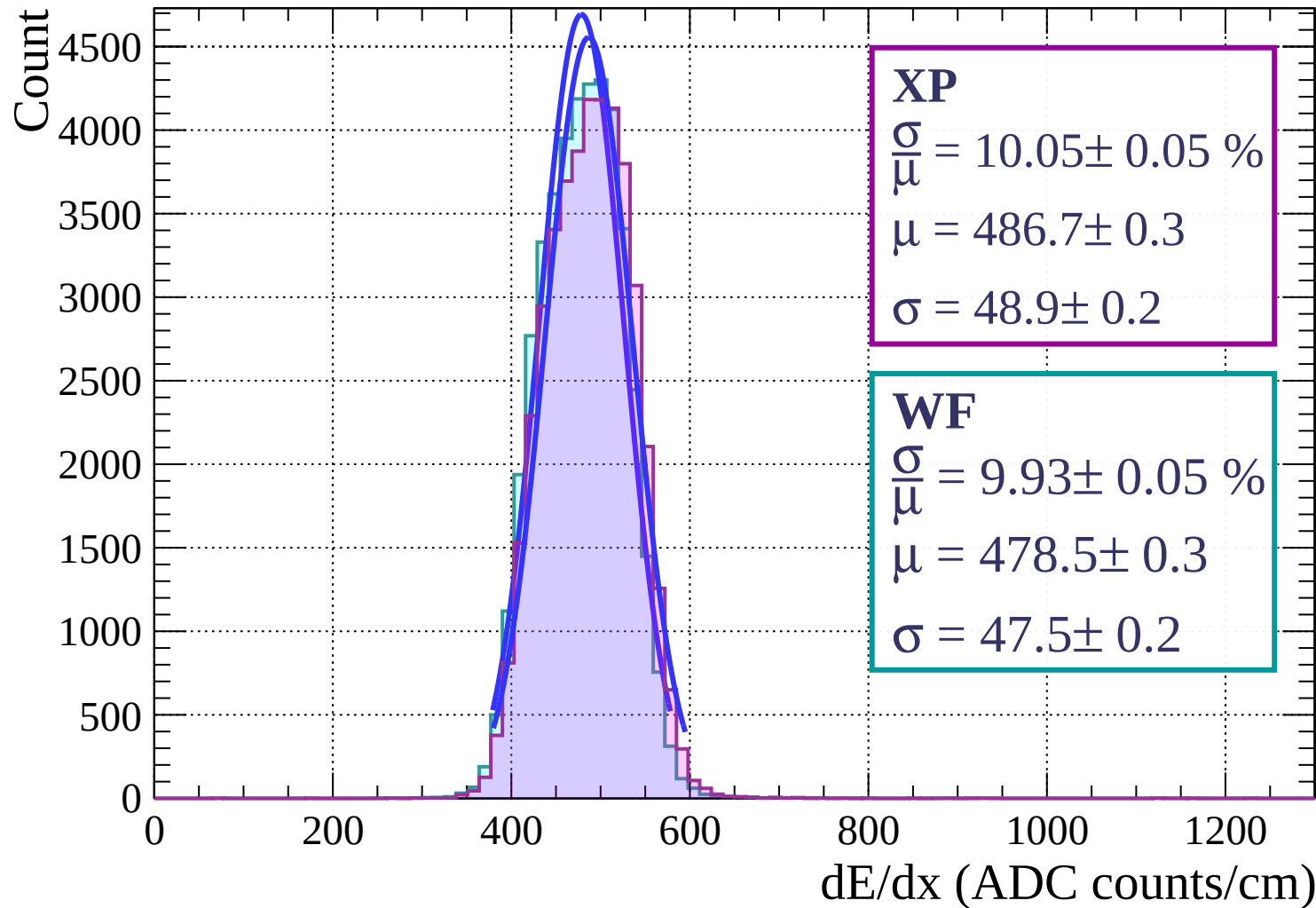
$$\frac{\sigma}{\mu} = 8.48 \pm 0.04 \%$$

$$\mu = 496.4 \pm 0.2$$

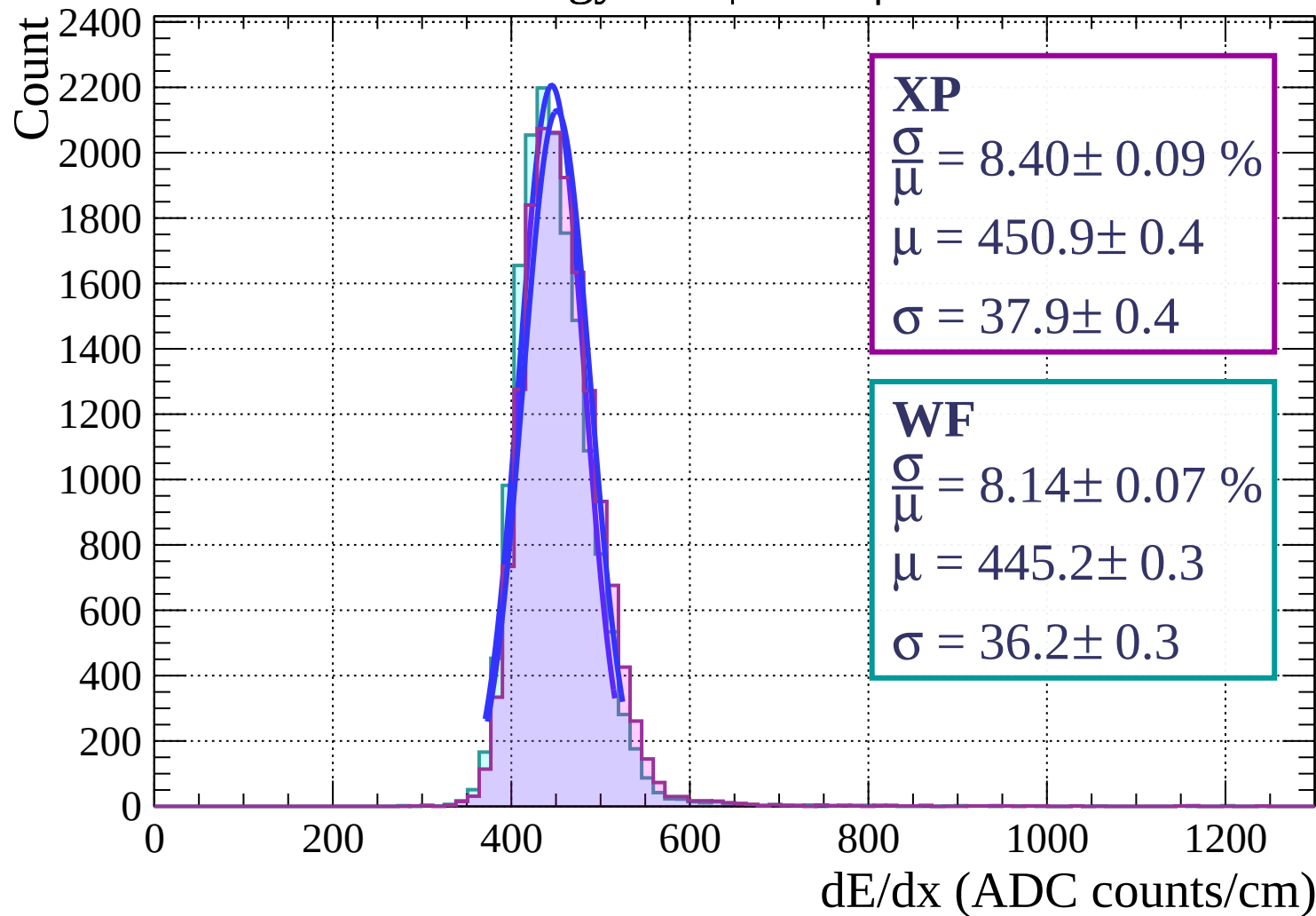
$$\sigma = 42.1 \pm 0.2$$



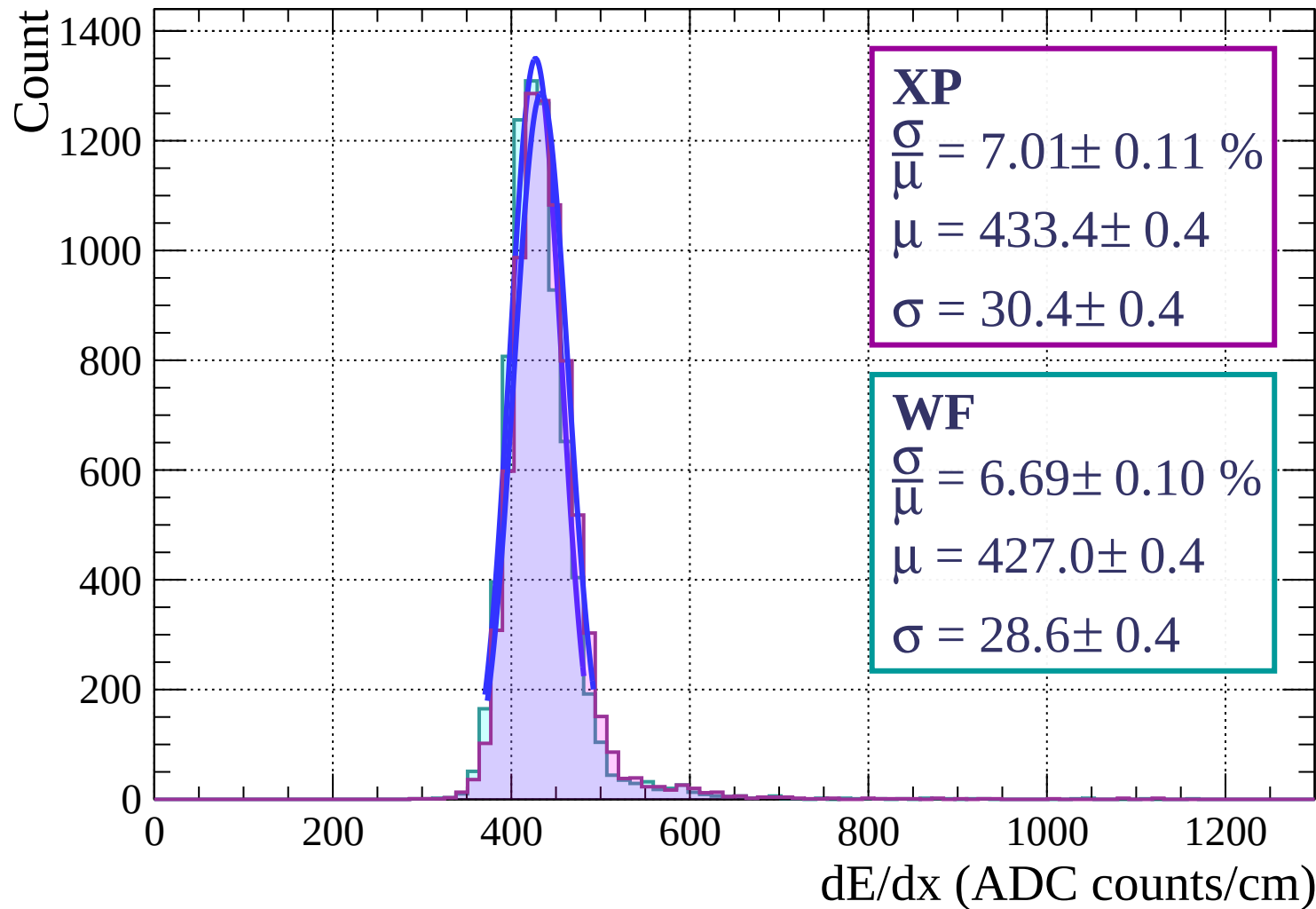
Energy loss | $9 < \phi < 12$



Energy loss | $12 < \phi < 15$

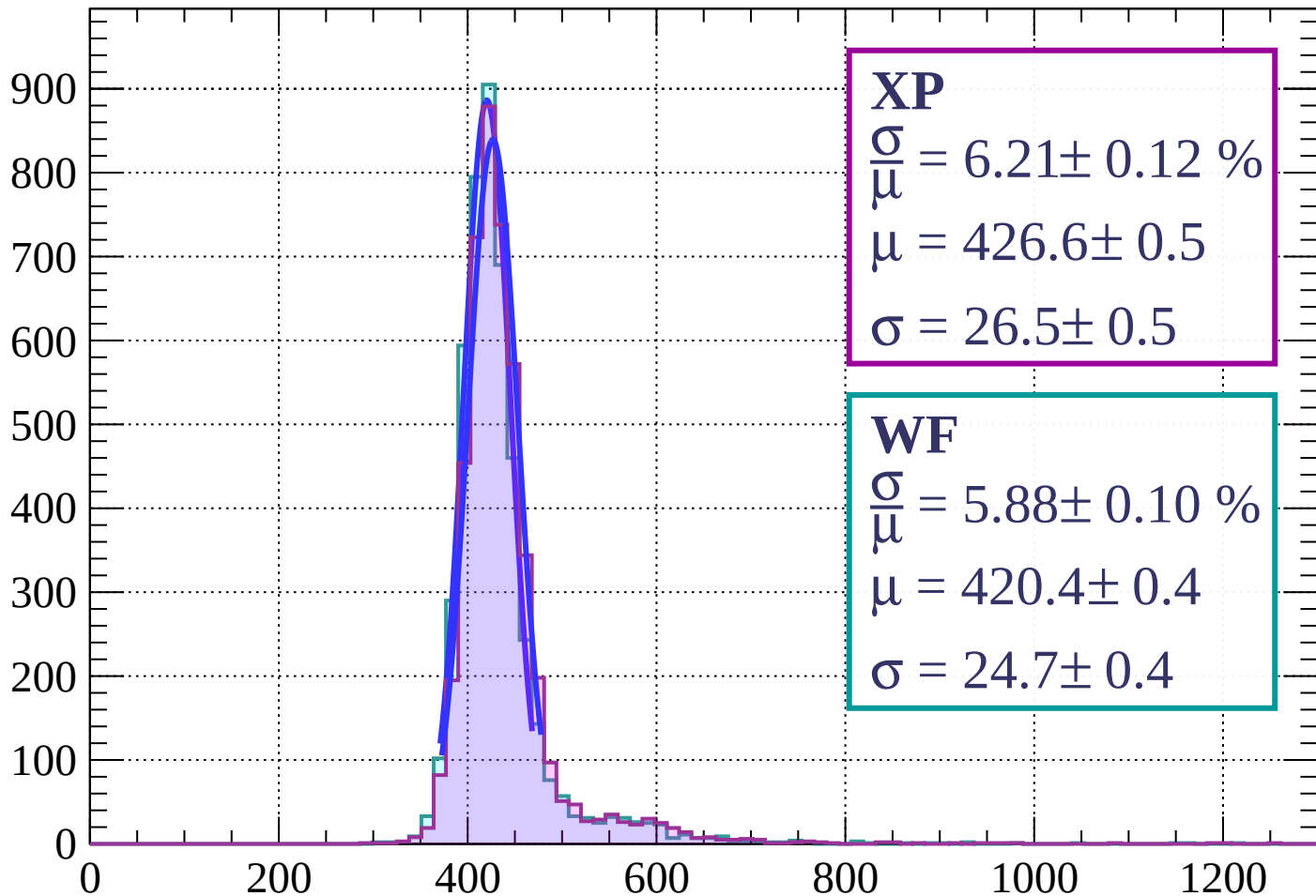


Energy loss | $15 < \phi < 18$



Energy loss | $18 < \phi < 21$

Count



XP

$$\frac{\sigma}{\mu} = 6.21 \pm 0.12 \%$$

$$\mu = 426.6 \pm 0.5$$

$$\sigma = 26.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.88 \pm 0.10 \%$$

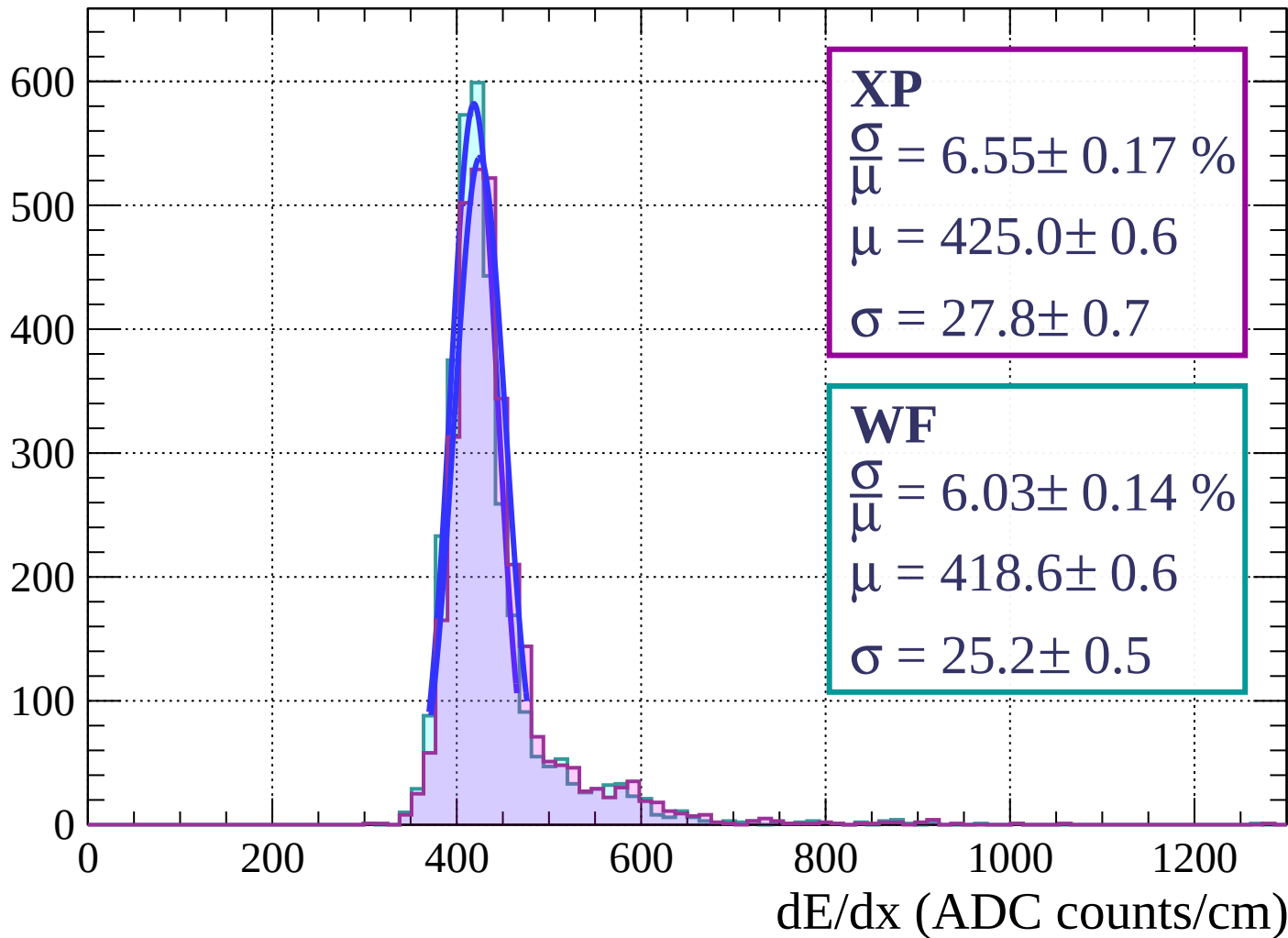
$$\mu = 420.4 \pm 0.4$$

$$\sigma = 24.7 \pm 0.4$$

dE/dx (ADC counts/cm)

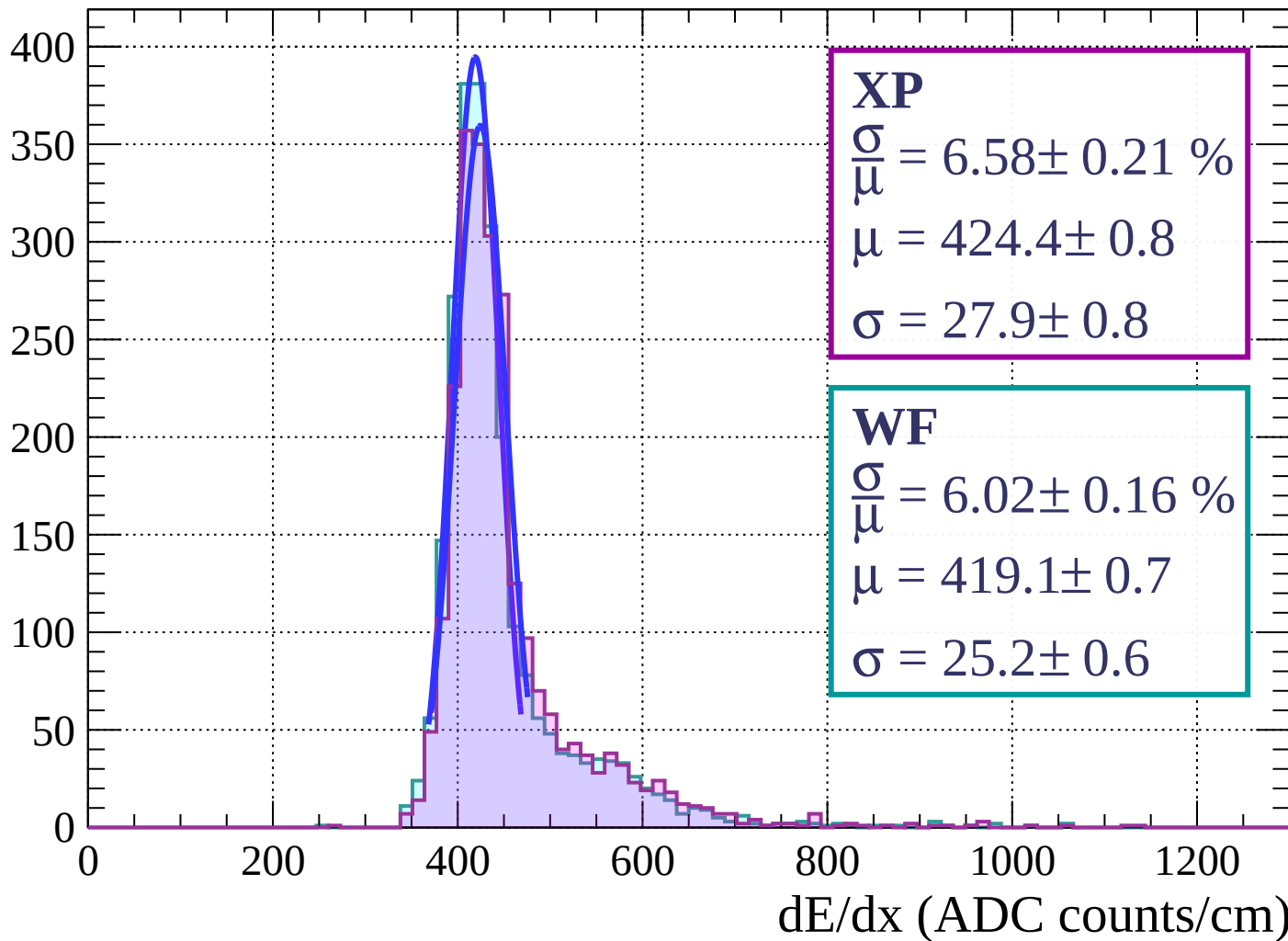
Energy loss | $21 < \phi < 24$

Count



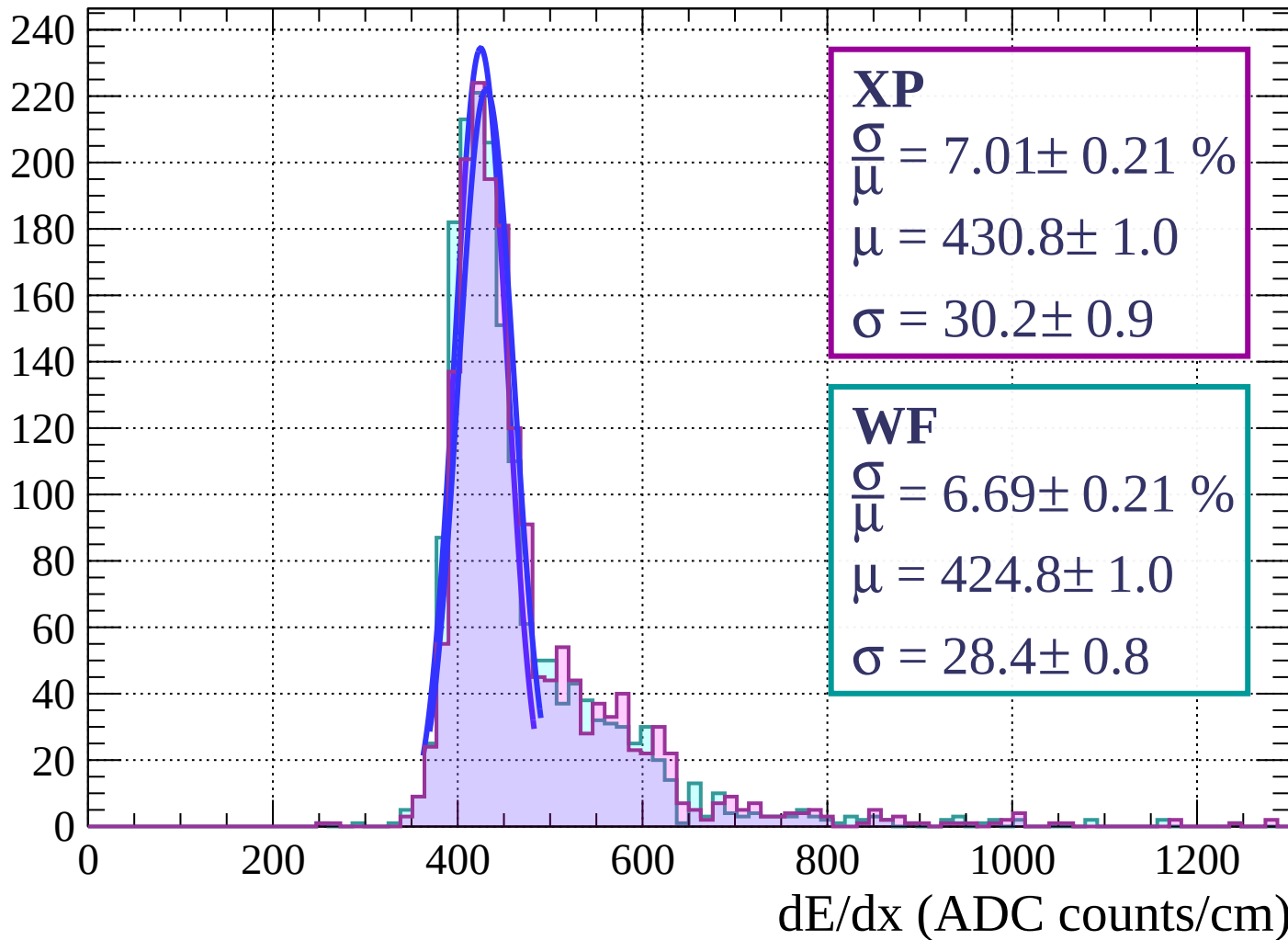
Energy loss | $24 < \phi < 27$

Count



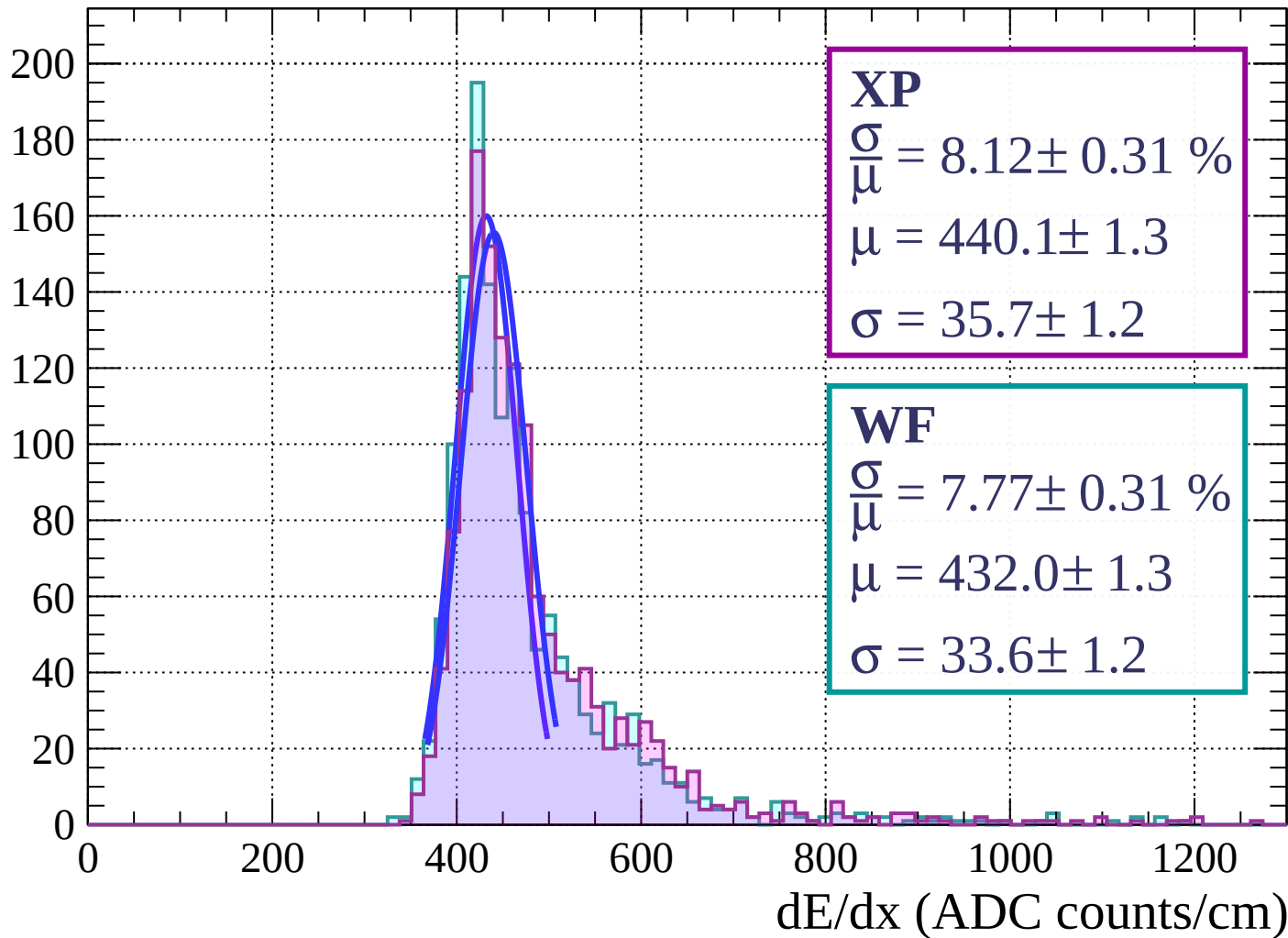
Energy loss | $27 < \phi < 30$

Count



Energy loss | $30 < \phi < 33$

Count



Energy loss | $33 < \phi < 36$

Count

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 10.99 \pm 0.63 \%$$

$$\mu = 462.5 \pm 2.9$$

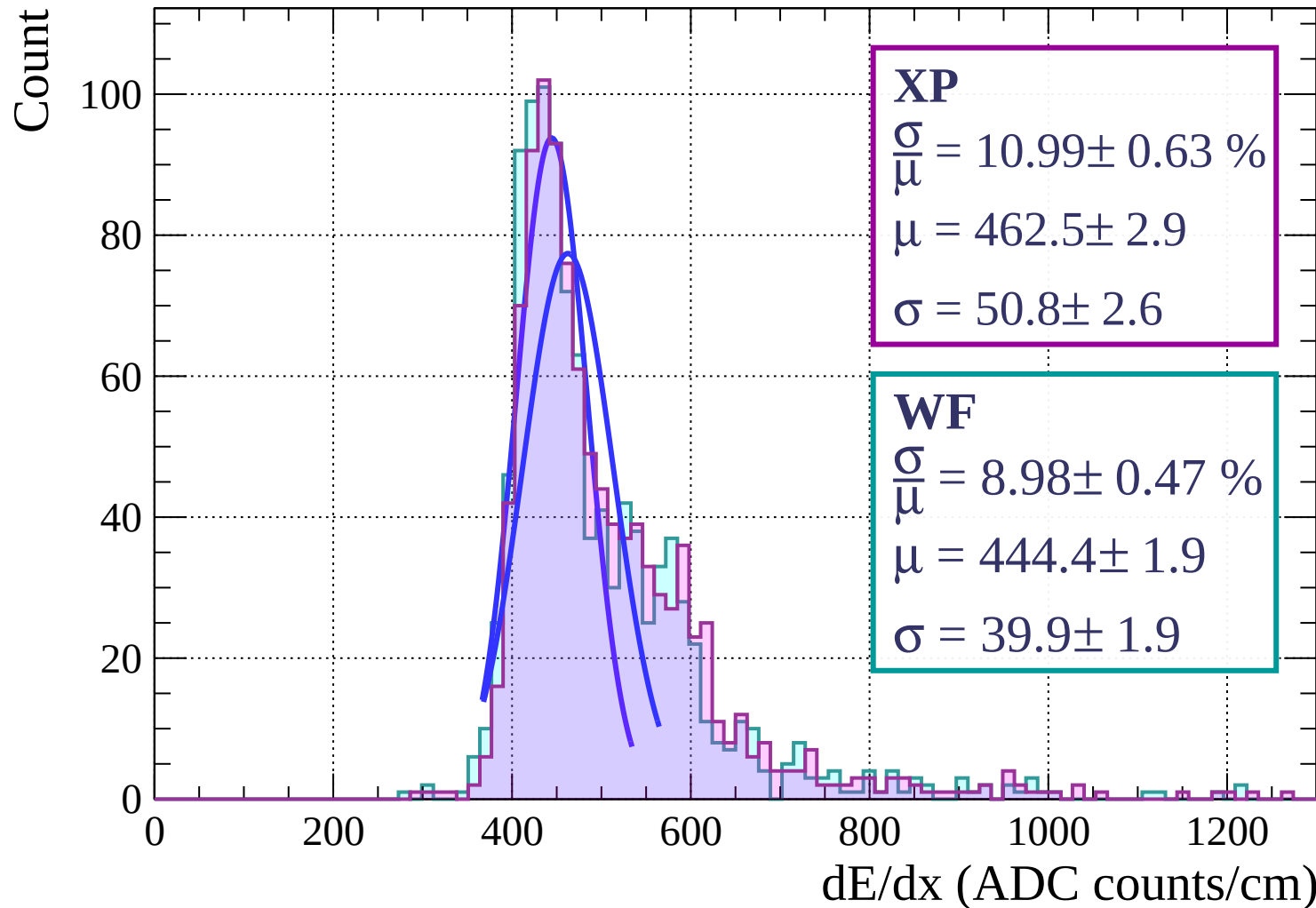
$$\sigma = 50.8 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.47 \%$$

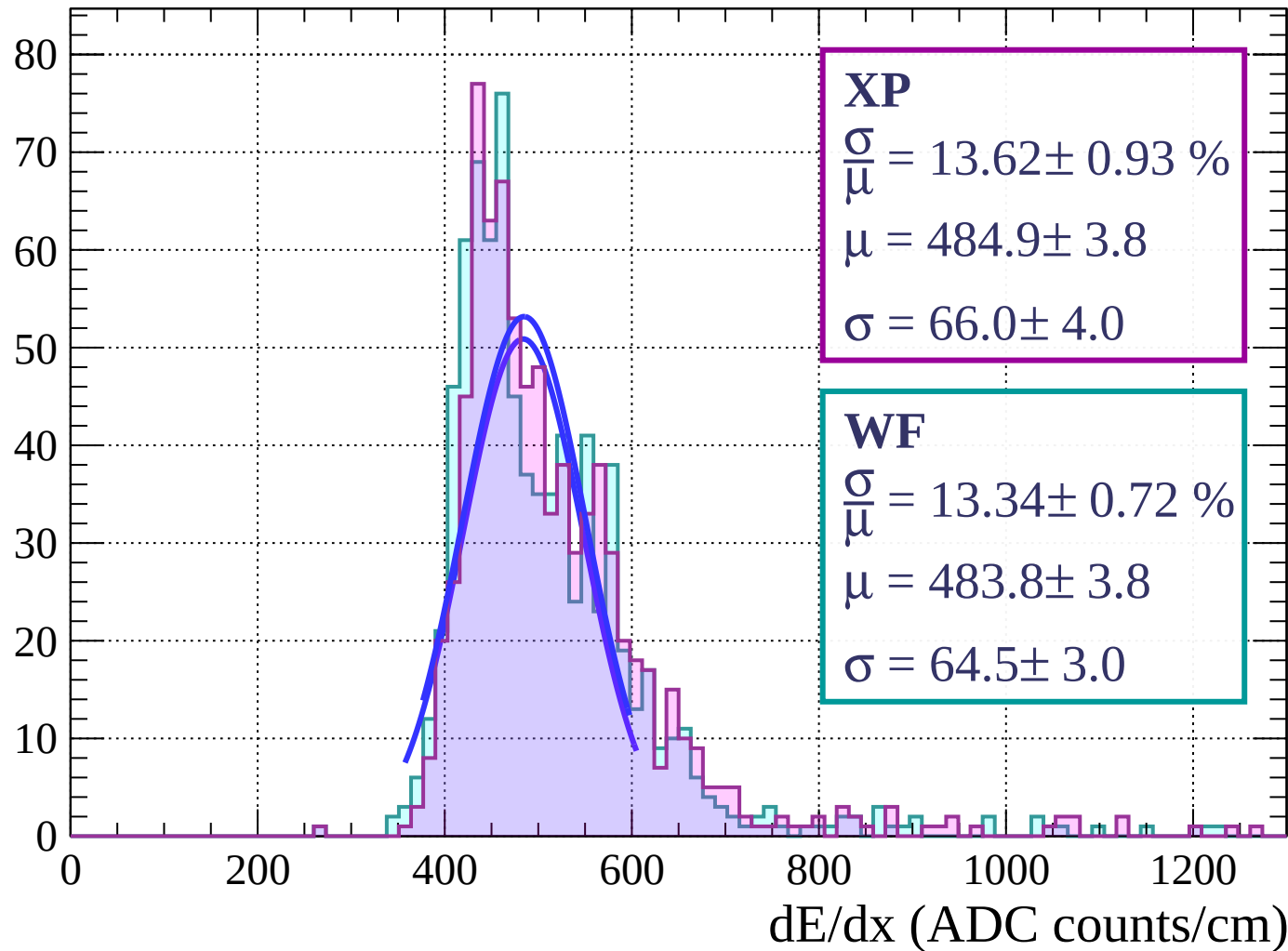
$$\mu = 444.4 \pm 1.9$$

$$\sigma = 39.9 \pm 1.9$$



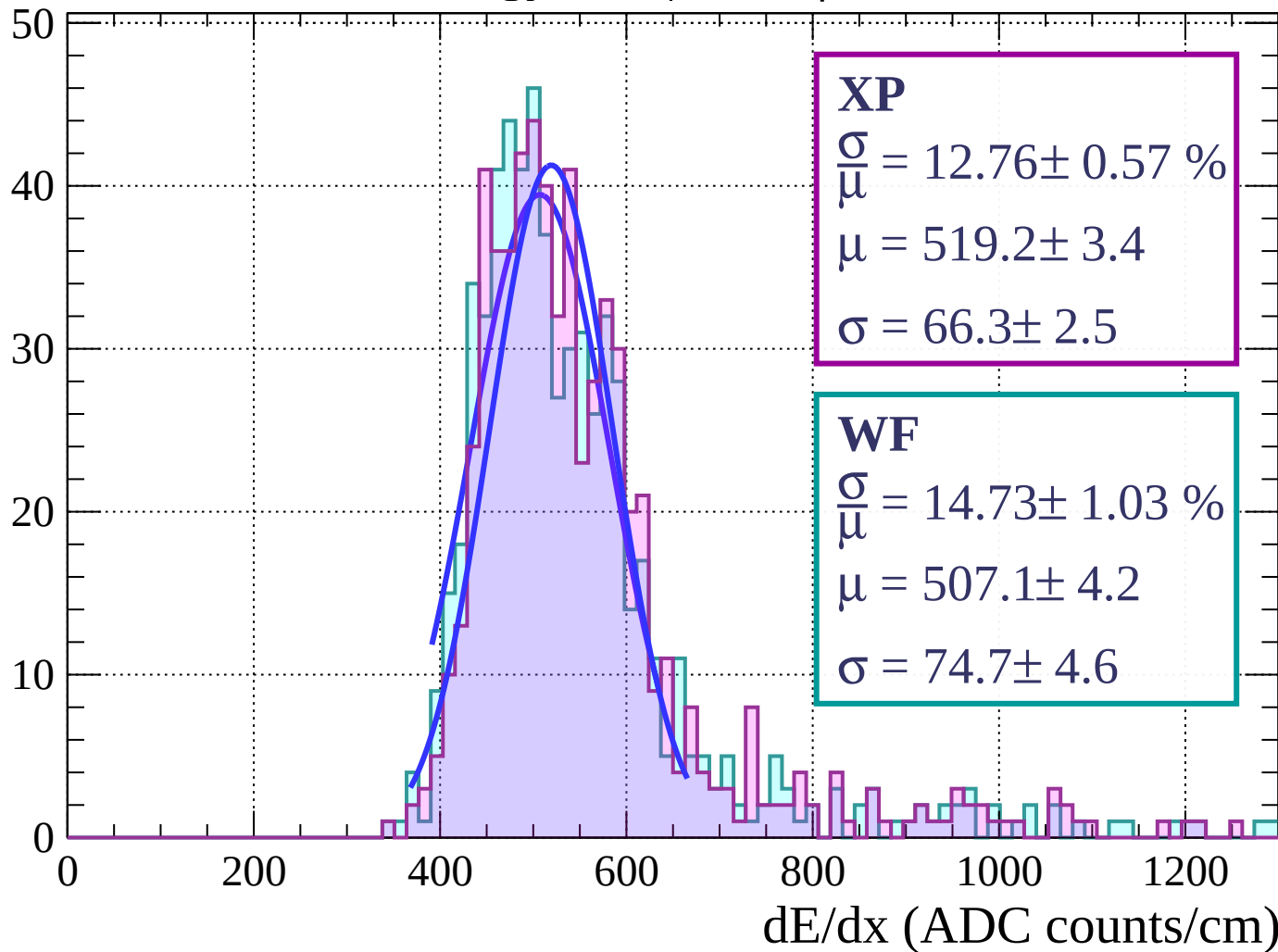
Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count



Energy loss | $42 < \phi < 45$

Count

XP

$$\frac{\sigma}{\mu} = 12.95 \pm 0.78 \%$$

$$\mu = 540.8 \pm 4.1$$

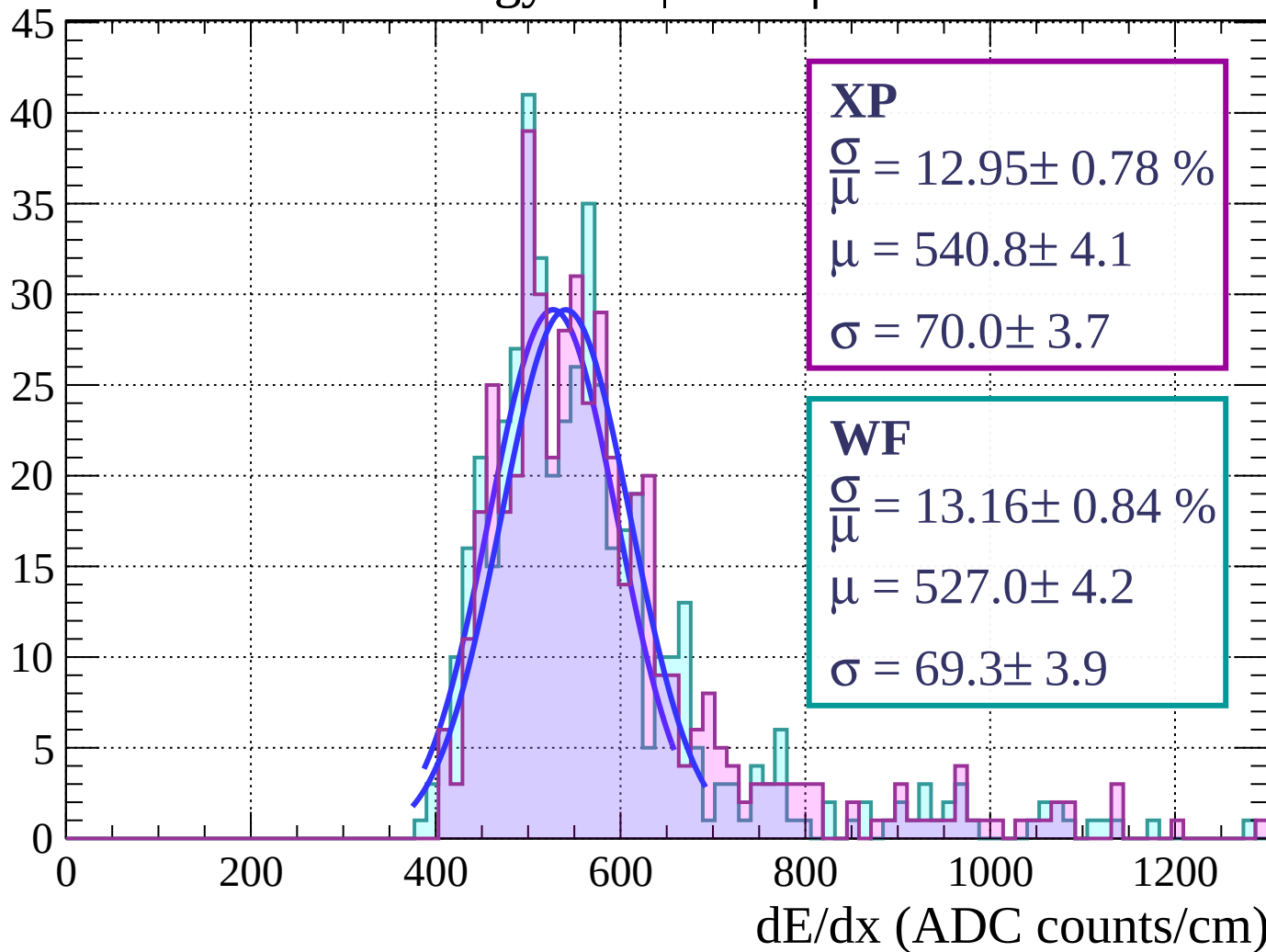
$$\sigma = 70.0 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 13.16 \pm 0.84 \%$$

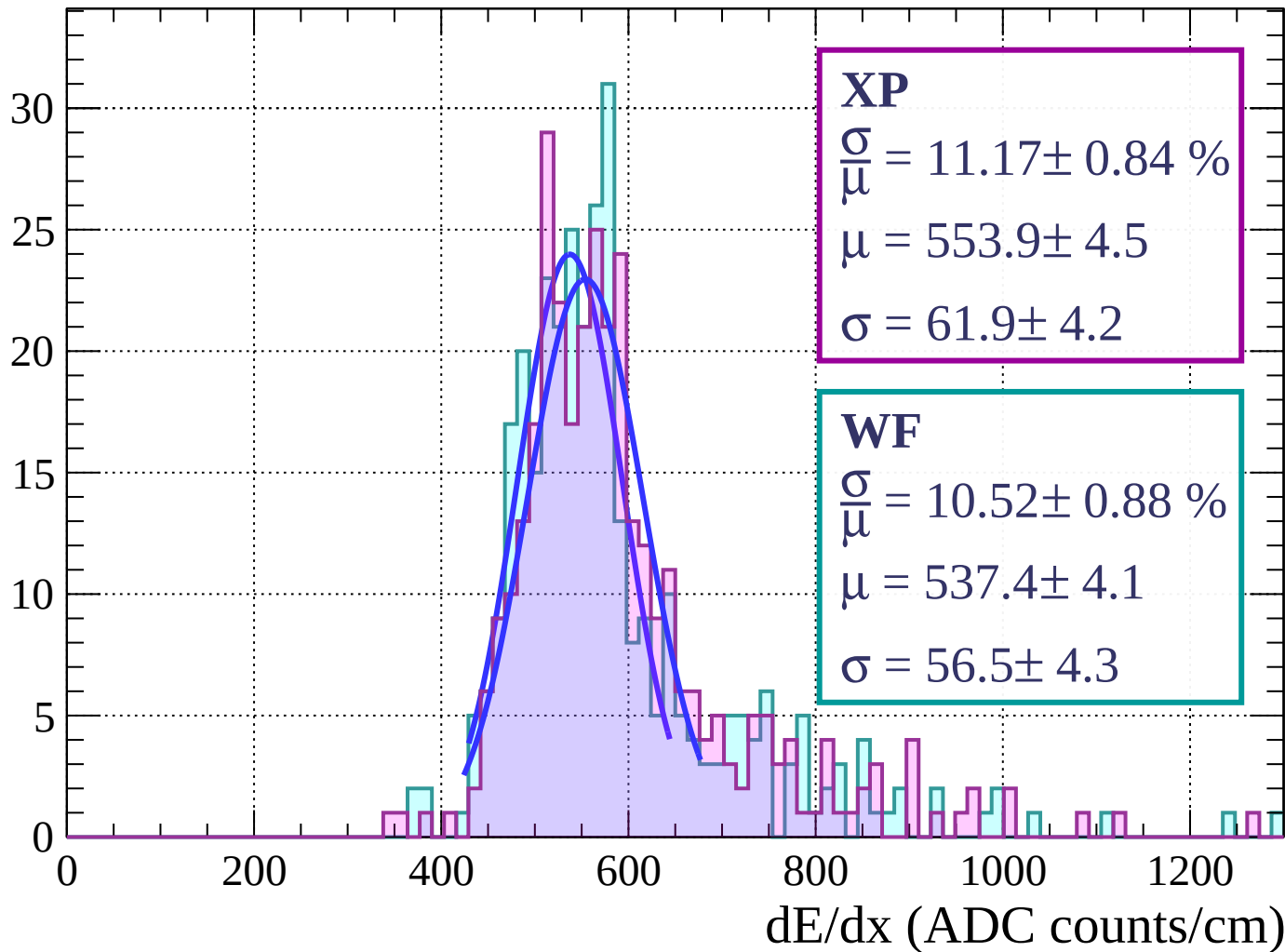
$$\mu = 527.0 \pm 4.2$$

$$\sigma = 69.3 \pm 3.9$$



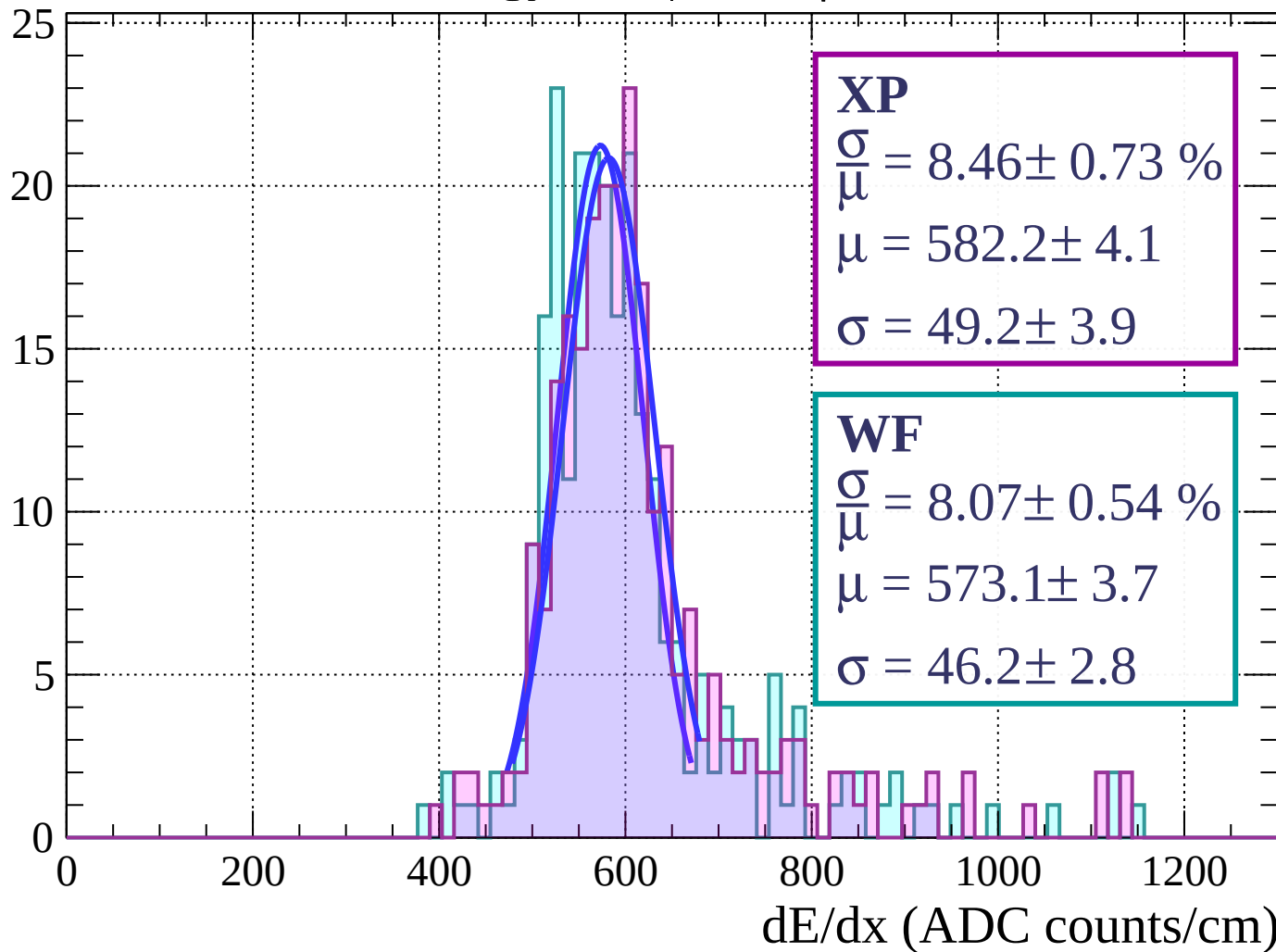
Energy loss | $45 < \phi < 48$

Count



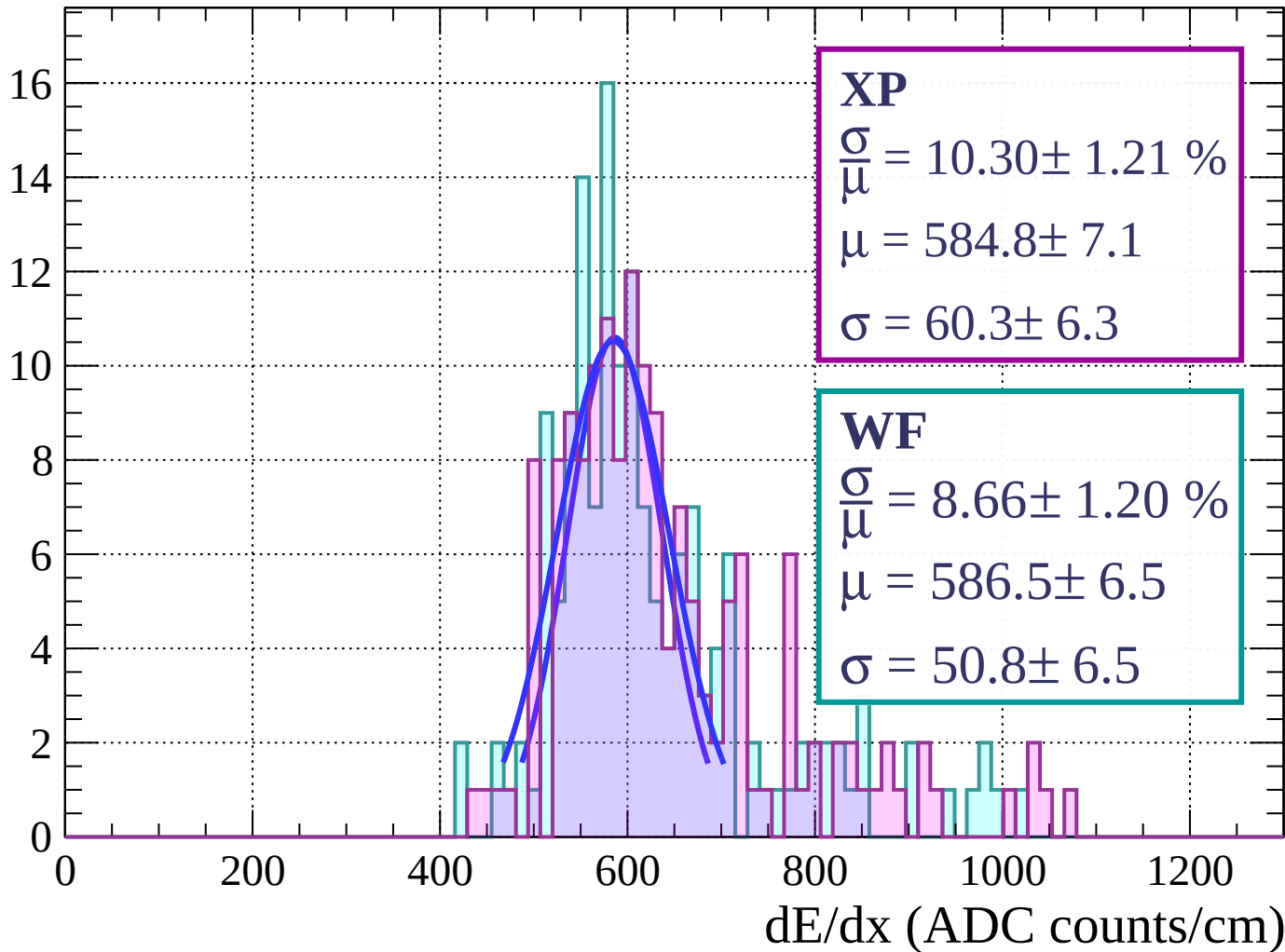
Energy loss | $48 < \phi < 51$

Count



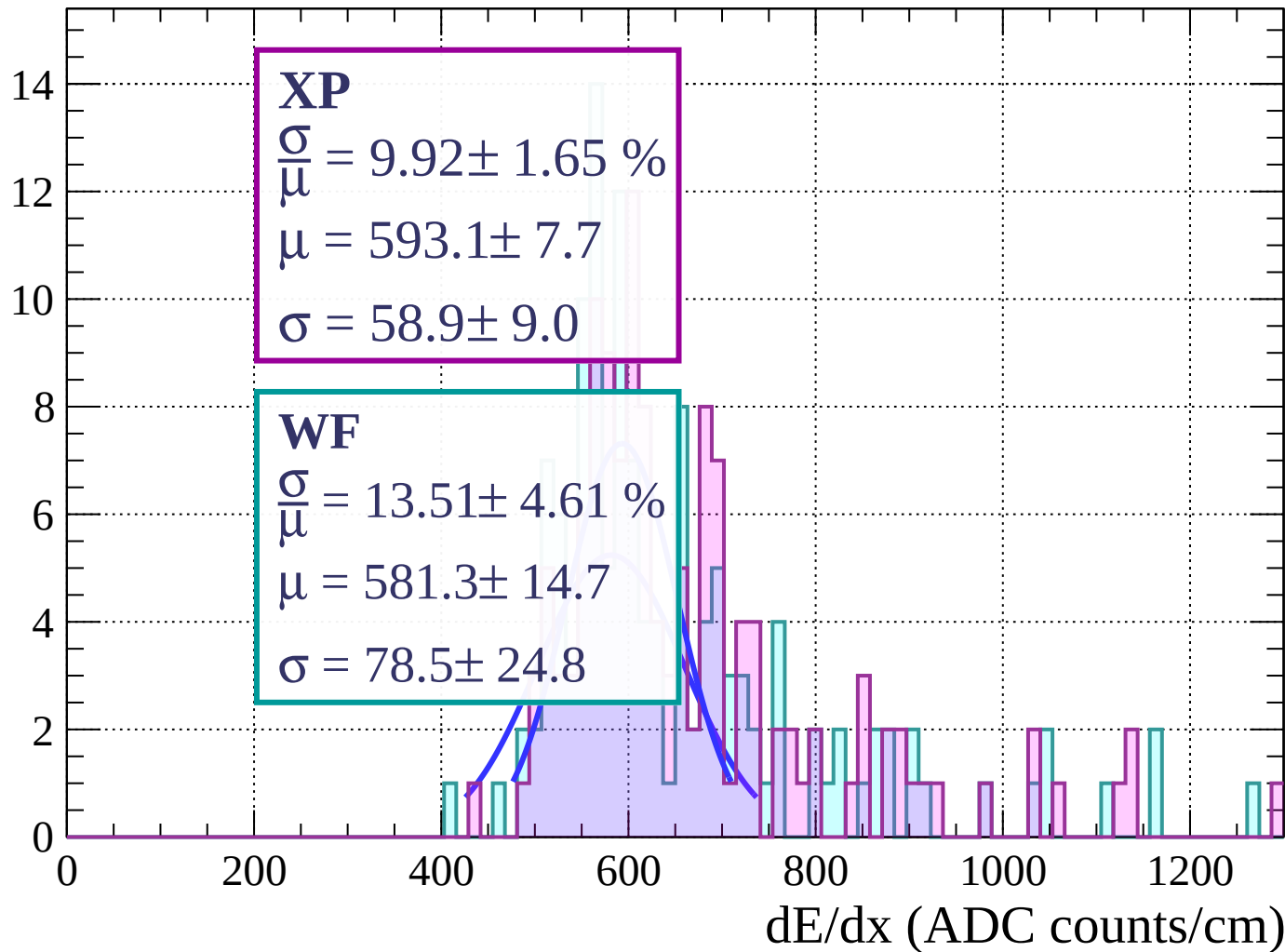
Energy loss | $51 < \phi < 54$

Count



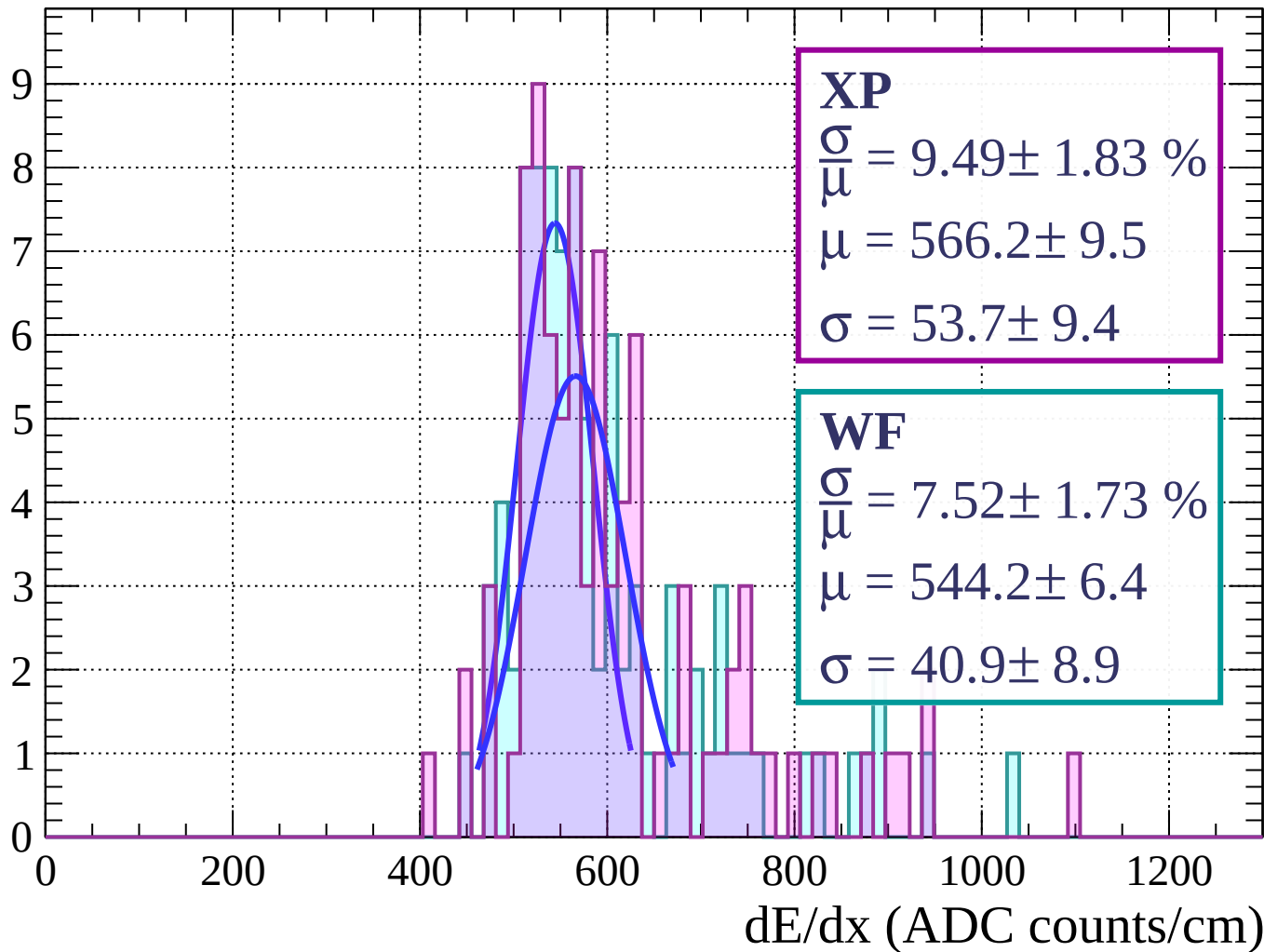
Energy loss | $54 < \phi < 57$

Count



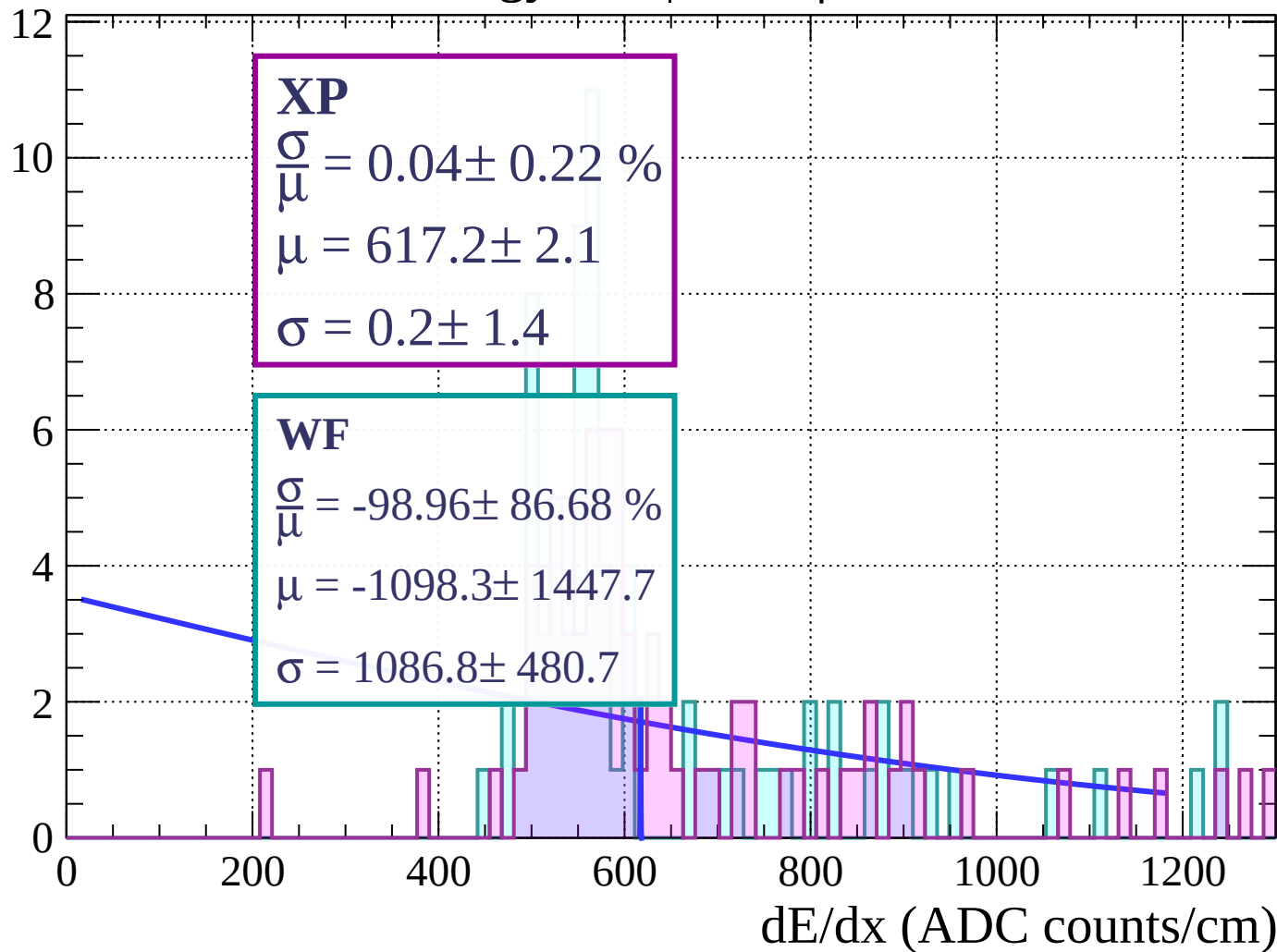
Energy loss | $57 < \phi < 60$

Count



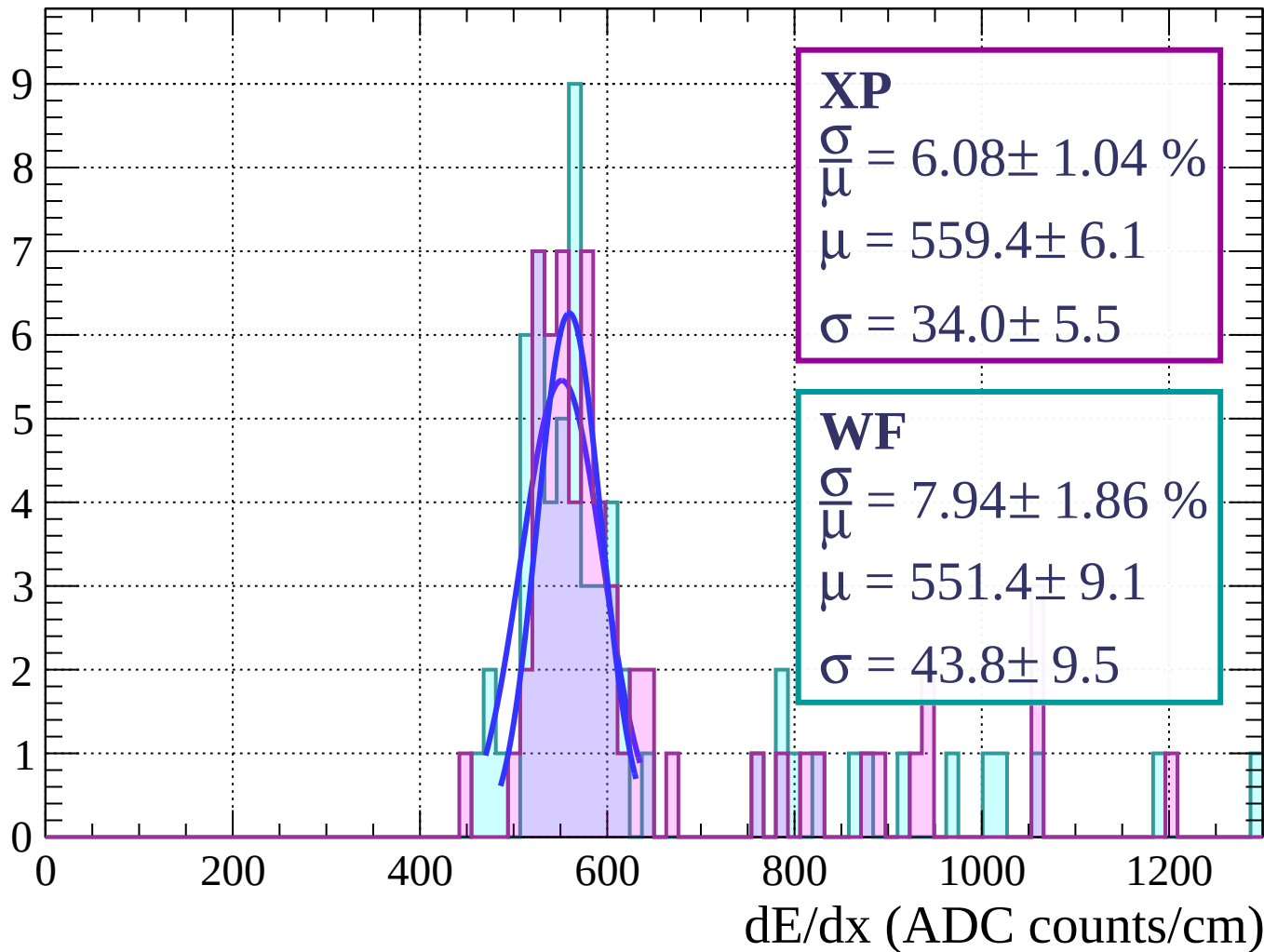
Energy loss | $60 < \phi < 63$

Count



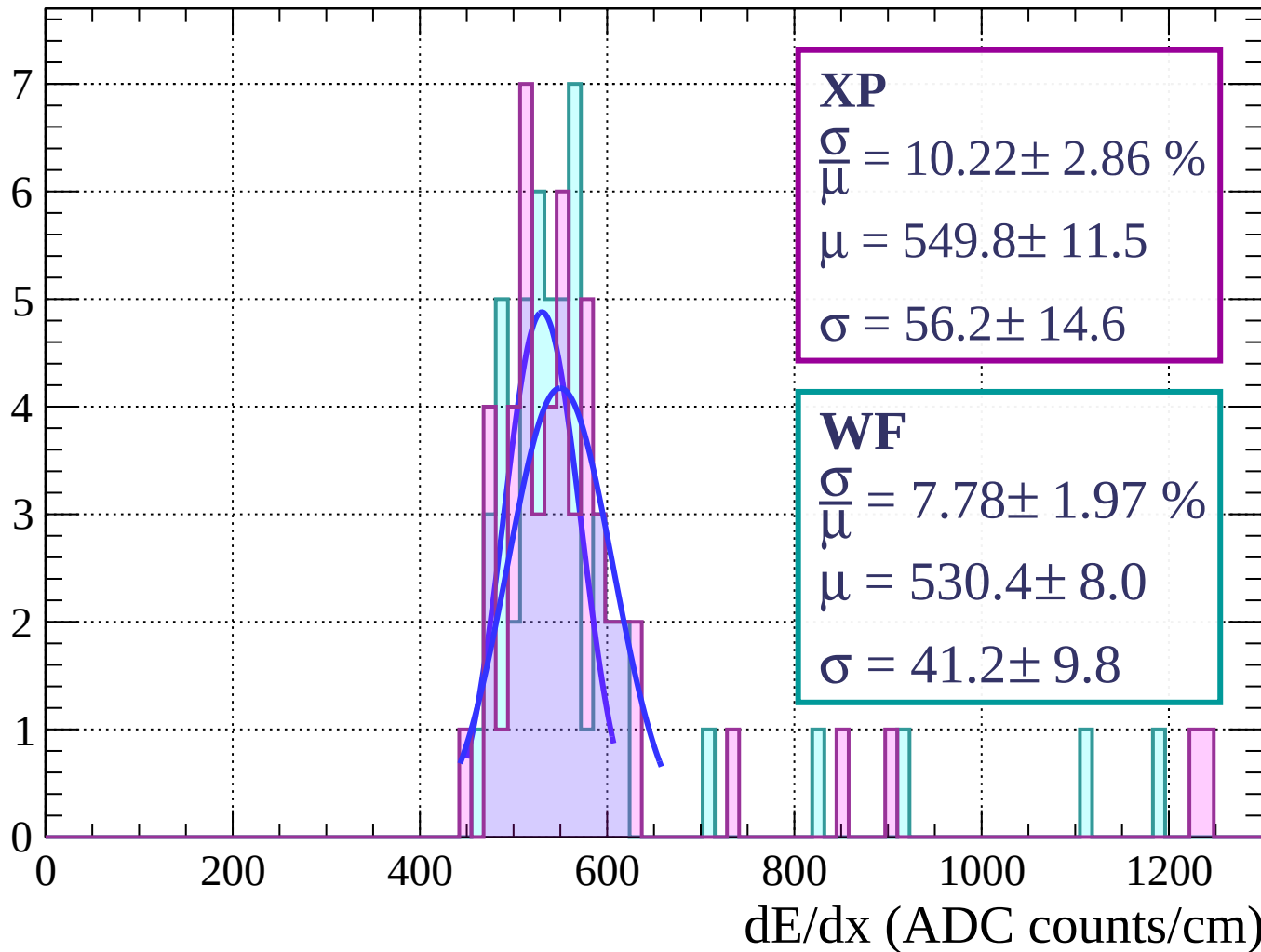
Energy loss | $63 < \phi < 66$

Count



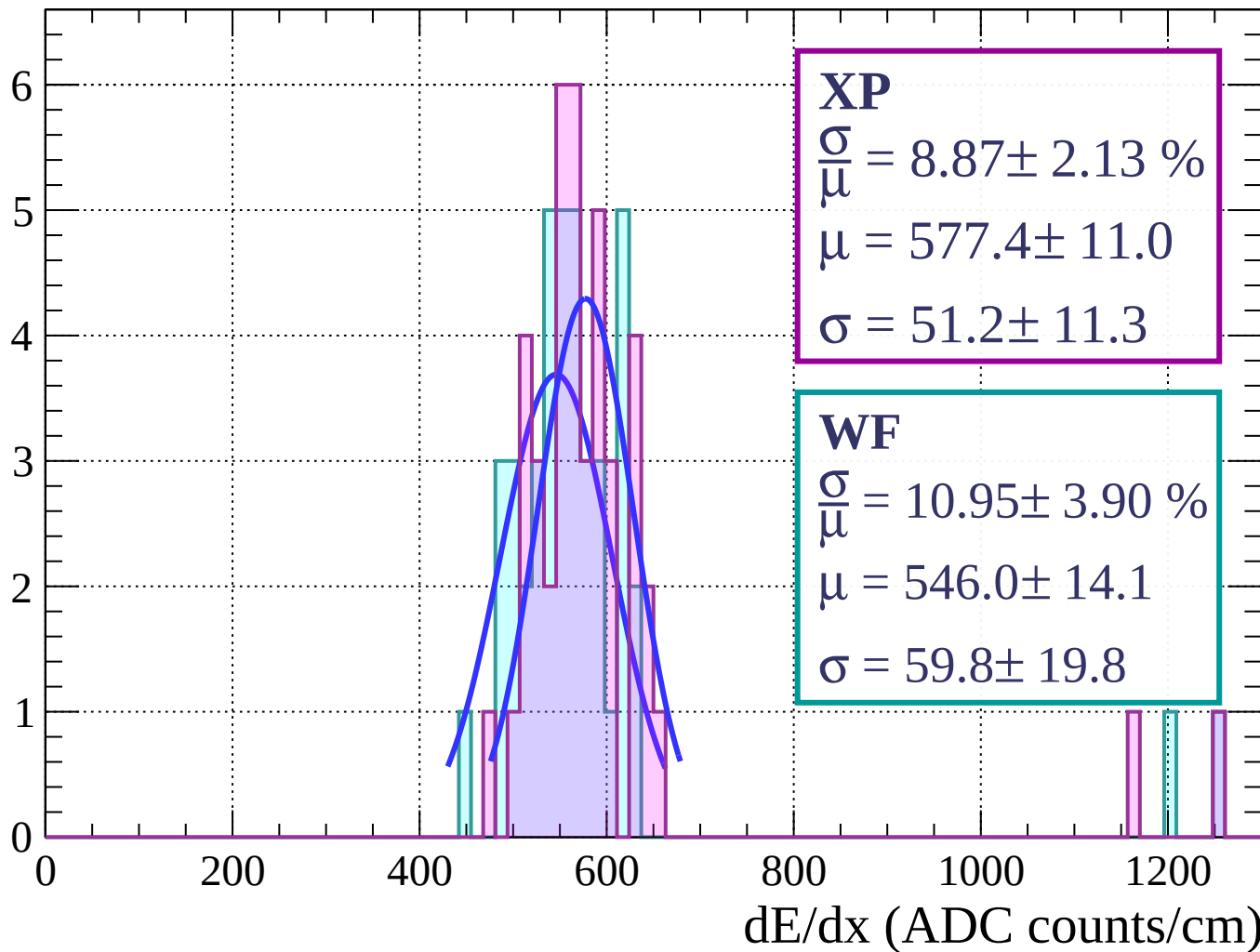
Energy loss | $66 < \phi < 69$

Count



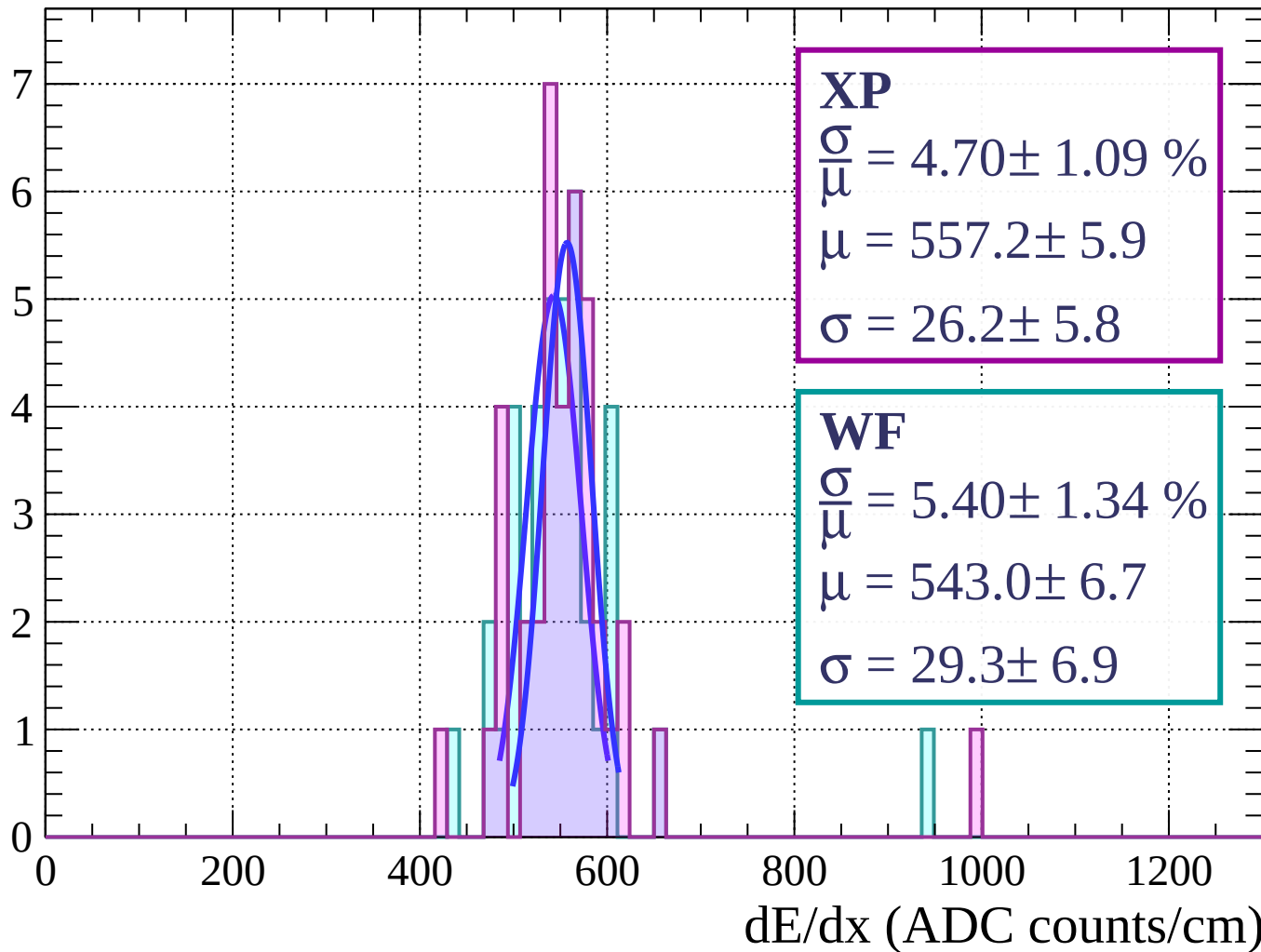
Energy loss | $69 < \phi < 72$

Count



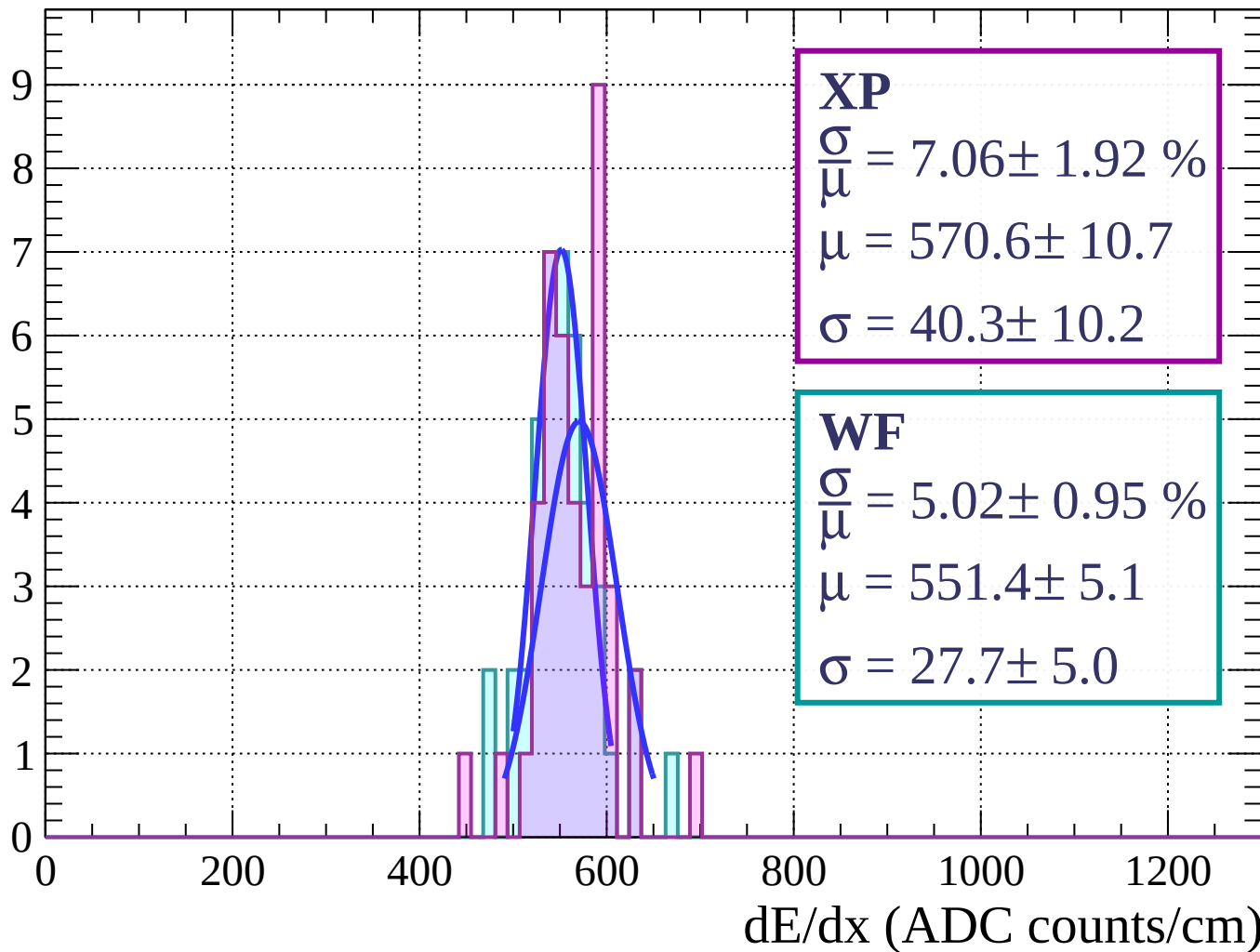
Energy loss | $72 < \phi < 75$

Count



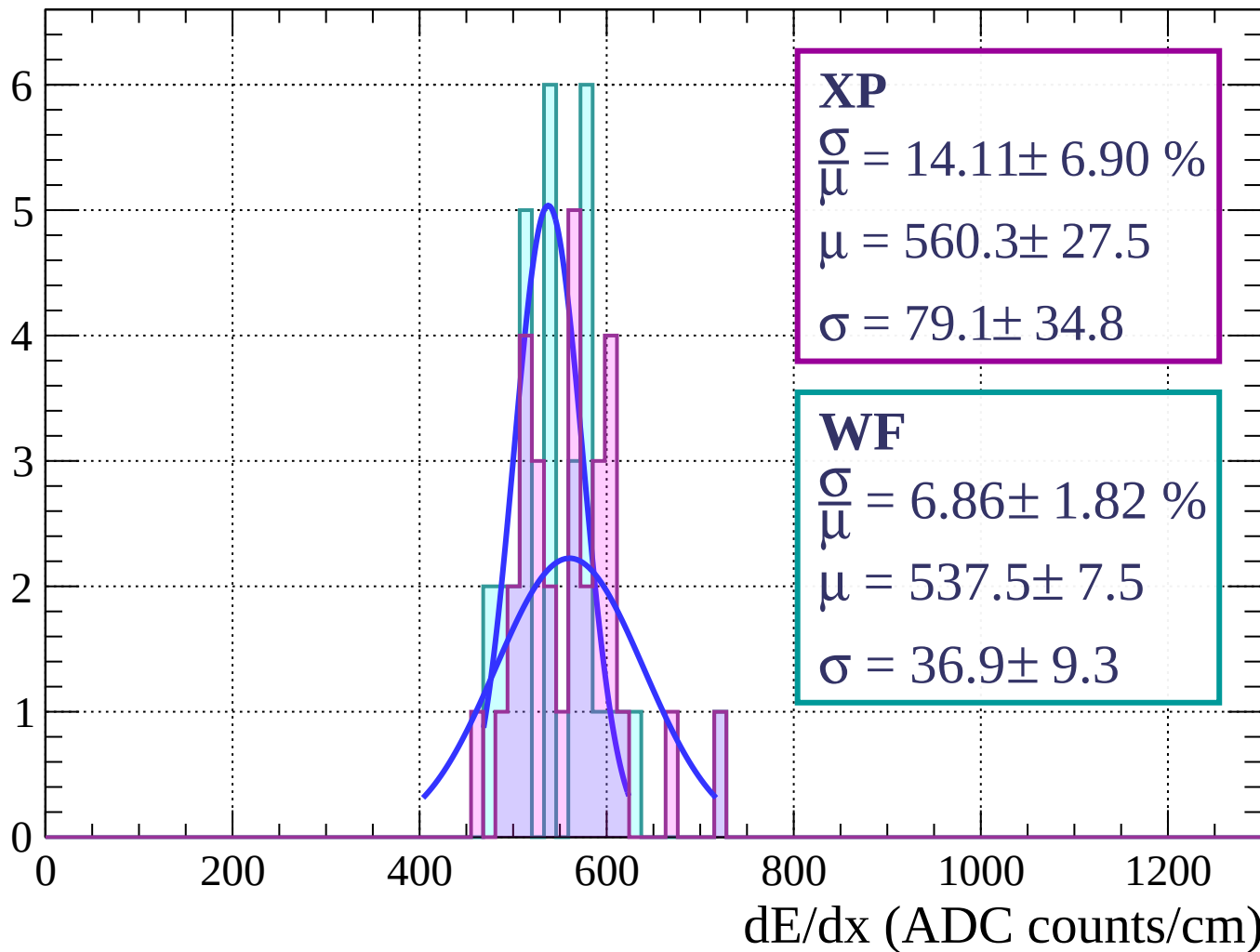
Energy loss | $75 < \phi < 78$

Count



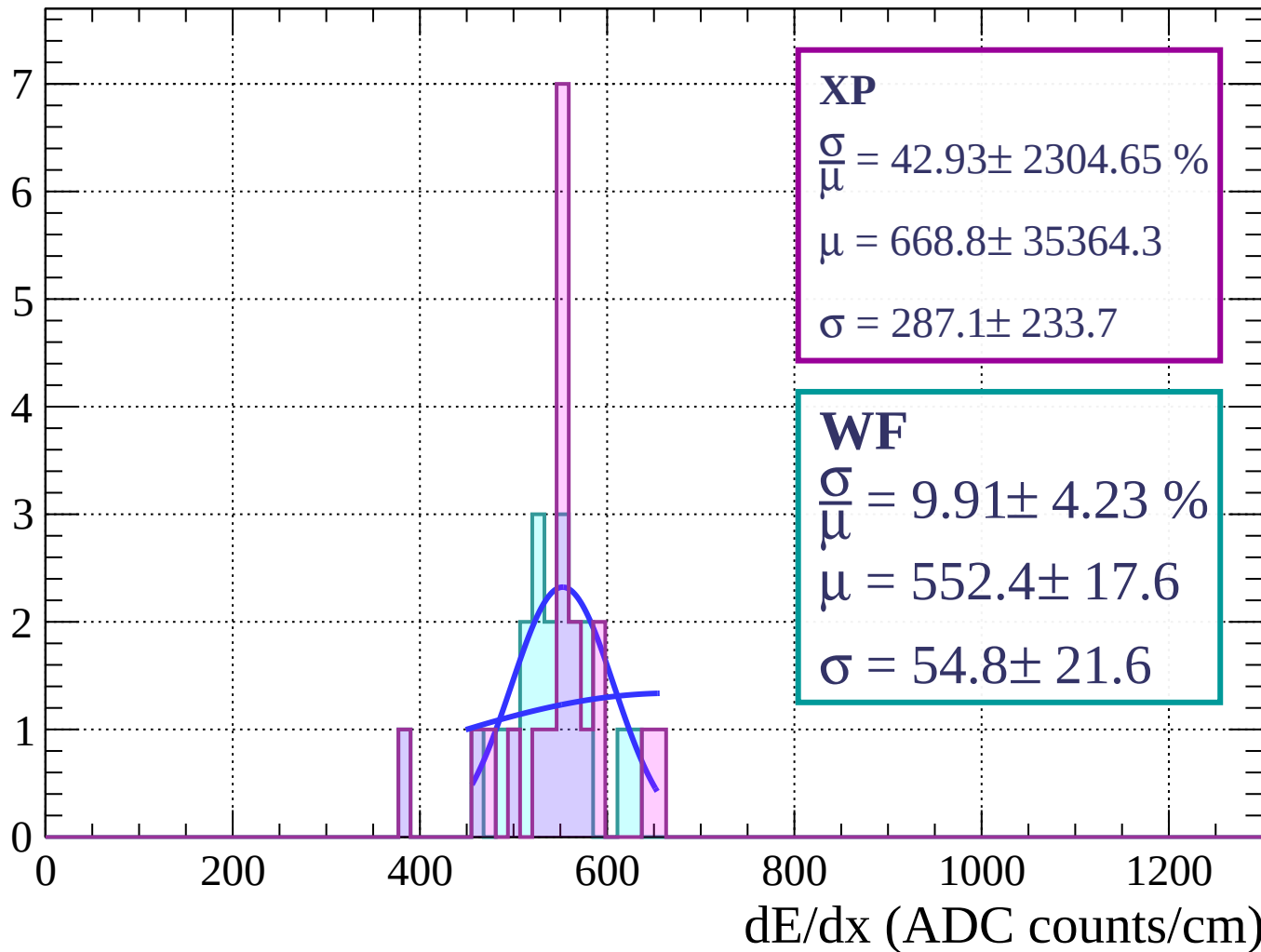
Energy loss | $78 < \phi < 81$

Count



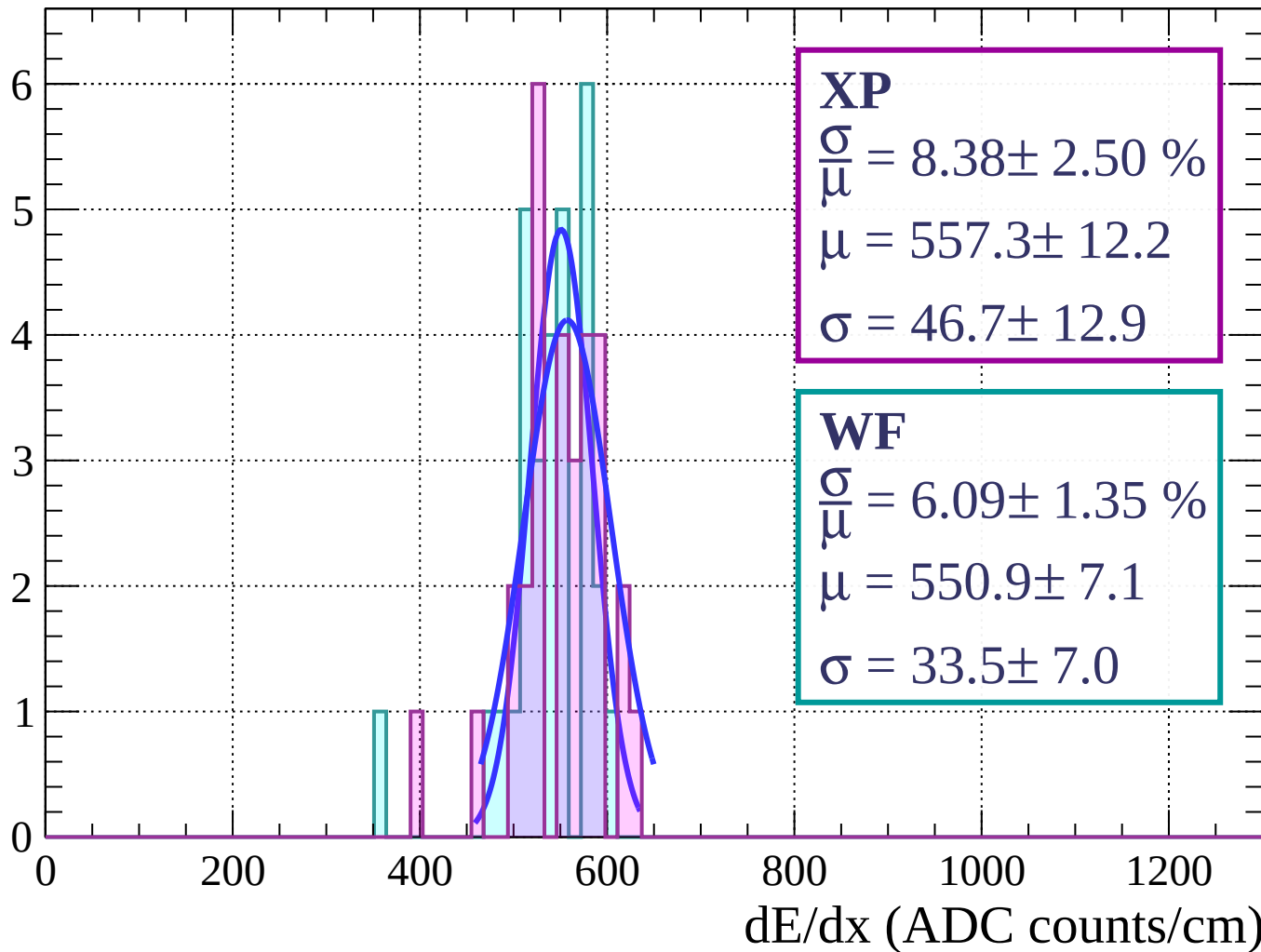
Energy loss | $81 < \phi < 84$

Count



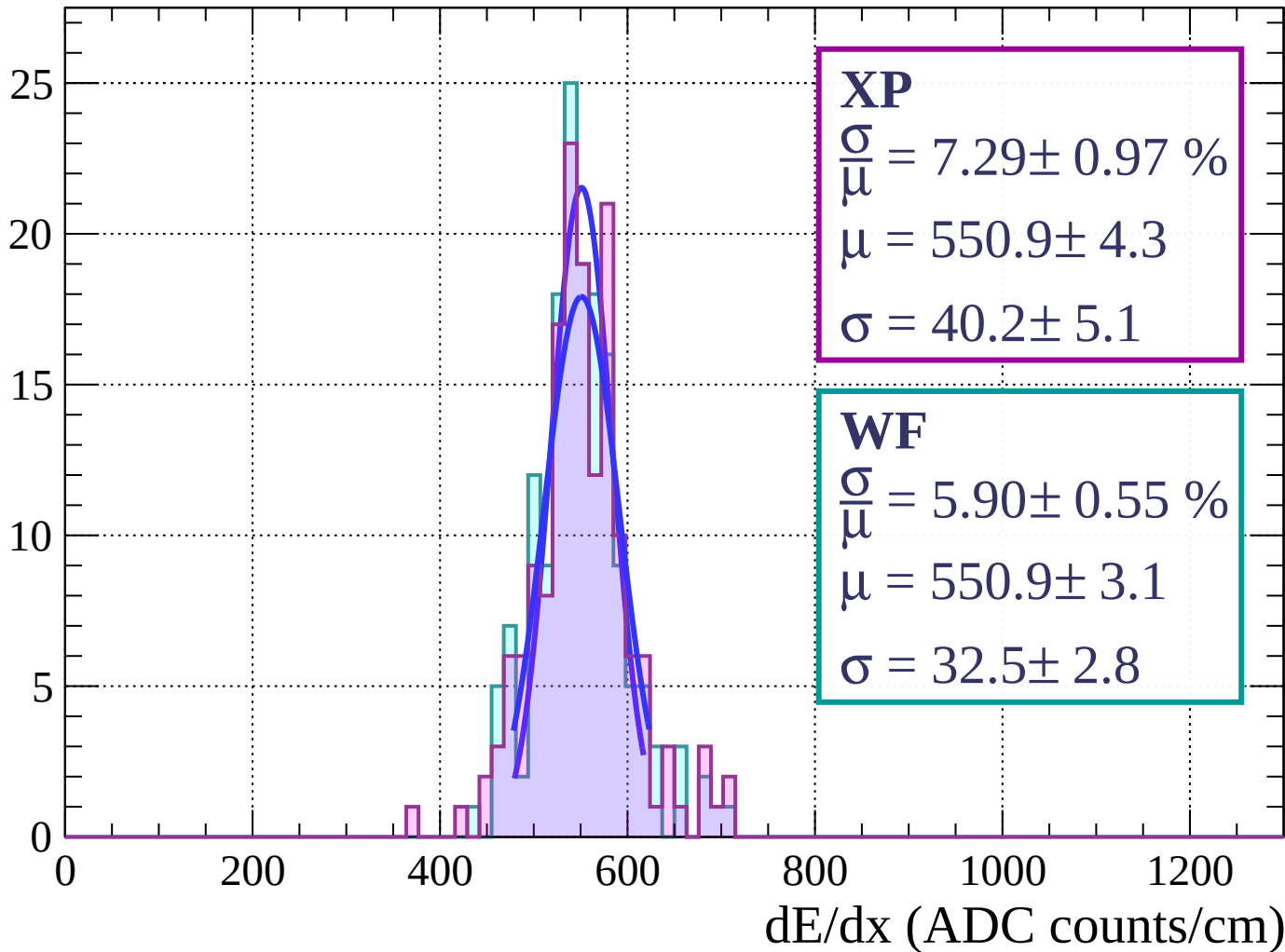
Energy loss | $84 < \phi < 87$

Count



Energy loss | $87 < \phi < 90$

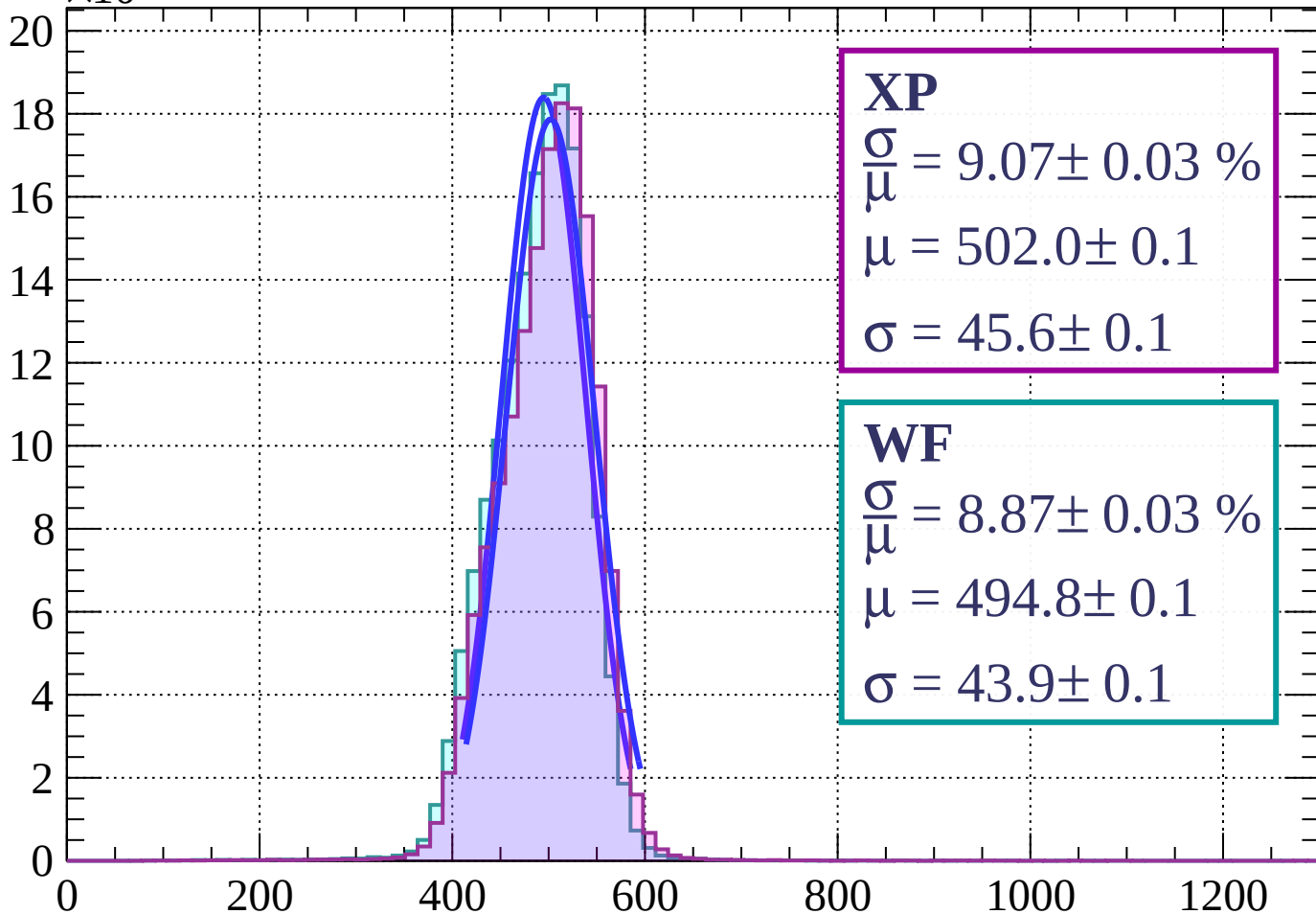
Count



Energy loss | $0 < \theta < 3$

Count

$\times 10^3$



XP

$$\frac{\sigma}{\mu} = 9.07 \pm 0.03 \%$$

$$\mu = 502.0 \pm 0.1$$

$$\sigma = 45.6 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 8.87 \pm 0.03 \%$$

$$\mu = 494.8 \pm 0.1$$

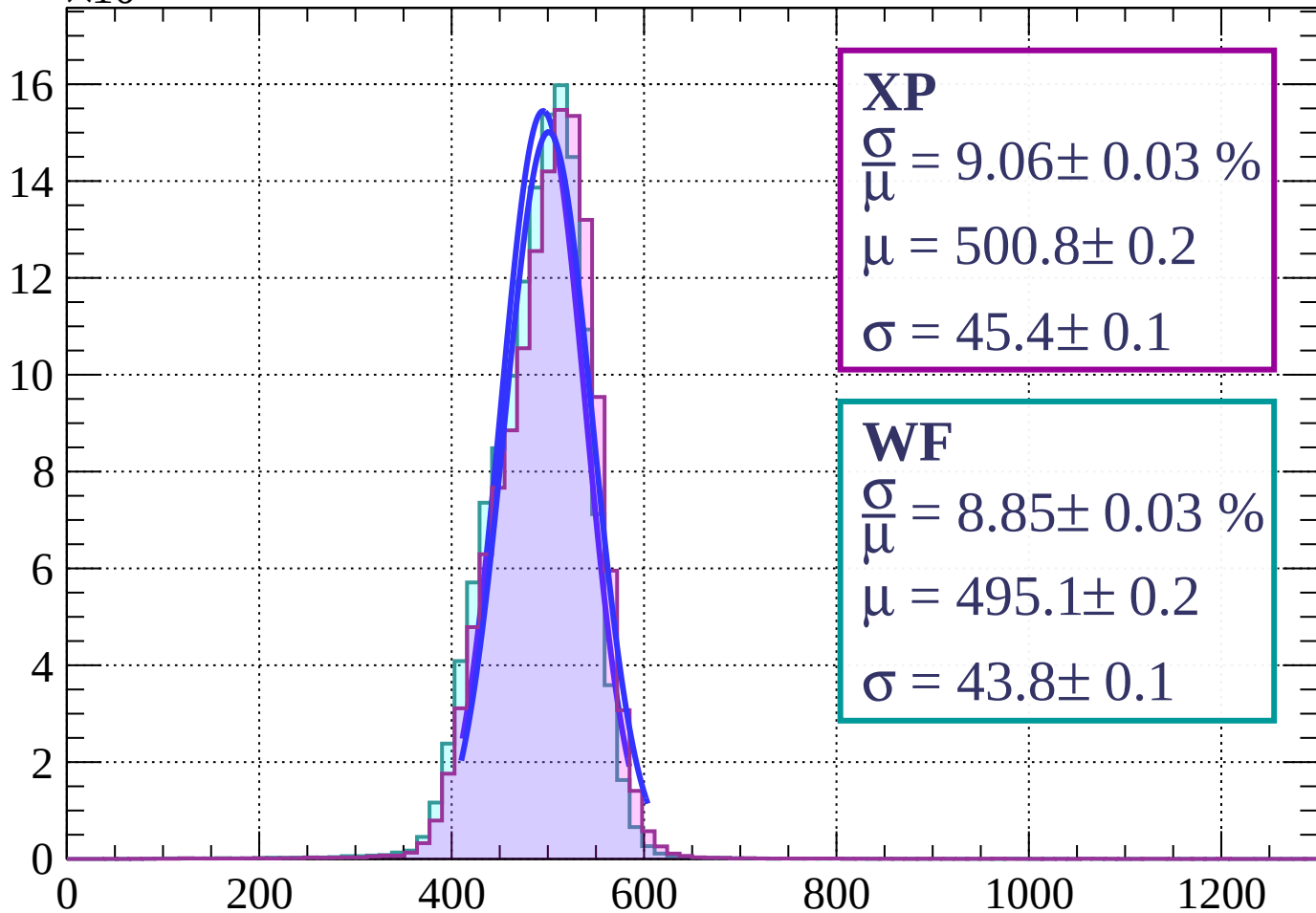
$$\sigma = 43.9 \pm 0.1$$

dE/dx (ADC counts/cm)

Energy loss | $3 < \theta < 6$

Count

$\times 10^3$



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.03 \%$$

$$\mu = 500.8 \pm 0.2$$

$$\sigma = 45.4 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 8.85 \pm 0.03 \%$$

$$\mu = 495.1 \pm 0.2$$

$$\sigma = 43.8 \pm 0.1$$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 9$

Count

$\times 10^3$

XP

$$\frac{\sigma}{\mu} = 9.12 \pm 0.04 \%$$

$$\mu = 502.7 \pm 0.2$$

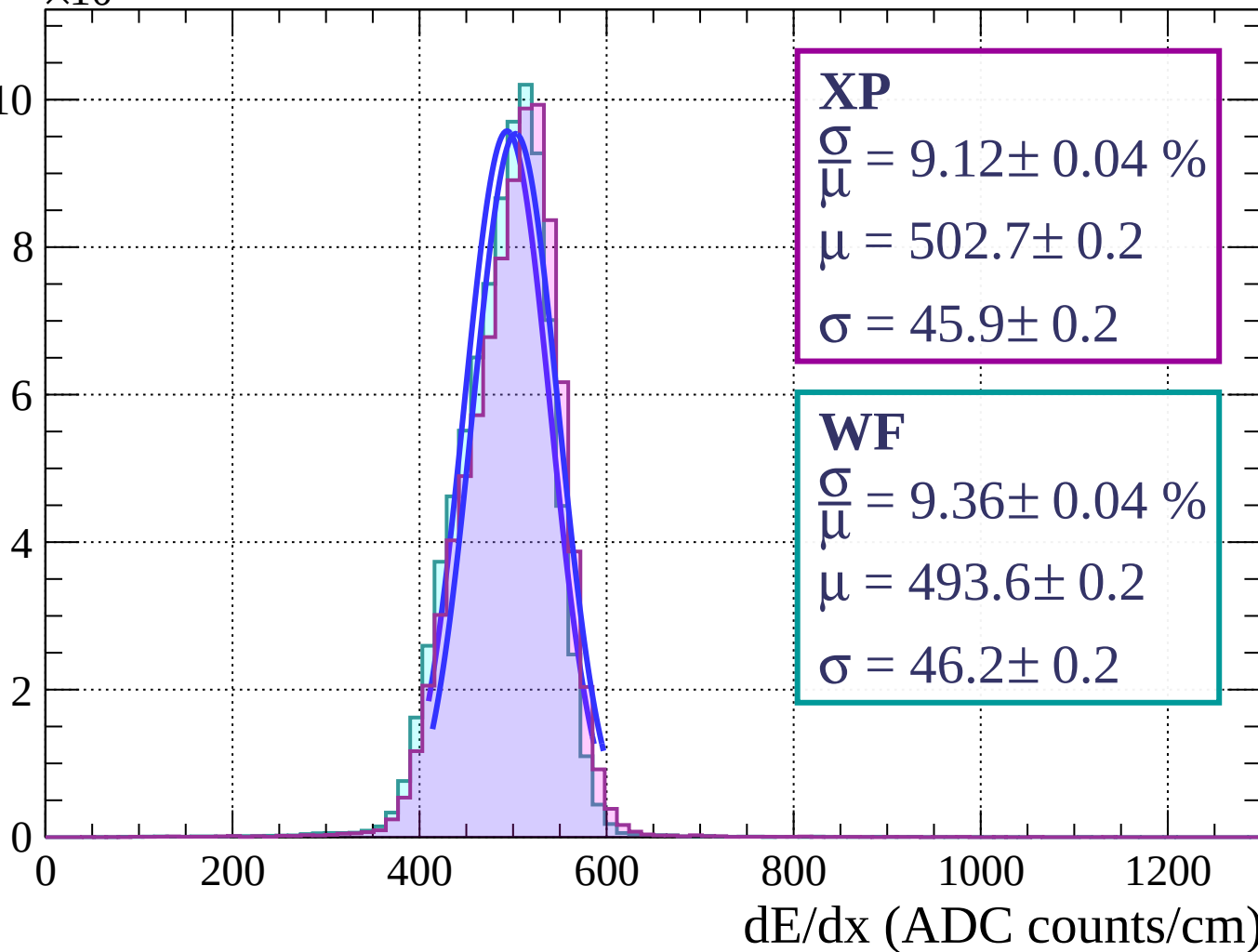
$$\sigma = 45.9 \pm 0.2$$

WF

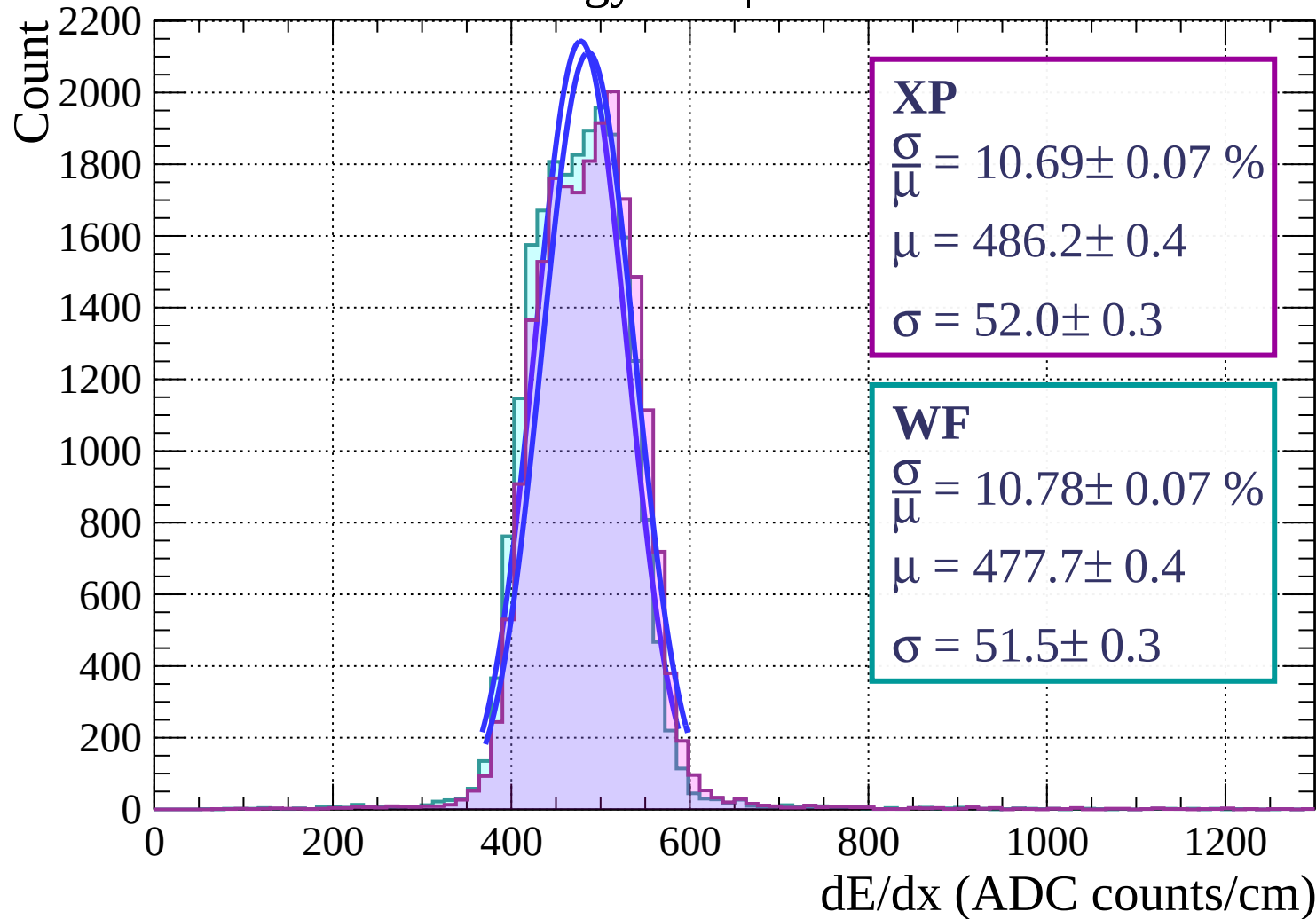
$$\frac{\sigma}{\mu} = 9.36 \pm 0.04 \%$$

$$\mu = 493.6 \pm 0.2$$

$$\sigma = 46.2 \pm 0.2$$

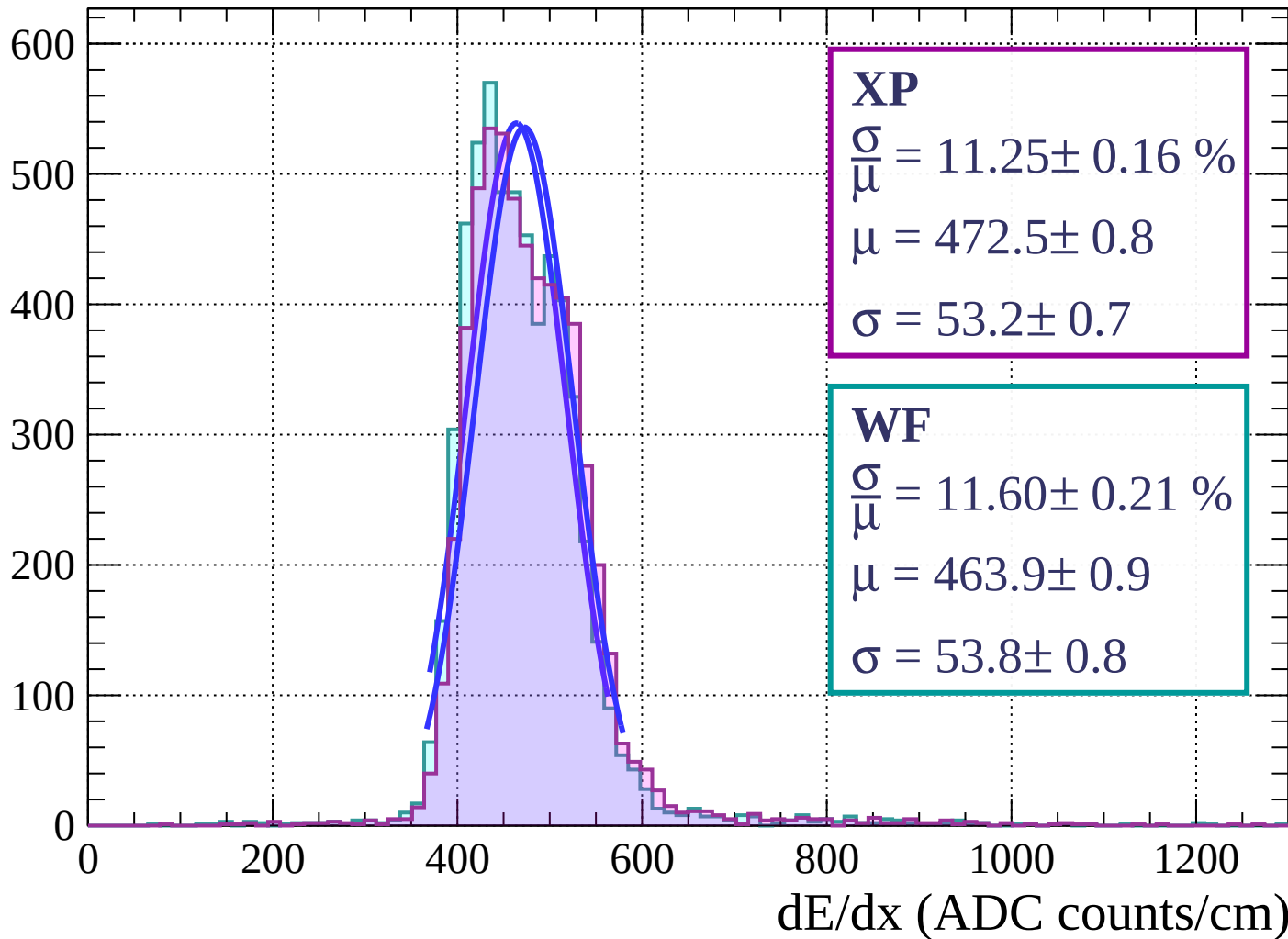


Energy loss | $9 < \theta < 12$



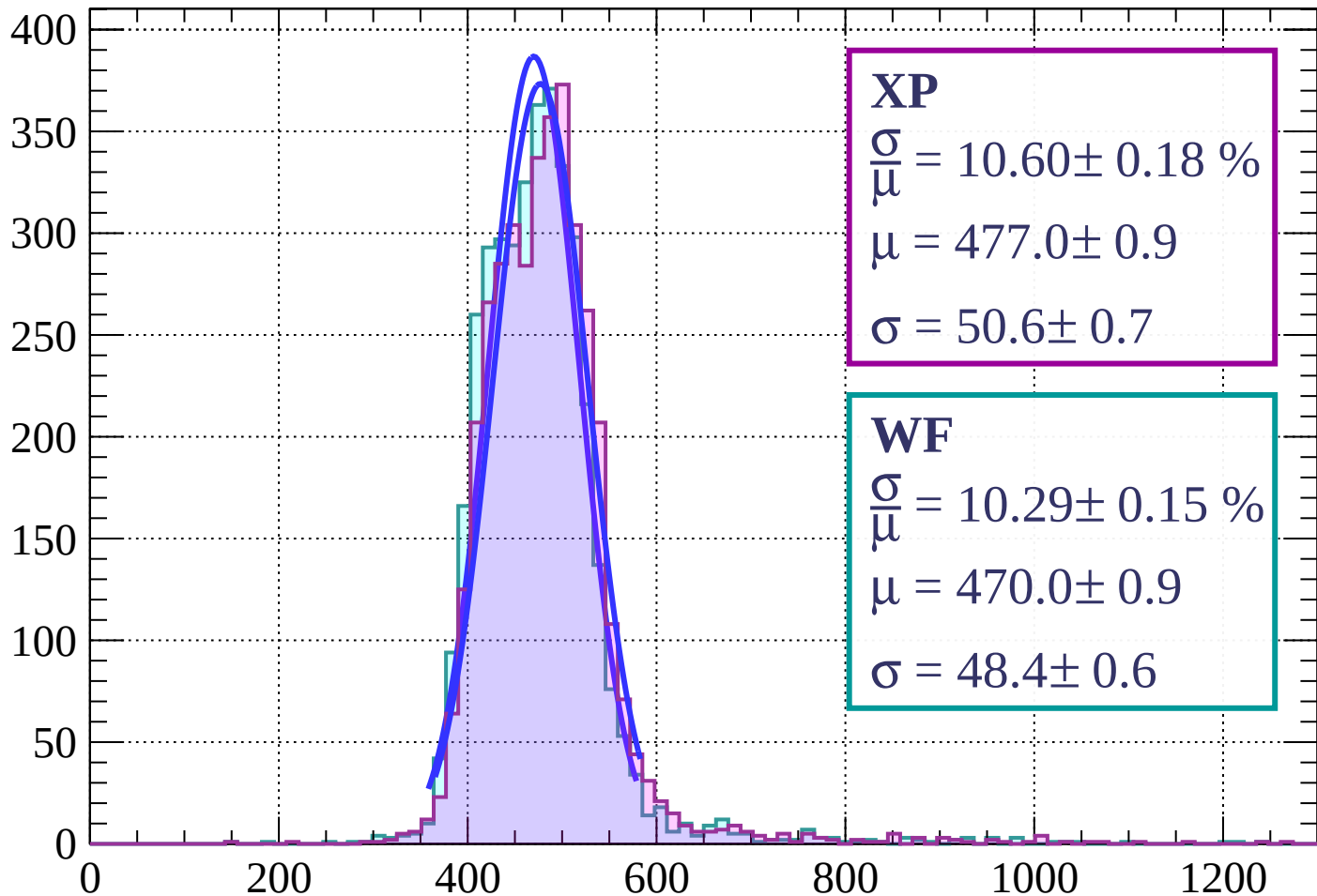
Energy loss | $12 < \theta < 15$

Count



Energy loss | $15 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 10.60 \pm 0.18 \%$$

$$\mu = 477.0 \pm 0.9$$

$$\sigma = 50.6 \pm 0.7$$

WF

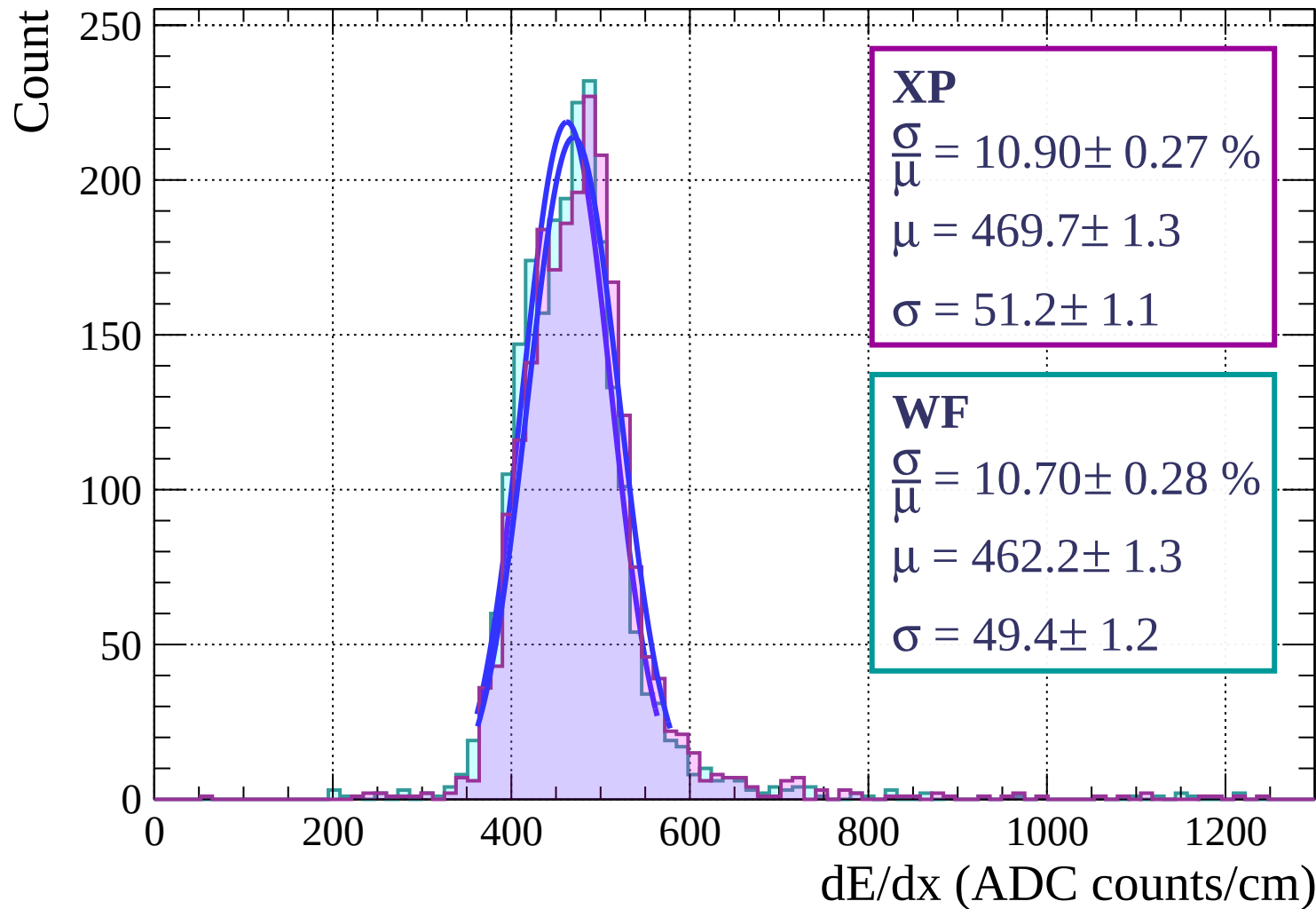
$$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$$

$$\mu = 470.0 \pm 0.9$$

$$\sigma = 48.4 \pm 0.6$$

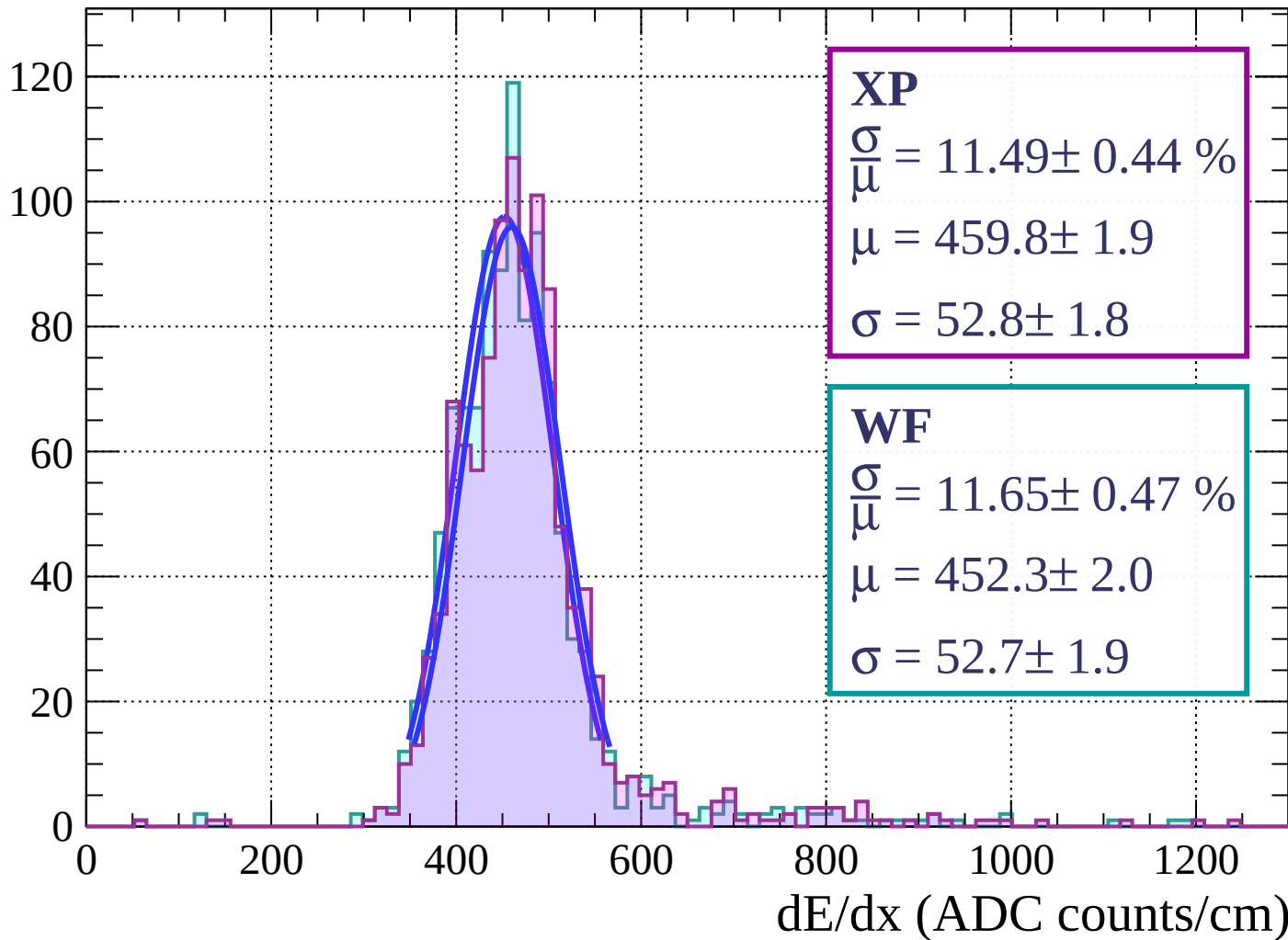
dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 21$



Energy loss | $21 < \theta < 24$

Count



Energy loss | $24 < \theta < 27$

Count

XP

$$\frac{\sigma}{\mu} = 12.50 \pm 0.77 \%$$

$$\mu = 450.5 \pm 3.1$$

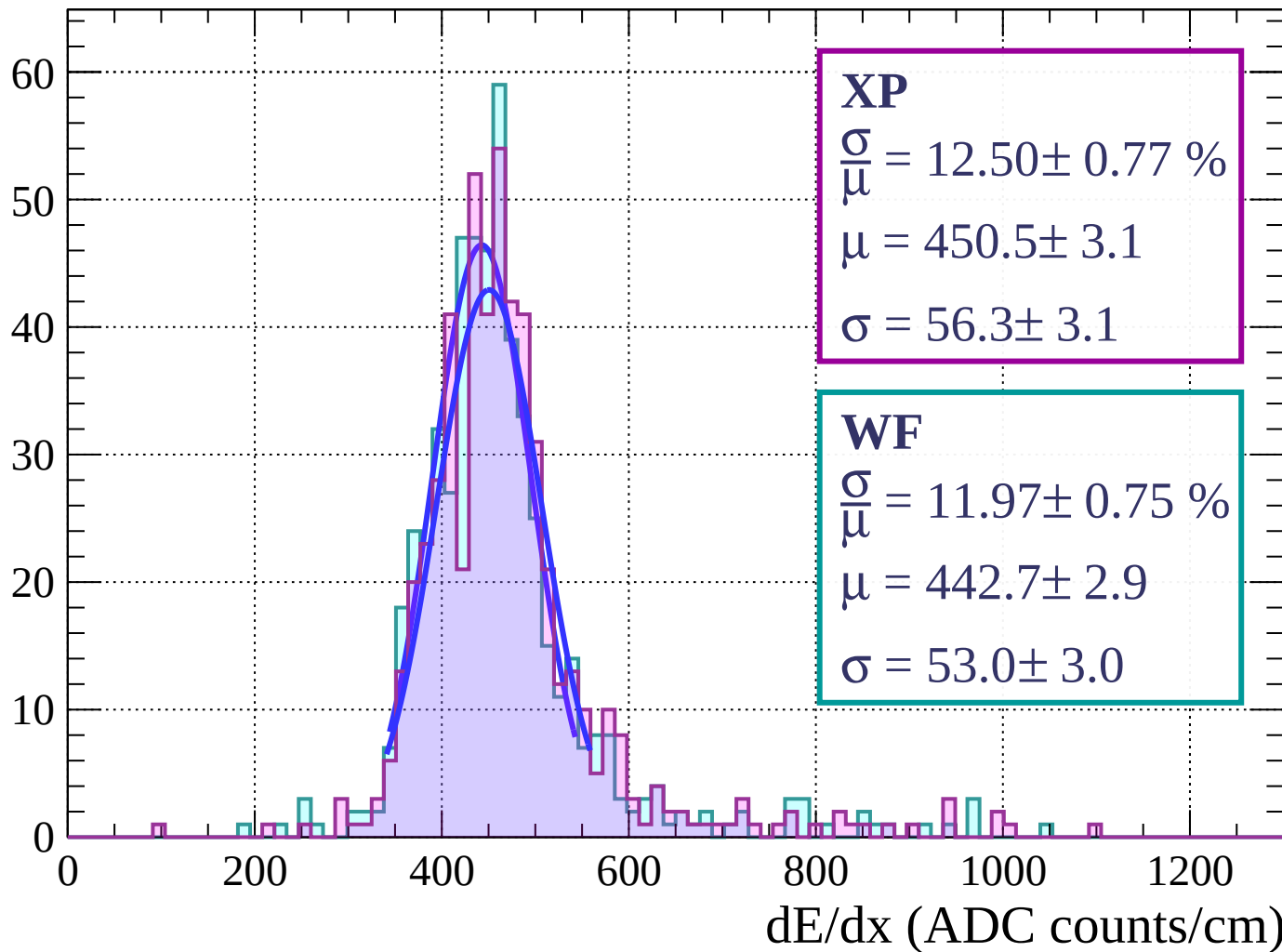
$$\sigma = 56.3 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 11.97 \pm 0.75 \%$$

$$\mu = 442.7 \pm 2.9$$

$$\sigma = 53.0 \pm 3.0$$



Energy loss | $27 < \theta < 30$

Count

XP

$$\frac{\sigma}{\mu} = 11.41 \pm 0.80 \%$$

$$\mu = 435.9 \pm 3.4$$

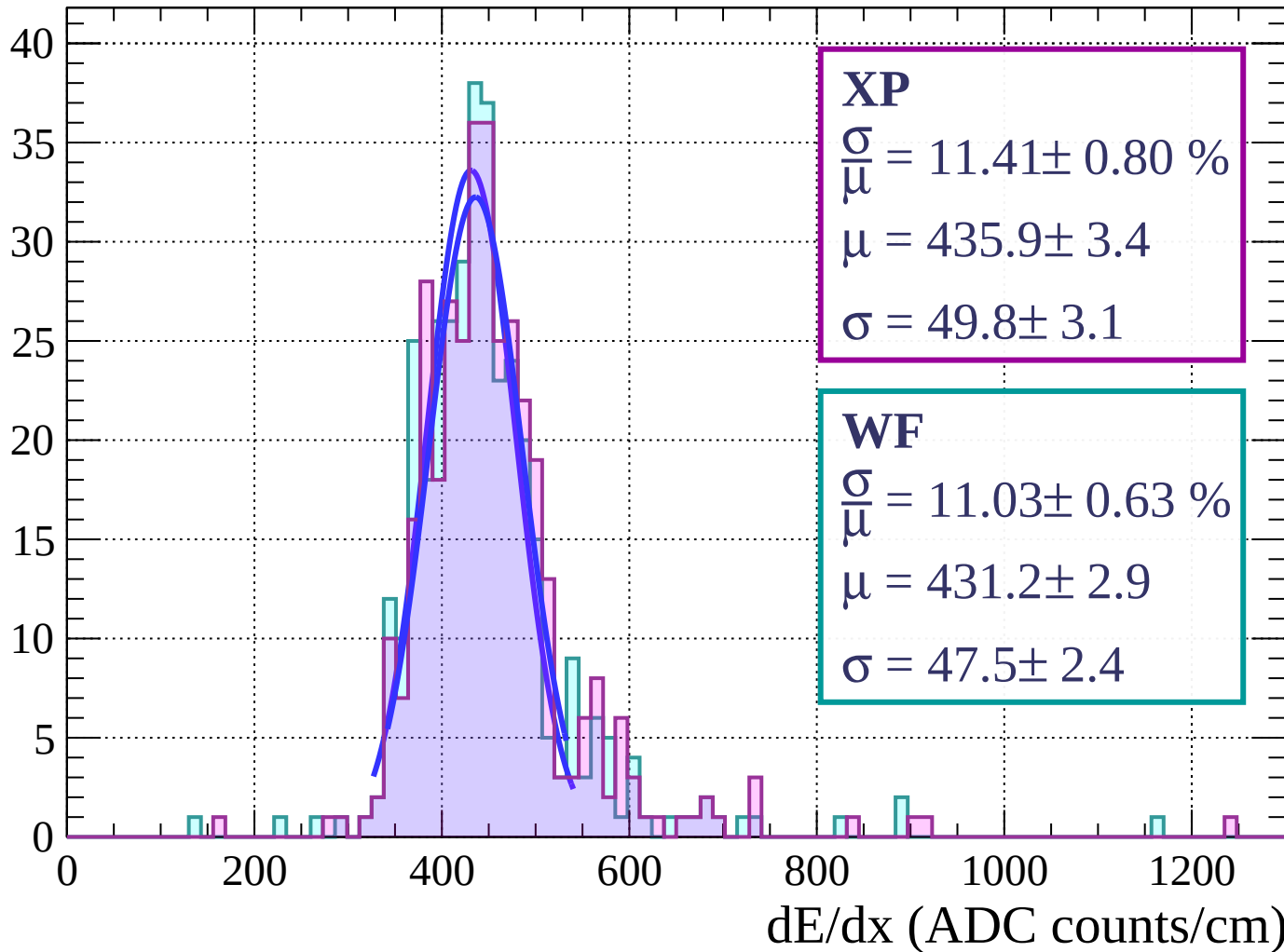
$$\sigma = 49.8 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 11.03 \pm 0.63 \%$$

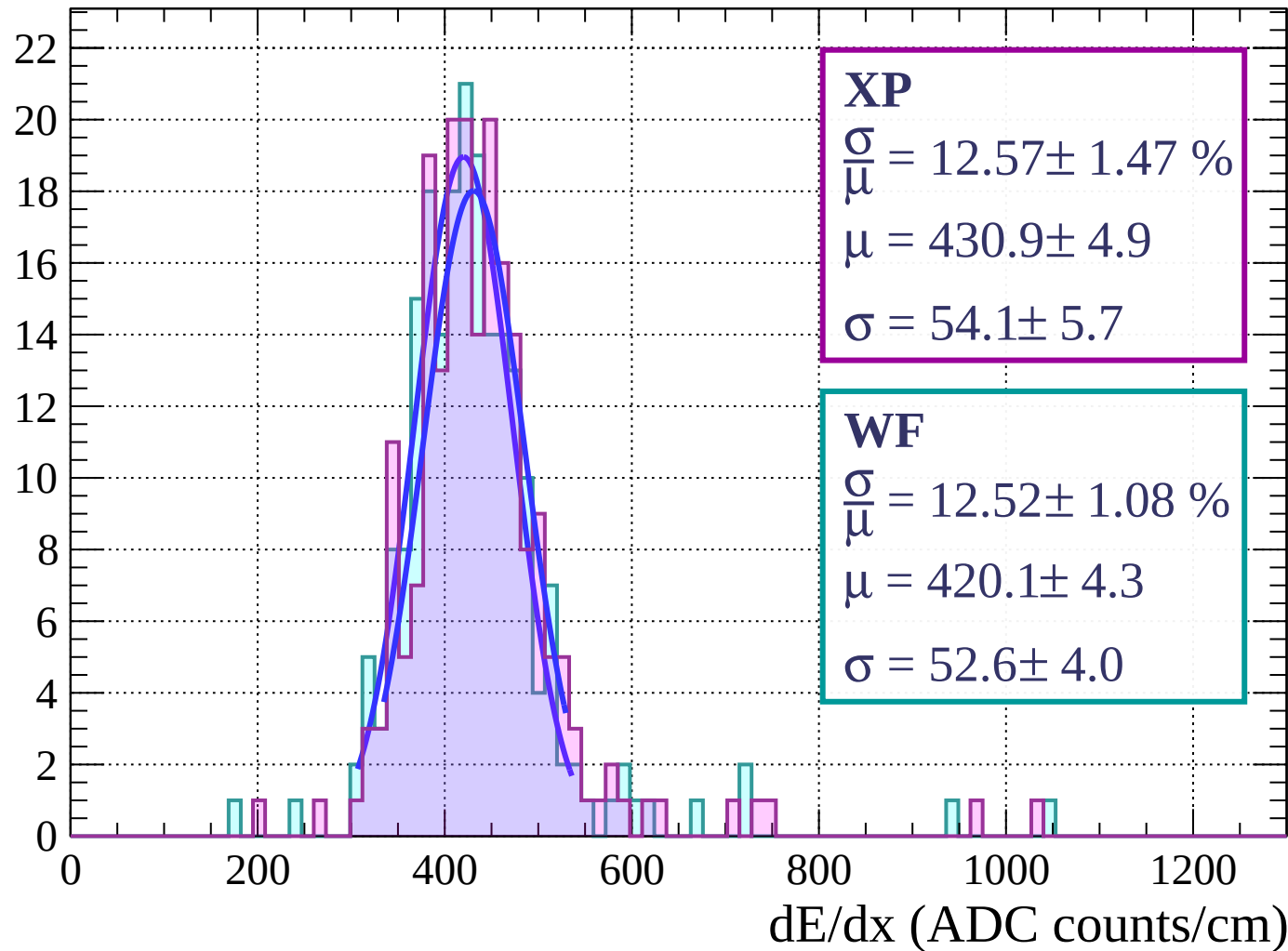
$$\mu = 431.2 \pm 2.9$$

$$\sigma = 47.5 \pm 2.4$$



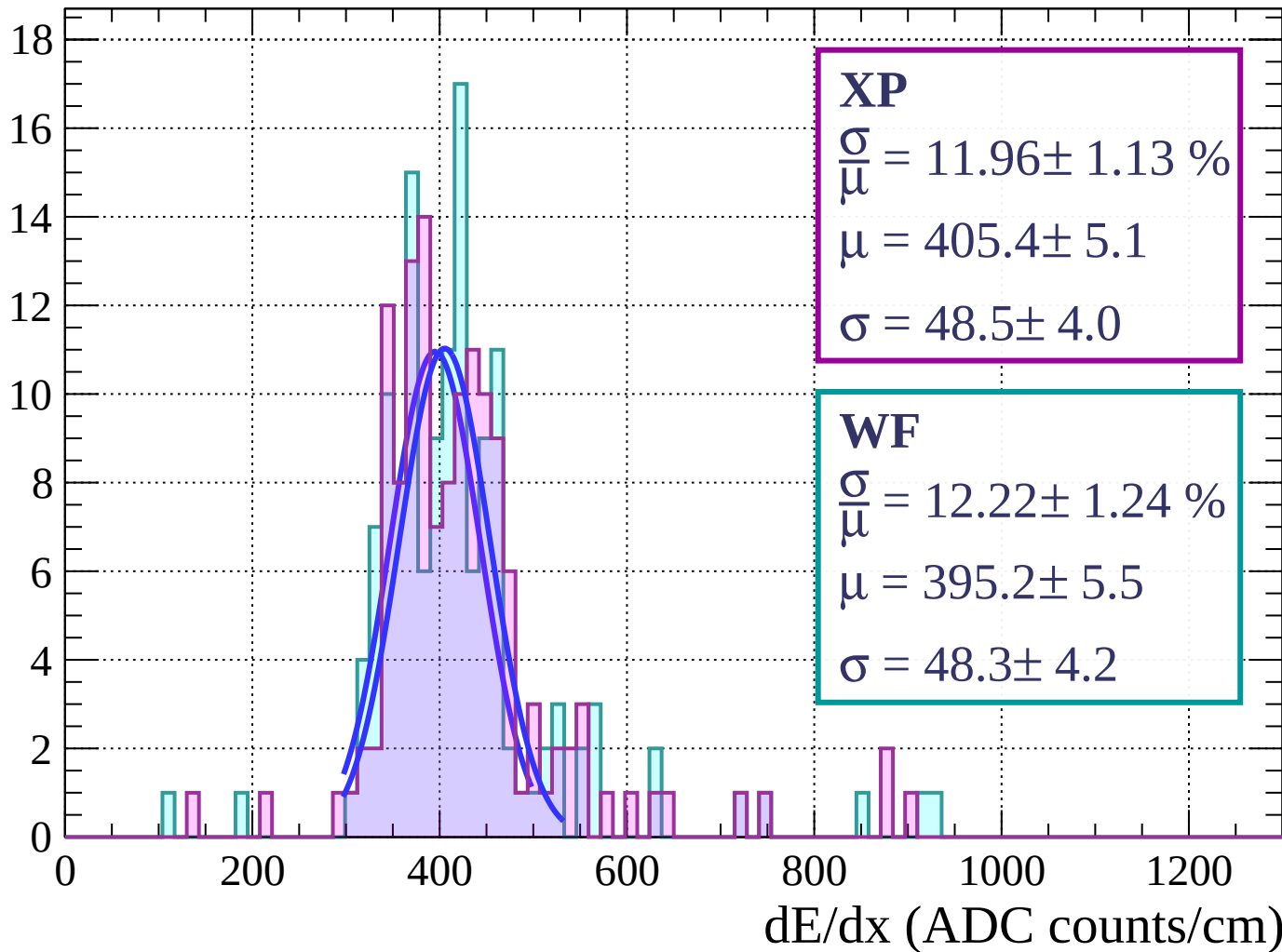
Energy loss | $30 < \theta < 33$

Count



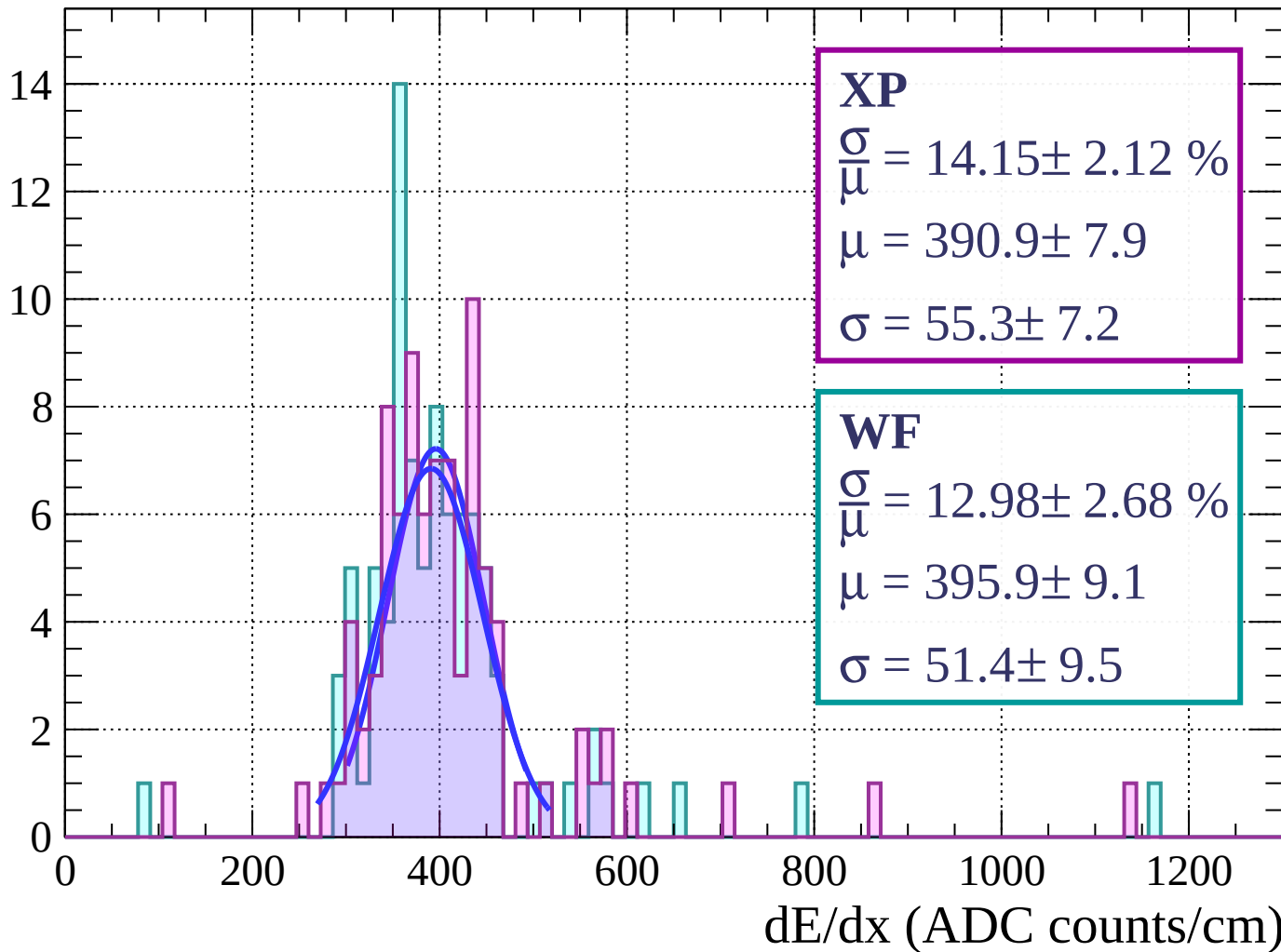
Energy loss | $33 < \theta < 36$

Count



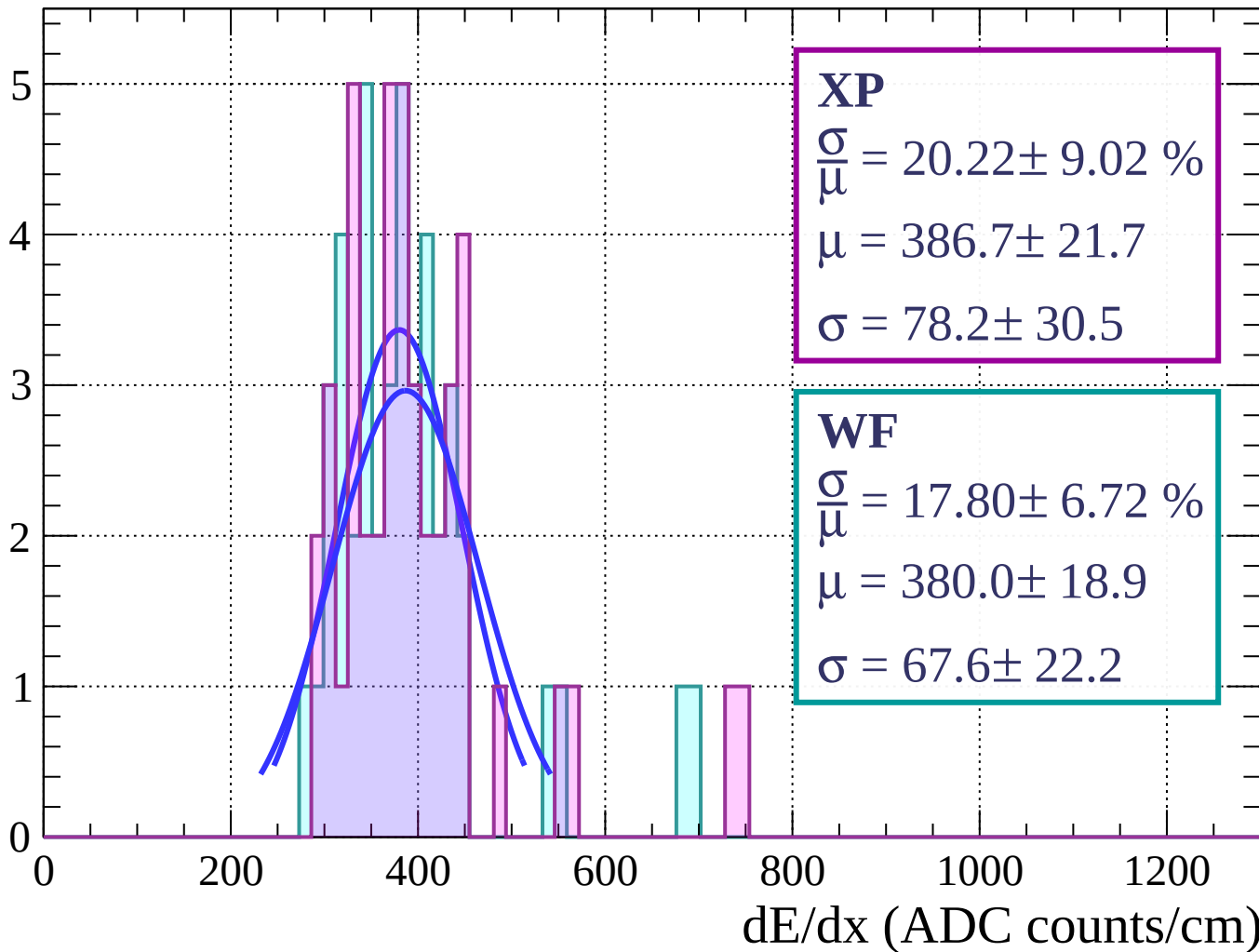
Energy loss | $36 < \theta < 39$

Count



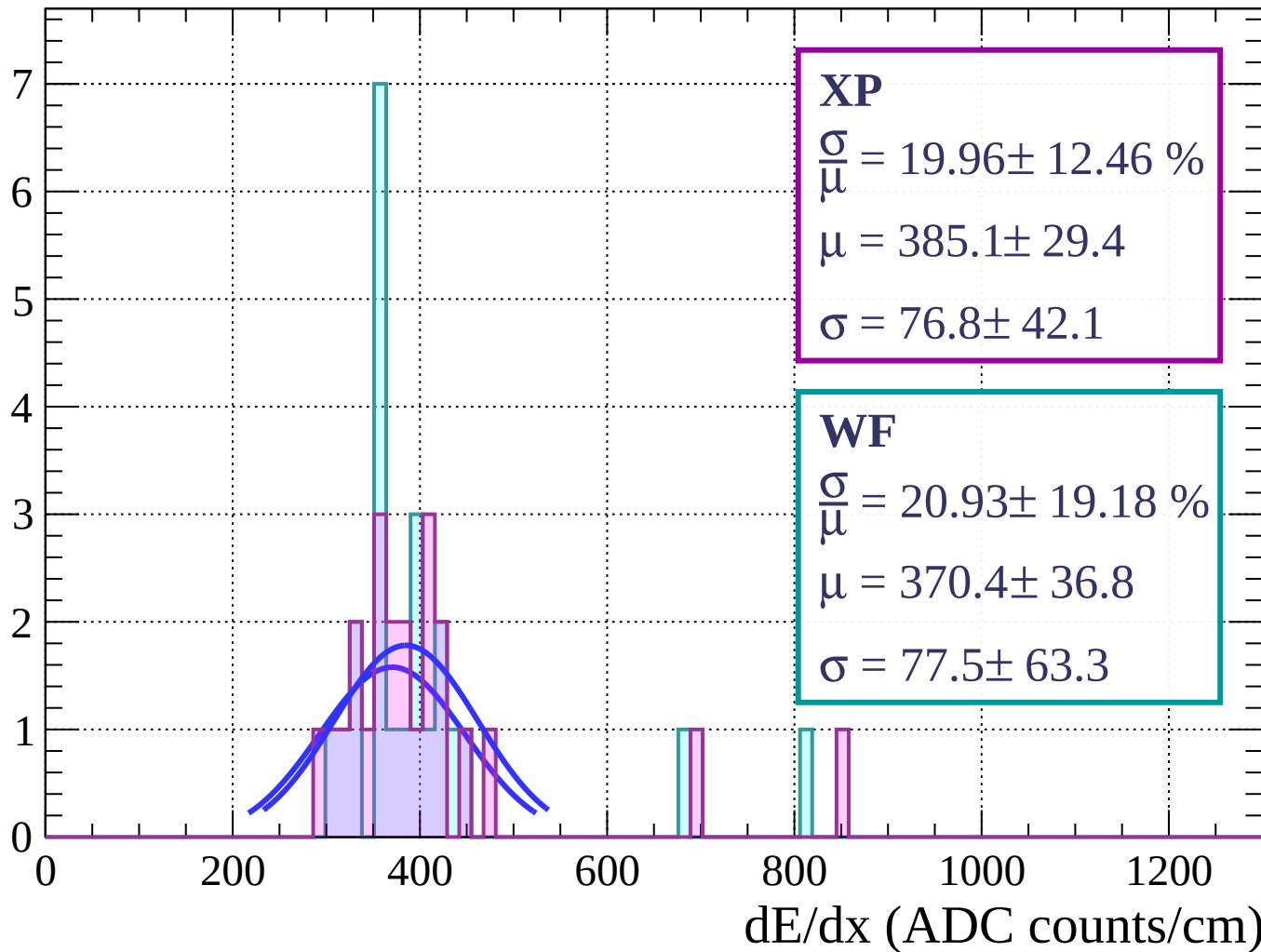
Energy loss | $39 < \theta < 42$

Count



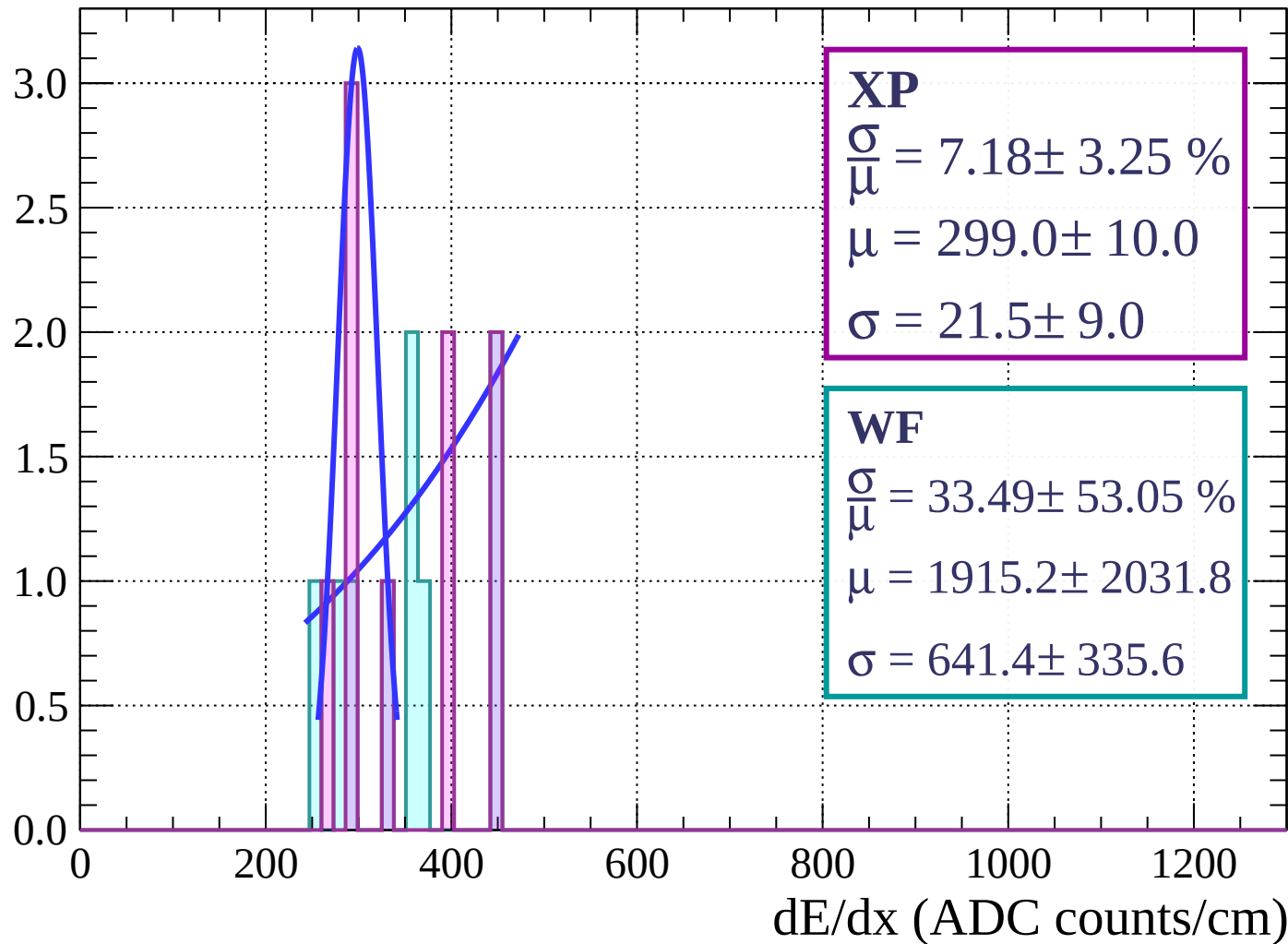
Energy loss | $42 < \theta < 45$

Count



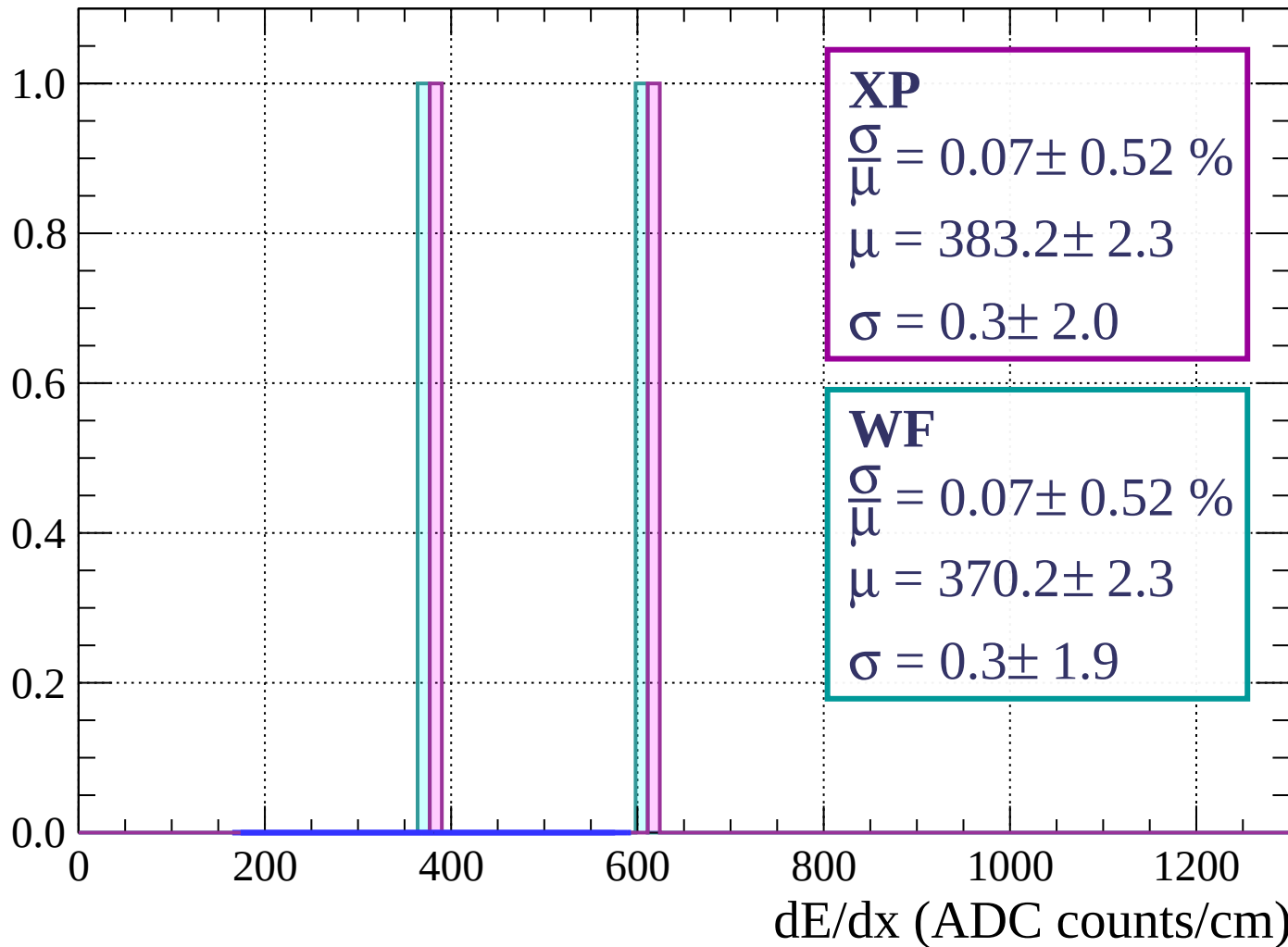
Energy loss | $45 < \theta < 48$

Count



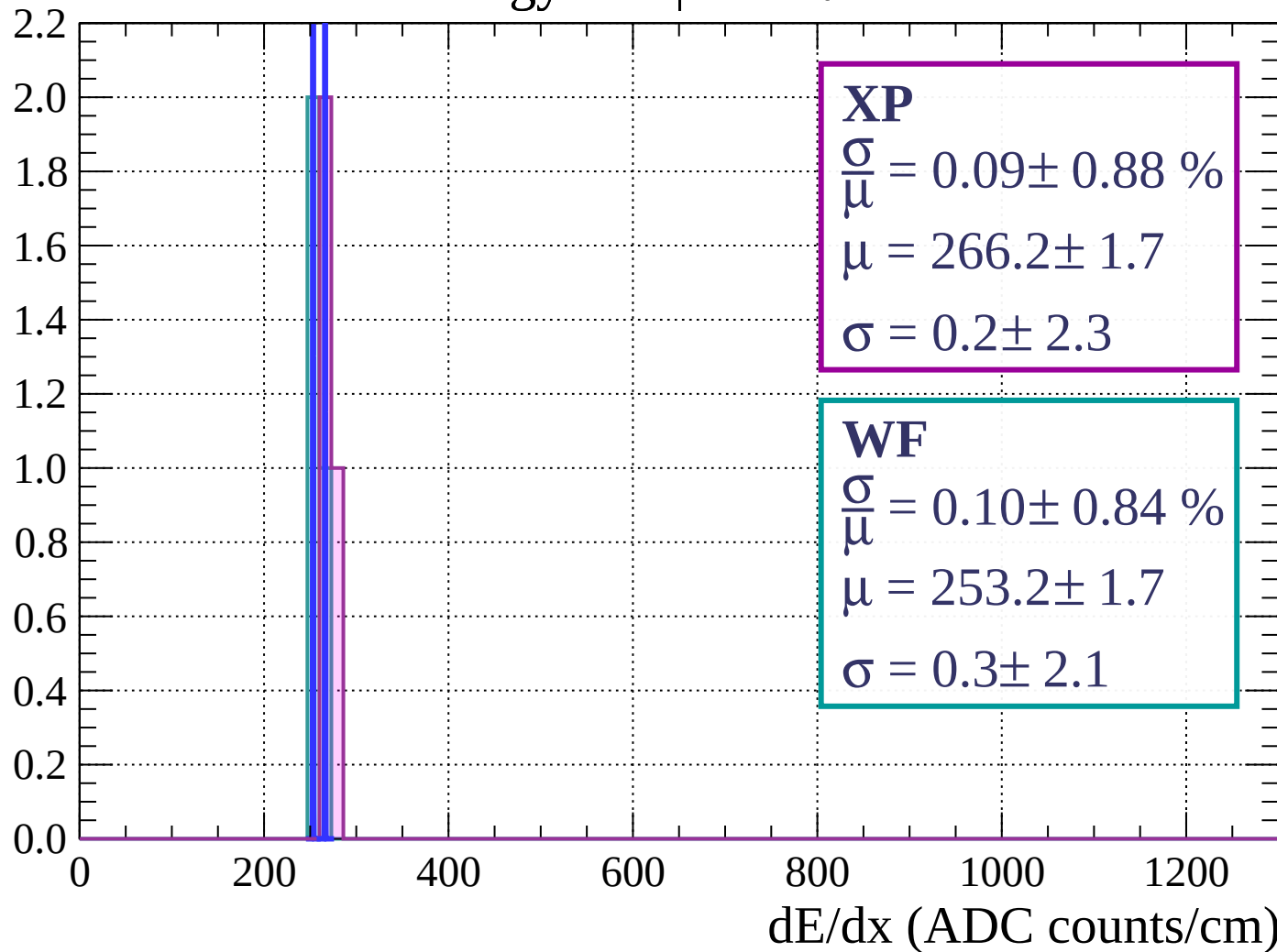
Energy loss | $48 < \theta < 51$

Count



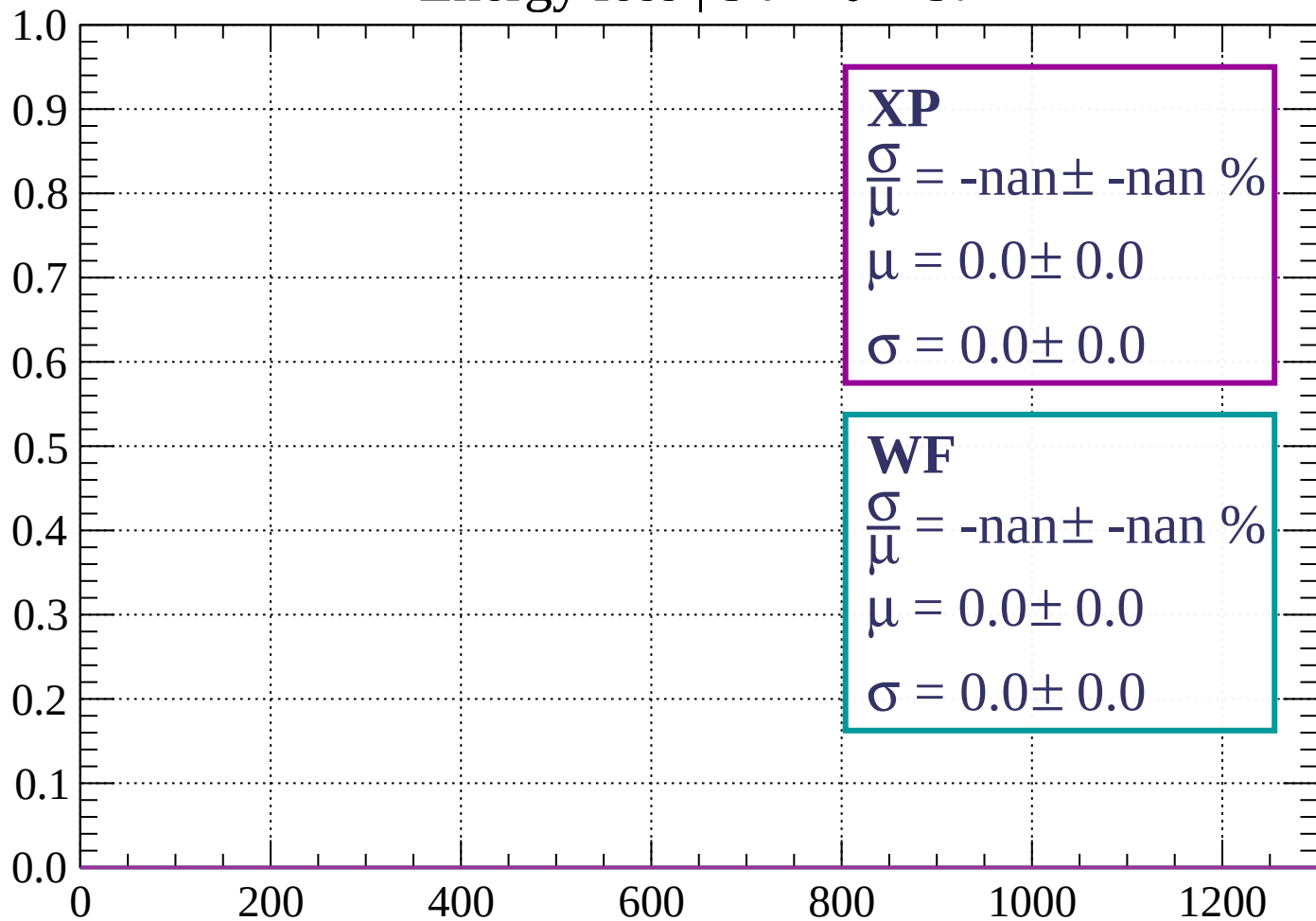
Energy loss | $51 < \theta < 54$

Count



Energy loss | $54 < \theta < 57$

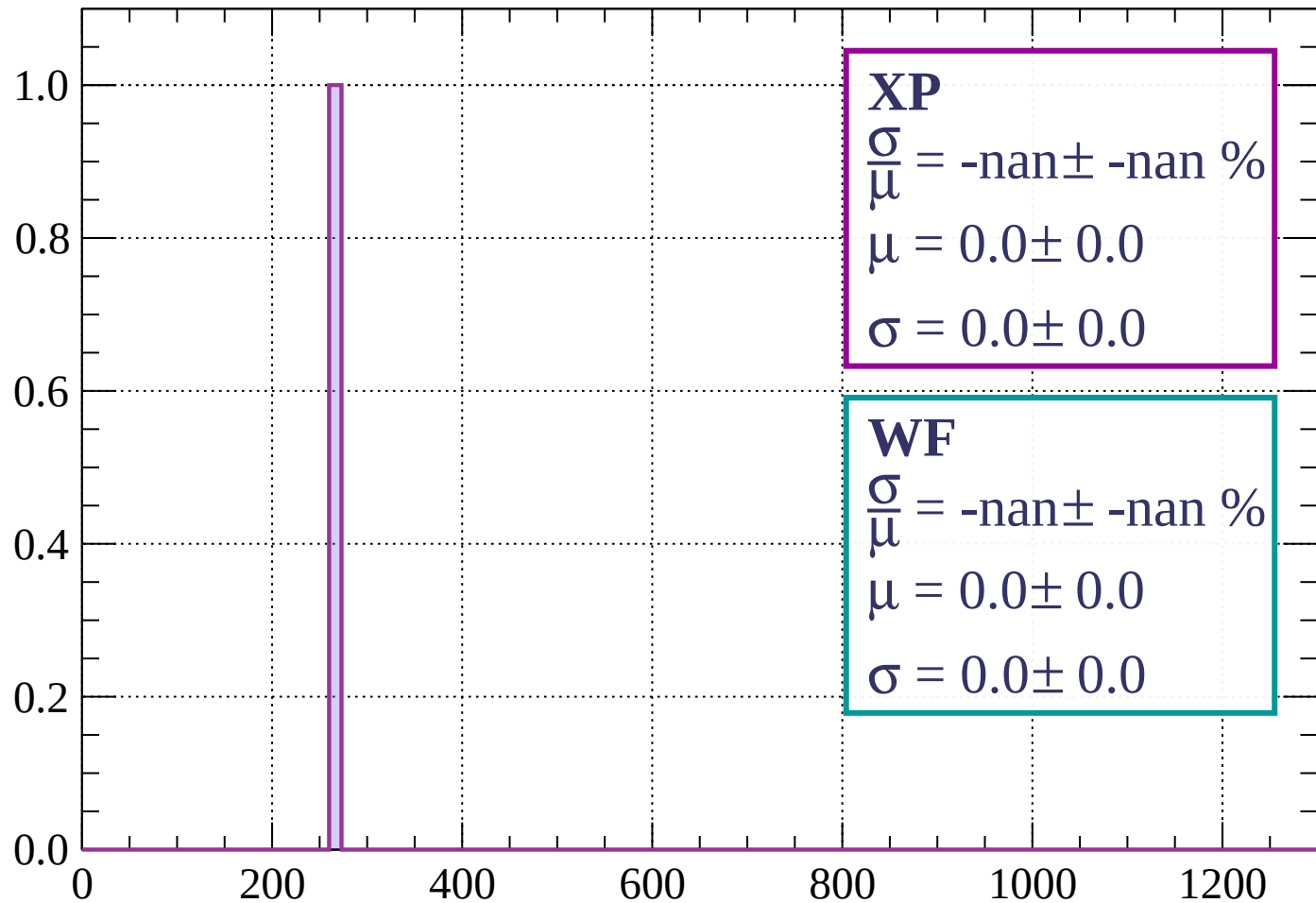
Count



dE/dx (ADC counts/cm)

Energy loss | $57 < \theta < 60$

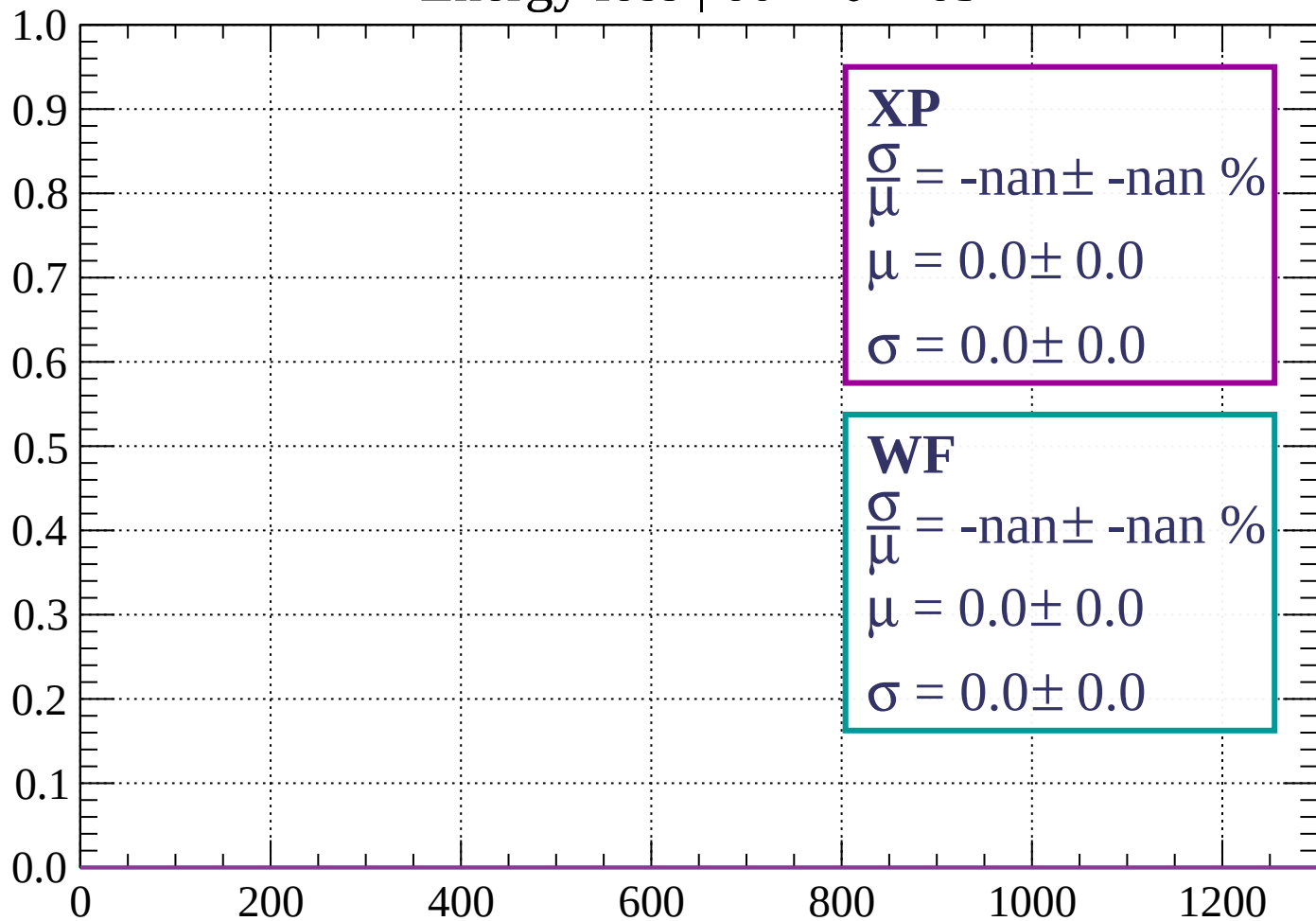
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

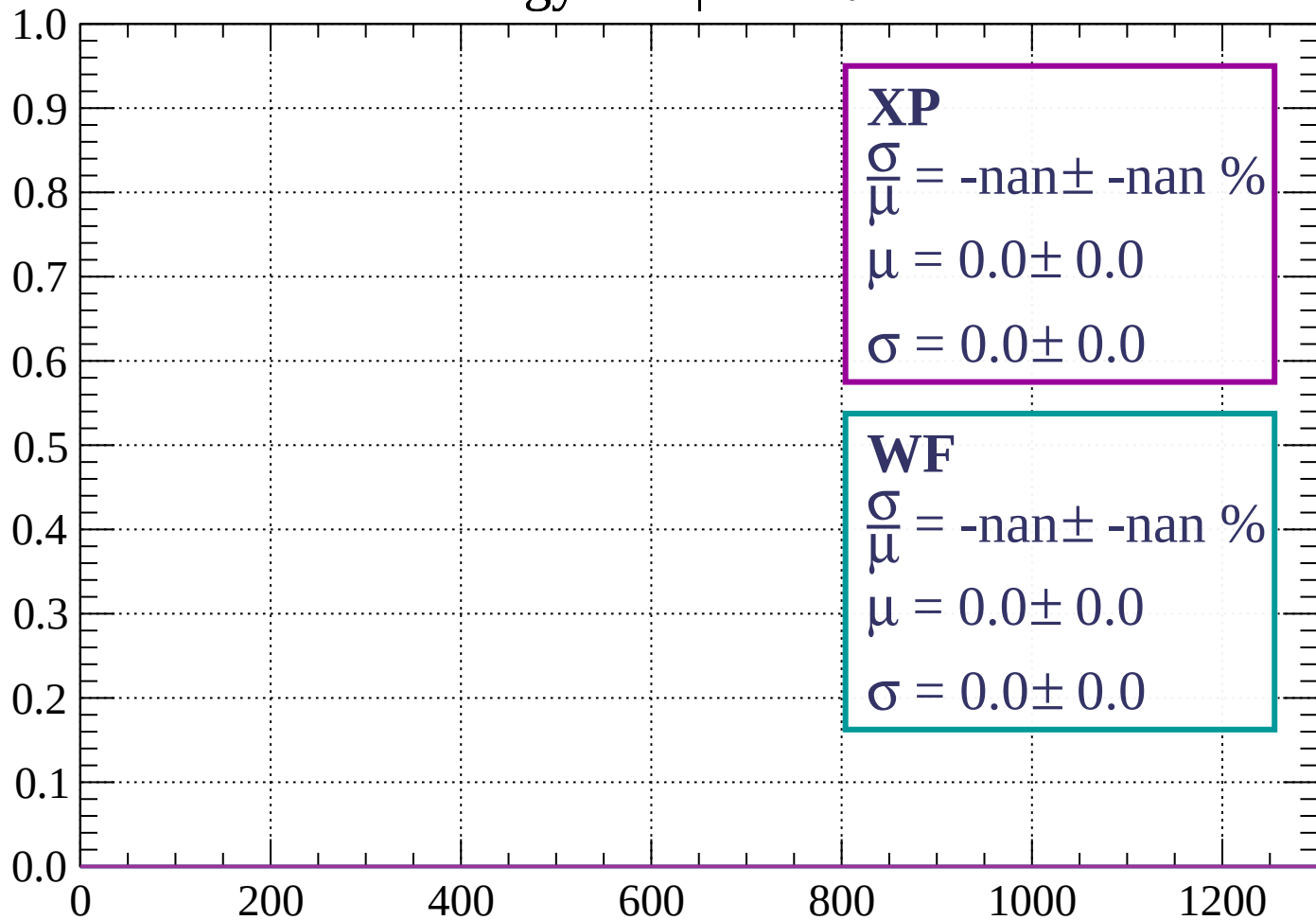
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

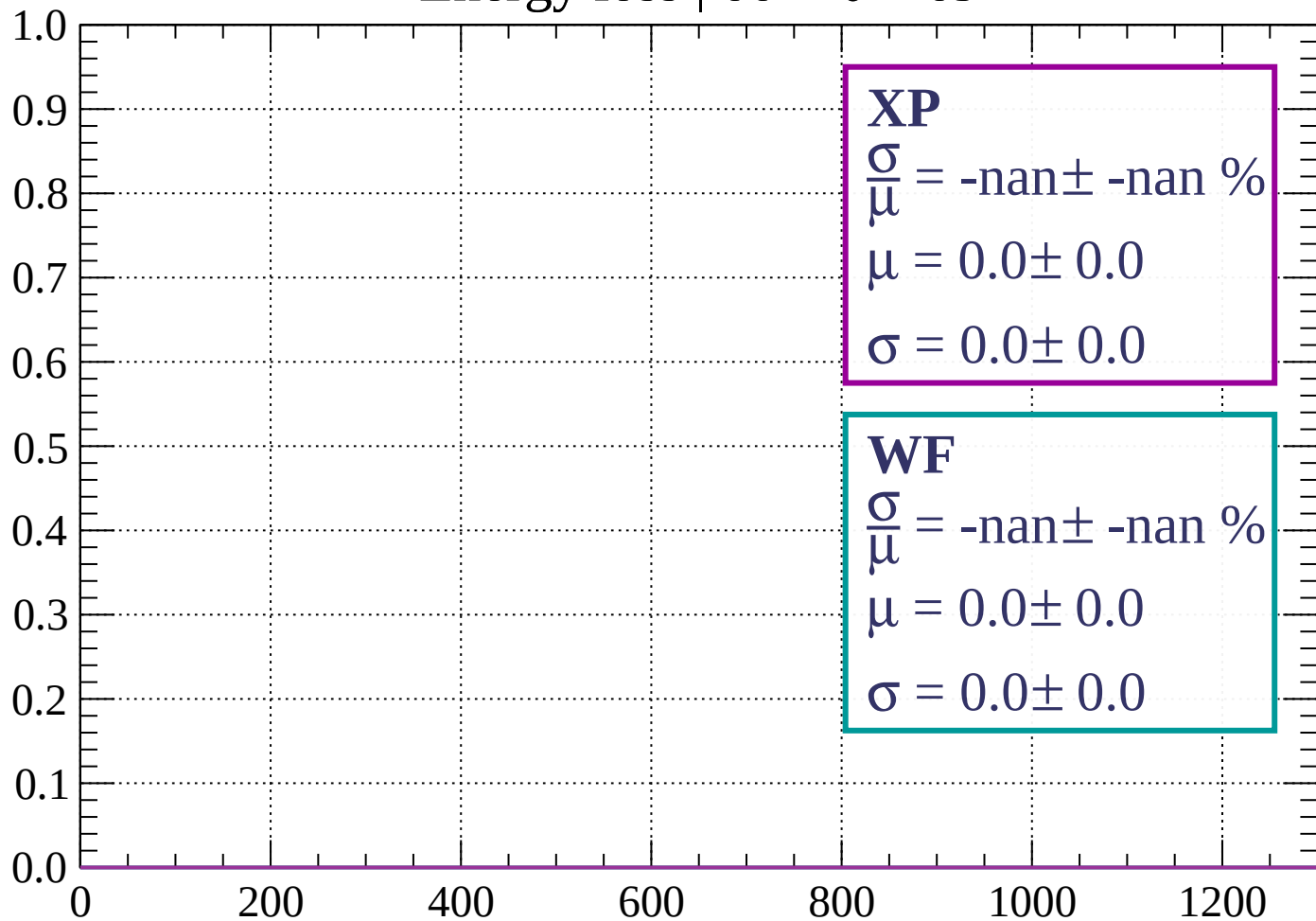
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

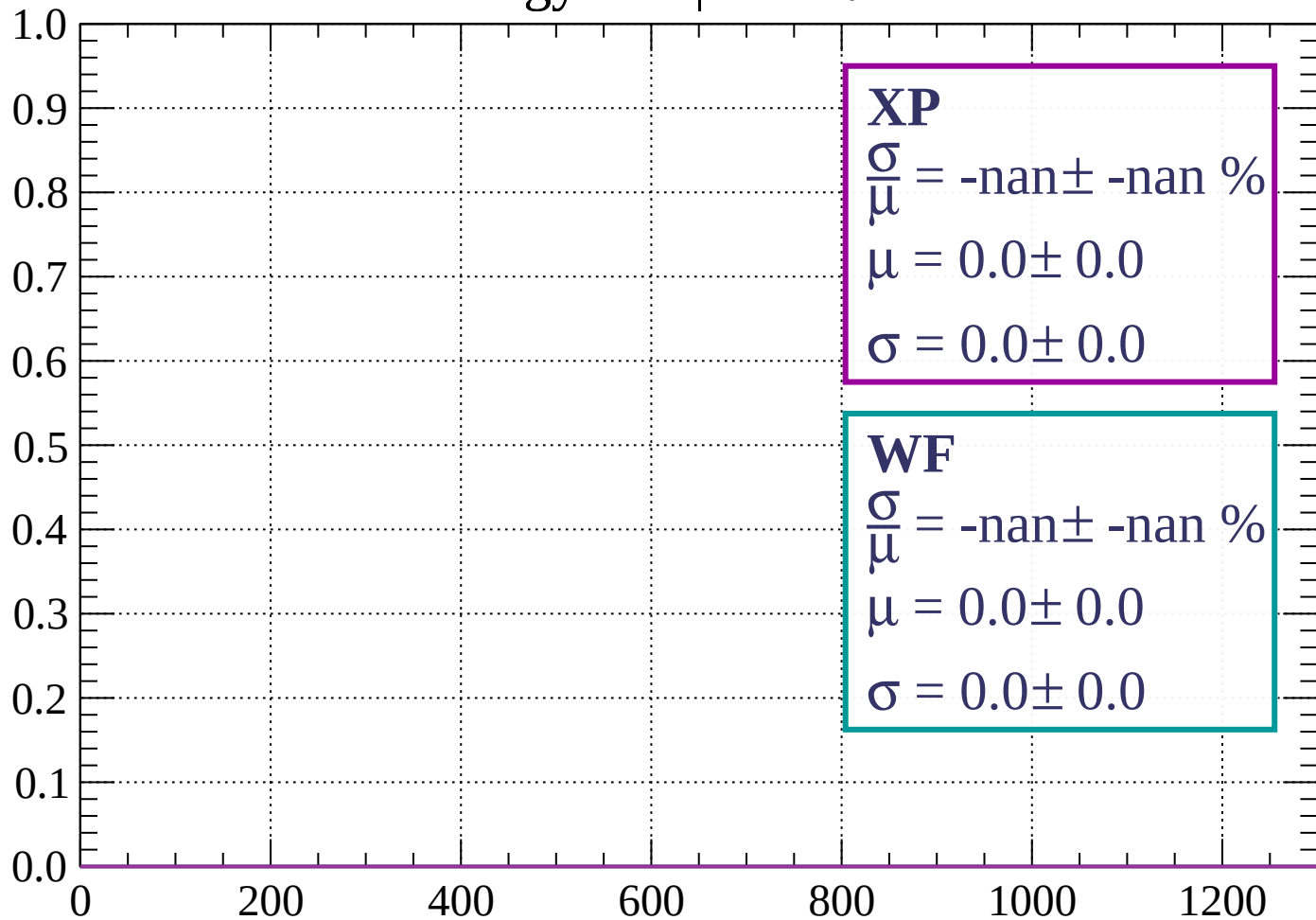
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \theta < 72$

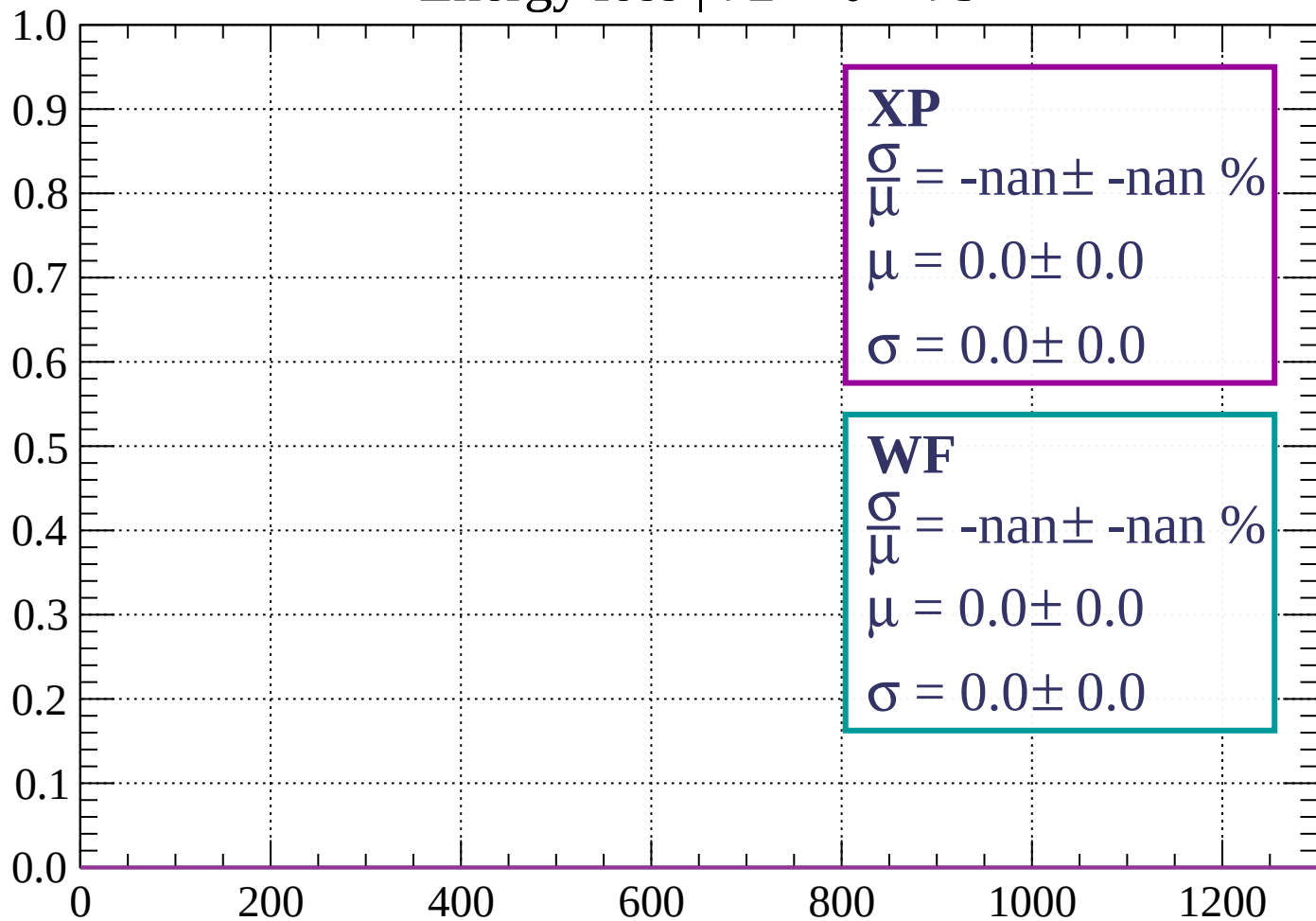
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

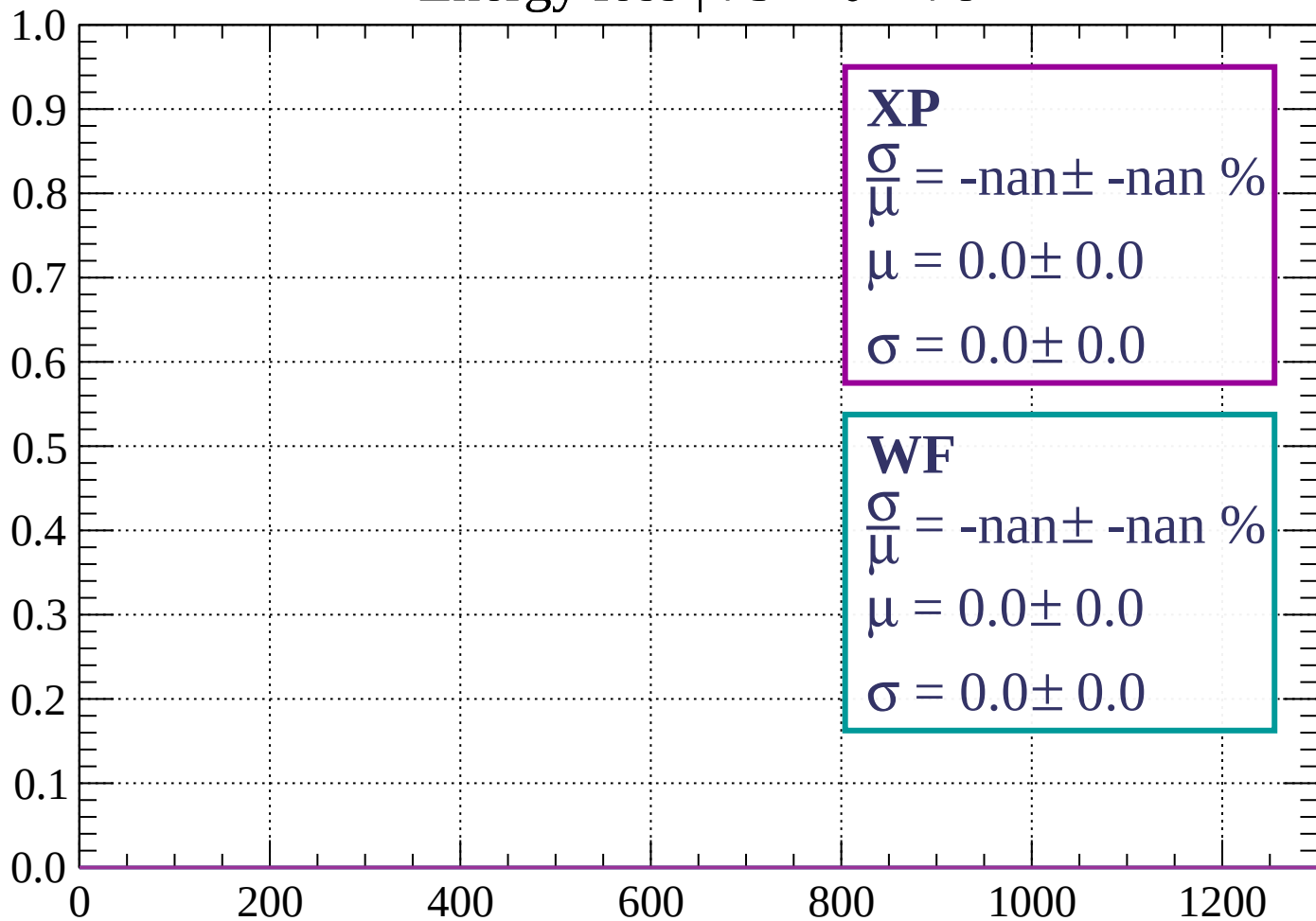
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

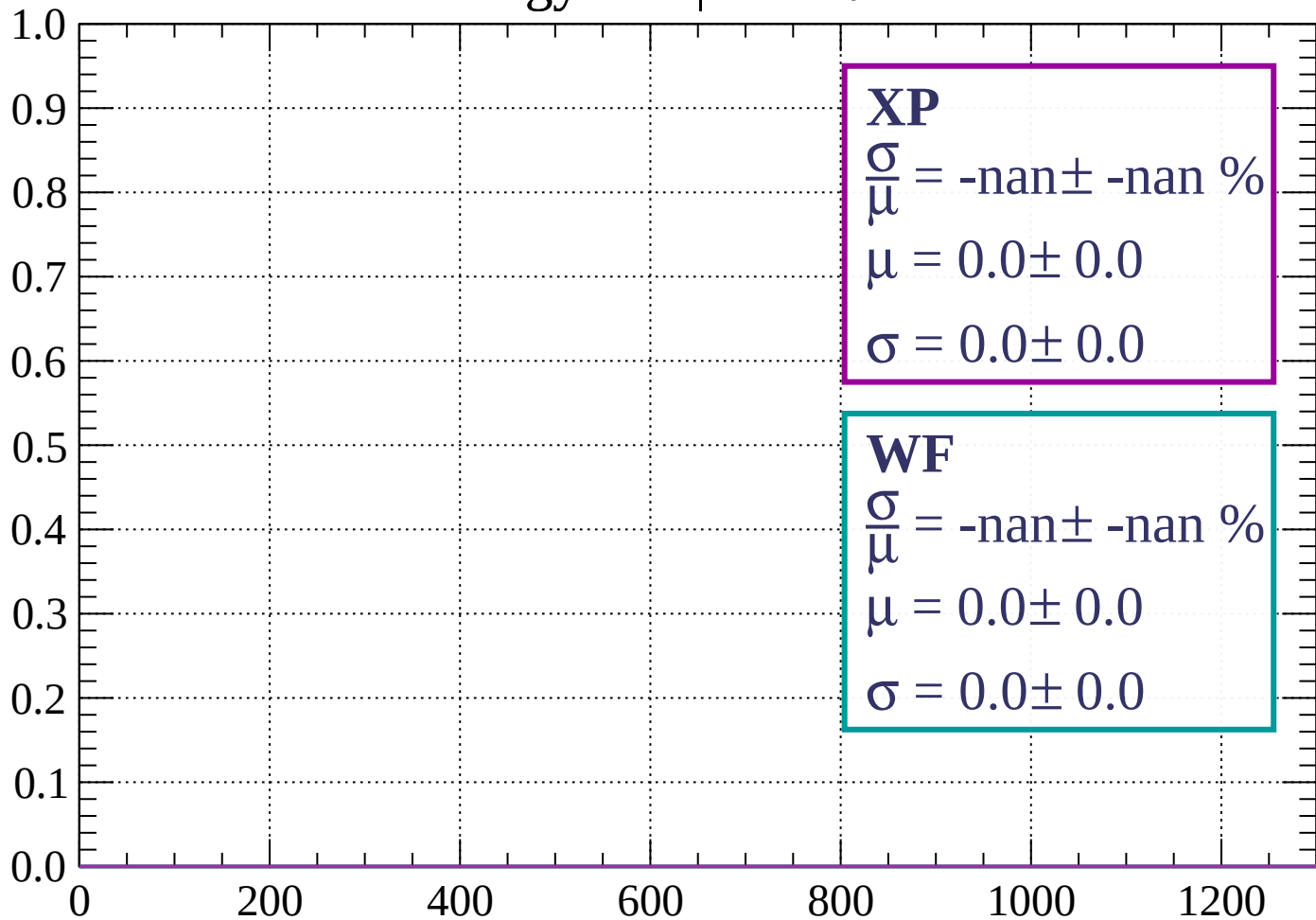
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

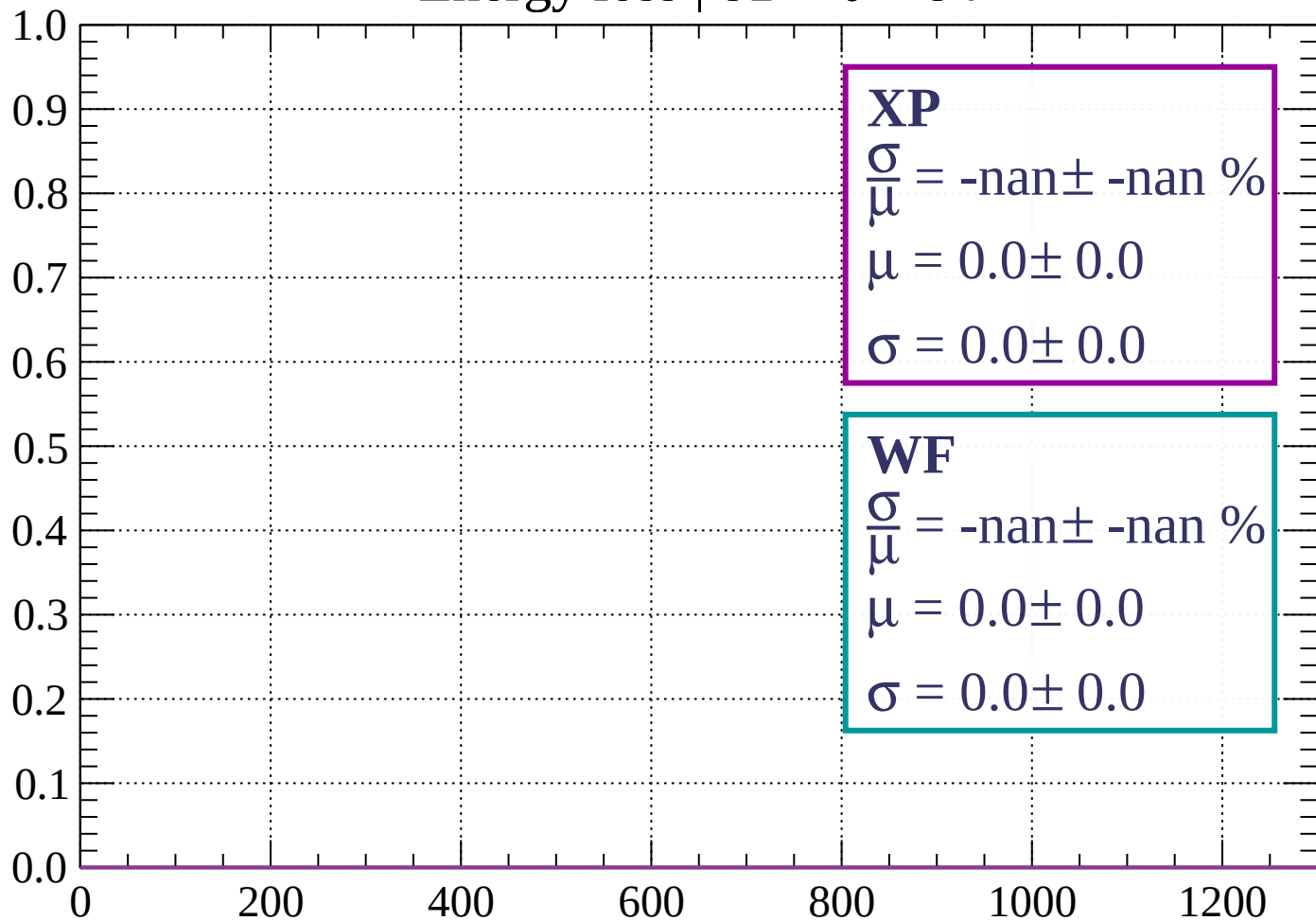
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

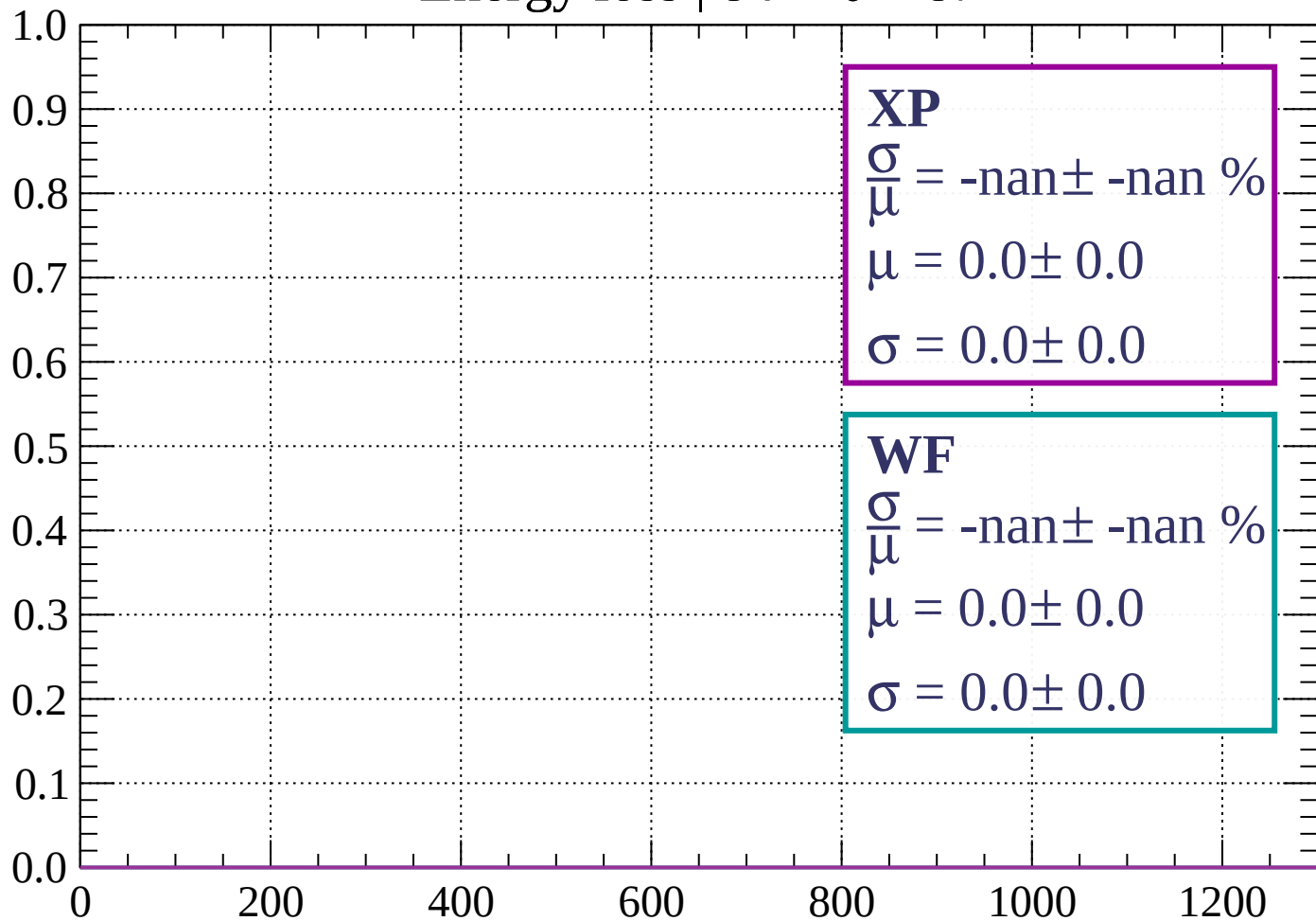
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

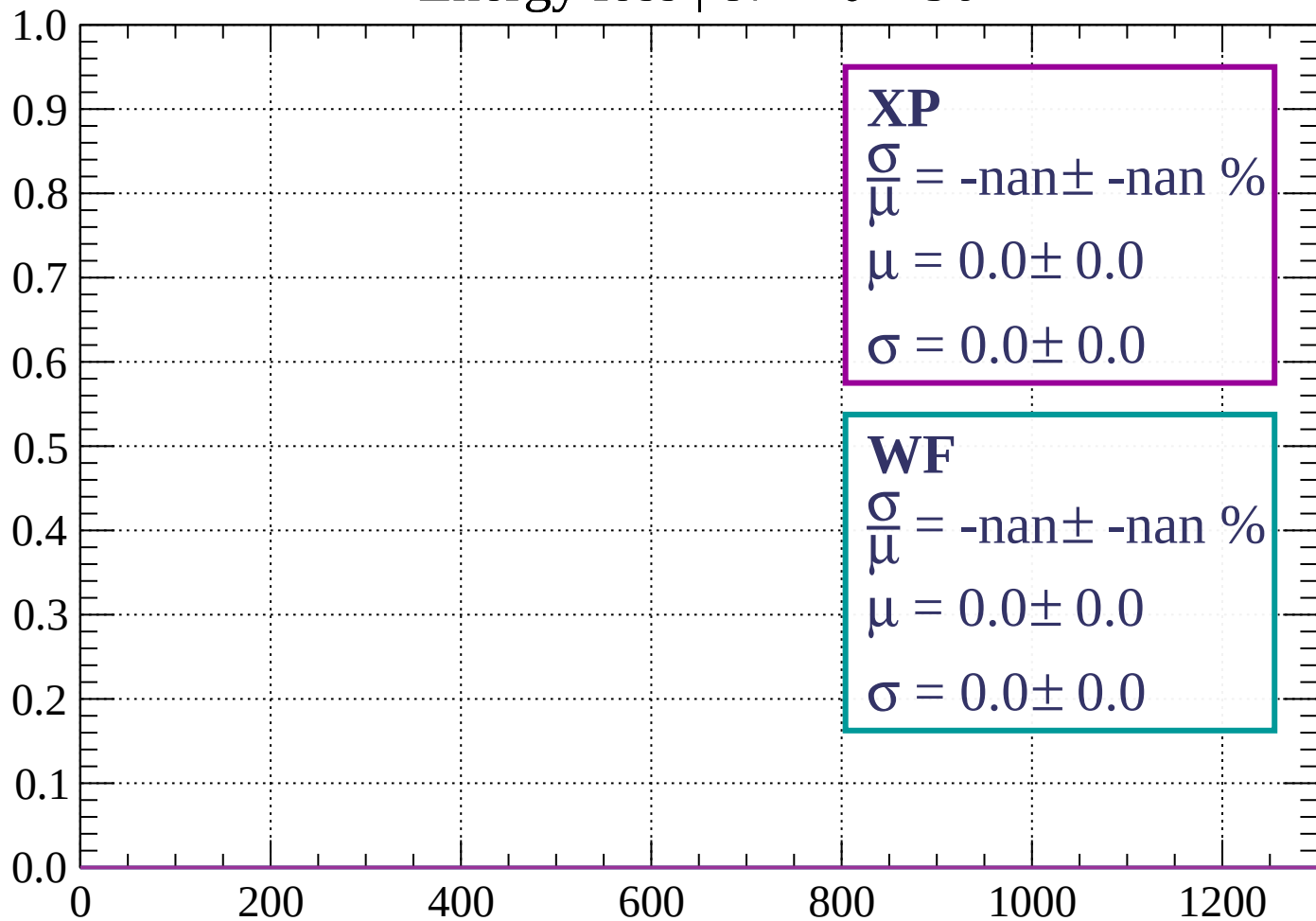
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

